

Important notice

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TABLE OF CONTENTS

Important notice	1
TABLE OF CONTENTS.....	3
1 General information.....	8
1.1 About this Manual.....	8
1.2 Additional ML STAR Line Manuals.....	8
1.3 Intended use of the ML STAR Line	9
1.4 Operation	9
1.5 Safety Precautions and Hazards	9
1.5.1 General Precautions.....	9
1.5.2 Biohazard and Chemical Precautions	13
1.5.3 Electrical Safety Precautions	13
1.5.4 Computer Precautions.....	14
1.5.5 Hazards	15
2 Description of the ML STAR Line	16
2.1 ML STAR Line	16
2.1.1 The Air Displacement Pipetting Principle	16
2.1.2 Tip Pick-up with the CO-RE Technology.....	17
2.1.3 Tip Recognition	18
2.1.3.1 Tip Recognition using the Autoload Function.....	19
2.1.3.2 Tip Recognition using the VENUS Tip Type Recognition Feature	19
2.1.4 Liquid Level Detection: LLD	19
2.1.5 Monitored Air Displacement: MAD	20
2.1.6 Capacitance-based Clot Detection.....	21
2.1.7 Anti-Droplet Control: ADC	21
2.1.8 Total Aspiration and Dispensing Monitoring: TADM	22
2.1.9 Availability of Technologies on the Pipetting Tools	23
2.2 Platforms.....	24
2.2.1 Microlab STARlet	25
2.2.2 Microlab STAR	26
2.2.3 Microlab STARplus.....	26

2.3	Pipetting Arm Configurations	28
2.3.1	Modular Pipetting Arm.....	28
2.3.2	Modular Pipetting Arm MPH.....	29
2.3.3	Pipetting Arm MPH/iSWAP	30
2.3.4	Pipetting Arm MPH.....	31
2.3.5	iSWAP Arm	31
2.3.6	Dual-Arm Configurations	32
2.4	Options and Upgrades	33
2.4.1	1000µl-Channels	34
2.4.2	5ml-Channels	39
2.4.3	CO-RE 96 Probe Head.....	42
2.4.4	CO-RE 96 Probe Head TADM	47
2.4.5	CO-RE 384 Probe Head.....	48
2.4.6	Pin Tools.....	52
2.4.7	iSWAP	53
2.4.8	Tube-gripper	54
2.4.9	AutoLys Tube Channel.....	55
2.4.10	Tube-twister Channel and De/re-capper module	56
2.4.11	Imaging Channel	59
2.4.12	CO-RE Gripper	60
2.4.13	FlipTube Tool	61
2.4.14	CO-RE Lid Tool	64
2.4.15	Autoload Option.....	65
2.5	Accessories	66
2.5.1	Carriers.....	66
2.5.2	Hamilton Heater Shaker	67
2.5.3	Multiflex Carrier	70
2.5.4	Multiflex Heating and Cooling Module.....	72
2.5.5	MFX Cooling-Heating Module	73
2.5.6	CR Needle Wash Station	73
2.5.7	Wash Station 96/384 Dual.....	75
2.5.8	Temperature Controlled Carrier (TCC)	77
2.5.9	Crystal Vacuum System	78
2.5.10	HEPA Filter Hoods	82
2.6	Tips and Needles	83
2.6.1	Tip Packaging.....	85
2.6.2	Tips for Pipetting Low Volumes from 0.5-10µl	86

2.6.3	Tips for Pipetting Sample Volumes from 1-50µl.....	88
2.6.4	Tips for Pipetting Sample Volumes from 0.5-50µl Using 384 MPH	88
2.6.5	Tips for Pipetting Sample Volumes from 10-300µl.....	89
2.6.6	Tips for Pipetting Sample Volumes from 10-1000µl.....	90
2.6.7	Tips for Pipetting Sample Volumes from 50-5000µl.....	91
2.6.8	Special Tips: Rocket tip.....	91
2.6.9	Special Tips: Piercing Tip.....	92
2.6.10	Special Tips: Extra-long and Slim Tip	93
2.6.11	Special Tips: Wide Bore Tips	94
2.6.12	Needles	94
2.7	Computer Requirements	95
2.8	Microlab STAR Line VENUS Software.....	95
2.9	Electronics and Interfaces	96
2.9.1	Interfaces of Dual Processor Board	96
2.9.2	Interfaces of LAN Dual Processor Board	97
2.10	Power / Voltage.....	98
2.10.1	ML STAR Line Instrument.....	98
2.10.2	Needle Wash Station.....	99
2.10.3	Hamilton Heater Shaker	100
2.11	Training	102
3	Routine Use.....	103
3.1	Loading the ML STAR Instrument	103
3.2	Run Simulations	111
3.3	Run-time Error Handling	112
3.3.1	Examples.....	113
3.4	Walk-away (predefined) Error Handling	115
4	Maintenance	116
4.1	STARwatch Advanced Service	117
4.2	Maintenance Intervals	118
4.3	If Maintenance Fails.....	118
4.4	Preventative Maintenance.....	118
4.5	Materials Required.....	119
4.6	Instrument Maintenance Procedures	119
4.6.1	Daily Maintenance.....	121

4.6.2	Weekly Maintenance	123
4.7	Printing a Report	128
4.8	Needles.....	128
4.9	CO-RE 96 Probe Head TADM	128
4.10	CR Needle Wash Station	144
4.10.1	Daily Maintenance.....	144
4.10.2	Weekly Maintenance	145
4.11	Wash Station 96/384 dual.....	146
4.11.1	Daily Maintenance.....	146
4.11.2	Weekly Maintenance	147
4.12	Hamilton Heater Shaker	148
4.13	Crystal Vacuum System.....	148
4.13.1	Activating CVS Maintenance.....	148
4.13.2	Daily Maintenance.....	149
4.13.3	Weekly Maintenance	150
4.14	Temperature Controlled Carrier (TCC)	150
5	ML STAR Instrument Decontamination	151
5.1	Surface Decontamination using Liquids.....	151
5.2	Decontamination using Gases.....	152
5.3	Decontamination using UV Light.....	152
6	Verification	153
6.1	General Description.....	153
7	Technical specifications	154
7.1	Basic ML STAR Line Specifications	154
7.2	Hamilton standard PCs Specifications	159
7.3	Touch Screen Monitor Specifications for the ML STAR Line.....	159
7.4	CR Needle Wash Station Specifications	160
7.5	96/384 Wash Station Dual Specifications	161
7.6	iSWAP Specifications.....	162
7.7	Tube-gripper Specifications.....	163
7.8	CO-RE Gripper 1000µl Specifications.....	164
7.9	CO-RE Gripper 5ml Specifications.....	164
7.10	Autoload Option: Barcode and Reader Specifications.....	165
7.10.2	Hamilton Heater Shaker Specifications.....	173

7.10.3	Hamilton Heater Shaker Box (HSB) Specifications	175
7.10.4	Crystal Vacuum System (CVS) Specifications.....	176
7.10.5	Temperature Controlled Carrier (TCC) Specifications	177
7.10.6	Multiflex Heating Module Specifications.....	177
7.10.7	Multiflex Cooling Module Specifications	177
APPENDICES		178
A.	Chemical Compatibility	178
B.	Ordering Information.....	186
C.	Instrument Dimensions.....	219
D.	Regulatory Affairs.....	236
	Radio Interference (USA and Canada).....	236
	<i>In vitro</i> Diagnostics.....	237
	Declaration of Conformity - EN ISO 9001:2008	238
	Declaration of Conformity - EN ISO 13485:2003 + AC 2009	239
	Declaration of Conformity: CE, CSA.....	240
	Declaration of Quality	241
	Declaration of Cross Contamination Free Processing	243
	WEEE Declaration	244
	RoHS Compliance.....	246
	21 CFR Part 11 Compliance.....	247
E.	Glossary	258

1 General information

This Operator's Manual is designed to get the most out of a Microlab STAR Line pipetting workstation.

Hamilton's Microlab STAR, Microlab STARlet, and Microlab STARplus are in the following summarized as ML STAR.

Please read through the entire manual before beginning to operate the instrument. This first chapter should be read with particular attention. It contains important information about the use of the ML STAR instrument and this manual.

1.1 About this Manual

This manual is meant to help users operate the ML STAR Line instruments correctly and safely.

To achieve this goal, the manual will describe the different components of the ML STAR Line and how they work. The manual describes mainly the hardware of the ML STAR Line instruments in depth which enables the user to operate the instrument. The software is described in the Microlab STAR Programmer's Manual (P/N 624043).

The introduction to the various parts of the ML STAR Line is followed by a step by step description how to operate the instrument. Working through this manual provides all information to operate the ML STAR Line.

Warnings and **notes** are included in this manual in order to emphasize important and critical instructions. They are printed in italics; begin with the word "Attention" accompanied by the  symbol, or the word "Note" accompanied by the  symbol, as appropriate.

[...] Push buttons and their corresponding description.

"..." Description for all kinds of entry fields, control fields, check boxes, lists, etc.

___ References to manuals, figures, chapters, etc.

This manual refers to VENUS three Software for the ML STAR.

1.2 Additional ML STAR Line Manuals

For the programmer of the system, the Microlab STAR Programmer's Manual (P/N 624043) describes all the features of the Microlab STAR Line VENUS Software. A detailed software reference for the ML STAR Line can be found in the Help Function of the VENUS Software. This Help Function will answer questions about details of the VENUS Software.

The Total Aspiration and Dispensing Monitoring (TADM) is an additional safety tool for the pipetting processes. The TADM feature allows optimization of the entire pipetting process by reading out the pressure curves of the pipetting process and comparing it to a tolerance band in real-time. The description of its functions and how to work with TADM is described in the TADM Manual (P/N 624031).

VENUS Dynamic Scheduler is a software tool for organizing and controlling the workflows of a laboratory equipped with the Hamilton instruments and other manufacturers instruments. The use of this tool is described in the VENUS Dynamic Scheduler User Manual (P/N 624030).

VENUS Data Base Plus is a software tool that allows using additional functions on the VENUS Database. It allows tracking over multiple runs and the use of an SQL Server via the network. The use of this tool is described in the Help Function of the VENUS Database Plus.

Further manuals for specific applications, standard solutions, workstations, features, etc. come together with the dedicated ML STAR Line instrument. Please refer to these specific manuals for the respective application.

1.3 Intended use of the ML STAR Line

ML STAR Line is a robotic pipetting workstation which is classified as a general laboratory instrument and not as an *in vitro* diagnostic device. It is intended to automate routine pipetting tasks and the required transports of labware such as plates and disposable tips used in the pipetting assay.

1.4 Operation

The operator of the ML STAR must have attended an appropriate training course. The procedures contained within this manual have been tested by the manufacturer and are deemed to be fully functional. Any deviation from the procedures given here could lead to erroneous results or malfunction.

Training courses will be held by a Hamilton representative. Please feel free to contact a local representative to arrange for an operator training.

1.5 Safety Precautions and Hazards

The following section describes the main safety considerations, electrical and biological, in operating this product, and the main hazards involved.



ATTENTION

Read the following safety notices carefully before using the ML STAR instrument.

1.5.1 General Precautions

1.5.1.1 Instrument Installation and Relocation

The ML STAR instrument is installed or relocated by a Hamilton service engineer. Installations of instrument options and accessories are also done by a Hamilton service engineer.

The ML STAR Line instruments conform to European norms regarding the interference immunity. However, if the ML STAR Line is subject to electromagnetic RF fields, or if static electricity is discharged directly onto the ML STAR instrument, its Liquid Level Detection ability may be negatively affected. It is therefore recommended that the ML STAR instrument is kept

away from other equipment that emits electromagnetic RF fields in the laboratory, and that static electricity is minimized in its immediate environment.

During routine operation, the ML STAR instrument should be shielded from direct sunlight and intense artificial light.

The instrument should be positioned in the laboratory in a way permitting personnel to access the front and sides of the instrument in order to operate, maintain, open and close the protective covers. Accordingly, in order to calculate how much space is needed, consider the dimensions of the instrument (see [Chapter 7 "Technical Specifications"](#)) and sufficient room for a person to move and work comfortably.

Never lift a fully installed instrument for transportation from one location to another. It must be reinstalled in the new location by an authorized service engineer.

Depending on size, the instrument weighs 150 kg or more. Necessary precautions should be taken when transporting the instrument.

1.5.1.2 Maintenance, Service and Repair

Maintaining the instrument on a regular basis (e.g. cleaning operations) is a task of the instrument's operator (see [Chapter 4: "Maintenance"](#)).

Exchanging spare parts, repair and performing preventive maintenance on the ML STAR instrument are tasks of certified Hamilton service engineers or service engineers authorized by Hamilton.

For repair or shipment, all mechanical parts must be placed in their initial positions. An instrument sent away for repair also has to be decontaminated (see [Chapter 5 "ML STAR Instrument Decontamination"](#)) if it was in a laboratory environment with infectious or hazardous materials. The ML STAR instrument must be repacked in the original shipping crate by an authorized service engineer only; please contact a local Hamilton representative. No containers or tips on the ML STAR may be in the instrument during transportation.

The table below shows the Qualities of Hamilton Solutions which depicts the responsibilities of both Hamilton and the Customer.

Quality of Hamilton Solutions				
	DQ	IQ (Installation Qualification)	OQ (Operation Qualification)	PQ (Process Qualification)
Responsible	Hamilton: System Customer: Assay	Hamilton	Hamilton/Customer	Customer
			Verification of the customer's specifications Training	

Only original Hamilton ML STAR Line-specific parts and tools may be used with the ML STAR Line instrument, e.g. carriers, racks, tips, steel needles, and waste containers. Commercially available liquid containers, labware such as Microtiter Plates and tubes, may of course be used.

A breakdown of the power supply during a run may cause loss of data. If data loss is unacceptable, use an independent power supply.

1.5.1.3 Operating the Instrument

When using ML STAR Line instruments, Good Laboratory Practices (GLP) must be observed. Suitable protective clothing, safety glasses and protective gloves must be worn, particularly when interfering with the instrument's components and when there is a risk of contamination originating from spilled liquids.

Electrostatic discharge can cause damage to the instrument and can influence labware behavior and stability, therefore avoid any electrostatic charge onto labware and disposable tips during handling and manual loading of these to or from the ML STAR Line instrument.

During instrument operation, stand clear of all moving parts or the working deck of the instrument. In general, never lean over or into the instrument when it is working.

When using aggressive or corrosive liquid ensure that filter tips are being used. Do not re-use tips. Replace stop discs and O-rings more frequently; make sure to PM the instrument in short intervals since the system will need a higher level of support and maintenance. To avoid corrosion on the deck, use an exhaust suction device to remove corrosive fumes away from the deck.



ATTENTION

When using flammable or explosive fluids or vapors the necessary precautions must be taken. There is a high risk of damage to health and to the instrument. The instrument is not explosion-proof.

1.5.1.4 Method Programming

Perform test runs first with water and then with the final liquids, prior to routine use. Test for all the liquid classes which will be used. A newly programmed test method must first be run on the ML STAR instrument with the final liquids, prior to validation of the method and routine use. The method programmer should supervise this run.

Before using any newly created or modified method for routine test purposes, a comparison study between the method previously used and the new one must be carried out by the laboratory supervisor in order to ensure that the processing and data evaluation of both methods produce equal results.

When working with samples which will be used in particularly sensitive tests, take into account the evaporation and condensation that may occur while the method is running.

Use filter tips for aggressive liquids. Also use filter tips for tasks which are sensitive to cross-contamination (aerosols) or when working with liquids which are prone to clot formation. In these cases filter tips can prevent contamination of pipetting channels.

The inner diameter of sample tubes, reagent vessels, etc. must be greater than the channel diameter of the pipetting heads. For the 1000µl pipetting head, the inner diameter must be greater than 9mm. For the 5ml pipetting head, the diameter must be greater than 18mm.

Liquid Level Detection needs to be explicitly tested when working with foaming liquids. Foam may affect the accuracy of Liquid Level Detection.



ATTENTION

Never disable any security measure.

1.5.1.5 Loading

Do not exchange positions of sample and reagent tubes, or switch micro-plates after they have been identified by the barcode reader. This could result in incorrect test data or instrument crash.

Micro-plates must be placed on the carrier such that well A1 is in the position defined in the deck layout of the method.

When pouring liquid into the containers, ensure that there is no foam on the surface of the liquid. Note that foam may cause pipetting problems.

Do not overfill reagent containers, tubes, or other liquid containers.

Do not mix tip size and type (e.g. with or without filter, or different volumes) in the same tip rack. Make sure to match the tip type to be used with the particular method. Take care when using tips which cannot be distinguished by the tip recognition feature (refer to [Chapter 2.1.3 “Tip Recognition”](#) and [Chapter 2.6 “Tips & Needles”](#)).

Do not fill up partially consumed tip racks with tips from other racks. Tip racks should be loaded into the tip rack carriers as they are provided in the original package. The tip racks are individually labeled with a barcode for identification.

Work Routine

Do not try to open the front cover of the instrument during a run because the system will abort and this may cause a loss of data.

When the system is paused, do not wait too long before resuming the run. Loss of liquid from a full tip may result in invalid data.

Discard used tips. Do not reuse them.

Do not empty the tip waste during a run.

Do not leave tips on the pipetting channels for a long period of time (for example overnight). This may cause damage to the CO-RE O-rings. A daily maintenance procedure will remove the tips.

1.5.2 Biohazard and Chemical Precautions

If the ML STAR instrument becomes contaminated with biohazards or chemical material, it should be cleaned in accordance with the maintenance procedures. The procedures are described in Chapter 4 “Maintenance” and Chapter 5 “ML STAR Instrument Decontamination”. Observe and carry out the maintenance procedures given. Failure to do so may impair the reliability and correct functioning of the ML STAR instrument.

If working with bio-hazardous samples observe and carry out the maintenance procedures, paying particular attention to cleaning and decontamination. Wear gloves when handling the pipetting arm and channels, the carriers, racks, and containers, and the tips. Avoid touching tips discarded into the laboratory-supplied waste container. Any surfaces onto which liquid is spilled must be decontaminated.

Do not use disinfecting materials which contain hypochlorite (Javel water, Chlorox) or other bleaching fluids.

When working with biohazards or chemical materials, the user does not have to touch them. The instrument will drop its used tips into a waste container which should be emptied as soon as it is full.

When using an exhaust suction device in order to remove toxic, aggressive or corrosive gases, make sure to place them close to the source of gas or vapor in order to prevent contact of corrosives with ML STAR components, specifically pipetting channels.

1.5.3 Electrical Safety Precautions

Every installation and uninstallation of an electrically powered ML STAR Line component must be done by a trained service engineer.

1.5.4 Computer Precautions

Use the necessary precautions against software viruses. Use only manufacturer's original installation DVD/CD-ROM sets for the operating system, and the original Hamilton VENUS Software.

Running other software in parallel to the VENUS Software may negatively affect the operation of ML STAR instruments.

Any manipulation to the VENUS Software data files or other information determining or affecting ML STAR Line functions can result in erroneous test results or instrument failure.

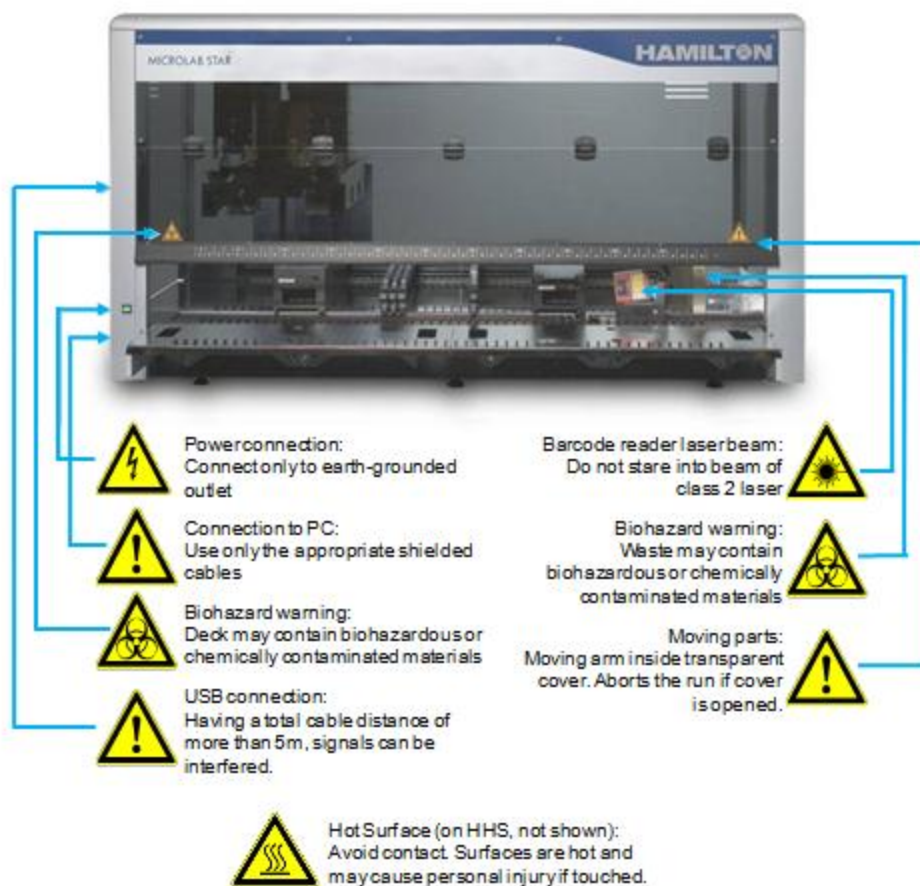
Only the VENUS Software may be used to control the instrument.

For reasons of data security and integrity the use of an uninterrupted power supply (UPS) is recommended, since a loss of power may cause data to be lost or corrupted.

To avoid computer breakdowns, configure a hard disk of sufficient space in the computer. Ensure that there is always enough storage capacity on the hard drive. Delete the log files from time to time. Generated data within the Log files directory, e.g. traces, TADM data and pipetting files, should be backed up on the laboratory's host device and deleted from the control PC's hard disk at weekly intervals.

1.5.5 Hazards

Location and explanation of warning and attention labels:



2 Description of the ML STAR Line

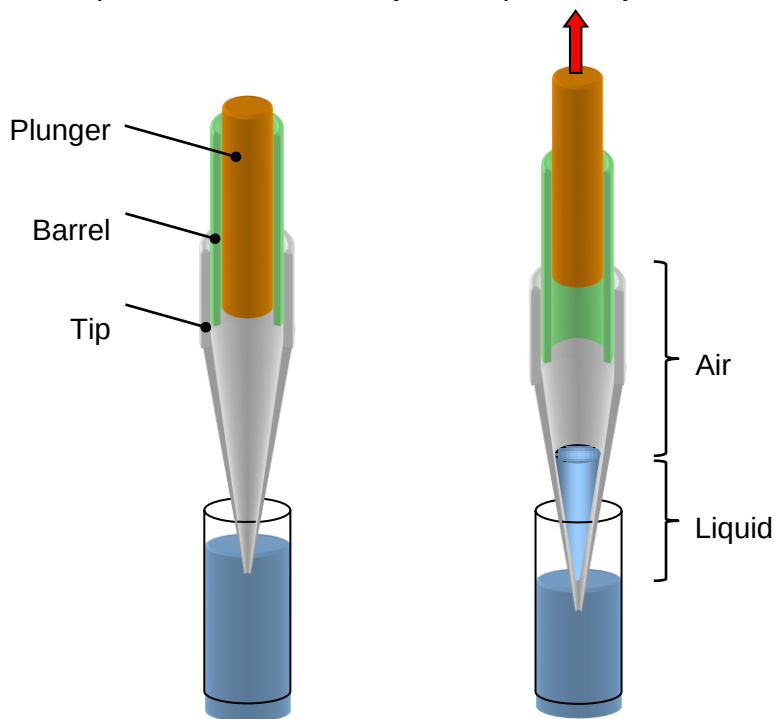
2.1 ML STAR Line

STAR is the Hamilton acronym for Sequential Transfer and Aliquoting Robot. The ML STAR instruments perform pipetting operations of liquids between containers and transferring of Microtiter Plates placed on their work surface.

Pipetting means transfer of small quantities of liquid from one container to another. A pipetting operation is achieved by aspirating (drawing) liquid from a source container, then transferring and dispensing (dropping) it into a target container.

2.1.1 The Air Displacement Pipetting Principle

The ML STAR instruments are based on the **air displacement pipetting** principle, comparable to the functioning of handheld pipettes. Air displacement means that the liquid is aspirated into and dispensed from a disposable tip or needle by the movement of a plunger. Between the plunger and the liquid surface is air. No system liquid of any kind is involved in the instrument.



The air displacement pipetting principle



NOTE

Pipetting with the ML STAR Line is identical to pipetting with a handheld plunger pipette. The pipetting head's barrel and plunger are not intended to be cleaned. We recommend using filter tips when working with infectious agents such as viruses etc.

2.1.2 Tip Pick-up with the CO-RE Technology

The first task for the ML STAR pipetting head is to pick up a Hamilton disposable tip or a reusable steel needle. Due to the unique CO-RE (Compression-induced O-Ring Expansion) technology precise tip attachment and positioning is achieved. Tip alignment is critical for positioning the orifice of the tip at the proper height of the liquid for accurate pipetting as well as for locating the tip precisely within the horizontal plane of the labware. The annular groove design of the Hamilton tip allows exact seating of the locking O-ring within the tip for proper vertical and horizontal alignment. The system requires no vertical force for tip attachment or tip ejection, thus eliminating mechanical stress and improving overall system reliability along with pipetting speed and dexterity. CO-RE technology in combination with the use of Hamilton tips facilitates maximum pipetting accuracy and reliability and guarantees for the pipetting specifications published.



Patent no: EP 1171240
US 7033543

Hamilton pipetting channels and CO-RE tips are designed to fit precisely into each other for maximum O-ring and tip life. The annular groove of the Hamilton tip ensures a proper seal between the tip and the pipette channel. This is important for pressure sensing which the pipettor uses for liquid level detection, anti-droplet control as well as proper aspiration and dispense monitoring and validation. Hamilton tips are dropped off by relaxing the O-ring and allowing gravity to pull the tip down. This way tips are gently released into the waste area minimizing the danger for liquid remaining on the tips to become airborne and to contaminate the pipetting workspace.



NOTE

The principle of the CO-RE technology has the following advantages:

- *Enables coupling of disposable tips or washable needles within the same run.*
- *Allows use of different sizes of tips on the same pipetting head in the same run.*
- *Picking up a gripper and other tools is possible with a pipetting channel.*
- *Prevents aerosols upon tip ejection.*

CO-RE technology is used on all Hamilton pipetting channels, including CO-RE 96 and 384 Probe Head.

2.1.3 Tip Recognition

The tips used in a pipetting procedure must be matched with the suitable pipetting channel or pipetting head in order to prevent damage to the device. Therefore the ML STAR Line instruments offer two ways to automatically recognize the tip type used. All tip trays and racks have color-coded barcode labels for automated recognition. The color helps distinguishing the tip volume by eye. The text on the barcode label enables a distinct identification by eye.

Tip volume	Barcode label	Color code
10µl		light orange
50µl		light red
300µl		yellow
1000µl		white
5000µl		green

2.1.3.1 Tip Recognition using the Autoload Function

2.1.4 **The Autoload** (see [Chapter 2.4.15 “Autoload Option”](#)) includes a barcode reader which identifies the barcodes on the tip trays. Barcodes of tip trays and racks are read automatically when loading tips and the risk of mixing up tip types is excluded. This method is suitable for all tip types, regardless of the pipetting tools to be used.



Tip carrier with five bar-coded tip trays loaded

2.1.3.2 Tip Recognition using the VENUS Tip Type Recognition Feature

The VENUS Software offers a tip type recognition feature for 1000µl-channels, based on the different tip geometries and independent from barcode (please refer to the [Microlab STAR Programmer's Manual](#)). It is available for both, disposable tips and needles and it is activated during installation by the service engineer. This feature increases the instrument's safety when no Autoload is used and when different tip types are used.

For distinguishing disposable CO-RE tips, 50µl and CO-RE tips, 300µl, a special library is needed. Please consult a local Hamilton representative.

Slim tips 300µl cannot be distinguished from CO-RE tips 1000µl by the tip type recognition feature. We recommend using the Autoload for tip loading in this case and in any case to implement a visual check to be sure.

2.1.5 Liquid Level Detection: LLD

The liquid level of the container to be aspirated from can be detected. This can be provided by ML STAR Line's Liquid Level Detection (LLD) feature, based on either capacitive (cLLD) or pressure (pLLD) signal detection. Normally capacitive LLD is used for conductive liquids. The sensitivity of the capacitive LLD that is to be used depends on the vessel size, volume and the conductivity (or polarity) of the liquid to be detected.

For non-conductive liquids, or in case of an insufficient coupling between container bottom and carrier, pressure LLD is used. Pressure LLD only works with new and empty tips for the aspiration of liquids. The pLLD is available on the individual pipetting channels only.

In case of detecting under demanding circumstances, e.g. foaming liquids, the capacitive and the pressure Liquid Level Detection can be used at the same time.

**NOTE**

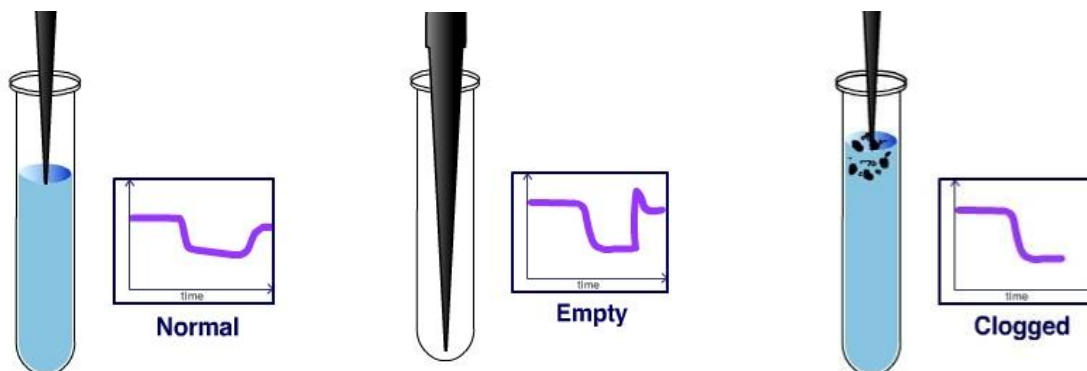
We recommend always using a defined sensitivity, such as 'low', 'medium', 'high' or 'very high'. If the value in the labware is 'Off' it causes bad results.

When selecting 'very high' all the time liquid may be detected in a wrong position, cLLD can for example be triggered by vapor of volatile liquids or by foam.

2.1.6 Monitored Air Displacement: MAD

The ML STAR Line is equipped with an aspiration monitoring feature. During the aspiration process, the pressure within the pipetting channel is measured in real time. Analyzing the shape of the $p(t)$ curve, the system can distinguish the following situations:

- A correct aspiration takes place.
- Air is aspirated into the tip (because, for example, the container has not been filled properly).
- A clot blocks the tip.



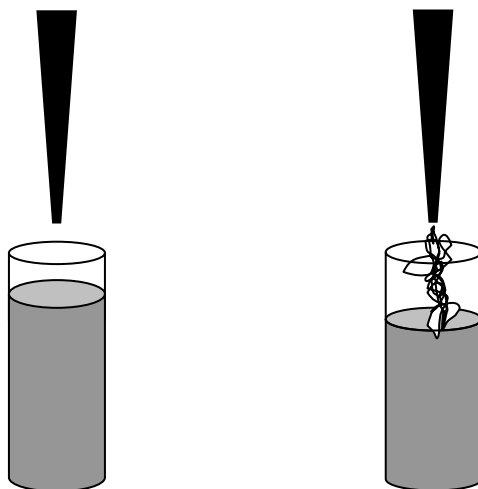
Aspiration monitoring based on pressure.

The Monitored Air Displacement feature is available on the individual pipetting channels only.

The diagram shows the functioning of Monitored Air Displacement based on pressure.

2.1.7 Capacitance-based Clot Detection

In addition to pressure-based clot detection, the ML STAR Line is equipped with capacitance-based clot detection. This detection approach works in the case of aspiration with capacitance Liquid Level Detection switched on. The system measures the conductive signal when the tip leaves the liquid after aspiration. Due to the air gap between the tip and the liquid, the capacitance signal will vanish once a given height is reached. If a clot is present, it bridges the distance and the signal will remain, resulting in an error message. This clot detection is independent of pressure-based monitoring.



2.1.8 Anti-Droplet Control: ADC

Does the application require pipetting of volatile solvents? The Anti-Droplet Control (ADC) enables doing this with highest process security. In conventional pipettes the high evaporation pressure of volatile solvents causes immediate dripping from the tip. The ML STAR Line uses Monitored Air Displacement Technology and can therefore detect pressure changes following aspiration and compensate for them in real time.

The principle is shown in the illustration on the next page: as the evaporation causes a pressure increase (red line) the pipetting unit detects the changes and compensates for them with plunger movements (blue line). The liquid remains in the tip.

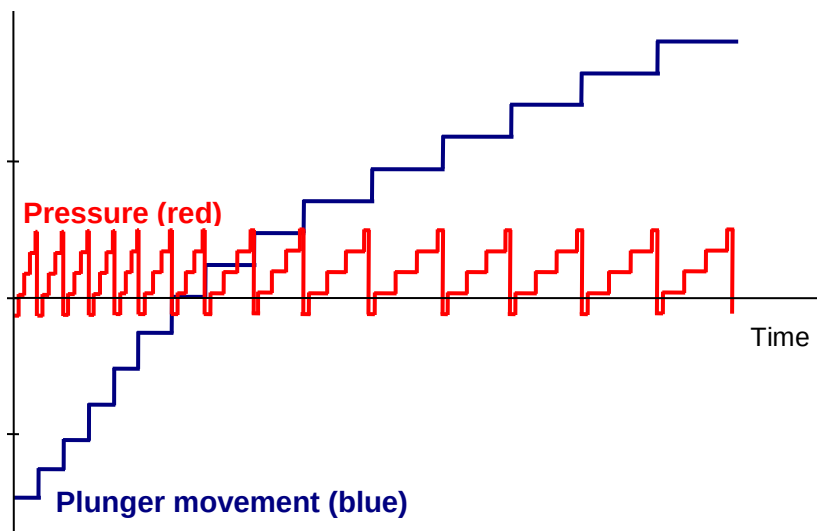
The Anti-Droplet Control (ADC) has to be activated in the 'HSLML_STARLib' command library of the VENUS Software. It is available on the individual pipetting channels only.



ATTENTION

Use ADC only when pipetting liquids that ask for this mode.

There are two reasons that make a continuous use of ADC not practical. First, one might run into a position in which the plunger is no more able to handle the upcoming pressure difference and will therefore trigger an error. The second reason is that there might be a pressure interference which cannot be compensated by the plunger while the instrument is not pipetting.

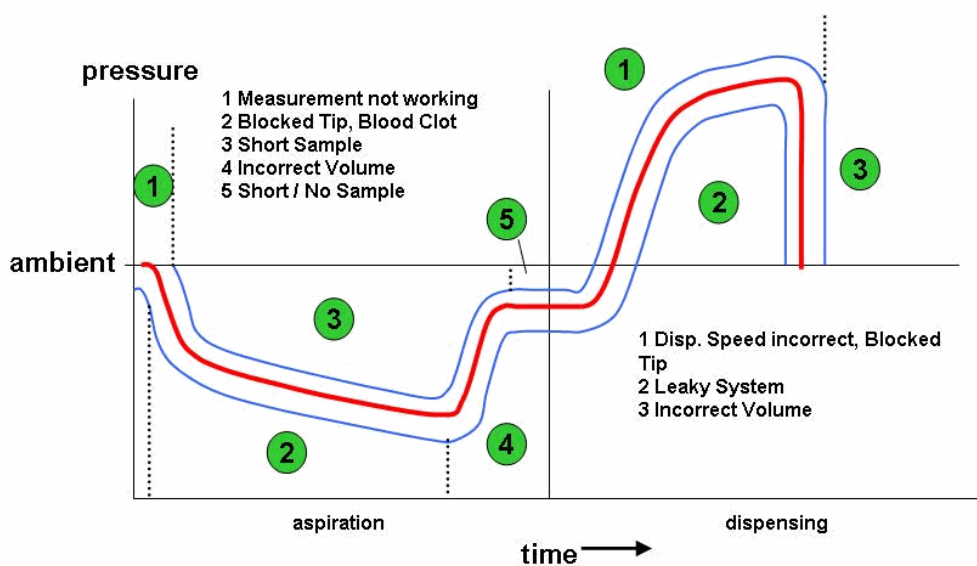


Patent no: EP 1614468
US 2007102445

2.1.9 Total Aspiration and Dispensing Monitoring: TADM

During crucial sample transfers, the ML STAR Line can monitor aspiration and dispense steps in real time. TADM verifies that a sample has been transferred by using a traceable digital audit trail. The function of TADM is a software option and therefore not available by default.

TADM is available for individual pipetting channels and with the CO-RE 96 Probe Head 1000µl TADM.



Patent no: EP 1412759
US 6938504

2.1.10 Availability of Technologies on the Pipetting Tools

The following table provides an overview over availability of the individual technologies on the various pipetting tools.

Pipetting tool	Technology	CO-RE	cLLD	pLLD¹⁾	MAD	TADM	ADC
1000µl-channel		✓	✓	✓	✓	✓ ²⁾	✓
5ml-channel		✓	✓	✓	✓	✓ ²⁾	✓
CO-RE 96 Probe Head		✓	✓ ³⁾	-	-	-	-
CO-RE 96 Probe Head TADM		✓	✓ ³⁾	-	-	✓ ²⁾	-
CO-RE 384 Probe Head		✓	✓ ⁴⁾	-	-	-	-

¹⁾ Available only for use with new disposables; not available for needles

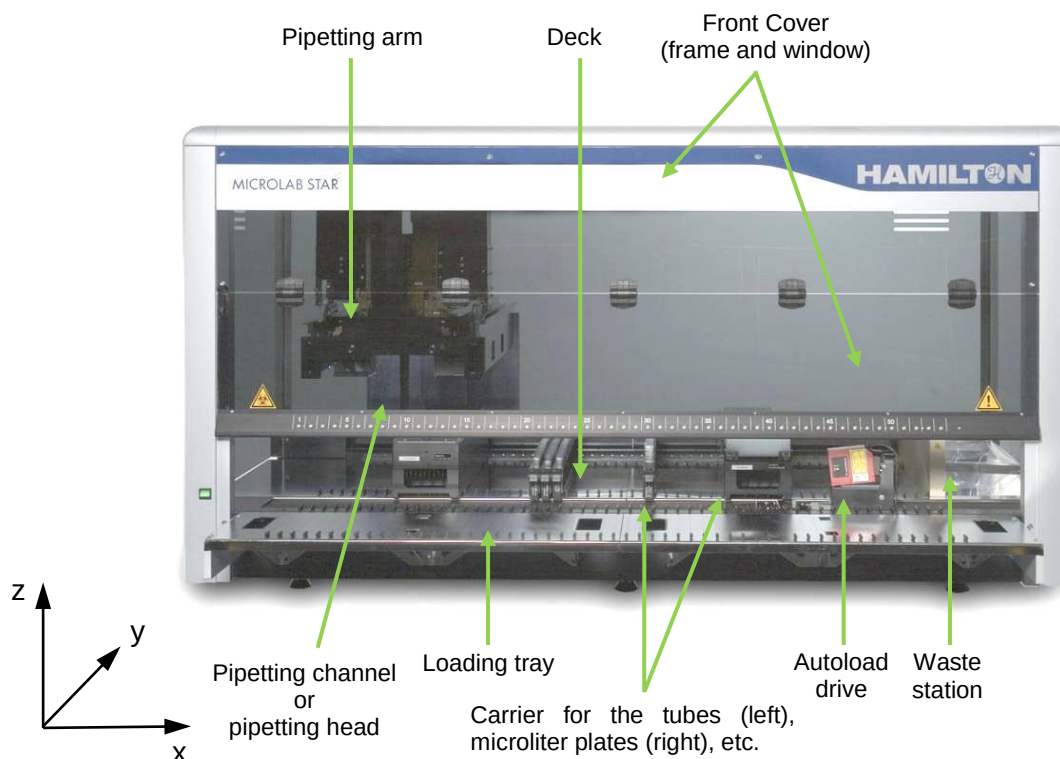
²⁾ Available with VENUS ONE TADM FEATURE (P/N 911099)

³⁾ Available on channels A1 and H12

⁴⁾ Available on channels A5 and P24

2.2 Platforms

The ML STAR Line instrument work surface, called a “deck”, for placing loadable carriers is available in three different sizes. These carriers hold reagent containers, such as tubes, Microtiter Plates, or other kinds of labware.



The Microlab STAR

The instrument is fully covered by an acrylic glass hood. The front cover consists of a hinged transparent window made of acrylic glass. This window is equipped with a magnetic switch that is monitored during a run. Opening the cover will abort the running method.



ATTENTION

An aborted running method (stopped by opening the front cover) cannot be restarted. For opening the window during a run, click [Pause] in the run screen, wait until the instrument stops and then open the window.

The instrument deck is divided into equal tracks (T) for loading carriers in predetermined positions. This eliminates the need for precise measurement of positions. The deck has partitions of 22.5mm, which is equivalent to 1-T (track). The labware carriers are adapted to those partitions, e.g. 1-T carriers for sample tubes, or 6-T carriers for Microtiter Plates or CO-RE tips, etc. An additional partition provides space for the tip waste.

The instrument's internal coordinate system is shown in the picture above, located at its origin. Please note that the ZERO position is 100mm below the metal deck sheet.

A Touch Screen Monitor is available to be attached to the right side of the ML STAR Line instruments. It is the perfect interface to start methods, enter data and to make decisions during runtime. For ergonomic reasons it is recommended to use a desktop computer with monitor, keyboard and mouse for extended programming work.



2.2.1 Microlab STARlet

The compact version of the ML STAR Line is called Microlab STARlet. The Microlab STARlet is the ML STAR instrument with the smallest deck width for loading carriers. The deck has partitions for a maximum of 30 1-T carriers for sample tubes, or a maximum of five 6-T carriers for Microtiter Plates and CO-RE tips. This means that a total of 25 SBS (Standard format of the Society for Biomolecular Screening) positions can be placed onto a STARlet deck.



Microlab STARlet

2.2.2 Microlab STAR

The ML STAR's deck is divided into 54 equal tracks (T) for loading carriers in predetermined positions. The deck has partitions for a maximum of 54 1-T carriers for sample tubes, or a maximum of nine 6-T carriers for Microtiter Plates and CO-RE tips. This means that a total of 45 SBS (Standard format of the Society for Biomolecular Screening) positions can be placed onto a ML STAR deck.



Microlab STAR

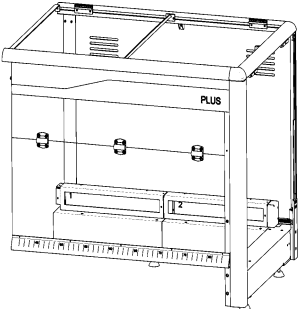
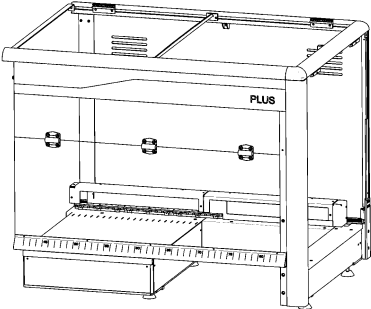
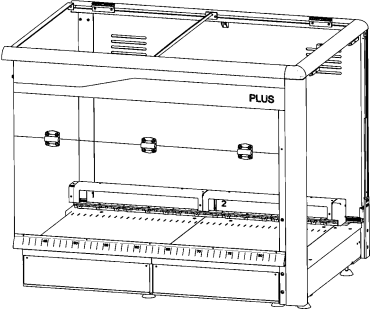
2.2.3 Microlab STARplus

The large version of the ML STAR Line is called Microlab STARplus. If equipped with a full deck extension the Microlab STARplus has a deck divided into 71 equal tracks for loading carriers. The deck has partitions for a maximum of 71 1-T carriers for sample tubes, or a maximum of eleven 6-T carriers for Microtiter Plates and CO-RE tips. This means that a total of 55 SBS (Standard format of the Society for Biomolecular Screening) positions can be placed onto a ML STARplus deck.



Microlab STARplus

The Microlab STARplus is the compact Microlab STARlet extended on site by additional workspace. The workspace can be filled by additional tracks. The other possibility is to integrate e.g. 3rd party devices, as readers, washers, centrifuges, etc. The extension part of the Microlab STARplus is available in 3 versions:

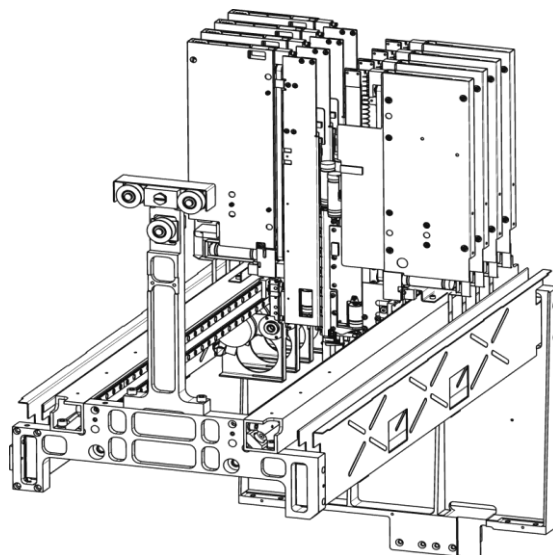
 Without deck extension: • ML STARplus has 30 tracks, • Total of 25 SBS positions	 With left deck extension: • ML STARplus has 50 tracks, • Total of 40 SBS positions	 With complete deck extension: • ML STARplus has 71 tracks, • Total of 55 SBS positions
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2.3 Pipetting Arm Configurations

The ML STAR Line offers a selection of a variety of arms, depending on the pipetting units and plate handling modules chosen. The pipetting arm moves in x-direction. Whenever higher throughput is required, it is possible to have two arms on the system working in parallel. E.g. while one arm is reserved for the pipetting tasks, the other can transfer plates on the deck or to/from a peripheral device. There are further channel-based tools which can be mounted on the respective arms. Some example configurations are given below.

2.3.1 Modular Pipetting Arm

The modular pipetting arm typically contains a set of pipetting channels, which work independently. It can be equipped with: up to 16 channels with 1000µl pipetting head, up to eight channels with 5ml pipetting head, a plate handling tool (iSWAP), a tube handling tool and an imaging channel.



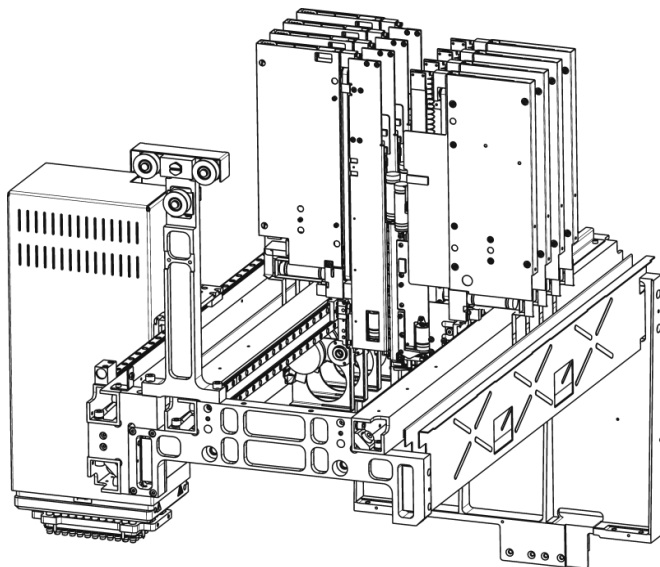
A possible configuration including all the tools is: six channels with 1000µl pipetting head, two channels with 5ml pipetting head, the tube handler, the imaging channel, with or without the plate handling tool (iSWAP).

The minimum distance between two 1000µl pipetting channels on this arm is 9mm.

The minimum distance between two 5ml pipetting channels on this arm is 18mm.

2.3.2 Modular Pipetting Arm MPH

The modular pipetting arm MPH may also hold the CO-RE 96 Probe Head, the CO-RE 96 Probe Head TADM or the CO-RE 384 Probe Head as an option paired with up to 12 channels with 1000µl pipetting head, up to eight channels with 5ml pipetting head, a plate handling tool (iSWAP), a tube handling tool and an imaging channel.



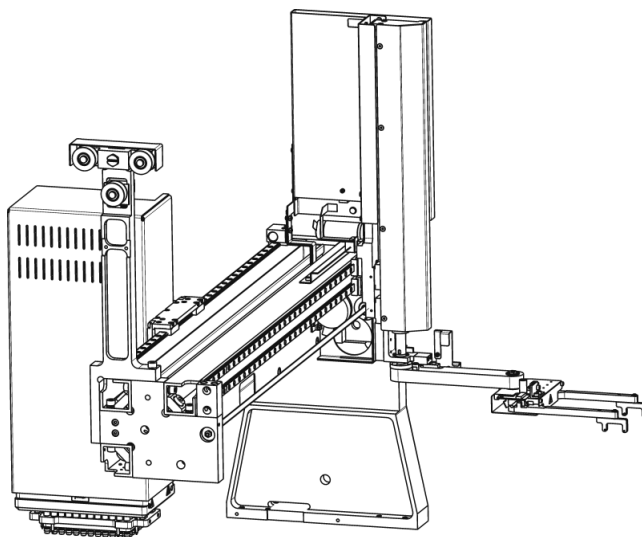
A possible configuration including all the tools is: one CO-RE multiprobe head (CO-RE 96 Probe Head, CO-RE 96 Probe Head TADM or CO-RE 384 Probe Head), six channels with 1000µl pipetting head, two channels with 5ml pipetting head, the tube handler, the imaging channel, with or without the plate handling tool (iSWAP).

The minimum distance between two 1000µl pipetting channels on the modular pipetting arm MPH is 9mm.

The minimum distance between two 5ml pipetting channels on the modular pipetting arm MPH is 18mm.

2.3.3 Pipetting Arm MPH/iSWAP

The pipetting arm MPH/iSWAP is fitted with the CO-RE 96 Probe Head, the CO-RE 96 Probe Head TADM or the CO-RE 384 Probe Head on the left. The right side of this arm can be equipped with up to eight channels with 1000 μ l pipetting head, up to four channels with 5ml pipetting head, a plate handling tool (iSWAP), a tube handling tool and an imaging channel.



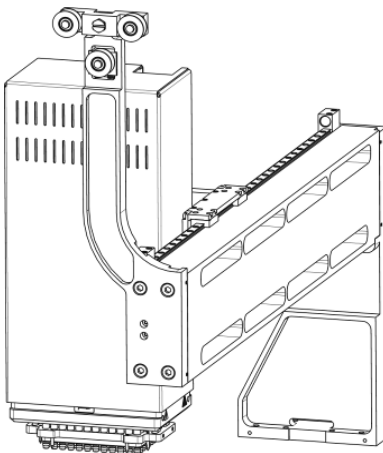
A possible configuration including all the tools is: one CO-RE multiprobe head (CO-RE 96 Probe Head, CO-RE 96 Probe Head TADM or CO-RE 384 Probe Head), 2 channels with 1000 μ l pipetting head, 1 channel with 5ml pipetting head, the tube handler, and the imaging channel, with or without the plate handling tool (iSWAP).

The minimum distance between two 1000 μ l pipetting channels on the pipetting arm MPH/iSWAP is 18mm.

The minimum distance between two 5ml pipetting channels on the pipetting arm MPH/iSWAP is 36mm.

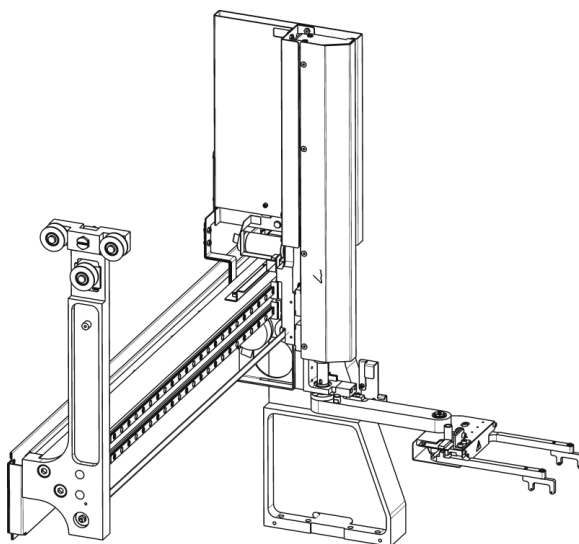
2.3.4 Pipetting Arm MPH

This arm is equipped with the CO-RE 96 Probe Head, the CO-RE 96 Probe Head TADM or the CO-RE 384 Probe Head.



2.3.5 iSWAP Arm

The iSWAP arm is typically fitted with the iSWAP for dual-arm configurations. This arm can be equipped with up to 8 channels with 1000 μ l pipetting head, up to 4 channels with 5ml pipetting head, a plate handling tool (iSWAP), a tube handling tool and an imaging channel.



A possible configuration including all the tools is: 2 channels with 1000 μ l pipetting head, 1 channel with 5ml pipetting head, the tube handler, the imaging channel, with or without the plate handling tool (iSWAP).

The minimum distance between two 1000 μ l pipetting channels on the iSWAP arm is 18mm.

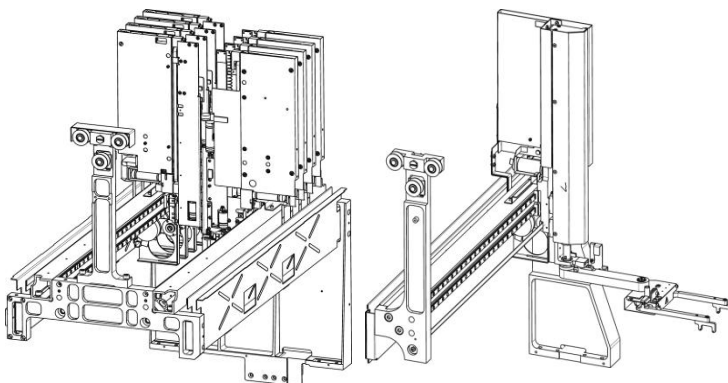
The minimum distance between two 5ml pipetting channels on the iSWAP arm is 36mm.

2.3.6 Dual-Arm Configurations

When higher throughput is required, it is possible to equip the ML STAR Line with two arms working in parallel. The preferential platform for dual-arm configurations is the Microlab STARplus. The following examples show some typical dual arm configurations. Several other combinations are possible.

Dual Arm Assay Workstation 8+iSWAP

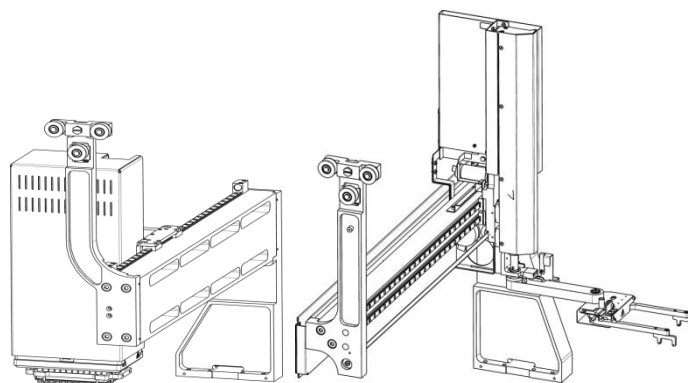
Modular pipetting arm with 8 channels combined with the iSWAP arm.



This workstation for instance makes sense if the sample preparation task (processed by the 8-channels) is isolated from the assay (as incubation tasks, plate washing tasks, analyzing task, etc). The iSWAP is used to transfer the processed Microtiter Plate to/from the 3rd party equipment as reader, incubator, plate washer, etc.

Dual arm assay workstation 96+iSWAP

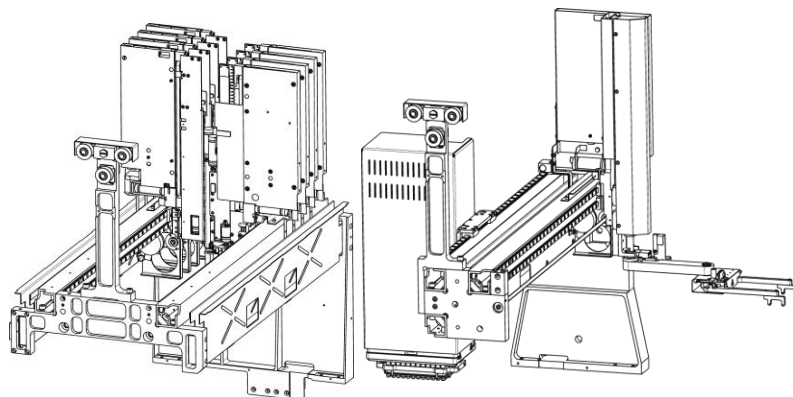
Pipetting arm 96 with the CO-RE 96 Probe Head combined with the iSWAP arm.



This workstation for instance makes sense if the pipetting tasks (plate copy, plate reformation, serial dilution) take part in the format of Microtiter Plates only, and the assay needs no further pipetting steps. In addition, plate transfer from and to other devices is required.

Dual arm assay workstation 8+96/iSWAP

Modular pipetting arm with 8 channels combined with a pipetting arm 96/iSWAP.



This workstation for instance makes sense if the sample preparation task (processed by the 8-channels) is isolated from the assay part (as incubation tasks, plate washing tasks, analyzing task, etc). The assay part needs pipetting in the pattern of the CO-RE 96 Probe Head (one volume per plate, row or column). The iSWAP is used to transfer the processed Microtiter Plate to or from 3rd party equipment, such as reader, incubator, plate washer, etc.

2.4 Options and Upgrades

Options are defined as components or configurations that are part of the instrument. Predefined options are as follows: Manual load or Autoload, the type and quantity of pipetting arms, the quantity of pipetting channels from 1 to 16 max, for 1000µl channels or 1 – 8 max, for 5ml channels, the CO-RE 96 Probe Head, the CO-RE 384 Probe Head, the plate handling tool (iSWAP), the tube handling tool and the camera channel.

Accessories include assemblies such as wash stations, temperature controlled carrier, vacuum system, labware carrier, etc.

Pipetting channels, CO-RE 96 Probe Head, CO-RE 96 Probe Head TADM, CO-RE 384 Probe Head and accessories may be ordered as an option for new installation, or added as upgrades to an existing instrument.

The instrument's configuration is set within the configuration editor of the VENUS Software by a Software Engineer.

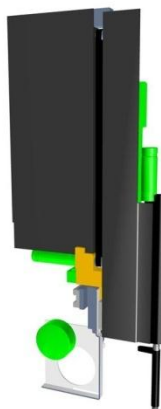


NOTE

An option may require a library, labware definitions etc. for use with the VENUS software. Please consult a local Hamilton representative.

2.4.1 1000µl-Channels

The ML STAR instrument comes with 1 to 16 pipetting channels working in parallel for simultaneous transfer of liquids. Each pipetting channel moves independently on the y-axis, as well as the z-axis. Each channel uses its own high-precision motors and electronics to reach any position on the deck without the need of teaching.



The 1000µl-channels support pipetting with disposable tips or with needles.

The following table gives an overview of disposable tips which can be used with the 1000µl-channels for pipetting from/to the respective plates. Other combinations may be possible, but require special attention.

1000µl-channel	Tips for working volume and purpose					
Plate type	10µl Low vol. CO-RE tip	50µl	300µl CO-RE tip	1000µl CO-RE tip	250µl Robotic piercing tip	300µl Slim tip
 96-well						

2 Description of the ML STAR Line

 96-well archive						
 384-well						

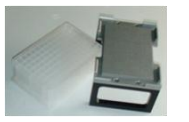
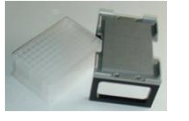
1000µl-channel	Tips for working volume and purpose					
Plate type	10µl Low vol. CO-RE tip	50µl	300µl CO-RE tip	1000µl CO-RE tip	250µl Robotic piercing tip	300µl Slim tip
384-well archive 						
1536-well 						

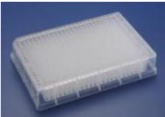
The minimum distance between two 1000 μ l-channels on the modular pipetting arms is 9mm, on all other arms it is 18mm.

The pipetting channels have a set “traverse height” of 245mm above the origin, or 145mm between the top of the disposable tip and the deck of the instrument. This means that when a channel moves from one location on the deck to another, it automatically does so at that particular height. This is a safety precaution to prevent the channels from colliding with items that may be on the deck.

Instruments with 4, 8, or 16 channels are best operated with carriers holding micro-plates and tip racks in landscape orientation, whereas carriers with portrait orientation for micro-plates and tips are best suited for the 12-channel ML STAR instrument.

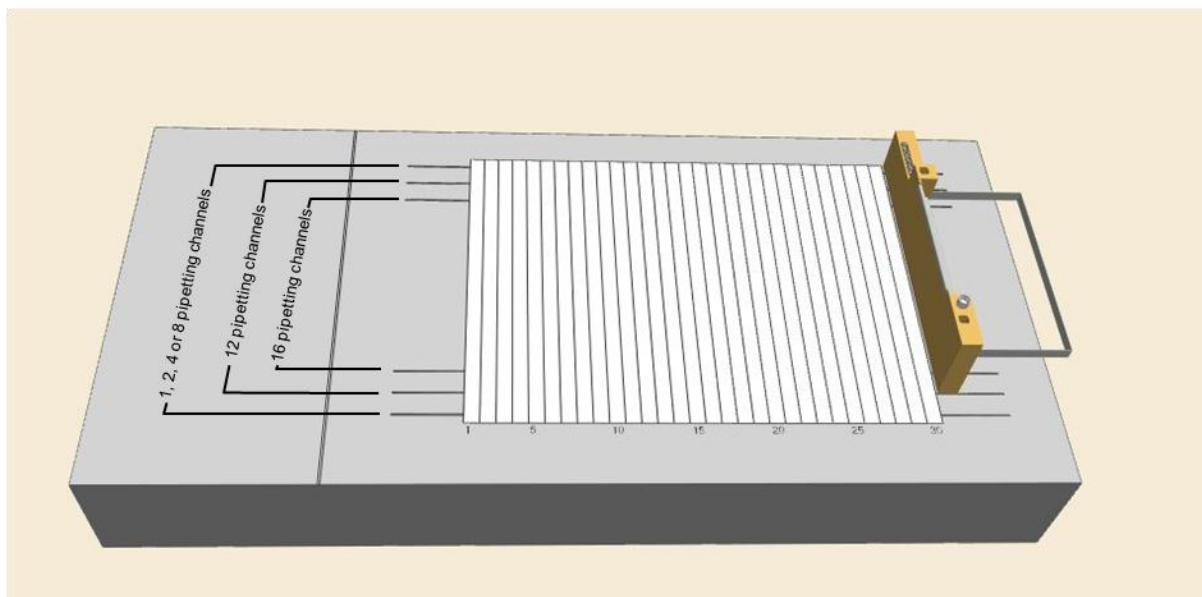
The following table gives an overview of carriers and modules which can be used with the 1000 μ l-channels for pipetting from/to the respective plates. Other combinations may be possible, but require special attention.

1000 μ l-channel	Carriers and modules			
 96-well	 Plate carrier for 96/383 well plates or  Deep Well Plates	 MFX DWP nest or  MFX MTP nest	 Plate carrier for 1536-well plates	 Multiflex MTP module or  Multiflex DWP module
 96-well archive	 Plate carrier for Deep Well Plates	 MFX DWP nest or  MFX MTP nest		 Multiflex DWP module
 384-well	 Plate carrier for 96/383 well plates or	 MFX DWP nest or	 Plate carrier for 1536-well plates	 Multiflex MTP module or

1000µl-channel	Carriers and modules			
	 Deep Well Plates	 MFX MTP nest		 Multiflex DWP module
 384-well archive	 Plate carrier for Deep Well Plates	 MFX DWP nest or MFX MTP nest		
 1536-well		 MFX DWP nest or MFX MTP nest	 Plate carrier for 1536-well plates	

An instrument equipped with 1 to 8 pipetting channels on the modular arm has the largest "random access space" which is the area, where any of the pipetting channels is able to reach. For a 12 or 16 channel equipped instrument, the "random access space" of the instrument is reduced.

As shown in the screenshot below, the "random access space" of the different numbers of channels is indicated by lines at the left and right side of the deck layout of the VENUS Software.



To guarantee random access to sample tube carriers, only the inner tube positions should be used. The number of blank positions in front and rear, to be used for the different number of 1000 μ l-channels and the different tube carriers, is listed in the following table. The deck layout shows the relevant carriers directly above the columns of the table in each case:



NOTE

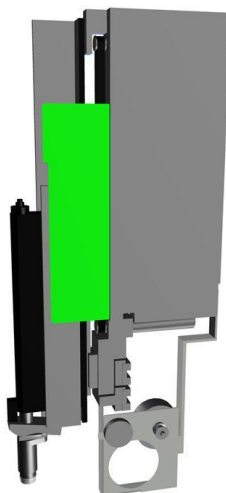
If pipetting positions outside the access range of the instrument are used, the system reports an error. However, a strictly batch-type process can eliminate these problems.

Low-volume tips do not reach the deck, even at the lowest z-position of the channel.

2.4.2 5ml-Channels

The ML STAR Line instrument comes with 1 to 8 pipetting channels working in parallel for simultaneous transfer of liquids. Each pipetting channel is moved independently on the y-axis, as well as the z-axis. Each channel uses its own high-precision motors and electronics to reach any position on the deck without the need of teaching.

The 5ml-channels support pipetting with disposable tips.



The following table gives an overview of disposable tips which can be used with the 5ml-channels. Other combinations may be possible, but require special attention.

5ml-channel	Tips for working volume and purpose	
Plate type	4ml	5ml
 96-well	 CO-RE tip, 4000µl	 CO-RE tip, 5000µl

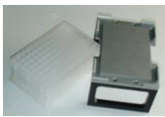
5ml-channel	Tips for working volume and purpose	
Plate type	4ml	5ml
 96-well archive	 CO-RE tip, 4000µl	 CO-RE tip, 5000µl

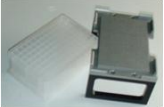
The minimum distance between two 5ml-channels on the modular pipetting arms is 18mm, on all other arms it is 36mm.

The pipetting channels have a set traverse height of 245mm above the origin, or 145mm between the top of the disposable tip and the deck of the instrument. This means that, when a channel moves from one location on the deck to another, it automatically does so at that particular height. This is a safety precaution in order to prevent channels from colliding with items that may be on the deck.

Instruments with 4 or 8 channels are best operated with carriers holding micro-plates and tip racks in landscape orientation, whereas carriers with portrait orientation for micro-plates and tips are best suited for the 6-channel instrument.

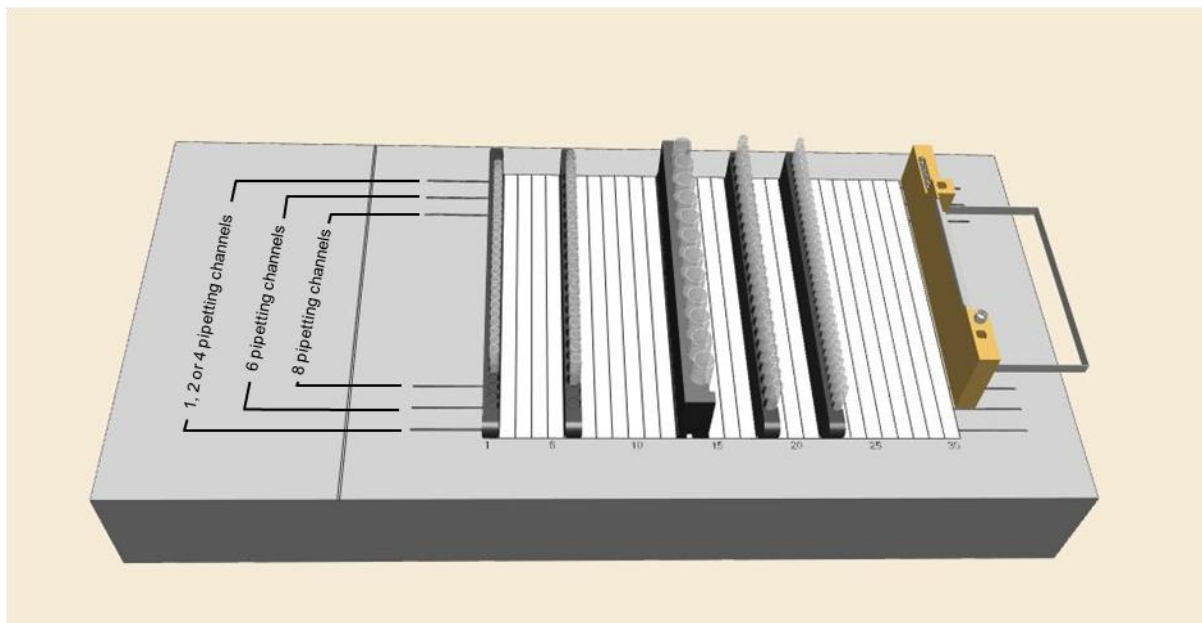
The following table gives an overview of carriers and modules which can be used with the 5ml-channels for pipetting from/to the respective plates. Other combinations may be possible, but require special attention.

5ml-channel	Carriers and modules			
 96-well	 Plate carrier for 96/383 well plates or  Deep Well Plates	 MFX DWP nest or  MFX MTP nest	 Plate carrier for 1536-well plates	 Multiflex MTP module or  Multiflex DWP module

5ml-channel	Carriers and modules			
 96-well archive	 Plate carrier for Deep Well Plates	 MFX DWP nest or  MFX MTP nest		 Multiflex DWP module

An instrument equipped with 1 to 4 pipetting channels on a modular arm has the largest "random access space" which is the area, where any of the pipetting channels is able to reach. For a 6 or 8 channel equipped instrument, the "random access space" of the ML STAR Line instrument is reduced.

The "random access space" of the different numbers of channels is indicated by lines at the left and right side of the deck layout of the VENUS Software, as shown in the screenshot:



To guarantee random access to sample tube carriers, only the inner tube positions should be used.



NOTE

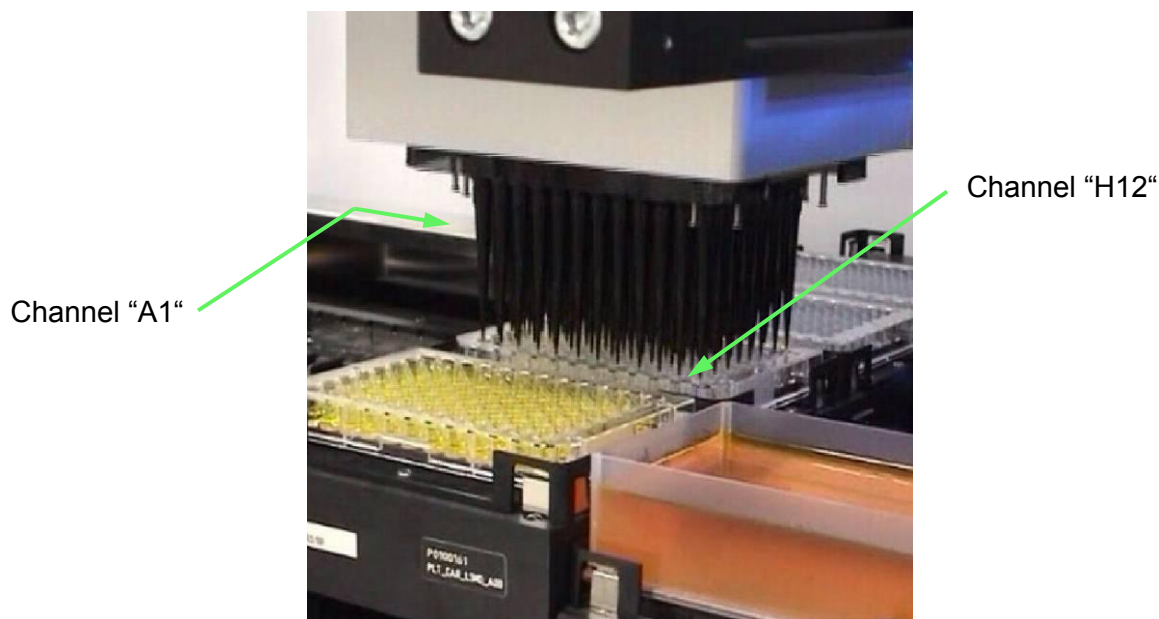
If pipetting positions outside the access range of the instrument are used, the system reports an error. However, a strictly batch-type process can eliminate these problems.

2.4.3 CO-RE 96 Probe Head

The CO-RE 96 Probe Head is a high-throughput pipettor built with the CO-RE technology as an ML STAR pipetting channel. The CO-RE technology guarantees fast and accurate pick-up and release of disposable tips.



cLLD (capacitive Liquid Level Detection) is available on the two special channels A1 and H12.



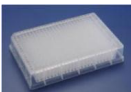
View from right to left side of the instrument

The CO-RE 96 Probe Head is available with 96x 1000µl-channels. The 96 channels always work simultaneously with the same volume.

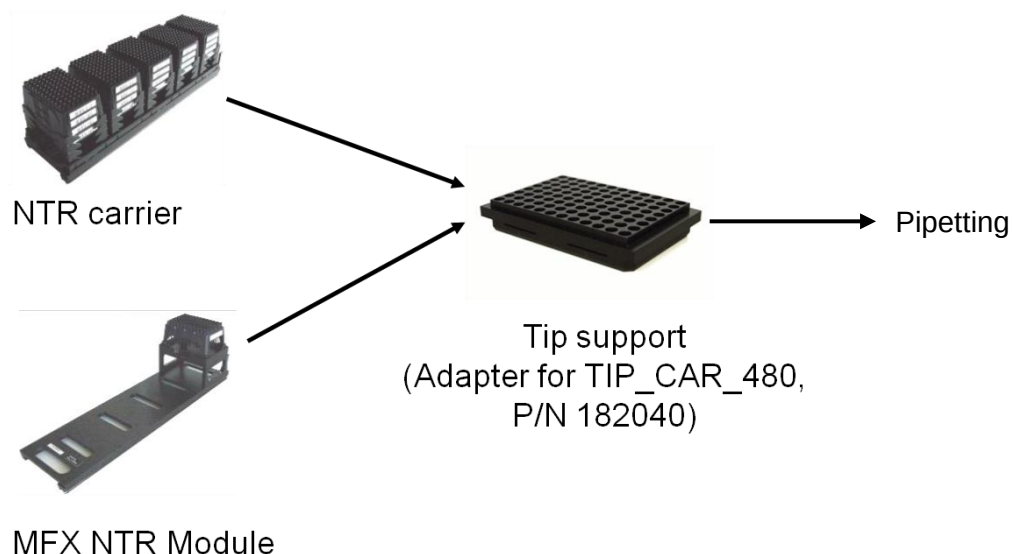
2 Description of the ML STAR Line

The CO-RE 96 Probe Head supports pipetting with low-volume (10µl), intermediate volume (50µl), standard (300µl), high volume (1000µl), piercing and slim disposable tips. The following table gives an overview of disposable tips which can be used with the CO-RE 96 Probe Head for pipetting from/to the respective plates. Other combinations may be possible, but require special attention.

CO-RE 96 Probe Head / CO- RE 96 Probe Head TADM	Tips for working volume and purpose					
	10µl Low vol. CO-RE tip	50µl	300µl CO-RE tip	1000µl CO-RE tip	250µl Robotic piercing tip	300µl Slim tip
 96-well						
 96-well archive						

CO-RE 96 Probe Head / CO- RE 96 Probe Head TADM	Tips for working volume and purpose					
Plate type	10µl Low vol. CO-RE tip	50µl	300µl CO-RE tip	1000µl CO-RE tip	250µl Robotic piercing tip	300µl Slim tip
 384-well						
 384-well archive						

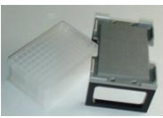
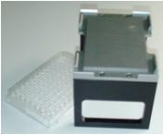
The use of a tip support makes picking up one or more columns, rows, a single quadrant or even a single tip feasible from a NTR with the CO-RE 96 Probe Head.

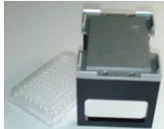
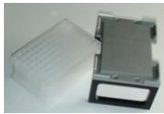
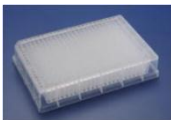


NOTE

For shifted tip pickup, make sure to use a tip support (P/N 182040) as an intermediate storage position for tips. It allows picking up of a single tip/row/column using the CO-RE 96 Probe Head.

The following table gives an overview of carriers and modules which can be used with the CO-RE 96 Probe Head for pipetting from/to the respective plates. Other combinations may be possible, but require special attention.

CO-RE 96 Probe Head / CO-RE 96 Probe Head TADM	Carriers and modules			
 96-well	 Plate carrier for 96/383 well plates or  Deep Well Plates	 MFX DWP nest or  MFX MTP nest	 Plate carrier for 1536-well plates	 Multiflex MTP module or  Multiflex DWP module

CO-RE 96 Probe Head / CO-RE 96 Probe Head TADM	Carriers and modules			
 96-well archive	 Plate carrier for Deep Well Plates	 MFX MTP nest		 Multiflex DWP module
 384-well		 MFX DWP nest or  MFX MTP nest	 Plate carrier for 1536-well plates	
 384-well archive		MFX MTP nest 		

2.4.4 CO-RE 96 Probe Head TADM

Hamilton's proprietary TADM feature is available with the CO-RE 96 Probe Head TADM. All other features of this head are identical to those of the CO-RE 96 Probe Head. The CO-RE 96 Probe Head TADM is compatible with MPH ready ML STAR Line instruments and runs with VENUS TADM feature as well as with VENUS Three.



The CO-RE 96 Probe Head TADM supports pipetting with low-volume (10µl), intermediate volume (50µl), standard (300µl), high volume (1000µl), piercing (limited) and slim disposable tips.

The CO-RE 96 Probe Head could be used with piercing tips but does not work in all labware, thus makes pipetting with piercing limited. The CO-RE 96 Probe Head does not have enough force in the z-drive to pierce a 96 rack format occupied by tubes with caps. It is only possible to pierce a 96-well plate with a thin aluminum foil on the CO-RE 96 Probe Head therefore piercing tips are used to pierce a 96-well plate with the aluminum foil or the tubes with caps.

2.4.5 CO-RE 384 Probe Head

The CO-RE 384 Probe Head is a highly parallel pipettor. It uses the CO-RE technology as the individual channels and the CO-RE 96 Probe Head do, offering accurate and gentle tip pickup. The positioning accuracy allows pipetting into 96, 384 and 1536 well Microtiter Plates. The 384 channels always work simultaneously with the same volume. The pipetting range is between 0.1µl and 50µl. The 384 channels always work simultaneously with the same volume.



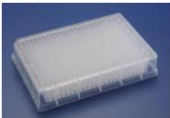
NOTE

Special 50µl tips were designed for pipetting with the CO-RE 384 Probe Head.



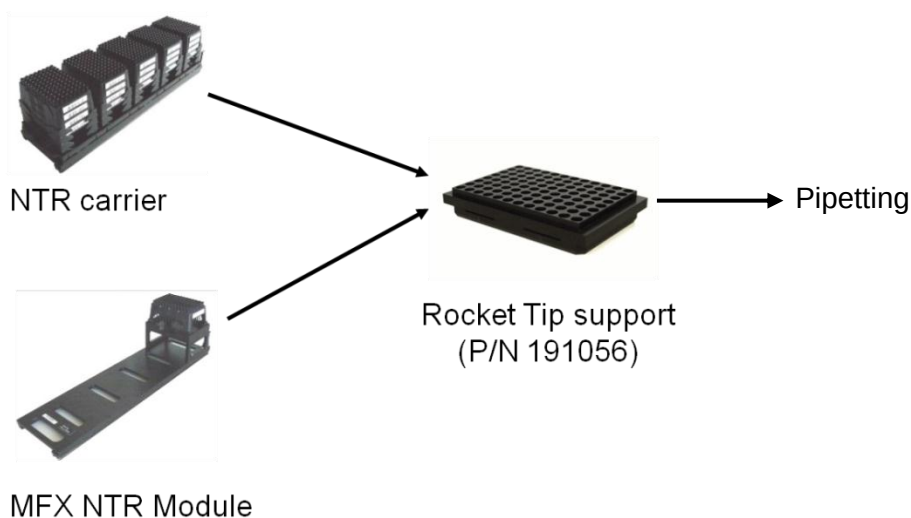
cLLD (capacitive Liquid Level Detection) is available on two special channels (A5 and P24).

The following table gives an overview of disposable tips which can be used with the CO-RE 384 Probe Head for pipetting from/to the respective plates. Other combinations may be possible, but require special attention.

CO-RE 384 Probe Head	Tips for working volume	
Plate type	50µl	300µl
 96-well	 384 Probe Head CO-RE tips 50µl	 Rocket Tip
 96-well archive	 384 Probe Head CO-RE tips 50µl	 Rocket Tip
 384-well	 384 Probe Head CO-RE tips 50µl	
 384-well archive	 384 Probe Head CO-RE tips 50µl	

CO-RE 384 Probe Head	Tips for working volume	
Plate type	50µl	300µl
 1536-well	 384 Probe Head CO-RE tips 50µl	

The use of a tip support makes picking up one or more columns, rows, a single quadrant or even a single tip feasible from a NTR with the CO-RE 384 Probe Head.



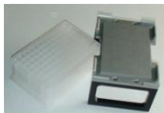
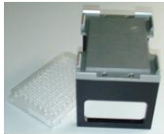
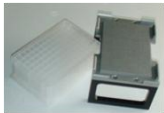
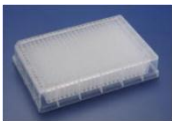
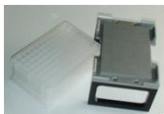
NOTE

For shifted tip pickup, make sure to use a tip support (P/N 191055 / 191056) as an intermediate storage position for tips. It allows picking up of a single tip/row/column using the CO-RE 384 Probe Head

The use of special NTRs equipped with 96 tips per rack (P/N 235993 / 235447; 96 of the 384 positions filled with tips) converts the CO-RE 384 Probe Head into a 96 channel pipetting head with a volume range of 50µl.

The use of Hamilton's Rocket Tips converts the CO-RE 384 Probe Head into a 96 channel pipetting head with a volume range of up to 300µl. The use of a tip support makes picking up one or more columns, rows, a single quadrant or even a single Rocket Tip feasible from a NTR with the CO-RE 384 Probe Head.

The following table gives an overview of carriers and modules which can be used with the CO-RE 384 Probe Head for pipetting from/to the respective plates. Other combinations may be possible, but require special attention.

CO-RE 384 Probe Head	Carriers and modules			
 96-well	 Plate carrier for 96/383 well plates or Deep Well Plates	 MFX DWP nest or MFX MTP nest	 Plate carrier for 1536-well plates	 Multiflex MTP module or Multiflex DWP module
 96-well archive	 Plate carrier for Deep Well Plates	 MFX MTP nest		 Multiflex DWP module
 384-well		 MFX DWP nest or MFX MTP nest	 Plate carrier for 1536-well plates	
 384-well archive		 MFX MTP nest		
 1536-well		 MFX DWP nest or MFX MTP nest	 Plate carrier for 1536-well plates	

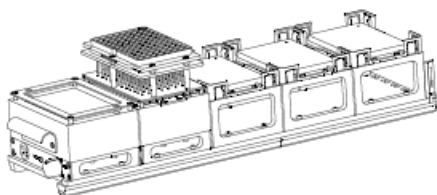
CO-RE 384 Probe Head	Carriers and modules			
		 MFX MTP nest		

2.4.6 Pin Tools

Pin tools from a 3rd party provider can be used with the ML STAR Line instruments. A pin tool is an array of metal pins for the transfer of small (nanoliter) amounts of liquids into high density formats. Hamilton offers an adaptor and accessories for pin tools from V&P Scientific, Inc. (www.vp-scientific.com) to be used with the CO-RE 96 Probe Head TADM or the CO-RE 384 Probe Head.

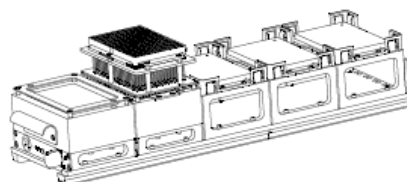
The adaptor plate (ADAPTOR PLATE PINTOOL96/384 P/N 188154APE) couples the pin tool 96/384 with the CO-RE 96 Probe Head or CO-RE 384-Probe Head. It will be produced according to the available type of pin tool and depends on the head type (CO-RE 96 or 384 Probe Head).

Service carriers (without pin tool) exist for pin tool 96 (PINTOOL 96 SERVICE CARRIER, P/N 188145APE) and for pin tool 384 (PINTOOL 384 SERVICE CARRIER, P/N 188146APE) on a Multiflex base:



Service carrier (without pin tool or pin tool adapter) on Multiflex base including:

- 1x Multiflex base plate*
- 1x parking positions for pin tool 96*
- 1x heated air drying station*
- 3x Deep Well Plate positions for troughs with wash solutions*



Service carrier (without pin tool or pin tool adapter) on Multiflex base including:

- 1x Multiflex base plate*
- 1x parking positions for pin tool 384*
- 1x heated air drying station*
- 3x Deep Well Plate positions for troughs with wash solutions*

The service carriers come with labware definitions for the pin tool.



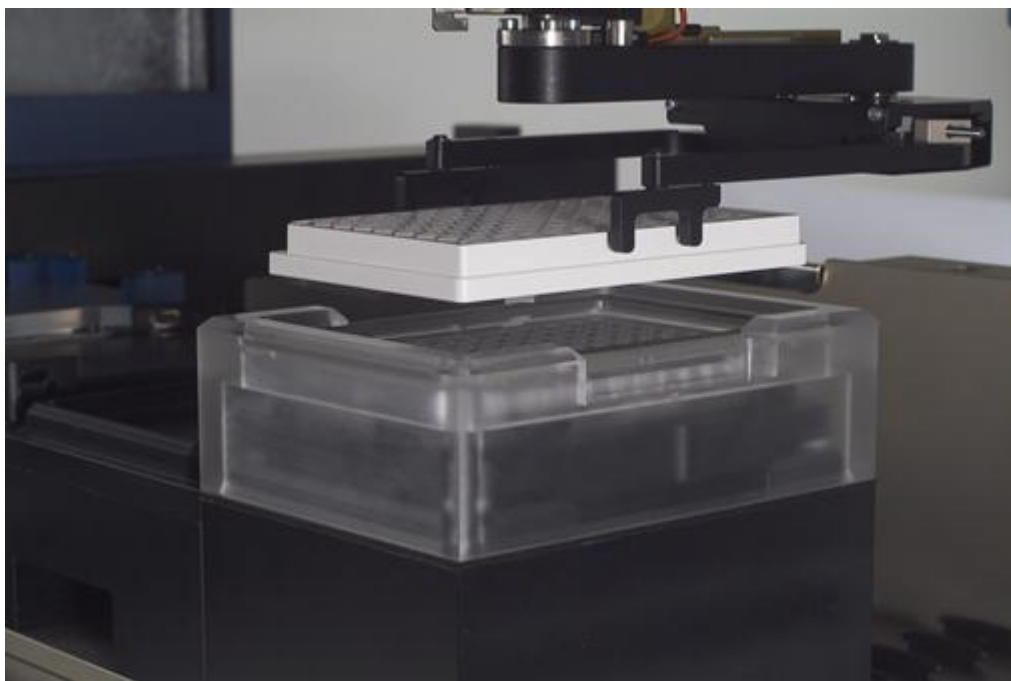
NOTE

Using a pin tool requires the pin tool itself (order from V&P Scientific, Inc.), the pin tool 96/384 adapter plate (which will be produced at Hamilton according to the specific pin tool), and the pin tool service carrier.

2.4.7 iSWAP

iSWAP (internal Swivel Arm Plate Handler) is a robotic arm that transports micro-plates, lids of micro-plates, archive plates, filter plates to and from positions on the instrument deck. Plates can be placed in landscape or portrait orientation or rotated. It is possible to pick up labware from and place it to labware any position inside or even outside of the deck. Typical handling tasks like loading/unloading of plates outside the instrument becomes very simple with the iSWAP. The iSWAP also can stack plates or tips with specially provided carriers.

Among the special features of iSWAP is a sensor which signals to the robot if it is gripping a labware object. The newest generation of iSWAP monitors the presence of an object. Like ML STAR pipetting channels, iSWAP has a “traverse height” of 145mm above the deck (245mm above the origin).



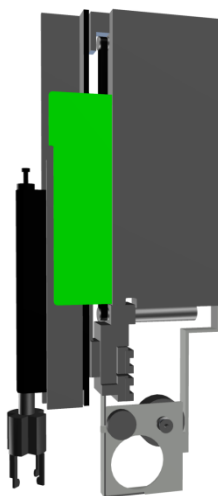
iSWAP robotic hand at work transporting micro-plates to CVS.

The iSWAP is mounted on the pipetting arm. It does not affect the movement of the pipetting tools.

2.4.8 Tube-gripper

The tube-gripper transports tubes to and from positions on the instrument deck. The tube-gripper tool is mounted to a separate channel. It can handle tubes with diameters from 8mm up to 20mm and up to a length of 120mm.

The tube-gripper is moved independently on the y-axis, as well as the z-axis. The use of high-precision motors and electronics allows the tube-gripper to reach any position on the deck without the need of teaching.



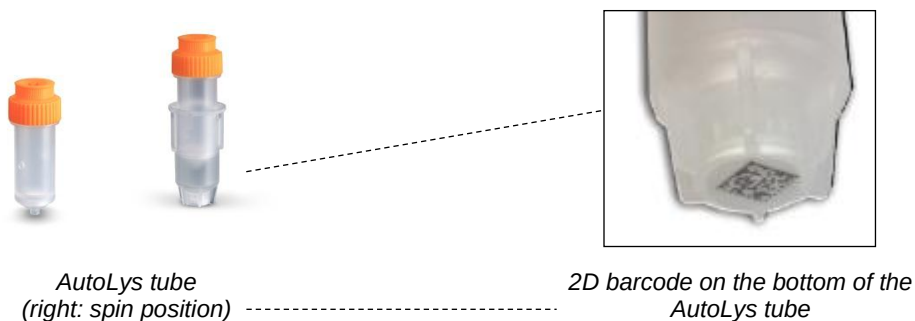
Like ML STAR pipetting channels, the tube-gripper has a “traverse height” of 145mm above the deck (245mm above the origin) between the tube and the deck of the instrument. This means that when a tube is moved from one location on the deck to another, it automatically does so at that particular height. This is a safety precaution, so that the transported tube will not collide with any items that may be on the deck.

2.4.9 AutoLys Tube Channel

The AutoLys channel handles AutoLys tubes. It is mounted to a separate channel.

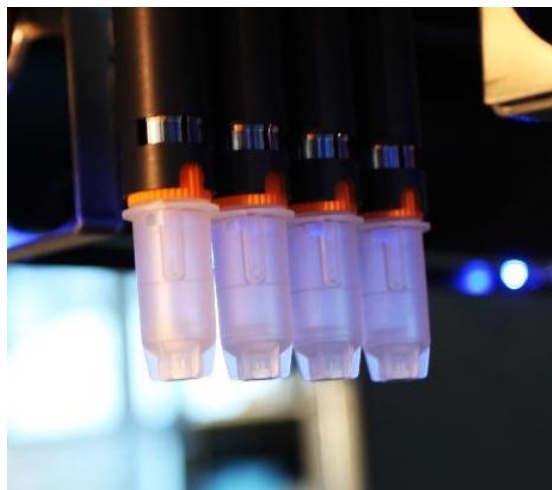


The AutoLys tube is an automation-friendly consumable designed for sample lysis and extraction. It is a 3-part assembly consisting of an outer tube with 2D barcode, an inner tube with flow-thru filter and a threaded cap. The AutoLys tube facilitates processing and filtering of up to 500µl of lysate. The lysate is usually captured by centrifugation.





Four AutoLys tool channels



Four AutoLys tool channels holding AutoLys tubes

The AutoLys A tool channel is moved independently on the y-axis, as well as the z-axis. The use of high-precision motors and electronics allows the AutoLys tool channel to reach any position on the deck without the need of teaching. It automates the entire process of AutoLys tube handling during a lysis procedure:

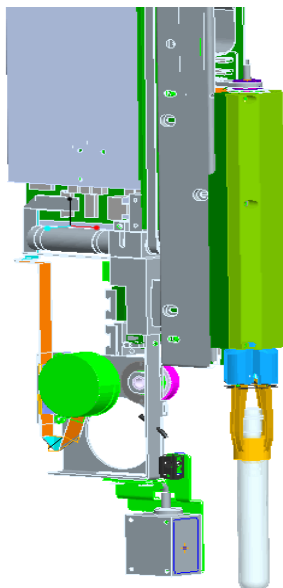
- De-capping the AutoLys tube, holding the cap and re-capping the tube.
- Transfer the AutoLys tube to and from racks
- Lifting the inner tube and locking it in the “spin position”
- Removing and holding the inner tube post centrifugation
- Replacing the inner tube in the original outer tube

2.4.10 Tube-twister Channel and De/re-capper module

2.4.10.1 Tube-twister

The tube-twister can grip and rotate tubes, read their barcodes and transport the tubes to and from positions on the instrument deck. The tube-twister is mounted to a separate channel. It can handle tubes with diameters from 15mm up to 38mm and caps from 18mm up to 44mm. The maximum tube length including cap is 120mm.

The tube-twister channel is moved independently on the y-axis, as well as the z-axis. The use of high-precision motors and electronics allows the tube-twister channel to reach any position on the deck without the need of teaching. The attached tube-twister head is specific for the tube-twister channel and includes two drives. One twister drive is able to grip tubes with its four fingers. The other twister drive rotates the gripper tool. This motion can be used to rotate the tube for barcode identification and for alignment, placing the tube in a carrier with barcode in correct position. A 1D barcode reader is attached to the tube-twister channel. Barcode reading is executed after gripping the tube. The motion can also be used for mixing liquid inside the tube.

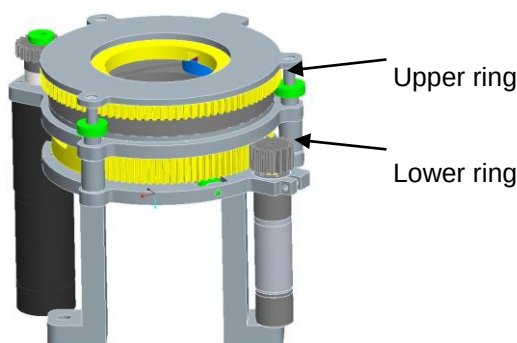


Like ML STAR pipetting channels, the tube-twister has a “traverse height” of 145mm above the deck (245mm above the origin) between the tube and the deck of the instrument. This means that when a tube is moved from one location on the deck to another, it automatically does so at that particular height. This is a safety precaution, so that the transported tube will not collide with any items that may be on the deck.

In combination with the de/re-capper module the tube-twister can also be used for de-capping and recapping of tubes.

2.4.10.2 De-capper

In combination with the tube-twister, the de/re-capper module allows opening and closing tubes with screw-top caps. It consists of two rings. The upper ring clamps and turns caps, the lower ring clamps and moves tubes. For reliability and increased process control, the de/re-capper is equipped with independent upper ring drive and lower ring drive.



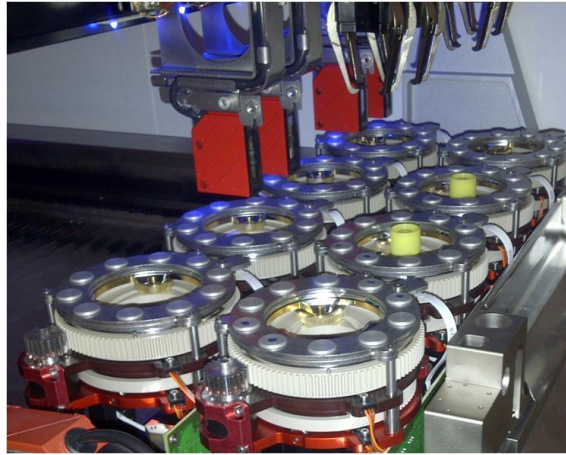
De/re-capper module

For de-capping a capped tube is presented to the de/re-capper using the tube-twister. The lower ring of the de/re-capper is closed in order to hold the tube. The upper ring is closed to hold the cap and disengage the cap from the tube. The tube-twister unscrews the cap and throws it into the waste or holds it. The lower ring of the module releases the tube. At this stage the pipetting

channels have access to the liquid in the tube. Alternatively, the tube can be transported to a different location by the tube-twister.

For re-capping an uncapped tube is presented to the de/re-capper using the tube-twister. The lower ring of the de/re-capper is closed in order to hold the tube. The tube-twister gets a cap from a cap supply and screws the cap with tube minimal force. The cap is then finally locked using the upper ring of the de/re-capper module. The capped tube is released and can be transported by the tube-twister.

The de/re-capper module is mounted on the deck on a special carrier. Several modules can be installed in parallel (see below).



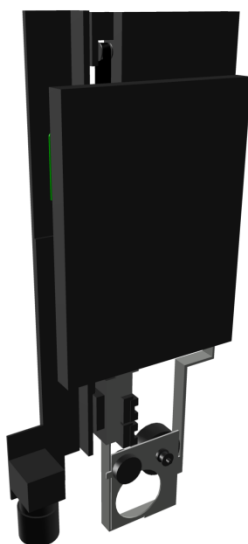
*Eight de/re-capper modules mounted on the deck.
Four tube-twister channels (above).*

2.4.11 Imaging Channel

The purpose of the imaging channel is to capture the digital image from any object all over the instrument deck. A high-resolution CCD-Camera is fixed on a separate channel and allows acquiring images from a target which is as big as the SBS format. The images are sent to the image analyses software then for further investigation.

A typical application is the image analysis of bacteria colonies. In the proprietary easyPick software, the user defines parameters and determines the weighting for the typical criteria to identify colonies: coordinates, size, shape (circularity), color, proximity to next colony, distance to the margin, etc.

Reliable automation requires proper quality of the analyzed images. Appropriate lighting of the object of interest is fundamental. In case of image analyses of bacteria colonies a back-light table is used in the system. For other applications different lighting options can be applied.



2.4.12 CO-RE Gripper

The CO-RE gripper is a plate handling tool picked up by two pipetting channels during a run. This tool is only available if the system has at least two individual pipetting channels.

The CO-RE gripper transports micro-plates, lids for micro-plates, archive plates, filter plates, etc. to and from positions on the instrument deck. Plates can be gripped in landscape or portrait format within the working area. Rotation of plates is not an option.

The traverse height of the channels with the gripping jaws is the same as with tips: 145mm above the deck. Given that there is no sensor working; one must ensure that CO-RE gripper does not grip plates too tightly causing them to rattle. On the other hand, a loose CO-RE gripper can result in a step loss.



CO-RE gripper for the 1000 μ l-channels(inner positions) and the 5ml-channels (outer positions): the “paddles” in their holder



CO-RE gripper on the 1000 μ l-channels

The holder for the two gripping paddles can be mounted on any standard plate carrier and needs one position which can be freely selected. Another position for the holder of the two gripping jaws is on the waste block. In this case, the CO-RE gripper does not consume one plate position.



NOTE

When using plate stacks or filter plate systems with an elution plate on the CVS, the stacked plates and elution plates are only accessible by the CO-RE gripper, NOT with the iSWAP.

2.4.13 FlipTube Tool

Hamilton FlipTubes are 1.5ml reagent tubes with a lid attached which can close the tube tightly for convenient and reliable preparation, centrifugation and storage of small volume sample. The FlipTube tool is used for opening and closing Hamilton FlipTubes. It is picked up by a 1000µl-channel during a run. Up to four FlipTube tools can be used in parallel when picked up by four 1000µl-channels, if available. The traverse height of a channel with the FlipTube tool is the same as with tips: 145mm above the deck.



A Set of FlipTube tool (consisting of four fingers) within the holder



Hamilton FlipTube (P/N 235454)



NOTE

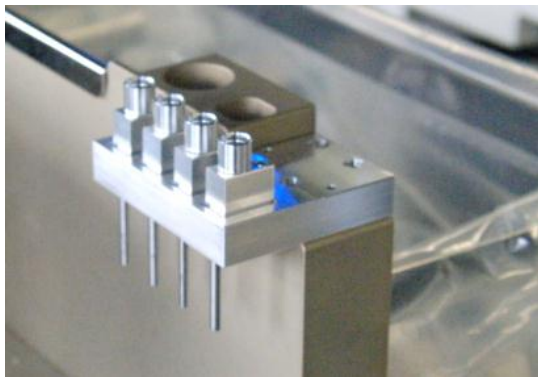
The FlipTube tool can be used for Hamilton FlipTubes only, not for other locked tubes.

Park positions (holders) for FlipTube tools are available for a Multiflex carrier or for the waste block. The park position on the waste block has the advantage that it does not consume a plate position.

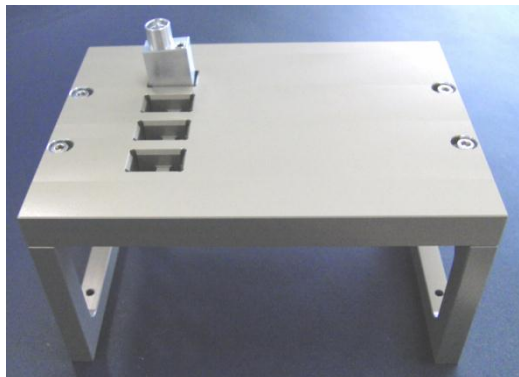


NOTE

FlipTube tools have to be aligned on the deck such that the long rod is on the left hand side on the park position (see picture on the next page).



*FlipTube tool park position on the waste block.
The long rod faces to the left hand side on the deck.*



*FlipTube tool park positions for a Multiflex carrier
in their correct orientation on the deck,
FlipTube tool positioned on the left hand side,
long rods facing left.*

FlipTubes must be fixed tightly in a special carrier or adapter for opening and closing by a FlipTube tool. This is achieved by a specific sample carrier, an adapter for a Multiflex Module or an adapter for the Hamilton Heater Shaker.



Sample carrier for 32 FlipTubes SMP_CAR_32FT (P/N 809030)



NOTE

The Multiflex adapter for FlipTubes and the HHS adapter for FlipTubes have to be aligned on the deck such that the lid of the tube faces to the left hand side (see pictures on the next page).



*Multiflex adapter for FlipTubes (P/N 809300).
Mount such that the lids of the tubes face left.*



*HHS adapter for FlipTubes (P/N 809031).
Mount such that the lids of the tubes face left.*



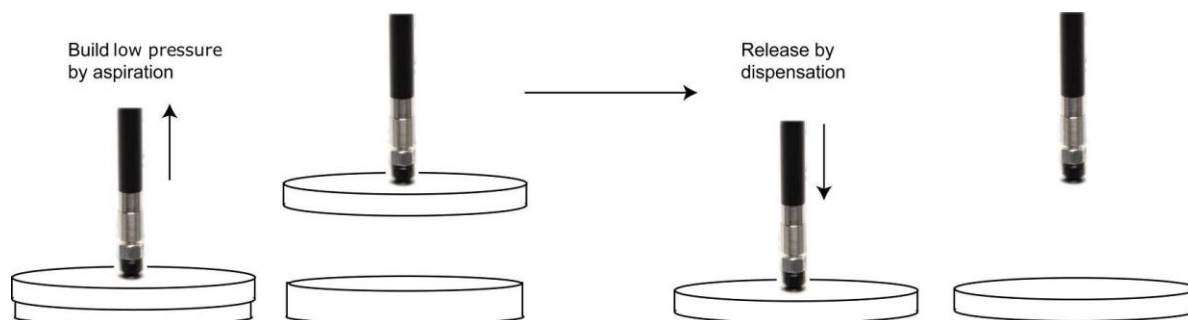
NOTE

FlipTubes are normally mounted on the waste block or in the SBS position; however when the waste block is occupied by the CO-RE gripper, either the SBS position or the FlipTube tool will be used instead. For more information about FlipTubes, please refer to the FlipTube Manual.

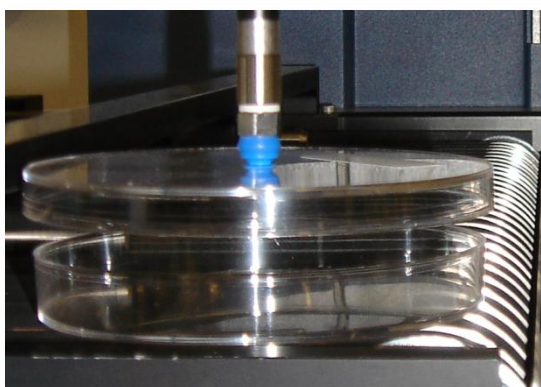
There is a spring found on the carrier or adapter which can be adjusted when necessary.

2.4.14 CO-RE Lid Tool

The CO-RE lid tool (CLT) is a suction cup device for picking up and transporting lightweight lids of petri dishes, plates, troughs and reservoirs. It can be picked up by a 1000µl- channel. The plunger drive generates a negative pressure (vacuum) on the surface of flat clean labware objects so that they can be lifted. When not in use, the CO-RE lid tool is parked on the waste block in one of the teaching/maintenance needle positions.



Mode of operation of the CO-RE lid tool



CO-RE lid tool picking up the lid of a petri dish



CO-RE lid tool on its park position on the waste station

For operation of the CO-RE lid tool a special library is required. The labware pick-up and eject positions, the tool park position and the number of the channel to be used have to be defined.



NOTE

Please consult the local Hamilton representative for the specific library to use the CO-RE lid tool.

2.4.15 Autoload Option

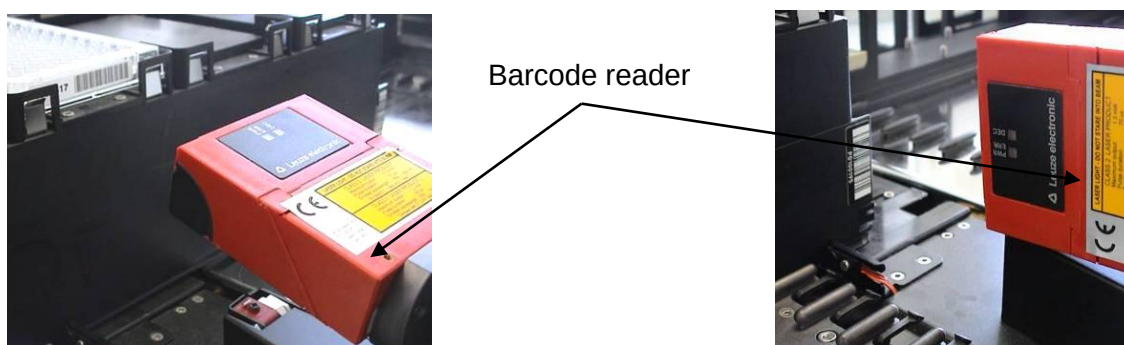
The Autoload option of the ML STAR configuration must be specified by the customer at the time of purchase of the instrument. To retrograde at a later time cannot be done.

The Autoload drive is a device enabling automatic loading of carriers onto the ML STAR instrument deck.

- moves in x-direction,
- shunts carriers onto and off the deck, and
- reads the barcodes of carriers, tubes, and Microtiter Plates.

There is a presence sensor that identifies the tubes present on a sample carrier.

Carrier identification by barcodes, and reading of barcodes on plates and tubes, is only possible in combination with the Autoload option.



The Autoload option including the barcode reader, which is shown reading horizontally for plates and vertically for carriers and tubes.

The following barcode types can be read by the Autoload option:

- ISBT Standard
- Code 128 (Subset B and C)
- Code 39
- Codabar
- Code 2 of 5 interleaved
- UPC A

Barcodes must be black bars on white background. Hamilton recommends using the barcode type Code 128 (Subset B and C).



NOTE

In addition, barcodes must have a minimum readability (i.e., good contrast, size, correct orientation and distance between bars) to be fully functional.

Ensure the correct barcode orientation for tubes and plates (see specifications).

For details of barcodes, see the specifications given in Chapter 7.10 "Autoload Option: Barcode and Reader Specifications".

2.5 Accessories

Accessories are defined as additional automation components. They provide a high degree of adaptability and permit customizing for multiple applications. These components can be ordered later by the customer and installed by Hamilton-authorized personnel.



NOTE

An accessory may require a library, labware definitions etc. for use with the VENUS software. Please consult a local Hamilton representative.

Note that the availability of accessories is subject to change. Please consult a local Hamilton representative or visit www.hamiltonrobotics.com.

2.5.1 Carriers

The labware such as plates or tubes is placed on special carriers that are loaded onto the ML STAR instrument deck. Hamilton provides a wide range of standard carriers for Microtiter Plates, tubes, tips etc. All standard carriers can be loaded to the deck manually or by the Autoload option.

The naming of carriers follows a systematic nomenclature “**X_CAR_Y_Ann**” where

- X** represents the type of labware placed on the carriers, e.g., TIP (= tips), PLT (=plates), SMP (=samples), RGT (=reagent)
- CAR** carrier
- Y** describes the labware details, e.g.
 - L**: landscape orientation
 - P**: portrait orientation
 - Number**: number of items placed on the carrier (plates or tips)
 - MD**: medium density (96- or 384-well micro-plates)
 - HD**: high density micro-plates (1536)
 - AC**: 96-well archive plates
- Ann** identifies the part number revision (e.g. A00)
 - A**: variant
 - nn**: revision

Example: PLT_CAR_L5MD_A00 is a plate carrier for 5 medium density (96- or 384-well) micro-plates in landscape orientation.

A carrier must always be identified in the VENUS Software (e.g. in deck layouts and methods) by the unique description with which it is tagged.

2.5.2 Hamilton Heater Shaker

The HHS is designed to heat and / or shake standard Microtiter Plates in SBS format. Plates must be placed on the HHS *without* lids since a lid is not locked during a shaking process.

The maximum shaking speed depends on the labware used. Shaking can be performed clockwise or counter-clockwise.

Before heating or shaking is started, the plates are locked and positioned in the centre of the HHS. When heating or shaking has finished the plates are unlocked and can then easily be removed from the HHS.

The HHS can be heated to temperatures up to 105°C. The temperature is constantly measured by two sensors located on the adapter plate.

The HHS is a compact module designed for operation on the ML STAR Line robotic pipetting workstation. It is not suitable as standalone product. The HHS is used for heating and shaking of standard Microtiter Plates in SBS format. In order to get efficient and evenly distributed heating of plates, specific adapters are required.



NOTE

The HHS is only intended to be used in combination with the ML STAR Line. A standalone operation of the HHS is not foreseen.

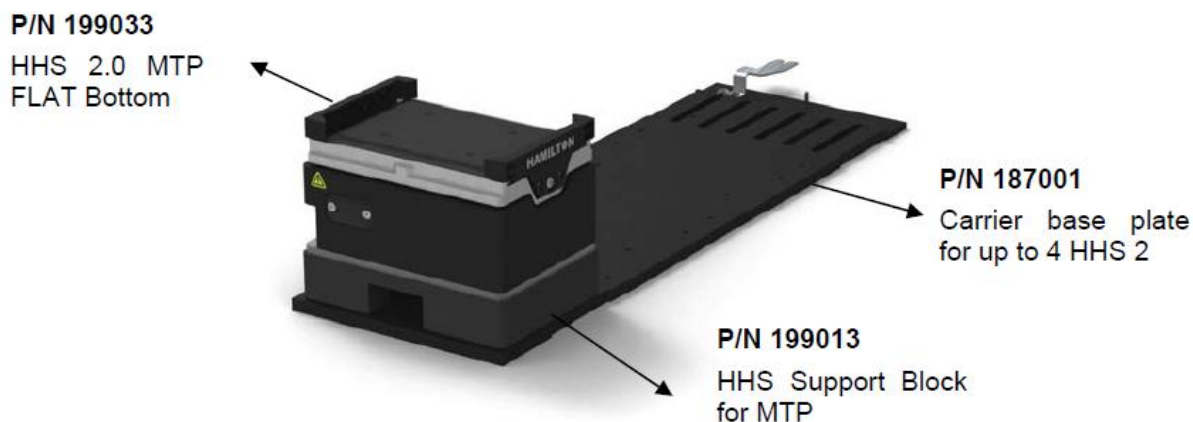
Please also refer to the [HHS User Manual \(P/N 624175\)](#).

2.5.2.1 Installation of the Hamilton Heater Shaker

Generally; the HHS will be installed by trained service engineers.

Up to 8 HHS modules can be integrated into one ML STAR Line instrument. When integrating more than 2 heater shakers, the additional Heater Shaker Box is required for control.

Up to 4 HHS can be mounted onto one carrier base plate for heater shakers. A HHS for Deep Well (archive) Plates should be positioned directly onto the carrier base plate; the position of a HHS for standard plates can be elevated by using the HHS support block.



Ordering example for HHS with flat bottom



NOTE

Carrier base plate and support block have to be ordered separately.

For Ordering information see [Appendix B "Ordering Information"](#)

Instead of micro-plate shakers, however, the shaker carrier base can be equipped with ordinary micro-plate modules. The list of available modules can be found in [Appendix B "Ordering Information"](#).

This carrier base will occupy 7 tracks of your ML STAR instrument deck.



2.5.2.2 Operation of the Hamilton Heater Shaker

For operation of the HHS, a specific library is required (please refer to the Microlab STAR Programmer's Manual, P/N 624043 and the HHS User Manual, P/N 624175).



ATTENTION

When working with samples to be used in particularly sensitive tests, take into account the evaporation that may occur while the HHS is running. Be aware that heating of liquids will affect the pipetting process, as well as shaking the labware can influence the precision of pipetting.



ATTENTION

When shaking labware, the speed has to be adjusted to the type of labware and the volume within the labware to prevent cross-contamination due to spillages.



ATTENTION

Do not touch the HHS during run time and also not shortly after finishing a run, as it might be hot. Wait until the HHS has cooled down to room temperature, which can take up to 1 hour.



NOTE

The maximum shaking speed of the HHS depends on the orbit and the type of labware used (see Chapter 7.10.2 "Hamilton Heater Shaker Specifications").



NOTE

An efficient heating of the liquid is only achieved if the appropriate adapter for the labware is used. The heating time and the maximum temperature, which can be reached for the liquid inside the plate, is dependent on the amount of liquid, the size / weight of the used labware, the size / weight of the used adapter and the connection between the plate and the adapter.

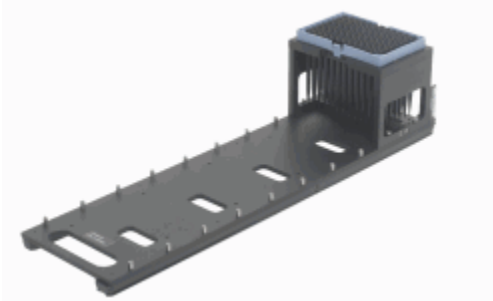
Further information about the maximum shaking speed for MTPs and DWPs can be found in Chapter 7.10.2 "Hamilton Heater Shaker Specifications".

2.5.2.3 Loading of Labware

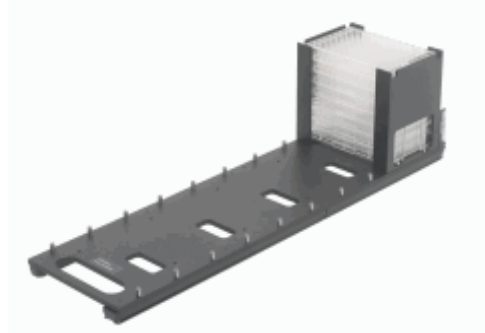
The appropriate labware can either be loaded manually or automatically by using the iSWAP or the CO-RE gripper from the ML STAR Line instrument or by other labware handling devices which can reach the labware on the HHS device.

2.5.3 Multiflex Carrier

The Multiflex carrier consists of a multiple-use carrier base offering space for up to five modules. Several modules can be chosen, such as: Tip rack module, micro-plate module, plate stacker module, module for heating or cooling labware, reagent trough module, tube or cup module, etc. The list of available modules can be found in [Appendix B “Ordering Information”](#).



Multiflex tip Module on Multiflex carrier



*Multiflex stacker Module (landscape)
on Multiflex carrier*



NOTE

For configuring Multiflex Carriers a Multiflex carrier assistant software tool is required. Please consult a local Hamilton representative.

Note that the availability of accessories is subject to change. Please consult a local Hamilton representative for an up-to-date list, or consult www.hamiltonrobotics.com.

The modular design of the Multiflex carrier allows space optimizing and customizing the instrument. This carrier will occupy 6 tracks of the instrument deck.

2.5.3.1 Installation of the Multiflex heating Modules, Cooling Modules and the MFX Cooling-Heating Module

Generally; the Multiflex heating and cooling Module will be installed by trained service engineers.



ATTENTION

When placing the Multiflex heating/cooling Module on the deck, one track position must remain empty on each side of the module for air circulation. The modules must not be placed next to each other on the deck. Extra spacing between the modules is necessary to allow proper ventilation.

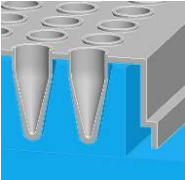
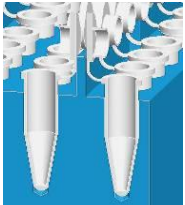
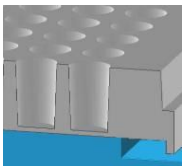


NOTE

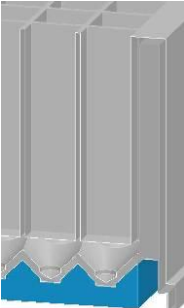
An efficient heating/cooling of the liquid is only achieved if the appropriate adapter for the labware is used. The heating/cooling time and the maximum/minimum temperature which can be reached for the liquid inside the labware is dependent on the amount of liquid, used type labware, the used adapter and the connection between the labware and the adapter.

The temperature is measured on the labware adapters. Because of the liquid containers and the ambience there is a temperature deviation (ΔT) between the labware adapter and the liquid.

Typical cooling / heating deviation in the liquid (water) compared to the desired temperature:

Labware	Adapter	ΔT cooling (0 °C)	ΔT heating (99 °C)
PCR plate		$\leq 0.5^{\circ}\text{C}$	$\leq 1.0^{\circ}\text{C}$
Eppendorf tubes		$\leq 0.5^{\circ}\text{C}$	$\leq 1.0^{\circ}\text{C}$
MTP (flat bottom)		$\leq 3.0^{\circ}\text{C}$	$\leq 5.0^{\circ}\text{C}$

Typical time (at 40% rel. humidity, $T_{\text{ambient}} \approx 22^{\circ}\text{C}$, $T_{\text{liquid}} \approx 22^{\circ}\text{C}$) to heat (ambient $+5^{\circ}\text{C}$ to 65°C) and cool (15°C to 4°C) is between 5 and 20 minutes.

Labware	Adapter	ΔT cooling (0°C)	ΔT heating (99°C)
DWP (U-bottom)		$\leq 5.0^{\circ}\text{C}$	$\leq 7.0^{\circ}\text{C}$

Typical time (at 40% rel. humidity, $T_{\text{ambient}} \approx 22^{\circ}\text{C}$, $T_{\text{liquid}} \approx 22^{\circ}\text{C}$) to heat (ambient $+5^{\circ}\text{C}$ to 65°C) and cool (15°C to 4°C) is between 5 and 20 minutes.

The Multiflex lid park Module can be used for parking the heating/cooling Module lid.

2.5.4 Multiflex Heating and Cooling Module

The Multiflex Heating and Cooling Module heats or cools plates. The temperature range for the heating module is from ambient $+5^{\circ}\text{C}$ to 65°C . The temperature range for the cooling module is from 15°C to 4°C . A temperature selection dial is located on the top of the module for setting the desired temperature. The status LED “OFF” indicates the Multiflex heating/cooling Module is coming to temperature. When the LED is “ON”, the temperature of the Labware adapter is within $\pm 1^{\circ}\text{C}$ of the desired temperature. It is possible to place up to four Multiflex carriers on one ML STAR instrument.



ATTENTION

Do not touch the Multiflex heating and cooling Module during run time and also not shortly after finishing a run. Wait until the module has cooled down to room temperature approximately up to 1 hour.

2.5.5 MFX Cooling-Heating Module

The MFX Cooling-Heating Module is a combined module that cools or heats plates. The temperature range for the cooling-heating module is 0°C to 99°C. An automated touch pad with a graphical user interface display is located on the top of the module to set the temperature. The touch pad illuminates once it has been turned on. MFX Cooling-Heating Module runs as a standalone unit however must be docked in the instrument. It is also possible to control the module by a library from the Venus Software. The module has a dry mode feature which dries out the instruments. Only one single labware is allowed for each degree.

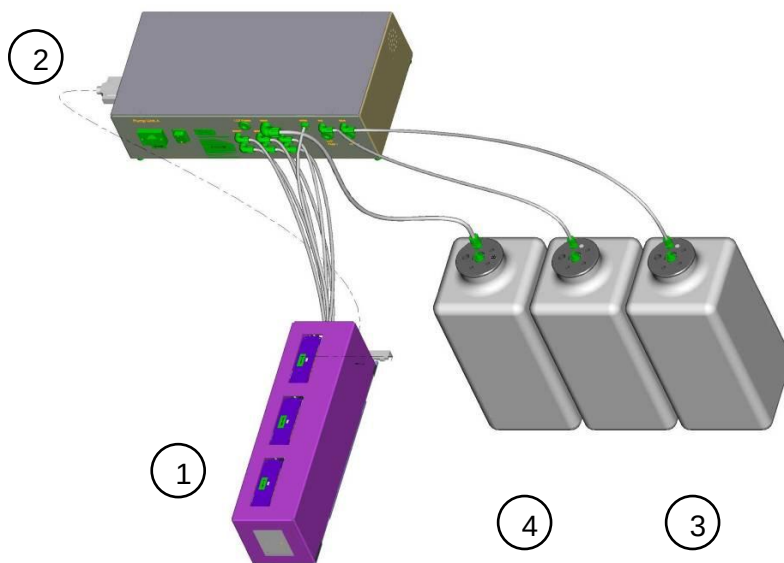


DB9 Connector Male Power

2.5.6 CR Needle Wash Station

The CR needle wash station is a device for the washing of up to 24 steel needles in parallel to the pipetting process. The wash station has the width of a standard micro-plate carrier (6 tracks), and up to two needle wash stations can be mounted on the instruments deck of the ML STAR. The carry-over of the wash station depends on the wash settings. Typical values are 10^{-5} to 10^{-6} . A set of default parameters is given for the wash process within the relevant dialog boxes of the VENUS Software. Wash parameters can be adapted.

Along with each wash station (item 1) comes a pump station (item 2) with two reservoir containers (item 3) for wash solutions and a waste container (item 4). The wash and the waste containers have a capacity of 12 liters and LLD. The pump station is placed under the bench of the instrument.

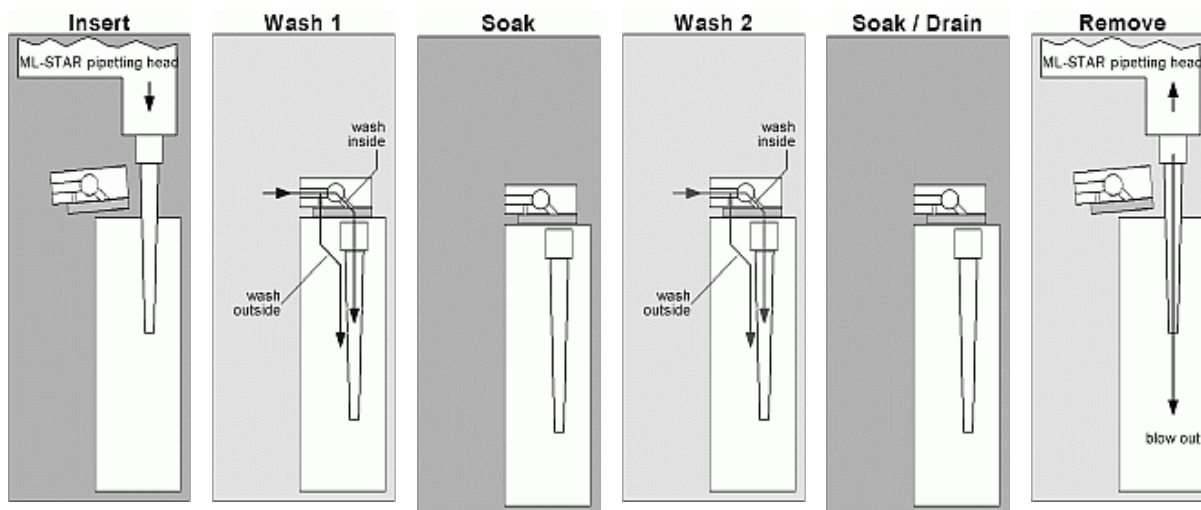


The wash station consists of three individual 8-fold wash modules. The modules are able to wash the 1000µl needles, 300µl needles or the 10µl needles. Within one wash station module all 3 needle types can be washed.

The wash cycle works in parallel to the pipetting steps. In case of an 8-channel instrument, the following steps can take place:

1. The needles from the first module are picked up; they are used for pipetting, and then placed back into the same module. Needle washing starts.
2. While the needles of the first module are washed, the needles from the second module are picked up. After pipetting these needles are placed back into the second module. The wash cycle of this module is started parallel to the next steps.
3. While the instrument starts pipetting with the needles of the third module, the first module has finalized needle washing. After pipetting the needles are placed back into the third module. Then the wash cycle of this module can start.
4. Clean needles are picked up from the first module again. The process is repeated again.

The principle of the wash station is illustrated in the following figure which shows a typical procedure.



Schematic drawing of needle wash process: needles are placed into the wash module, washed from inside and along the outside using 1) wash and 2) rinse solution. The pipetting channels blow air through the needles to expel any residual liquid.



NOTE

Disposable tips may not be used in the needle wash station.

Hamilton recommends replacing the needles every 6 months.

For a 16-channel instrument, two independent wash stations are necessary to enable high-throughput pipetting with one needle type.



NOTE

A table of chemical compatibilities with the wash station can be found in Appendix A "Chemical Compatibility". The information listed there is based on laboratory tests with raw materials and should be interpreted as a guideline only.

Consider local regulations for handling and storage of wash liquids as regards toxicity, contamination, fire protection, etc.



ATTENTION

The needle wash station for ML STAR is not explosion-proof. When working with flammable or explosive fluids or vapors, the necessary precautions must be taken.

A trained Hamilton service engineer must perform the installation of the Needle Wash Station.

2.5.7 Wash Station 96/384 Dual

The wash station 96/384 is an optional device for washing disposable tips of the CO-RE 96 and 384 Probe Heads. Low volume CO-RE tips, 50µl volume CO-RE tips, standard volume CO-RE tips, 300µl Rocket Tips, and high volume CO-RE tips without filter can be washed.



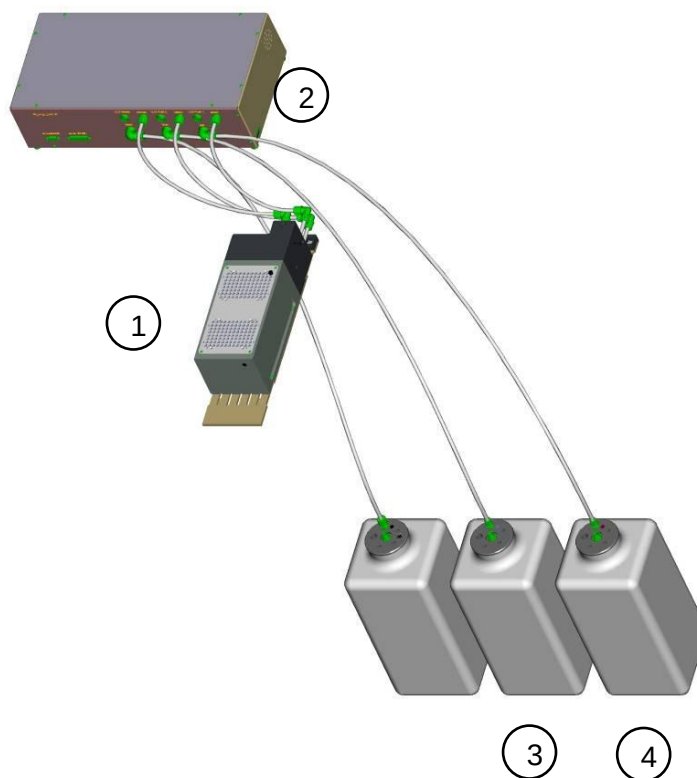
NOTE

Washing of filter tips is not recommended and not intended because there is a risk of contamination and a risk of damage to the pipetting device.

Washing takes place by aspiration/dispense cycles with the CO-RE 96 Probe Head or the CO-RE 384 Probe Head respectively.

The wash station has the width of 6 tracks and is mounted on the deck of the instrument. Two wash chambers make the process of washing tips efficient. Each chamber can be filled individually with wash liquid out of two source containers.

Along with each wash station (item 1) come a pump station (item 2) with two reservoir containers (item 3) for wash solution and a waste container (item 4). Both the wash and the waste container have a capacity of 12 liters LLD monitoring. The pump station is placed under the bench of the instrument.



The pump unit includes two wash pumps and a waste pump. The wash pumps fill the wash chambers in the washer unit. The waste pump empties the wash chamber and/or the overflow chamber.



NOTE

When re-filling the wash liquid container the waste liquid container must be emptied as well.

When re-using washed tips, pipetting precision may decrease.



ATTENTION

Only use water or DMSO max. 30% as wash liquid for the wash station.



ATTENTION

The wash station for ML STAR instruments is not explosion-proof. When working with flammable or explosive fluids or vapors, the necessary precautions must be taken.

**NOTE**

Older versions of wash stations, i.e. the 96 chamber wash station and the 384 dual chamber wash station are tip pattern specific!

If 50µl tips of the 384 Probe Head need to be washed in these older versions, the 384 dual chamber wash station must be used.

If Rocket Tips need to be washed in these older versions, the 96 chamber wash station must be used.

There is the possibility to field-upgrade an old version Wash Station to the Wash Station 96 / 384 Dual. Please consult a local Hamilton representative for this possibility.

2.5.8 Temperature Controlled Carrier (TCC)

The temperature controlled carrier (TCC) is a device for heating and cooling micro-plates. It is designed to function as a drawer. The TCC has a capacity of four positions for micro-plates. The proceeded temperature is the same on all positions. The heating function of the TCC reaches temperatures of up to 60°C, while cooling Microtiter Plates down to 22°C below ambient temperature is possible. Up to two TCCs can be installed on one instrument.

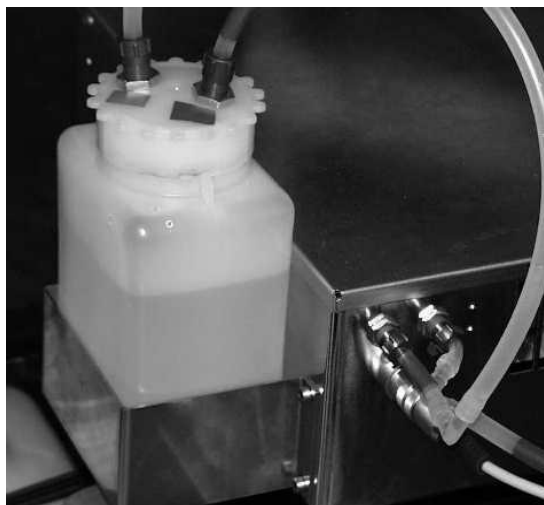
TCC consists of the heating/cooling plate carrier fixed on the instrument deck. The external heat exchanger device is placed below the bench. The working fluid in the refrigeration cycle is a synthetic liquid.

Typical times (at 40% rel. humidity) to heat and cool a Microtiter Plate are ($T_{\text{ambient}} \approx 20^{\circ}\text{C}$):

T_{ambient} to 60°C	~ 20 min
60°C to T_{ambient}	~ 20 min
T_{ambient} to 4°C	~ 15 min
4°C to 60°C	~ 25 min



The temperature controlled carrier (TCC)



The heat exchanger liquid reservoir.



NOTE

Ensure there is always enough liquid (minimum 1l) within the reservoir.

Allow air exchange between the exchanger and ambient air.

The default position of the TCC is loaded on the instrument deck. Always ensure that the TCC is loaded on the deck at the beginning of the loading process. Never leave the TCC unloaded on the loading tray.

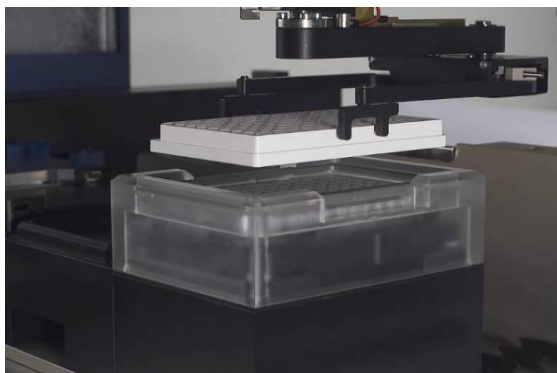
Given the TCC's fix-mounted chassis and movable carrier slide, do not use Smart Steps to load a TCC with the Autoload function. The Autoload slide may collide with the TCC carrier slide.

The installation of the temperature controlled carrier (TCC) must be performed by a trained Hamilton service engineer.

2.5.9 Crystal Vacuum System

The Crystal Vacuum System (CVS) replaces the older Basic Vacuum System (BVS) and allows automation of vacuum based kits for SPE, LC-MS, genomics and proteomics. CVS consists of a 7-track-wide carrier base equipped with the vacuum manifold, the park position for the manifold top and two Microtiter Plate positions. By default the manifold top and insert for standard filter plates are included. For processing deep well filter plates, the deep well kit is required. The DWP Kit can be found in [Appendix B "Ordering Information"](#).

The manifold top can be handled either by the iSWAP (see [Chapter 2.4.7](#)) or the CO-RE gripper (see [Chapter 2.4.12](#)). If working with the iSWAP, four tracks next to the CVS carrier have to be empty, usually on the right side.



The iSWAP in action to load a filter plate to the CVS

The CVS accommodates a wide variety of 96-well and 384-well filter plates. With the height adjustable inserts inside the vacuum chamber, almost any kind of elution plates can be used for the elution. Four sets of nuts and bolts are included for adjusting the height as required.

The CVS carrier is mounted on the instrument deck. It is possible to place up to four CVS carriers on one instrument.



NOTE

The Crystal Vacuum System is dependent on the elution plate's depth. If it happens to be plunged too deep, there is a need for it to be handled with the CO-RE grip.

“standard” formats:



deep formats:



2.5.9.1 Crystal Vacuum System Installation

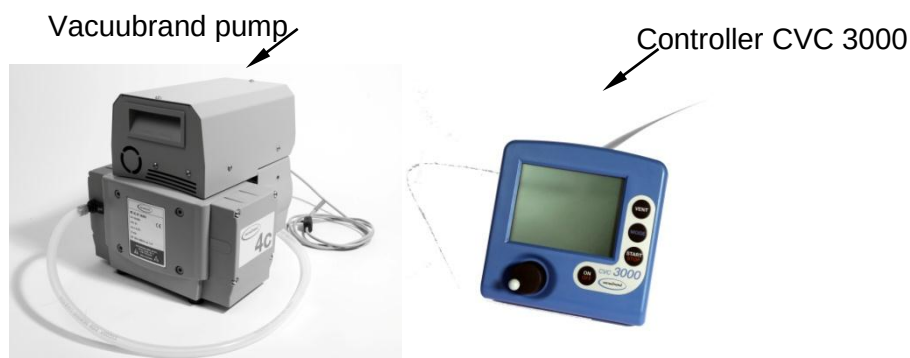
A trained Hamilton service engineer must perform the installation of the CVS Vacuum System.

The vacuum is generated with the Vacuubrand pump. The pump unit is fully integrated into the VENUS Software. For precise control the vacuum inside the chamber of the CVS can be monitored.

Vacuubrand pump

The vacuum inside the chamber of the CVS is generated with the Vacuubrand pump. The maximum flow rate of the pump is $\sim 5 \text{ m}^3/\text{h}$, and the possible final vacuum $\sim 80 \text{ mbar}$. There is a built-in sound absorber that reduces the noise level of the pump considerably.

The vacuum controller CVC 3000 regulates the pump. Communication with the computer is done via RS232 cable, and its software is seamlessly integrated into the VENUS Software.



The air-bleed valve and the pressure sensor are both mounted inside the CVS carrier.

Waste bottle for BVS / CVS

The CVS has to be connected either to a suitable in-house liquid disposal system or to a Hamilton waste bottle which collects liquid that is extracted during the vacuum step. The waste bottle is connected to the vacuum chamber and to the pump unit. Waste bottles come assembled with a bucket which prevents spillage in case of leakage.

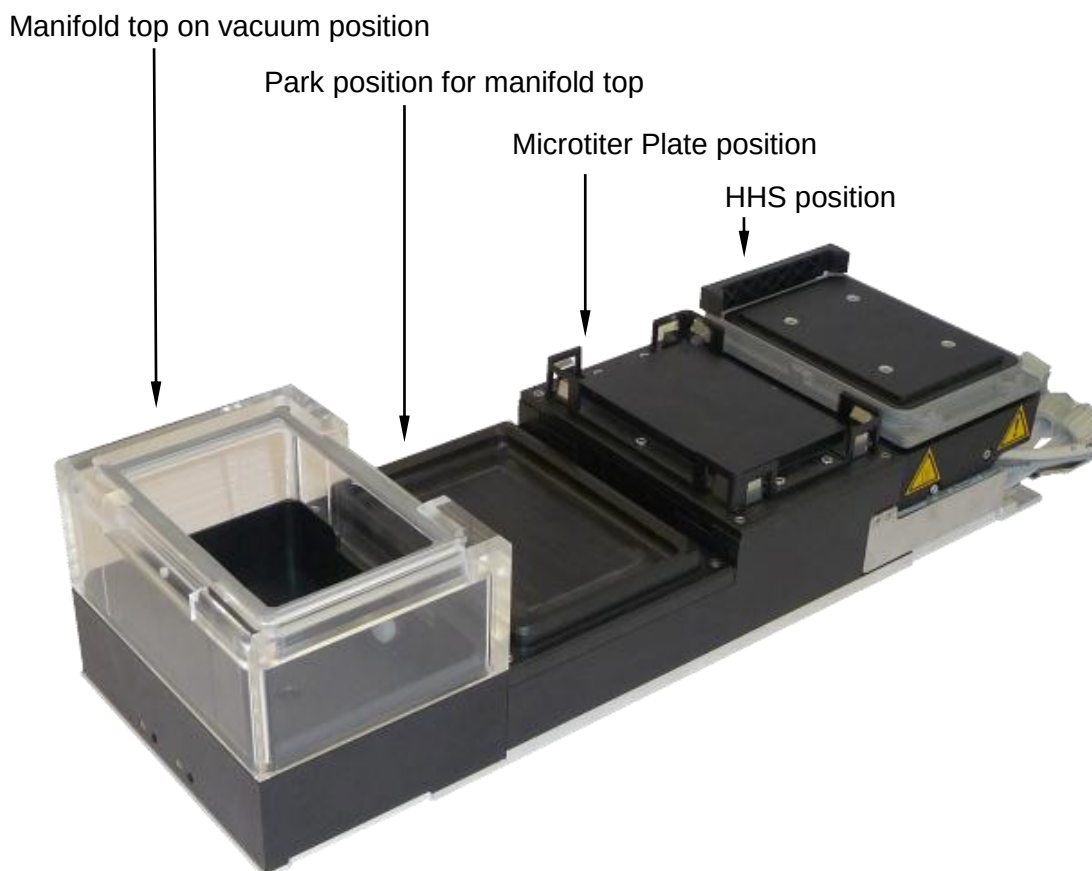


BVS / CVS waste bottle, 2L and 4L, in bucket

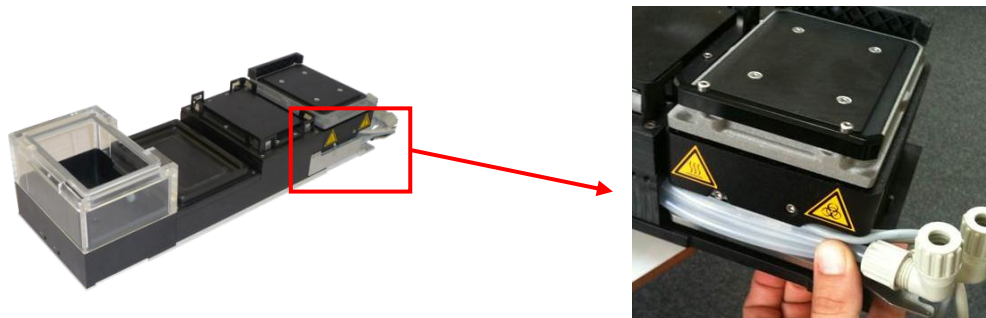
The BVS / CVS waste bottles are not part of a CVS, they have to be ordered separately. Please refer to the [Appendix B “Ordering Information”](#). Two sizes of waste bottles are available, with volumes of two or four liters. The waste bottle should be placed below the instrument.

2.5.9.2 Shaker on the CVS

Optionally, the rear Microtiter Plate position can be replaced by a Hamilton Heater Shaker or by third party heater shakers. The list of available modules can be found in [Appendix B “Ordering Information”](#).



The heater shaker will be installed by trained service engineers. The CVS comes with a metal plate for fixation of cables and tubing of a heater shaker, it must be retained in a safe place, including screws for mounting, for a later retrofit of a heater shaker.



Position of the metal plate on the CVS for fixation of shaker cables and tubing

HHS, tubing and cables, shown before fixation by the metal plate

2.5.10 HEPA Filter Hoods

The ML STAR Line instruments can be retrofitted with a HEPA (High Efficiency Particulate Air) Filter Hood for air filtration. The HEPA Filter Hood creates a laminar flow of filtered air from the top of the instrument to the deck surface. Room air is drawn in from the top and forced through a one stage filter. The air is discharged from the filters over the pipetting area inside the instrument and exits through the opening in front and vents in the acrylic glass cover on the back of the workstation to protect the assay from the environment outside the workstation. The flow rate of 0.3 m³/s is comparable to that of other commercially available flow cabinets making adaptations of the instrument for work under aseptic conditions possible, e.g. for mammalian cell culture.

Form fitting hoods are available for all three sizes of the STAR. The two HEPA Filter Hoods mounted on a ML STARplus can be controlled individually.



*Microlab STARlet
with HEPA Filter
Hood*



*Microlab STAR
with HEPA Filter Hood*



*Microlab STARplus
with HEPA Filter Hood*

Class II isolation enclosures are also available from third party supplies to enclose the entire workstation if the environment outside the workstation needs to be isolated from the assay.

Air flow volume can be controlled by the software running the workstation to alter conditions while running the assay. The on/off switch is mounted to the hood itself as well.

Volumetric flow	Software settings
89.39 cfm 2.53m ³ /min	20%
380.65 cfm 10.78m ³ /min	50%
832.63 cfm 23.58m ³ /min	100%



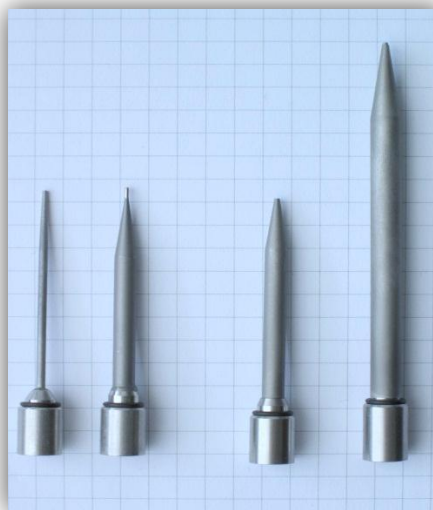
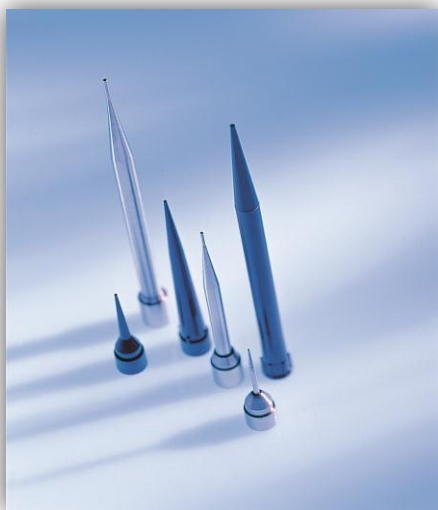
NOTE

The Hamilton HEPA Filter Hood is designed to draw ambient air in through the filter assembly and to provide an enclosed, 'clean air zone' within the ML STAR Line instrument workstation pipetting environment. For more information about the HEPA Filter Hood, please refer to the HEPA Filter Hood Manual.

A UV light option is also available to further reduce germ count inside the instrument, e.g. before and after a working day. When retro-fitting the UV option, the instrument covers should be exchanged from those made of standard acrylic glass to such made of Makrolon[®], due to its better resistance to UV irradiation.

2.6 Tips and Needles

Hamilton offers a large variety of disposable tips and steel needles for use with the channels and MPH heads.



Hamilton disposable tips and steel needles for the ML STAR Line.

Left to right: low volume (10µl), 50µl, standard-volume (300µl), and high-volume (1000 µl) disposable tips; 1000µl, 300µl, 10µl steel needles.

Hamilton's CO-RE tips are the only tips designed to work with the ML STAR Line instruments developed to meet the demands of the most flexible and reliable automated pipetting tasks. Channels lock on to the tip rather than the standard method of forcing a tip on a mandrel (see [Chapter 2.1.2](#) for the CO-RE technology). This design assures superior tip alignment, seals the tip to the pipette channel, eliminates tip distortion, and mitigates aerosol contamination.

The use of non-Hamilton tips is not supported on the ML STAR Line instruments. Non-Hamilton tips do not have the annular space to receive the O-ring. This may result in non-intended deformation of the O-ring and the tip material itself. It may lead to misalignment of the tip, improper sealing, reduced O-ring life, improper drop-off, and potential aborted runs from randomly dropped tips. Therefore, Hamilton does not support the use of non-Hamilton tips on the ML STAR pipetting platforms.



ATTENTION

Only Hamilton needles and disposable tips should be used for coupling to the pipetting channels and pipetting heads of the ML STAR instrument. Other tips may cause contaminated or damaged pipetting channels.



NOTE

Pipetting specifications are only guaranteed when using Hamilton Tips.

Hamilton's proprietary technologies for tip attachment (CO-RE) and air displacement pipetting together with the use of Hamilton tips facilitate maximum pipetting accuracy and reliability.

Hamilton disposable CO-RE tips are produced under clean room conditions (class 8), based on ISO 14644 standards. "Biological purity tested" tips are free of DNA, DNase/RNase, PCR inhibitors and endotoxin (non-pyrogenic). In addition to these criteria, "Biological purity^{PLUS}" tips are sterile, according to ISO 11135, and free of ATP.

Filter tips are available for preventing aerosol contamination.

Clear tips are available as well as conductive (black) tips specifically designed for cLLD.



NOTE

Make sure to match the tip type(s) used with the method and channel(s) / probe head(s). Please refer to Chapter 2.1.3 "Tip Recognition".



NOTE

Be aware that different tip classes (with and without filter, black or clear, non-sterile versus irradiated etc.) can have different performance when pipetting liquids. Pipetting parameters and liquid class settings may have to be adjusted when changing from one type to another in order to achieve accuracy and precision which is within published specifications.

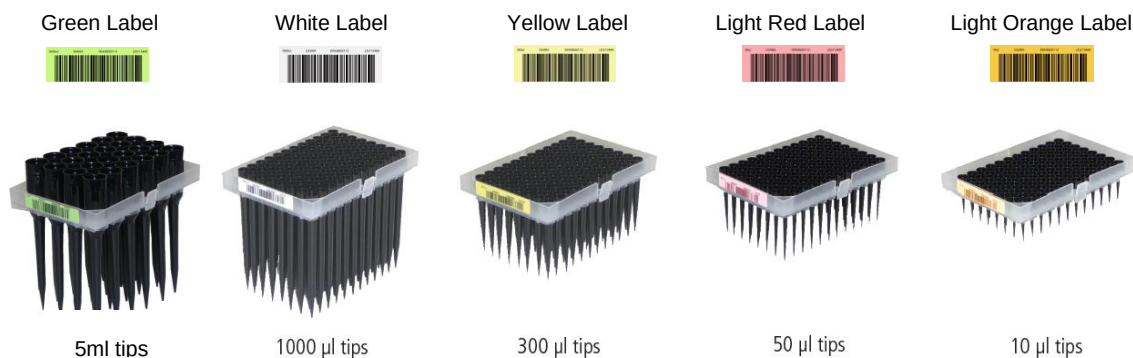
**NOTE**

All new or special tip types require additional settings (such as configuration file entries, liquid classes etc.) in the VENUS Software. Please consult a local Hamilton representative for implementation of non-standard tips.

For ordering information see [Appendix B “Ordering Information”](#), or contact a local Hamilton representative.

2.6.1 Tip Packaging

All CO-RE disposable tip types are available in trays with a sealed paper lid (blister pack). The tip trays are barcode labeled for automatic identification during the loading process. All instruments equipped with a barcode reader are able to check for the proper tip type loading.



Examples of barcode labeled tip trays

**NOTE**

The tip type is printed in plain text on the barcode label of the tip rack for visual identification. E.g. “5ml” for the 5ml CO-RE tips.

Generally, blister packs contain several trays of 96 tips each. 4000/5000µl tips come in blister packs containing trays of 24 tips each. CO-RE tips are available in boxes of 3840 high-volume tips and 5760 standard volume (300µl), 50µl or low-volume tips (10µl).

The trays are compatible with the respective tip carriers.



Tip carrier TIP_CAR_480 containing five tip trays

10µl, 50µl and 300µl CO-RE disposable tips without filter are stackable. These tip types are available in Nestable Tip Racks (NTR). This high packing density and high number of tips make longer runs possible without reloading procedures.



Nestable Tip Racks (NTRs)



Five stacks of four NTRs each

The Nestable Tip Rack (NTR) is designed to stack disposable tip racks which increase the amount of disposable tips by factor 4 per SBS position compared to one layer tip locations. NTRs come with 5 stacks of 4 layers each and have to be placed onto NTR Carriers of the ML STAR Line instrument.

Tip pick up directly from loaded NTR Carriers is basically possible. This eliminates the need for intermediate tip pickup positions since pipetting tools such as single channels 1000ul as well as MultiProbeHeads (MPH 96 and MPH 384) can pick up tips from the NTR stack in most cases.

A transport system (e.g. iSwap, CO-RE Gripper) is required to remove and dispose empty NTR frames away from NTR Stacks to allow access with the Pipetting Tools to the next lower layer with filled NTRs.

Forcing Pipetting Tools to go down to lower or to the lowest layer of the NTR stack may have limitations due to the required space between the Pipetting Tool and the neighbouring Carriers as well as its corresponding labware (also to other stacks, neighbour NTR's, etc.). In these cases, an intermediate Tip pick up position becomes necessary. This intermediate tip pickup

position is designed and available as an NTR MFX Module (MultiElexModule, MULTIFLEX NTR 96 MODULE, and MULTIFLEX NTR 384 MODULE).

The transport system as mentioned above must then be used to move NTRs to these intermediate Tip Pickup positions which enable constant tip pickup conditions in terms of the same X, Y and Z coordinates of a single NTR. Once the Tips out of the NTR frame of the intermediate tip pickup position are used up, the frame needs to be replaced by a new filled NTR.

When single-, row- or column wise Tip pickup is required with MultiProbeHeads, an additional tip pickup position equipped with a Tip adapter becomes necessary. The tip adapter may be positioned onto a Tip Rack Carrier as well as on a MFX Tip Module (MULTIFLEX TIP MODULE).

The recommended workflow would be:

1. Transportation of one NTR from a loaded NTR Carrier to the intermediate Tip pickup position
2. Tip pickup with MPH out of the intermediate Tip pickup position
3. Tip eject into the Tip adapter
4. As desired; single-, row- or column wise (shifted) Tip pickup out of the Tip adapter
5. Ready for pipetting

2.6.2 Tips for Pipetting Low Volumes from 0.5-10 μ l

The low volume CO-RE tip, 0.5-10 μ l, is available as clear tip and conductive (black) optic tip for use with cLLD. It can be used with 1000 μ l-channels and CO-RE 96 Probe Head / CO-RE 96 Probe Head TADM.



*Clear and conductive low volume CO-RE tip,
0.5-10 μ l*



Light Orange



*Tip tray and color code label
for low volume CO-RE tips, 0.5-10 μ l*

Low volume CO-RE tips, 0.5-10 μ l come in trays of 96 tips. One blister pack contains 5 trays. They are also available in the NTR Tip Rack Pack reload system with five stacks of four 96 tip racks in a blister pack.

2.6.3 Tips for Pipetting Sample Volumes from 1-50µl

The low volume CO-RE tip, 1-50µl, is available as conductive (black) tip for use with cLLD, with and without filter and also as clear tip. It can be used with 1000µl-channels and CO-RE 96 Probe Head / CO-RE 96 Probe Head TADM.



Clear and conductive low volume CO-RE tip, 1-50µl



Light Orange



Tip tray and color code label for low volume CO-RE tips, 1-50µl

Low volume CO-RE tips, 1-50µl come in trays of 96 tips. One blister pack contains 5 trays. They are also available in the NTR Tip Rack Pack reload system with five stacks of four 96 tip racks in a blister pack.



ATTENTION

Caution when using the CO-RE tips, 1-50µl and CO-RE tips, 10-300µl on the same system. Do not mix up one tip type with the other during the process of loading the deck. The wrong tip causes either bad pipetting results or damaged pipetting channels.

Use the Autoload function for loading tip carriers or the tip type recognition feature of the VENUS Software. Please consult the local Hamilton representative for the specific library which is needed.

2.6.4 Tips for Pipetting Sample Volumes from 0.5-50µl Using 384 MPH

The 384 head CO-RE tip, 0.5-50µl, is available as clear tip and conductive (black) tip for use with cLLD. It can be used with the CO-RE 384 Probe Head.



Clear and conductive 384 head CO-RE tip, 0.5-50µl



Light Red



NTR with 384 (left) and 96 (right) 384 head CO-RE tips, 0.5-50µl

Color code label (lower)

384 head CO-RE tips, 0.5-50 μ l come in NTRs of 384 or 96 tips per rack. The NTRs equipped with 96 tips per rack enable using the CO RE 384-Probe Head as a 96 channel pipetting head. The NTR pack reload system contains five stacks of four tip racks in a blister pack.



NTR pack containing five stacks of four 384 head CO-RE tip racks

2.6.5 Tips for Pipetting Sample Volumes from 10-300 μ l

The CO-RE tip, 10-300 μ l, is available as conductive (black) tip for use with cLLD, with and without filter and also as clear tip. It can be used with 1000 μ l-channels and CO-RE 96 Probe Head / CO-RE 96 Probe Head TADM.

It is compatible with Monitored Air Displacement (MAD) and Total Aspiration and Dispensing Monitoring (TADM)



*Clear and conductive CO-RE tip,
10-300 μ l*



*Tip tray and color code label
for CO-RE tips, 10-300 μ l*

CO-RE tips, 10-300 μ l come in trays of 96 tips. One blister pack contains 5 trays. They are also available in the NTR Tip Rack Pack reload system with five stacks of four 96 tip racks in a blister pack.



ATTENTION

Caution when using the CO-RE tips, 1-50 μ l and CO-RE tips, 10-300 μ l on the same system. Do not mix up one tip type with the other during the process of loading the deck. The wrong tip causes either bad pipetting results or damaged pipetting channels.

Use the Autoload function for loading tip carriers or the tip type recognition feature of the VENUS Software. Please consult the local Hamilton representative for the specific library which is needed.

2.6.6 Tips for Pipetting Sample Volumes from 10-1000µl

The CO-RE tip, 10-1000µl, is available as conductive (black) tip for use with cLLD, with and without filter. It can be used with 1000µl-channels and CO-RE 96 Probe Head / CO-RE 96 Probe Head TADM.

It is compatible with Monitored Air Displacement (MAD) and Total Aspiration and Dispensing Monitoring (TADM).



*Conductive CO-RE tip,
10-1000µl*



*Tip tray and color code label
for CO-RE tips, 10-1000µl*

CO-RE tips, 10-1000µl come in trays of 96 tips. One blister pack contains 5 trays.



ATTENTION

Caution when using the CO-RE tips, 10-1000µl and Slim tips, 10-300µl on the same system. Do not mix up one tip type with the other during the process of loading the deck. The wrong tip causes either bad pipetting results or damaged pipetting channels.

Use the Autoload function for loading tip carriers or take extra care and identify the tip type by reading the label on the barcode when loading tip carriers manually.

2.6.7 Tips for Pipetting Sample Volumes from 50-5000µl

The CO-RE tip, 50-5000µl, is available as conductive (black) tip with and without filter. It can be used with 5ml-channels.

It is compatible with Monitored Air Displacement (MAD) and Total Aspiration and Dispensing Monitoring (TADM).



*Conductive CO-RE tip,
50-5000µl*



*Tip tray and color code label
for CO-RE tips, 10-1000µl*



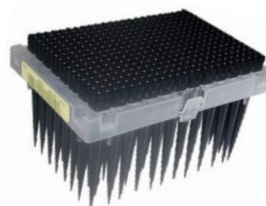
CO-RE tips, 50-5000µl come in trays of 24 tips. One blister pack contains 5 trays. In addition, individually wrapped, preloaded 1T disposable racks of 4 tips each are available in which the tips are physically separated by individual sheaths. They ensure contamination free tip loading.

2.6.8 Special Tips: Rocket tip

The unique Rocket tip, 2-300µl, fits to four channels of a CO-RE 384 Probe Head in parallel and turns it into a 96 pipetting head on the fly. It is available as conductive (black) tip for use with cLLD. Rocket tips can be used with Deep Well Plates and Microtiter Plates with well diameters $\geq 3.7\text{mm}$.



Rocket tip, 2-300µl



*Tip tray and color code label
for Rocket tips, 2-300µl*

Rocket Tips, 2-300µl come in trays of 96 tips per rack. One blister pack contains 5 trays.

2.6.9 Special Tips: Piercing Tip

The robotic piercing tip, 2-250µl is optimized for use with various sealed labware which can be pierced by applying a force of ≤ 10 N. The special tip design is compatible with most commercially available seals.

The tip is available as conductive (black) tip for use with cLLD, with and without filter. It is compatible with Monitored Air Displacement (MAD) and Total Aspiration and Dispensing Monitoring (TADM).



Piercing tip, 2-250µl



*Tip tray and color code label
for piercing tips, 2-250µl*

Piercing tips, 2-250µl come in trays of 96 tips per rack. One blister pack contains 5 trays.

For piercing of tubes additional equipment is required to hold the tubes in their position during the piercing process and to prevent them to be lifted up when retracting the piercing tip. The fixation frame for this purpose is dependent on the labware used. It is mounted onto a MTP park position (P/N 188041) on a Multiflex carrier (P/N 188039). The standard fixation frame (P/N 188313) is designed for 1.4ml matrix tubes. For all other tube types, a customized fixation frame is required.

For piercing of plates the Multiflex MTP fixation frame (P/N 188095) helps locking the sealed plates in their position on a Multiflex carrier during piercing.



*Fixation frame for 1.4ml matrix tubes
on a MTP park position*



*Multiflex MTP fixation frame
on a MTP park position*

**NOTE**

In general, piercing is a challenge. It requires a fixation frame for the labware to be pierced and it is highly dependent on the shape of the needle or tip and the material and thickness of the seal. It is crucial that any piercing method is tested first before routine application.

2.6.10 Special Tips: Extra-long and Slim Tip

Extra-long and slim tips minimize the risk of liquid overflow in cluster tubes, Deep Well Plates and two-phased liquid applications. The Slim tip, 10-300µl reduces the displaced liquid volume by 40%, compared to a CO-RE tip, 10-1000µl. The tip is available as conductive (black) tip for use with cLLD, with and without filter. It can be used with 1000µl-channels and CO-RE 96 Probe Head / CO-RE 96 Probe Head TADM.

The Slim tip is compatible with Monitored Air Displacement (MAD) and Total Aspiration and Dispensing Monitoring (TADM).



Slim tip, 10-300µl



Tip tray and color code label for Slim tip, 10-300µl

Slim tips, 10-300µl come in trays of 96 tips per rack. One blister pack contains 5 trays.

**ATTENTION**

Caution when using the CO-RE tips, 10-1000µl and Slim tips, 10-300µl on the same system. Do not mix up one tip type with the other during the process of loading the deck. The wrong tip causes either bad pipetting results or damaged pipetting channels.

Use the Autoload function for loading tip carriers or take extra care and identify the tip type by reading the label on the barcode when loading tip carriers manually.

2.6.11 Special Tips: Wide Bore Tips

For applications which require minimizing shear forces applied to the samples during pipetting, for highly viscous fluids or for particulate samples, Hamilton has designed CO-RE tips with various wide distal end orifices. The CO-RE tip wide bore 10-300 μ l with a length of 57.23mm which orifices size 0.71mm is a filter tip. The CO-RE tip wide bore 10-300 μ l with a length of 51.60mm which orifices size 1.55mm and the CO-RE tip wide bore 10-1000 μ l with a length of 80.14mm which orifices size 3.2mm are available with or without filter. Wide bore tips are conductive (black) for use with cLLD and they can be used with 1000 μ l-channels and CO-RE 96 Probe Head / CO-RE 96 Probe Head TADM.

Wide bore tips are compatible with Monitored Air Displacement (MAD) and Total Aspiration and Dispensing Monitoring (TADM).

For more information about the wide bore tips, please refer to the Disposables Catalog.



Wide bore tips come in trays of 96 tips per rack. One blister pack contains 5 trays.

2.6.12 Needles

Steel needles can be used for pipetting instead of disposable tips in combination with a needle wash station when the application does not require an exchange of a tip. Needles are rinsed between pipetting tasks and are resistant to many chemicals (refer to [Appendix A: Chemical Compatibility](#)).

10 μ l needles, 300 μ l needles and can 1000 μ l needles are available for use with 1000 μ l-channels.



NOTE

Hamilton recommends replacing steel needles every six months.

2.7 Computer Requirements

The ML STAR Line instruments are controlled by the dedicated VENUS Software which controls all functions for daily work routine, method programming, running methods, and other services.

Hamilton offers standard PCs for the VENUS Software. For specifications, please refer to [Chapter 7.2 “Hamilton standard PCs”](#). For ordering information, please refer to [Appendix B “Ordering Information”](#).



NOTE

Computer requirements as well as operating systems described in this manual are subject to change without notice, respectively may become obsolete.

Depending on the ML STAR instrument configuration and processor board implemented, the following interfaces are available for linking the control computer (see [Chapter 2.9 “Electronics and Interfaces”](#)):

- a serial interface (RS-232C, with dual processor board), or
- a Unified Serial Bus interface (USB, with dual processor board and LAN dual processor board), or
- Ethernet (with LAN dual processor board)

The communication interface used on the PC has to be set by the configuration editor. For further information about the recommended PC model refer to the [Chapter 7.1 “Basic ML STAR Line Specifications”](#)

To avoid data loss, using of a UPS (uninterruptible power supply) for the PC is recommended.

2.8 Microlab STAR Line VENUS Software

The VENUS Software provides everything to control the ML STAR instruments.

It is a WindowsTM-based, menu-driven interface allowing the user to define deck layouts and methods, and then to run the instrument.

The VENUS Software allows programming and running different methods for aspirating and dispensing liquids, also to control accessories such as a wash station, etc.

It is possible to upgrade from previous software versions to VENUS three. Software upgrade requires compatible firmware.



NOTE

ML STAR Line's functioning has been verified using Windows 7. Running the instruments under any other operating system may lead to severe problems and/or malfunction.

Each programmed method should be validated by the programmer.

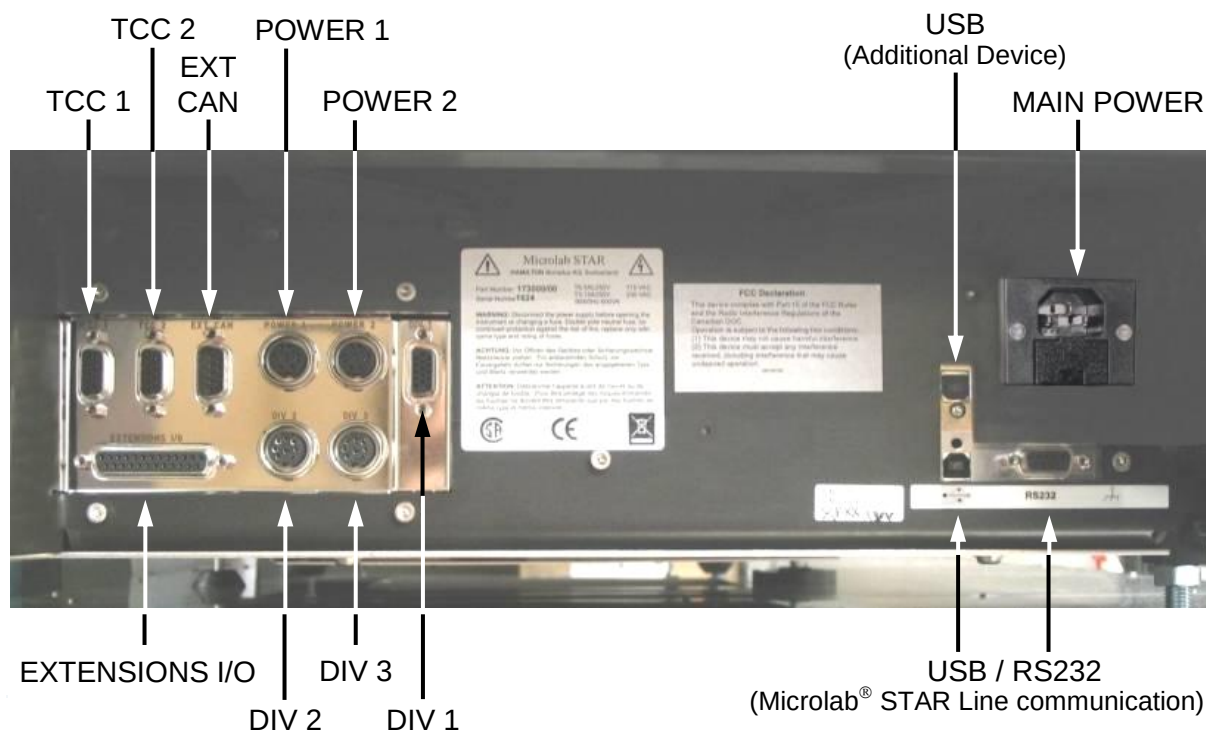
For more details refer to the [Microlab STAR Programmer's Manual \(P/N 624043\)](#).

If not using a Hamilton computer, the customer needs to ensure that a suitable control PC is available for installation of the VENUS Software. The ML STAR instrument will be unpacked and installed and the initial set-up will be performed by a trained Hamilton service engineer according to IQ form 610911.

2.9 Electronics and Interfaces

All the electrical connections are placed on the left side of the instrument, as shown below:

2.9.1 Interfaces of Dual Processor Board



The main power connection is located near the front of the instrument (on the right side in the picture above). The communication connections to the PC are positioned beneath this main power connection. The ML STAR Line can communicate either via USB (the preferred option) or via RS232.



ATTENTION

Never use both connections, USB and RS232 together!



ATTENTION

Do not attempt to install a 3rd party device via the electrical connections of the ML STAR instrument on your own! Always consult a local Hamilton representative for installation.

A second USB connector is reserved for an additional device, e.g. a CCD camera (optional).

Various connections for external devices are available. Up to two TCCs (temperature-controlled carriers) can communicate and be powered via the connectors labeled "TCC 1", "TCC 2" and also Hamilton Heater Shaker.

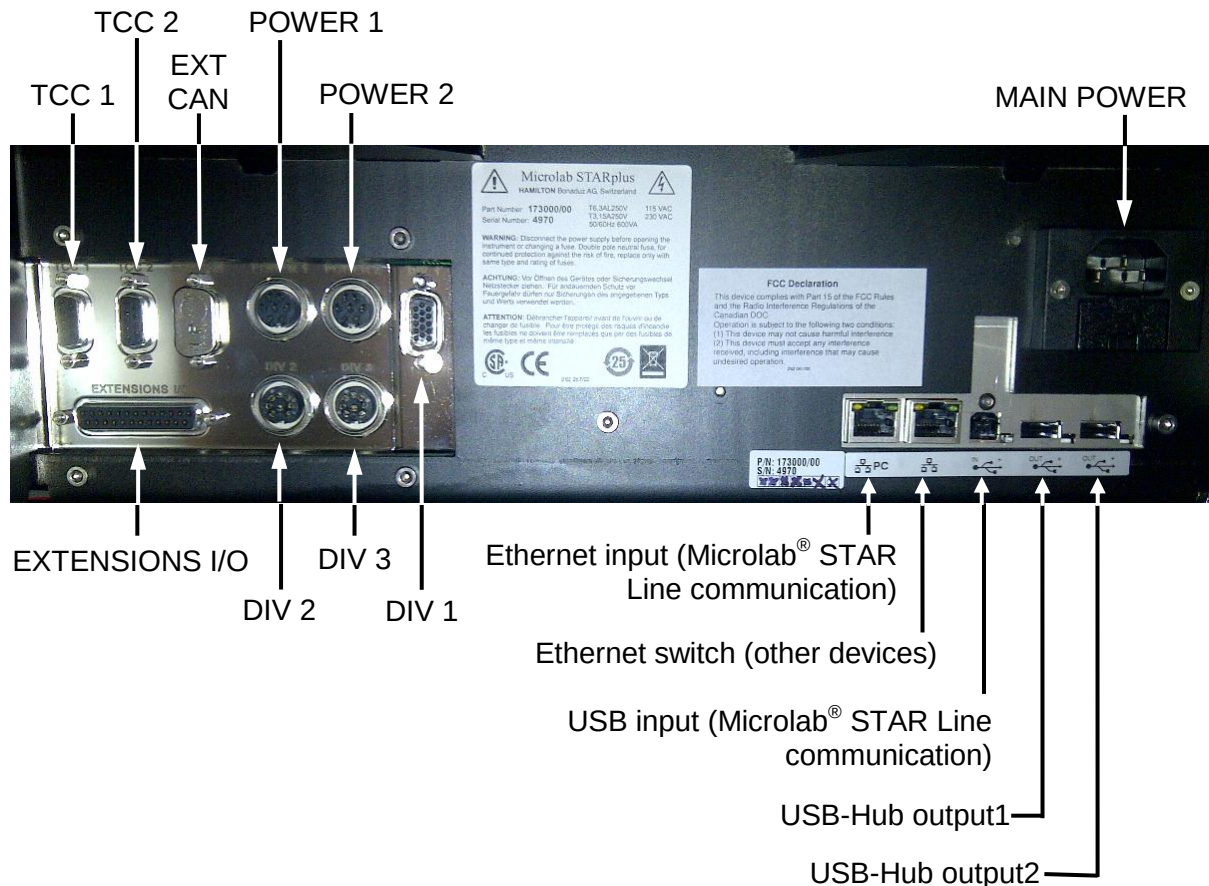
The CR needle wash station is being connected with the connector "EXT CAN".

The connectors labeled "POWER 1" and "POWER 2" provide different power supply voltages and also a CAN bus for communication.

The connectors "DIV 1", "DIV 2", "DIV 3", and "EXTENSIONS I/O" deliver several digital input/output signals as well as pulse-width-modulated (PWM) outputs, CAN bus and the TTL levels.

2.9.2 Interfaces of LAN Dual Processor Board

Depending on configuration, some ML STAR instruments come configured with a LAN dual processor board (e.g. in combination with a CO-RE 96 Probe Head TADM). The electrical connections of these instruments are as shown below:





NOTE

With CO-RE 96 Probe Head TADM installed there is no RS232 plug available anymore.

For further details on all connections, please consult a local Hamilton representative.

2.10 Power / Voltage

2.10.1 ML STAR Line Instrument

Make sure that the instrument is connected to a 115 or 230 VAC (50 or 60 Hz) socket. The instrument can be used with any voltage within that range.

We recommend using an uninterruptible power supply (UPS) for the instruments.

Ensure that the instrument is correctly grounded when connected to the electrical outlet.

The main plug is on the left-hand side of the instrument at the front.

The fuses for the instrument are placed in the main power socket (see picture below). During installation / IQ, the appropriate fuse is selected by the service engineer.

Plug the main cables for the computer and the instrument into the same electrical outlet. Connect them only to a grounded outlet.



ATTENTION

When replacing a fuse make sure to use the appropriate fuse (see [chapter 7](#) “[Technical Specifications](#)”) and place it in the main power switch before switching on the instrument.

2.10.2 Needle Wash Station

The pump unit has its own power supply. The main plug is on the rear of the pump station (see picture below).

Ensure that the needle wash station is correctly grounded when connected to the electrical outlet. Connect the wash station only to a grounded outlet.

Ensure that the voltage selector at the pump station of the needle wash station is correctly set before operating the needle wash station. The needle wash station does not work with varying voltage (115 – 230 VAC) automatically.

The fuses for the instrument are located in to the main socket (see picture below). The pump station has two fuses for the power supply which can be accessed by opening the cover underneath the main socket.

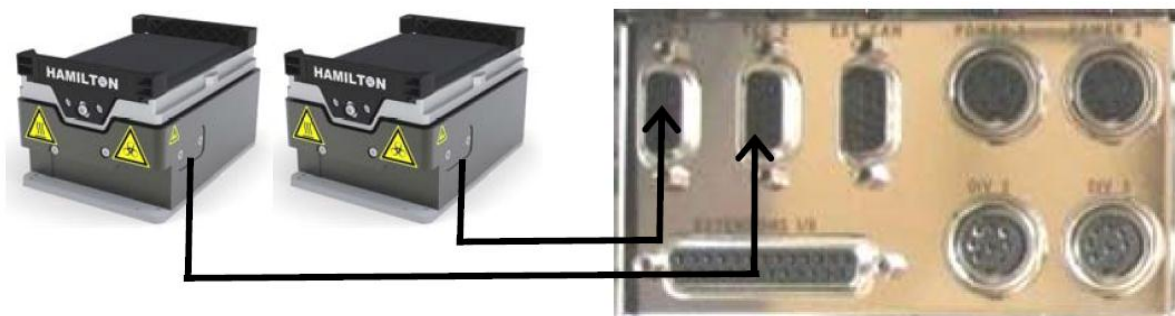


The technical specifications regarding electrical power and fuses to be used for the wash station are listed in Chapter 7.4 “CR Needle Wash Station”

2.10.3 Hamilton Heater Shaker

2.10.3.1 Up to 2 HHS

It is possible to integrate one or two HHS directly into a ML STAR instrument (via TCC connector, see picture below). The power and communication wire run inside the same cable. Both HHS are fully controlled by the ML STAR Software.



Up to two HHS connect to the TCC connector of the periphery connector board of the ML STAR instrument.



ATTENTION

Before connecting or disconnecting the HHS from or to a ML STAR instrument make sure that the ML STAR is switched off. Only when the ML STAR is switched off it is safe to connect or disconnect the HHS. Not obeying this instruction may result in damage of the HHS.

Do not touch the HHS by hand during run time and also not shortly after finishing a run, as it might be hot. Wait until the HHS has cooled down to room temperature.



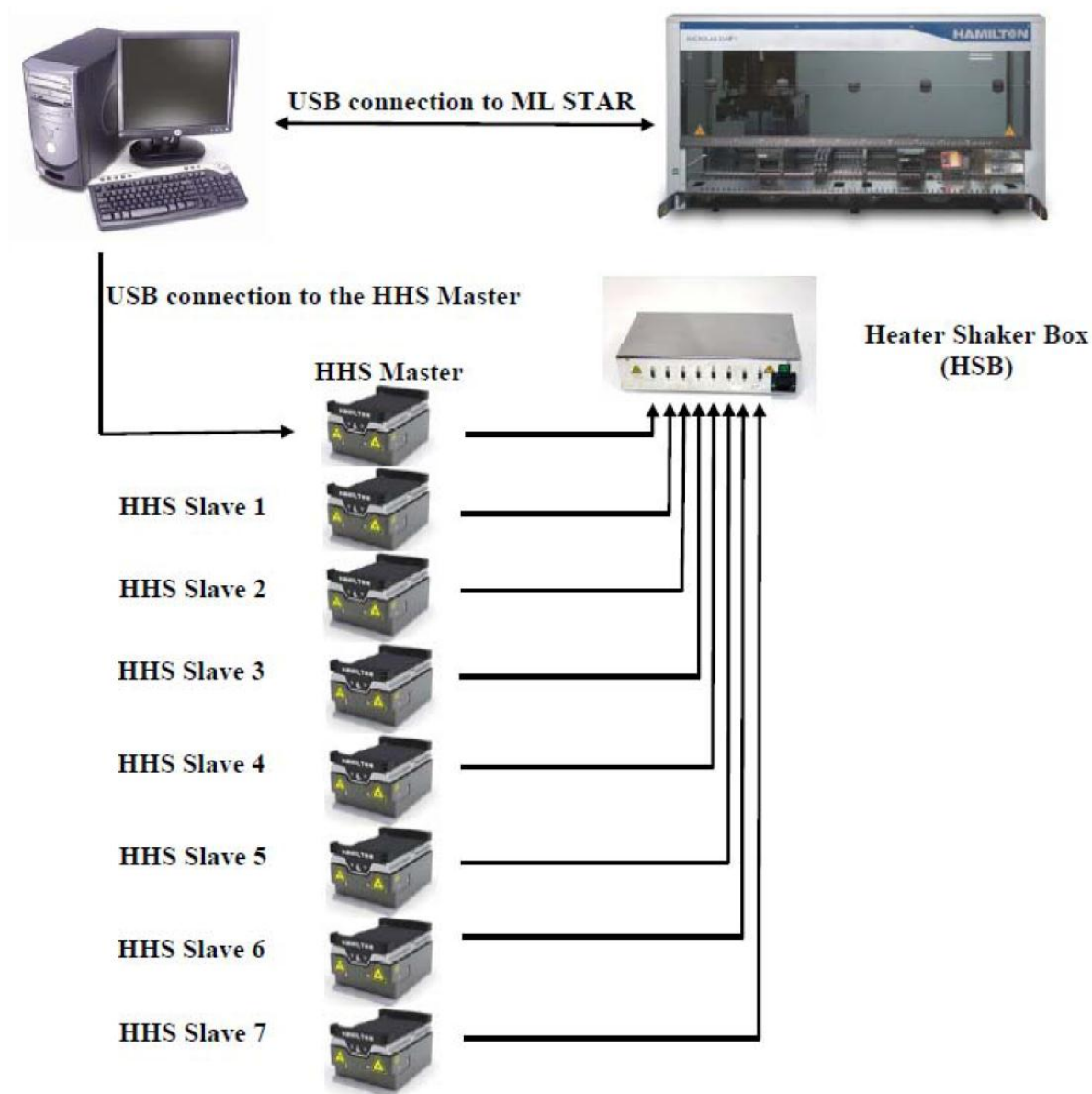
NOTE

The HHS is intended to be used within a ML STAR instrument only!

Do not use the HHS as a standalone device.

2.10.3.2 Up to 8 HHS via HSB and USB port

If more than two HHS are required, the Heater Shaker Box (HSB) is needed. One HHS (HHS master) is connected to the USB port of the PC and to the HSB and serves as master module. In addition to the master HHS, up to seven additional HHS (HHS slave 1.....7) can be connected to the HSB. The HSB serves a power supply and as a signal distributor for all the HHS slaves.



Up to eight HHS connect to the Heater Shaker Box and to the PC which is controlling the ML STAR instrument.

2.11 Training

Training in operation of the ML STAR instruments and general use of the VENUS Software will be provided by a Hamilton personnel upon initial set-up.

3 Routine Use

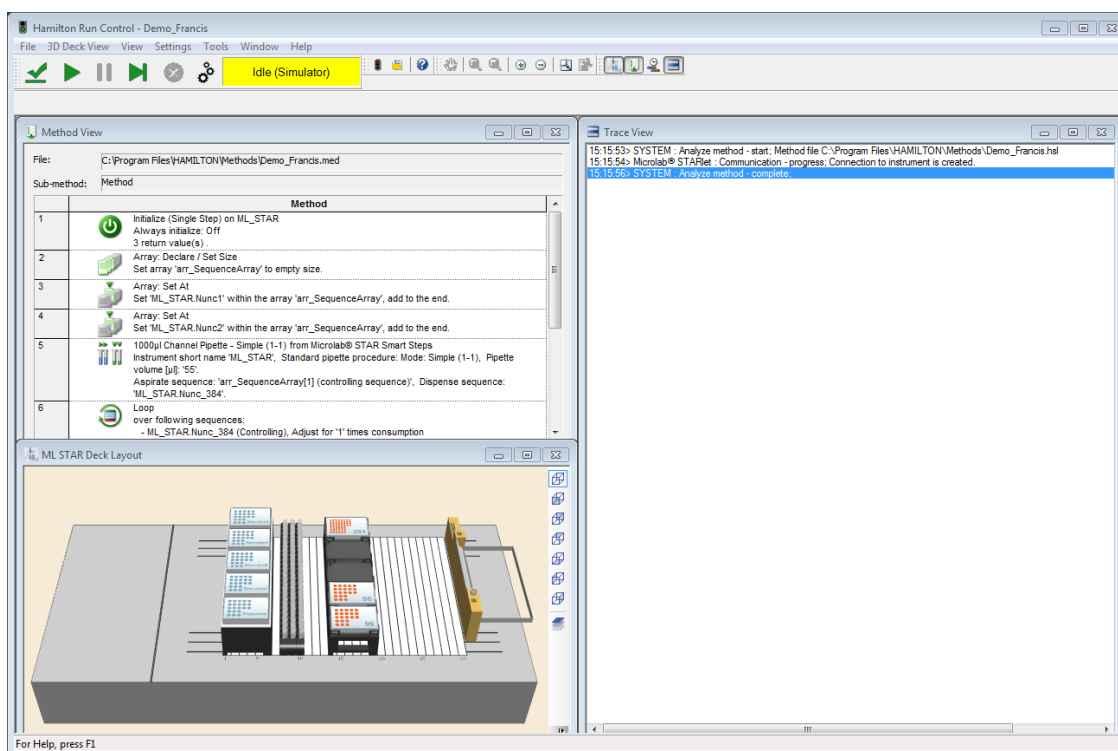
3.1 Loading the ML STAR Instrument

To run a method:

1. To access the Run Control, double-click the “Microlab STAR Run” shortcut on the desk top:



2. From the File menu of Run Control, select “Open”, to open a method (*.med). The method is now loaded:



The free area visible after starting “Run Control” can display up to three windows:

- Deck View
- Method View
- Trace View
- Schedule View
- HSL Debugger (if installed)
- Watch View (if HSL Debugger is installed)

The windows may be enabled/disabled by means of the menu “Run Views” (click the desired menu entry)

or by clicking on the appropriate icon in the “Run Views” toolbar  (if enabled).

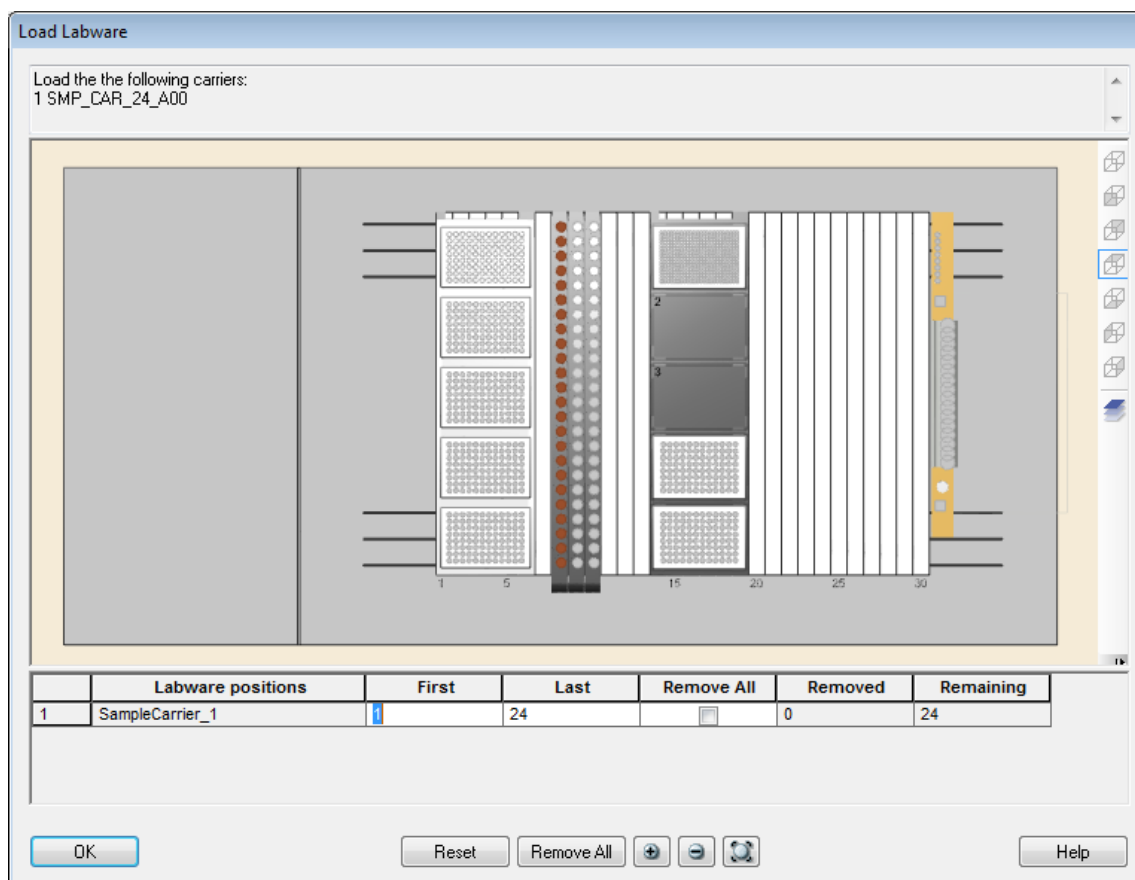
Arrange the windows according to preference using the menu "Window". In the window shown above, the deck layout and the method log (trace) of a run as it is generated are shown.

3. Press the [Start] button (the one which looks like a [Play] button)



to run the method. The steps in the method are traced to the log frame.

If a loading step is part of the method, a dialog appears that requests a reduction of the number of positions on e.g. the sample carrier, as well as a start position for the tips to be picked up. Both pieces of information are optional.



Loading Dialog

4. Enter the number of samples, tips or wells for this run, delete items graphically from the highlighted positions on deck (first select the appropriate item in the table), or accept the default.

Generally, whenever the system finds a 'Load Carrier' command in the method, the operator is requested to place the carrier holding the appropriate labware on the Autoload tray. The correct position is highlighted by blinking LEDs.

**ATTENTION**

It is important to ensure that adequate amounts of the correct liquids are placed in the correctly labeled containers.

When pouring reagents into the reagent containers, ensure that there is no foam on the surface of the liquid.

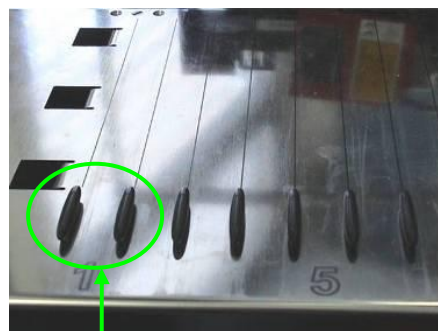
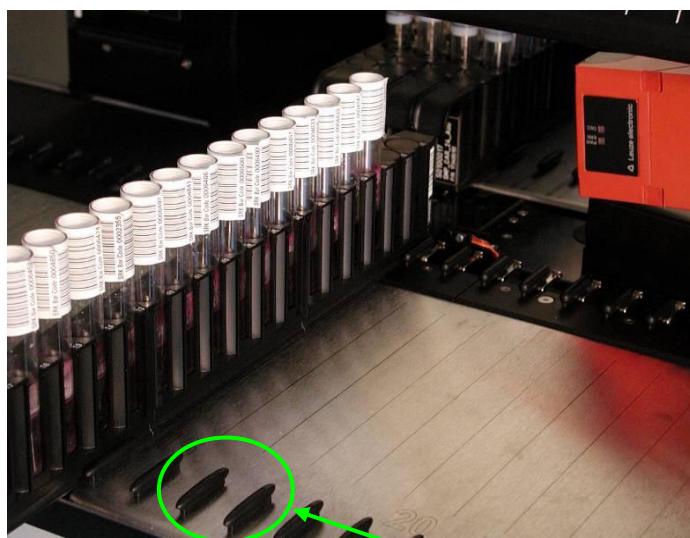
Do not overfill reagent containers: fill approx.10mm below the top of the container.

Always use the proper labware (tips, micro-plates, tubes, etc.) corresponding to the definitions of the method's layout.

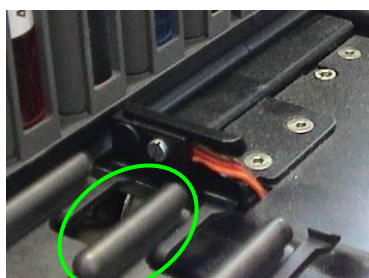
Position micro-plates correctly such that well #A1 is placed according to the deck layout.

Handle any 1-track carrier (such as a sample carrier) with particular care, as this type of carrier can fall over and cause injury or contamination. Position it on the Autoload tray (see figure below), or place several carriers together to minimize this risk.

5. Insert the carriers into the tracks between the front and rear slide blocks of the Autoload tray until they touch the stop hooks on the far side of the tray.



Slide Blocks for Carriers



Stop Hooks for Carriers

**ATTENTION**

Make sure the carriers are inserted completely as far as they will go on the loading tray.

6. Click **[OK]** in the dialog. The carriers are loaded onto the deck automatically by the 'Load Carrier' command in the method. During loading, the barcodes of carriers and labware are read and stored in a file.

Or:

Alternatively, load the carriers onto the defined positions of the Autoload tray before starting the method. Loading and barcode reading will then be performed without operator input. Still the dialog pops up:

Load Labware

Load the the following carriers:
1 PLT_CAR_L5AC_A00, 1 TIP_CAR_480_A00

	Labware positions	First	Last	Remove All	Removed	Remaining
1	Source	1	96	☐	0	96
2	Target	1	96	☐	0	96
3	Tips	1	480	☐	0	480

OK Reset Remove All + - ↺

Help

7. Enter the number of wells on e.g. the source plate for this run, delete wells graphically from the sequence or accept the default (copy the whole plate).

Manual Load System only:

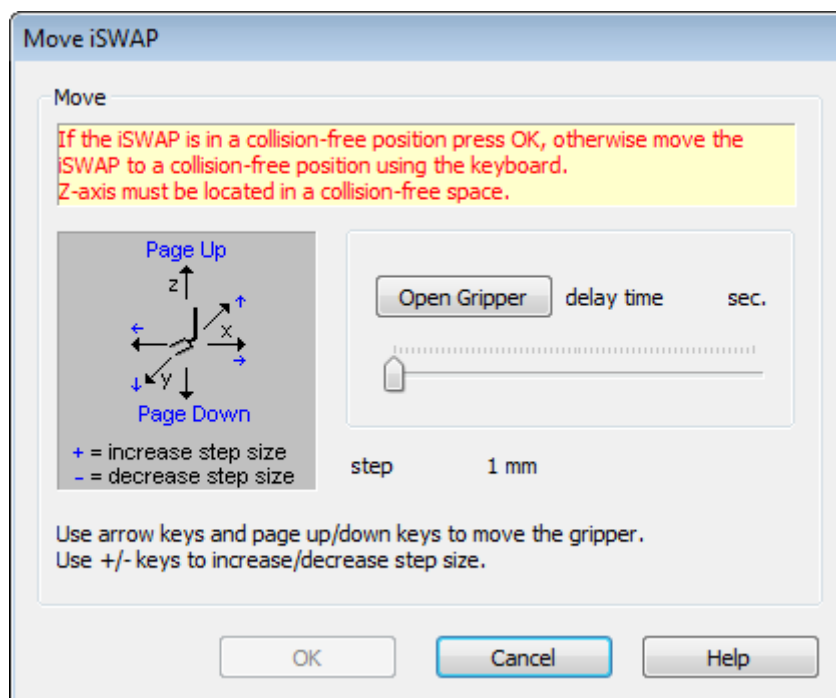
8. Load the deck with the carriers mentioned in the upper part of the dialog box (the two plate carriers and two tip carriers). Don't forget to place labware onto the correct positions.

Autoload System only:

9. Whenever the system finds a “Load Carrier” command in the method, the user is requested to feed the carrier holding the appropriate labware onto the Autoload tray. The correct position is highlighted by LEDs on the instrument. Alternatively, all carriers can be placed directly in their correct positions on the Autoload tray.
10. Click **[OK]** in the dialog box to start loading.

iSWAP only:

11. Usually, at the first run of a day, the iSWAP needs to be initialized. There is also a possibility that there is still a plate in the gripper of the iSWAP. In case a plate is caught by the gripper, it can be safely removed.

**Removing a gripped object:**

Use the “+”/”-” keys to define the distance (e.g. 10mm) the gripper shall be opened to release the plate. Define a “delay time” moving the pointer of the slide bar. Calculate the time (e.g. 5 sec) required for walking from the PC to the instrument and placing the hand underneath the plate. Open up the safety cover of the instrument and check if the plate can be reached. The pipetting arm can be moved using the arrow keys and page up/down keys for this purpose.

**ATTENTION**

Make sure the selected step size and moving direction will not crash the iSWAP.

Always wear gloves for manual manipulations inside the instrument.

If there is enough time and space to hold the plate and the gripper parameters are set press the **[Open Gripper]** button.

Removing a gripped object:

If the iSWAP cannot move up to the traverse height without a collision (e.g. shelving position where the iSWAP would collide to the shelf above it), it must be guided to a save area using the keyboard (arrow keys, page up/down keys, +/- keys).



ATTENTION

Make sure the selected step size and moving direction will not crash the iSWAP.

12. If the iSWAP is ready for initialization press the **[OK]** button.

If an unload step is part of the method, it is requested to unload the carriers from the deck. The following dialog box opens:

Unload Labware

The following carriers will be unloaded:
1 PLT_CAR_L54C_A00, 1 TIP_CAR_480_A00

	Labware positions	First	Last	Remove All	Removed	Remaining
1	Source	1	96	☐	0	96
2	Target	1	96	☐	0	96
3	Tips	1	480	☐	0	480

OK Reset Remove All Help

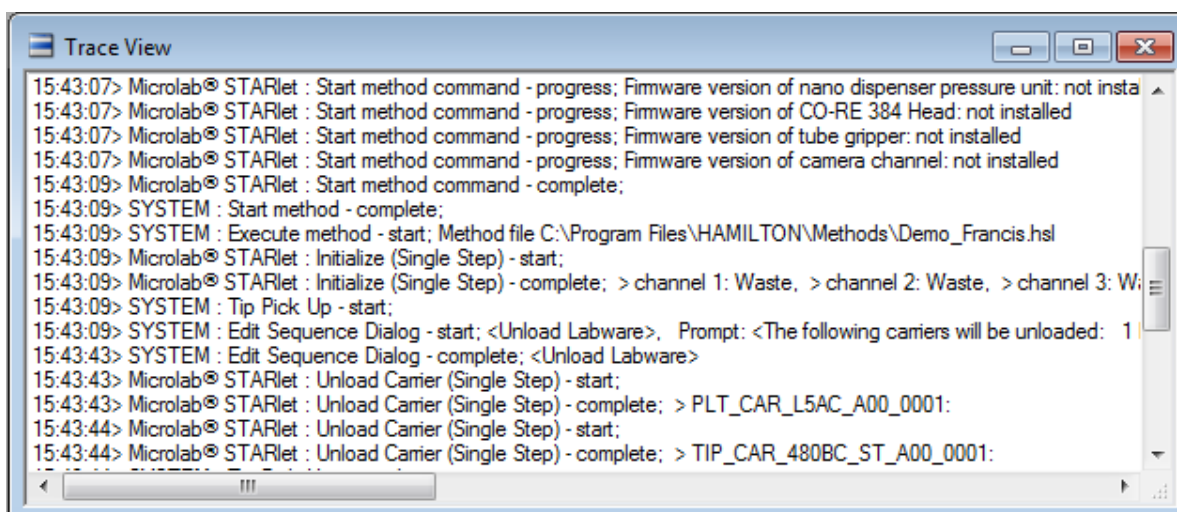
Manual Load System only:

13. Click **[OK]** in the dialog. Unload all the carriers manually. The unloading will be checked by the system.

Autoload System only:

14. Click **[OK]** in the dialog. The carriers are unloaded to the Autoload tray.

The method is now finished. Method completion information is visible in the “Trace View” window.

**Method trace file**

Every run creates a separate method trace file. The method trace file is stored under “{method name}\LogFiles” directory. Each method trace contains the method name, and a unique number consisting of 32 hexadecimal digits, here represented by “x32x”, within the file name. The method traces are not overwritten or appended.

Name	Größe	Typ	Geändert am
ComTrace_Simulator20070108.trc	83 KB	TRC-Datei	08.01.2007 11:24
Demo_a47818bacecf48ccb1d850084b0d6901_Trace.trc	6 KB	TRC-Datei	08.01.2007 11:24
Demo_Barcodes.txt	1 KB	Textdokument	08.01.2007 11:24

Communication trace file

The communication trace file is created every day or when the instrument is switched on. A communication trace file is also generated under the “communication name” (Simulator, HxUSBComm) and current date, where the communication of each run of the method will be appended. If barcode information is generated, its data will be stored in a file labeled with the method’s name followed by “*_Barcodes.txt”.

**NOTE**

From time to time all unused method traces and com traces should be deleted from the hard disk.

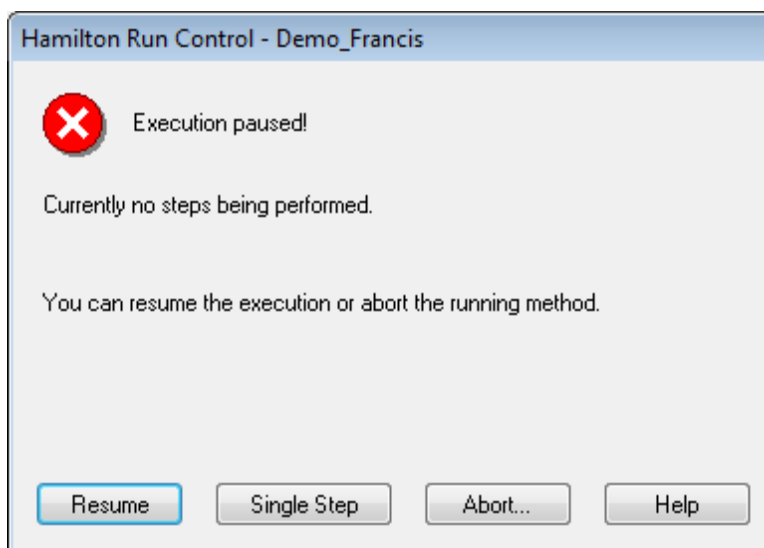
Single Step mode

The method can be tested using Single Steps. It is always possible to start a run in Single Step mode (like TipPickUp, Aspirate, Dispense, etc.).

1. Click the **[Single Step]** button, to start the method in single step mode.

Or:

2. Pause a method at any time by clicking on the **[Pause]** button of Run Control. The step currently running will be terminated, and the following message box appears:



3. Click anywhere in the pause dialog to stop the beeping.

During the pause, it is possible to open the front cover of the instrument.

4. Before continuing the method, make sure the cover is closed again.

5. Resume the paused method by clicking **[Single Step]**,

or

resume method execution by clicking **[Continue]**, to run the method without further breaks

or:

abort method execution by clicking **[Abort...]**. It will be prompted to confirm the abort.

**NOTE**

An abort may cause loss of data.

Aborted methods cannot be restarted again, unless explicitly programmed.

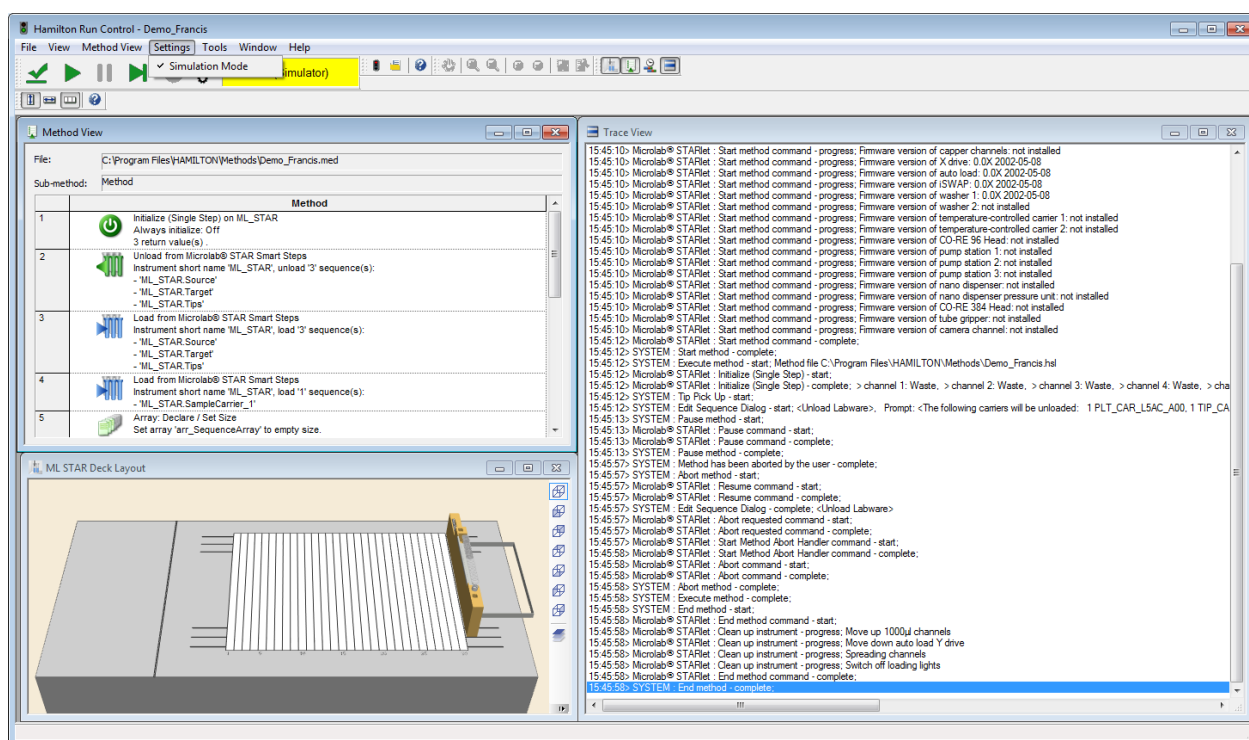
A fast abort can always be done by opening the front shield of the ML STAR Line instrument during a run.

*A method can be aborted even if **[Pause]** is active.*

*To prevent loss of data or bad pipetting result **[Pause]** will finish the proceeding step and stops then.*

3.2 Run Simulations

It is also possible to run a simulation instead of the actual run. It is recommended to always simulate a newly created method first, before running it on the instrument. The run simulation is switched on in the menu “Settings”:



Enable the simulation by clicking to **[Settings / Simulation Mode]**.

The run simulators check the consistency of each (single) step of the system, together with minimum aspiration and dispense volume management.

The speed for simulations is adjustable. If the default speed (at maximum is best guess under real runs) is too fast the speed can be reduced by the simulator delay via the configuration editor. Refer to the [MicroLab STAR Programmer's Manual](#).

3.3 Run-time Error Handling

Before using the run time error handling, several types of problems causing errors have to be solved first. Among these are

- Syntax errors when programming in HSL (e.g. forgotten “;”)
- Logical errors (e.g. “tip eject” before “pick-up”, or “asp 20µl”, “disp 100µl”)
- Semantic errors (e.g. wrong pipetting pattern)
- Method/deck interaction errors (e.g. dispense 100µl into the first well of a 1536-well MTP)
- Liquid handling/application errors (e.g. droplets, foam, non-pipetted wells)
- User-related errors (e.g. wrong deck loading)

These problems *cannot* be handled by any run time error handling. They have to be solved in advance, while or after the programming. Refer to the Microlab STAR Programmers Manual.

Problems that *can* be handled in run time are

- Not enough liquid
- Liquid level not found (if it occurs exceptionally)
- No tip picked up
- Clot detected
- Barcode unreadable (if it occurs exceptionally)
- Execution error (channel no. 1 has an error - e.g., not enough liquid -, then channel nos. 2-8 have an execution error because they have been stopped before completion of the step)



NOTE

In principle, each channel may have one or more different types of errors at a time.

If all channels have the same error at the same time, a collective recovery can be made.

3.3.1 Examples

We now focus on some important examples. A detailed description is available in the VENUS Help Function. Click on **[Error Settings]** within the single step dialogs of the ML STAR Line-specific commands and select **[Help]**.

Barcode reading error

In the case of an error, the process can be continued using the error handling procedure. If, for example, a barcode of a carrier cannot be read, a dialog window opens up:

Load Carrier - Error

Error

Description:

Barcode error.

	Position	Error	Assigned recovery
●	2	Barcode Error	
●	4	Barcode Error	
●	5	Barcode Error	
●	6	Barcode Error	
●	7	Barcode Error	
●	8	Barcode Error	
●	9	Barcode Error	
●	10	Barcode Error	

Position description:

Assigned barcode: "

Recovery

Repeat

Continue

Barcode...

Exclude

Unload carrier

Execute

Abort

Cancel

Help

The error recovery can be programmed to:

- **Continue** ignore the error message (here, failure of barcode reading)
[Continue] makes no sense in the case of a barcode reading error - at least a manual entry has to be made so that barcode data exists for further processing.
- **Repeat** try to read barcode once again.
Often a repetition of reading will solve the problem because the reading speed is reduced then. The selected action is displayed in the field **[Assigned recovery]**.
- **Barcode...** Enter barcode manually
Clicking **[Barcode]** opens an entry dialog box where a barcode can be entered (no entry is also allowed).

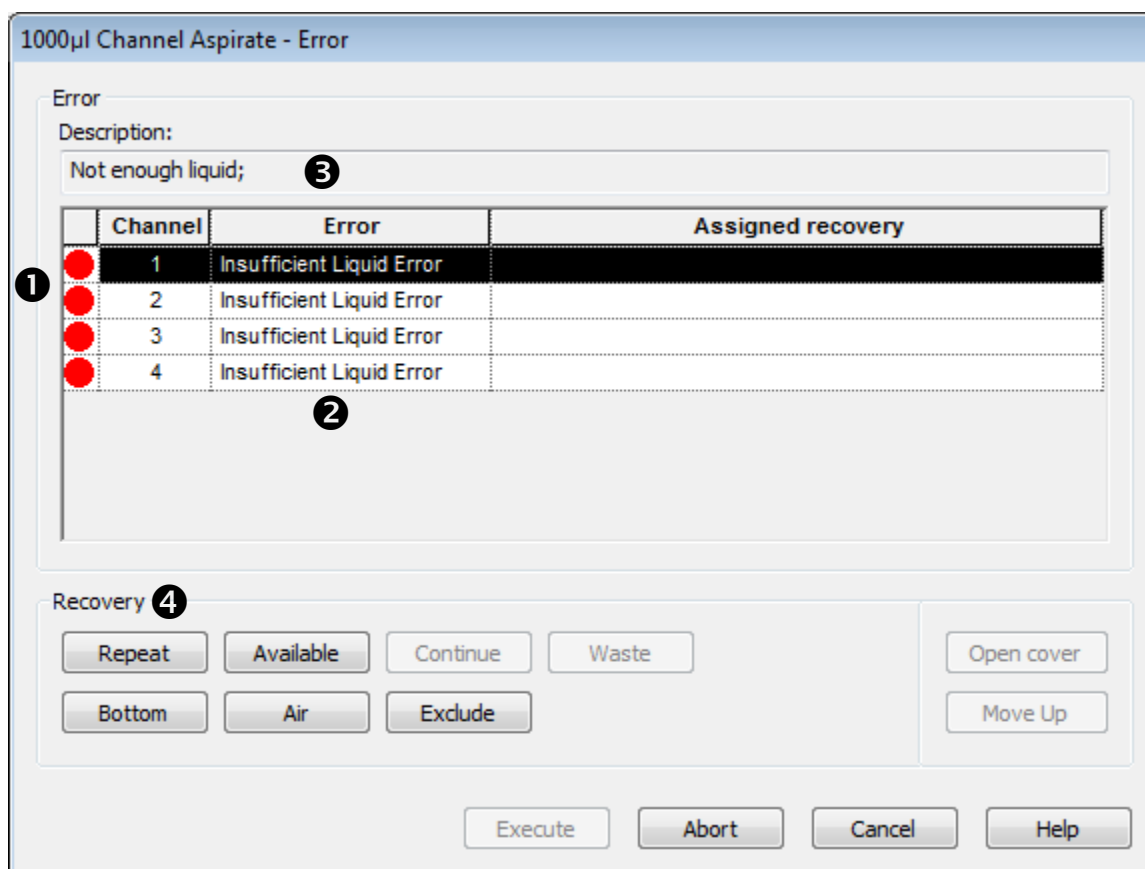
6. Assign a recovery option. The **[Execute]** button becomes active.

7. Click on **[Execute]**. The instrument proceeds with the selected recovery option.

Pipetting error

If an error occurs with the pipetting channels, a dialog shows its error state and its recovery options for every single channel (Different channels can have different errors.).

For example, in case of an LLD error, such as no liquid in the container while aspirating, a window similar to the following pops up:



All pipetting channels which produced an error are listed with a red dot on the left ❶. A short error description ❷ is given. A detailed error description is shown at the top text field ❸ labeled "Description:" for the selected channel. There may be different errors which lead to the same short error description.

The **[Recovery]** ❹ frame buttons offer several possible actions to solve the problem:

- **[Repeat]**: Executes the command which caused the error once again
- **[Available]**: Aspirates the available volume from source and fills up the missing volume with air
- **[Continue]**: Continues as if no error was recognized
- **[Waste]**: Means that the erroneous tip is ejected to the waste and the channel is excluded.
- **[Bottom]**: Activates the channel to move down to the bottom of the container, and the available volume is aspirated without LLD
- **[Air]**: Means air is pipetted instead of liquid and the method will continue

- **[Exclude]**: Allows disabling any further action on the error-affected channel
- **[Move Up]**: This is not a real error recovery procedure, but useful e.g. to manually remove a clot. This causes the following actions (this action can be repeated):
 - moves the barcode reader of the Autoload (if present) to the very right
 - moves the selected channel up by 10 mm
- **[Open cover / Close cover]**: this button does not start any error recovery actions either, but triggers the following actions:
 - **[Open Cover]**: enables opening the front cover during error recovery
 - **[Close Cover]**: enables closing the front cover before executing error recovery

An error recovery needs to be assigned for every error. Selecting a channel followed by any possible recovery procedure assigns the selected recovery procedure to all error-affected channels with the same error.

If any recovery procedure is assigned to a channel (even one which is not desired to be assigned), the dot ● in the first column (see picture above) changes color to green.

Some recovery buttons are disabled, to prevent further faulty steps, e.g. an error-affected aspiration step cannot be “Continued”, to prevent any later dispense with insufficient volume.

When the last faulty channel is assigned to a recovery procedure, the **[Execute]** button becomes active and the system can proceed.

In any case the method can be aborted without further recovery options.

3.4 Walk-away (predefined) Error Handling

The programmer can define a walk-away error handling which uses predefined default settings for different error situations. These settings can be customized for single steps and easy steps only. For Smart Steps, the error recovery is fixed.

For every instrument-specific single step of the method, an individual error recovery can be defined. The following can be configured:

- The appearance of the error recovery dialogs (which buttons are available).
- A timeout, after which the default recovery is carried out (the dialog automatically closes down).
- The default procedure (what should be executed if the timeout runs out).
- Which error is flagged in the trace file.

For this purpose, every instrument-specific Single Step and Easy Step has an **[Error Settings]** button. To deal with the subject of error handling consult the [Microlab STAR Programmer's Manual](#).

4 Maintenance

The ML STAR operator is instructed to maintain the instrument on a regular basis; this maintenance consists largely of surface cleaning and does not require opening up the instrument (i.e. any unscrewing, removing deck, covers, etc.).

The operator is responsible for changing consumable parts (disposable tips, needles, waste Bag, etc.). A service engineer generally changes spare parts (PCBs, cables, channels, etc.).

Periodic maintenance routines need to be run in order to ensure safe and reliable operation of the ML STAR instrument and the accessories. A service engineer will need to service (perform preventive maintenance) the instrument at least twice a year. In addition, the service engineer may be called to repair a damaged component of the instrument or to resolve a functional problem which the user cannot resolve himself (such as adjusting and calibrating the channels).



ATTENTION

When parts and accessories have application engineering (APE) modified parts adapted for a specific application, the original parts must be attached/used when performing adjustments/calibrations with the macro programs. After the adjustments/calibrations have been completed, re-install the modified parts.

Depending on how the VENUS Software is installed, the maintenance routines might be mandatory. If so defined – until they are completed, the user is reminded by a warning.

STARwatch advanced service is available increasing the uptime of STAR Line instruments.

4.1 STARwatch Advanced Service

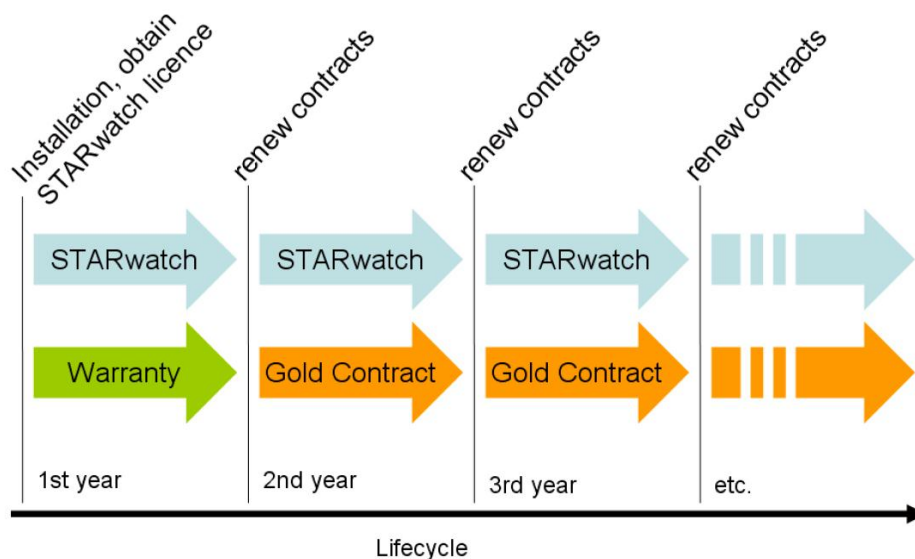


STARwatch is a pro-active service which increases the uptime of STARLine instruments.

STARwatch continuously monitors the condition of the dedicated instrument. The captured data is automatically analyzed, and when a critical pattern is recognized, the service organization is immediately informed to provide a proactive intervention.

STARwatch intended use is primarily for high throughput laboratories where a certain consistency of daily routine use is given and where a short down time is critical.

STARwatch is available in addition during the warranty period and later on in combination with a service contract. STARwatch has to be obtained on a yearly basis. Its license is valid for one year.



Please also refer to our Technical Note “STARwatch Condition Monitoring System”, available on our website www.hamiltonrobotics.com.

4.2 Maintenance Intervals

We recommend maintaining the instrument at the following intervals:

- daily recommended before the instrument is shut down.
- weekly recommended at the end of the week before the instrument is shut down.
- six-monthly preventive service maintenance carried out by a service engineer.



NOTE

If the operator decides not to run either daily or weekly maintenance before shut-down the instrument, these routines must be implemented at the next run start.

If any parts of the instrument, carriers or racks have become contaminated, the weekly maintenance procedure must be performed.

4.3 If Maintenance Fails

If an error is encountered during a maintenance procedure, try to rectify the problem and re-start the maintenance procedure. If this should fail, call a local service engineer.

4.4 Preventative Maintenance

Preventive maintenance including verification should be carried out at regular intervals by a trained Hamilton service engineer. A service agreement ensures regular maintenance and verification for a specified period of time. Hamilton recommends that maintenance and verification take place twice a year.

4.5 Materials Required

- Disposable latex gloves
- Protective glasses
- Lab coat
- Paper towels
- Lint-free cloths or Q-tips
- Set of 8 maintenance needles - please refer to [Appendix B, "Ordering information"](#)
- Ethanol (70%)
- De-ionized water
- Microlab Detergent & Disinfectant Kit (P/N 281242) (contains Deconex 61 DR) - please refer to [Appendix B, "Ordering information"](#) or Microcide SQ (only available in the USA).
- Microlab Disinfectant Spray Kit (P/N 281243) (contains Deconex SOLARSEPT) - please refer to the [Appendix B, "Ordering information"](#)
- Microlab Disinfectant STARTER Kit (P/N 281245) (contains Deconex 61 DR and Deconex SOLARSEPT) - please refer to the [Appendix B, "Ordering information"](#)



ATTENTION

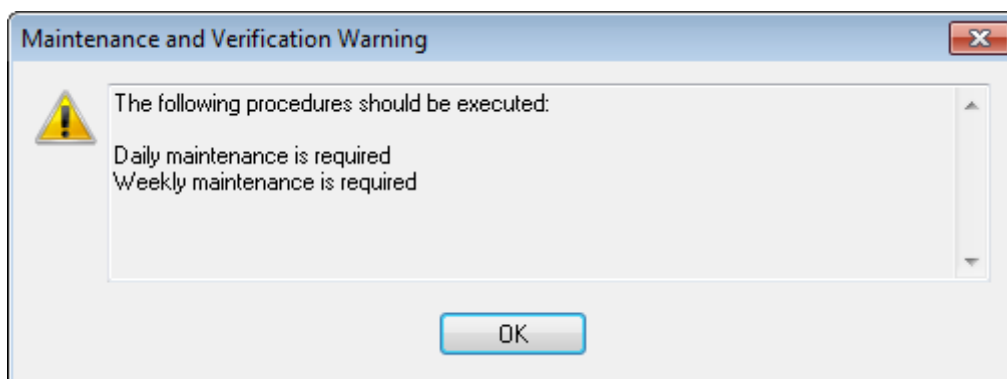
Use cleaning, disinfecting and decontaminating fluids in accordance with manufacturer's instructions. Do not use disinfecting materials which contain hypochlorite (Javel water, Chlorox) or bleaching fluids.

Prepare disinfectant fluids according to their labeling.

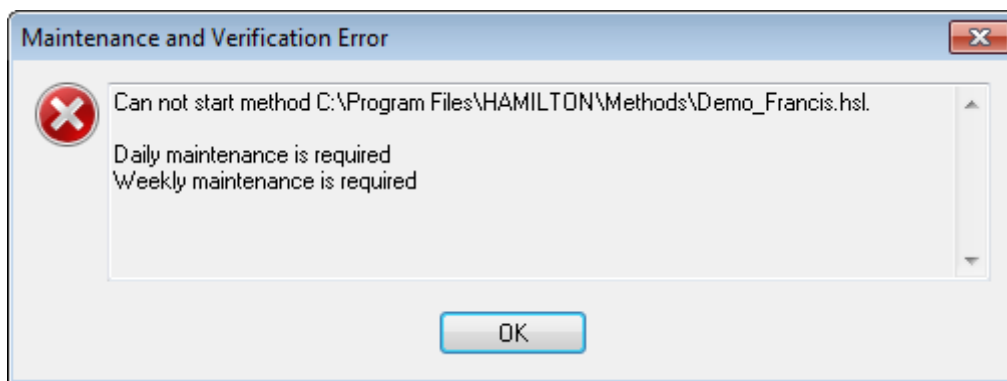
4.6 Instrument Maintenance Procedures

The operator will be guided by the VENUS Software through the regularly scheduled maintenance procedures.

Depending on the maintenance configuration settings (*optional*, *warning* or *mandatory*) the instrument will display messages when starting up the instrument:



Displayed warning if maintenance needed, config. setting = warning



Displayed error if maintenance needed, config. setting = mandatory

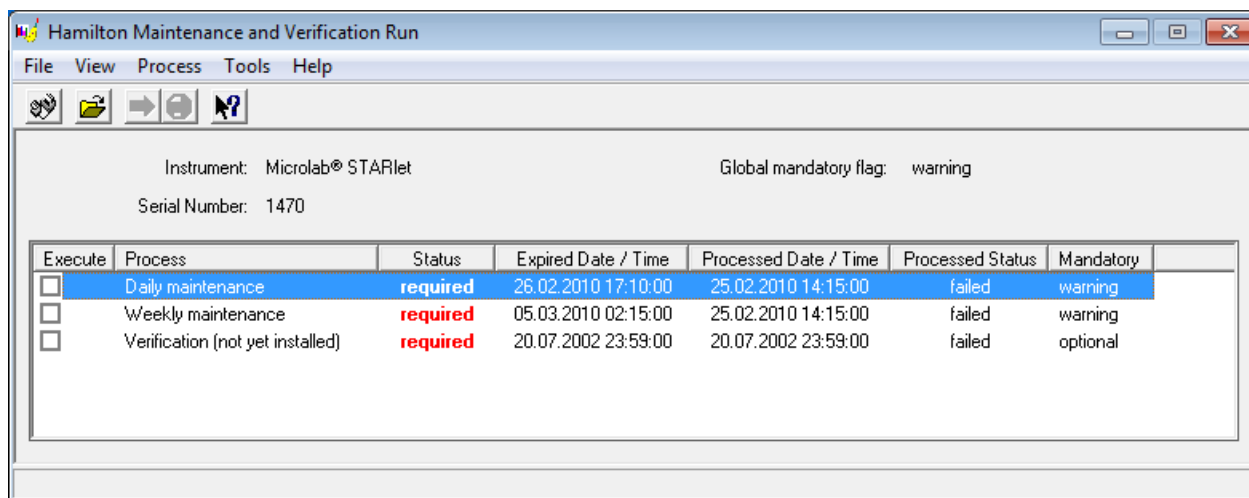
To execute a maintenance procedure:

1. Double-click the following icon on the desktop:



In the main window "Maintenance and Verification Run", the process status information view lists all maintenance and verification processes for the connected/selected instrument.

Here the operator can start the necessary commands to perform all maintenance-related routines.



Main window 'Maintenance and Verification Run'

2. Select the desired maintenance routine by clicking the specific check box and by pressing the **[Run Process]**  button. The VENUS Software will then issue on-screen instructions detailing all procedures required to perform the selected maintenance routine.

**ATTENTION**

Always wear disposable gloves during maintenance.

Do not clean the instrument in the vicinity of unshielded flames or devices which could create sparks. Do not use hot air blowers to dry the instrument. The liquids used for cleaning may be flammable.

Maintenance work on iSWAP is not permitted, because adjustment may be lost. Clean iSWAP only where necessary after spillage and be careful of sharp edges on the gripper's jaws.

Any regulations for waste disposal specific for the country of operation must be taken into account and observed.

Routine completion

A maintenance routine is completed once the procedure has been fully implemented and the results are within the specifications.

Aborting maintenance procedures

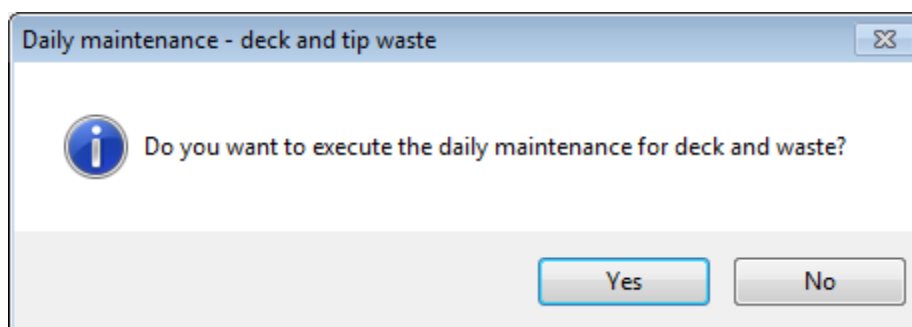
Aborting a maintenance procedure will lead to a 'failed' status, and maintenance will need to be started again.

4.6.1 Daily Maintenance

Daily maintenance is configuration dependant. The following tasks are examples belonging to the daily maintenance:

- Checking if deck is clean
- Empty the tip waste
- Checking the tightness of the 5ml/1000µl pipetting channel
- Verifying the functioning of the cLLD (5ml/1000µl pipetting channel)
- Rinse procedure of the wash station(s)
- Checking if the CVS Vacuum system is working

After initialization of the instrument, the operator will be asked to execute the daily maintenance:



After clicking **[Yes]** the daily maintenance procedure will be started. Pressing **[No]** will abort the procedure.

The front cover (the hinged acrylic glass window that shields the instrument in front) can be opened for user intervention.

1. Once the maintenance procedure has been started the pipetting arm moves to the left side. The operator now has access to the deck to check if cleaning is needed or not:
 - If the deck is clean continue with the daily maintenance.
 - If the deck needs to be cleaned the daily maintenance can be interrupted. Instead of the daily maintenance carry out the weekly maintenance.

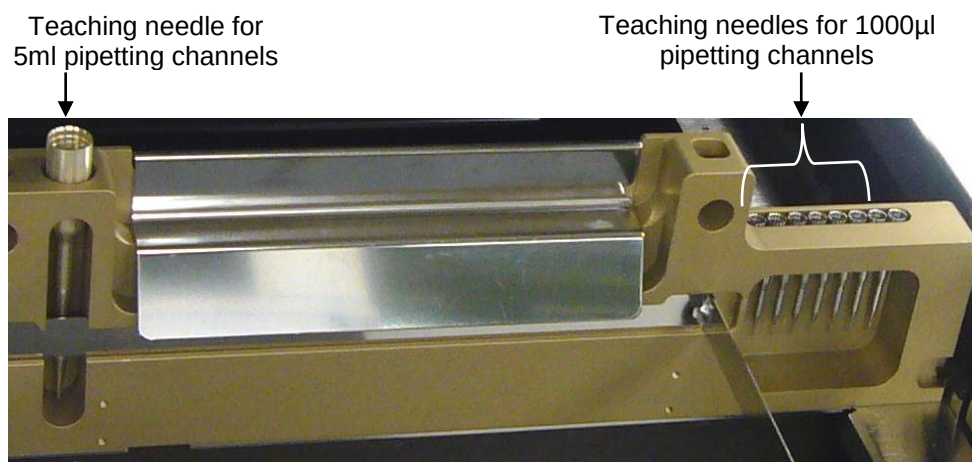
Continuing the daily maintenance procedure will lead the user to the next maintenance task. The tip waste needs to be emptied. Dispose of it with the rest of the laboratory's contaminated waste.



ATTENTION

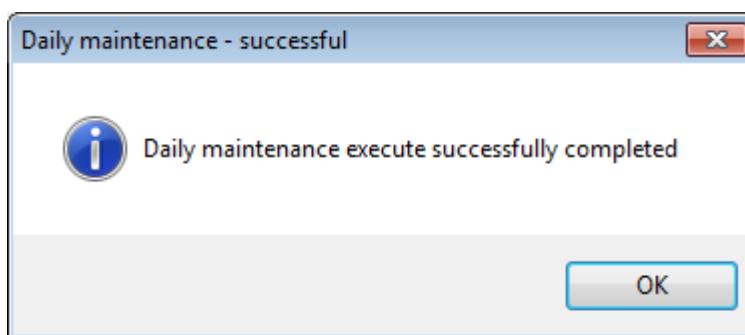
The tip waste is always to be regarded as contaminated.

2. For the next steps the teaching needles are required.



3. The procedure continues with the tightness check of the pipetting channels. The pipetting arm will travel to the right hand side to pick up the teaching needles.
Two checks are done with the pipetting channels, the over-pressure and the under-pressure check.
4. For the capacitive liquid level detection (cLLD) check the needles are picked-up again. One channel to the next is checked for the proper functioning of the cLLD.
5. If there is a CVS installed a rinse procedure is started. Refer to [Chapter 4.12](#).
6. If there is a needle wash station installed a rinse procedure is started. Refer to [Chapter 4.9](#).

7. The end of the daily maintenance is displayed:



The daily maintenance process status is saved on the instrument and a report file is created. Refer to [Chapter 4.7 "Printing a Report"](#).

**NOTE**

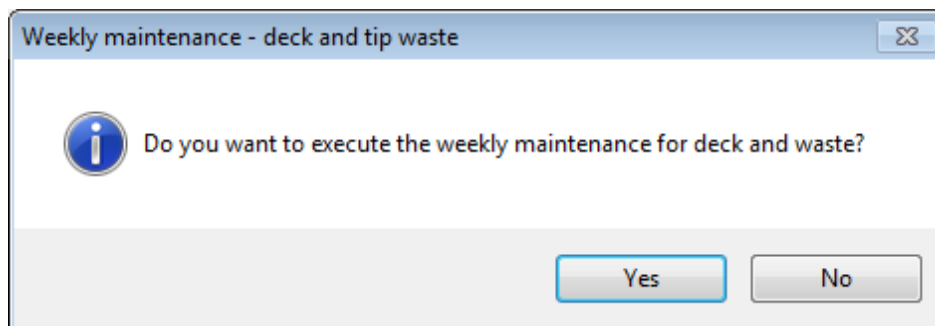
If any parts of the instrument, carriers or racks have become contaminated, the weekly maintenance procedure must be performed.

4.6.2 Weekly Maintenance

Weekly maintenance is configuration dependent. The following tasks are examples belonging to the weekly maintenance:

- Cleaning the deck and carriers
- Checking the condition of the carriers
- Emptying and cleaning of the tip waste
- Checking the tightness of the 5ml/1000µl-channel
- Verifying the functioning of the cLLD (5ml/1000µl-channel)
- Maintenance procedure for the wash station(s)
- Cleaning of the pipetting head: stop disk, O-ring, tip eject sleeve
- Cleaning of the covers, Autoload protecting ribbon
- Checking if the CVS vacuum is working

After initialization of the instrument, the operator will be asked to execute the weekly maintenance:



When the instrument is initialized, the weekly maintenance program advises the user to unload the deck manually. If the Autoload option is available, this step is carried out automatically.

1. Clean all carriers with Deconex® SOLARSEPT spray and leave them to dry. If they are heavily soiled, soak them afterwards in a solution of Deconex® 61 DR cleaning liquid (refer to the product data sheet for further information).

Examine each carrier for scratches on the barcode and any signs of damage. If damage is apparent, replace with new carriers.

2. Continuing the weekly maintenance program will advise the Autoload (if configured) to move to the right hand side of the instrument.

Open the front cover and wipe the deck with a cloth saturated with Deconex® SOLARSEPT. The slide blocks in particular must be checked for cleanliness. Close the front cover.



ATTENTION

Do not spray directly at the Autoload unit or at electrical boards or connectors.

3. The next step of the maintenance procedure will advise the Autoload (if configured) to move to the left hand side of the instrument. The tip waste needs to be emptied and cleaned. Dispose of it with the rest of the laboratory's contaminated waste.



ATTENTION

The tip waste, the tip eject plate, and the plastic bag are always to be regarded as contaminated.

Remove the tip eject plate of the tip waste station and clean it: spray Deconex® SOLARSEPT directly onto the surface and wipe. Remove the frame that holds the plastic bag in place, and discard the plastic in the laboratory's contaminated waste. Pull a new plastic bag over the frame and re-attach it. Put the clean tip eject plate back in place.

Remove the maintenance needle and clean it: spray Deconex® SOLARSEPT directly onto the surface and wipe. Put the clean and dry maintenance needle back in place.

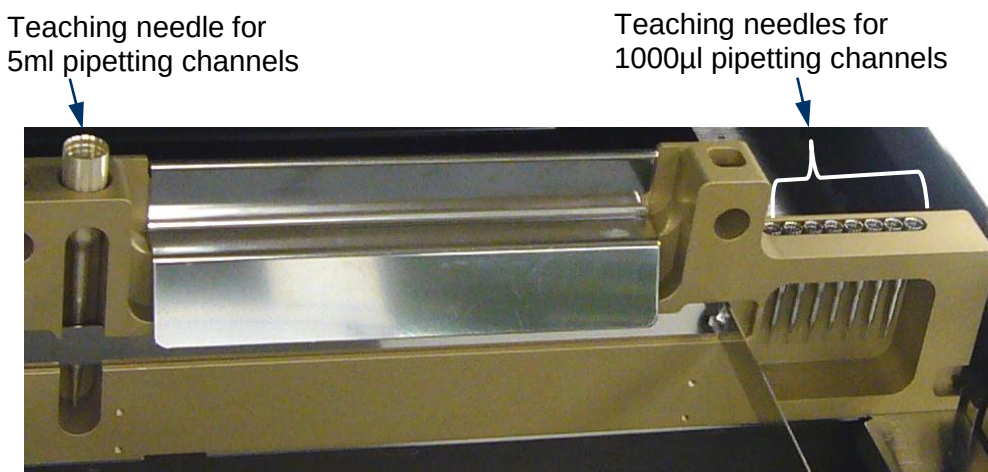
4. If the Autoload option is configured: to prevent unreliable barcode reading, check the laser scanner window of the barcode reader and clean it with a lint-free cloth or Q-tips lightly soaked in Ethanol (70%).



ATTENTION

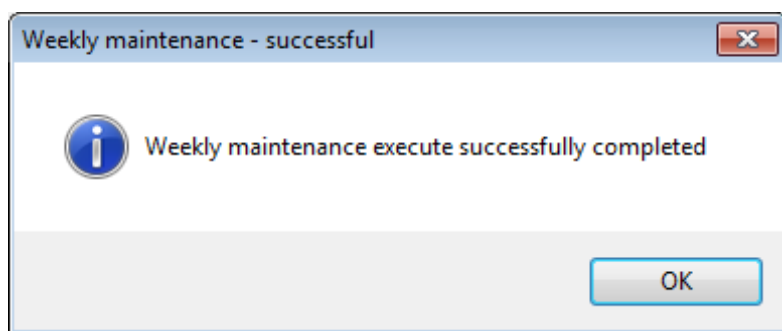
The laser scanner window must be completely dry and free from dust and fibers before the instrument can be reused.

5. For the next steps the maintenance needles are required.



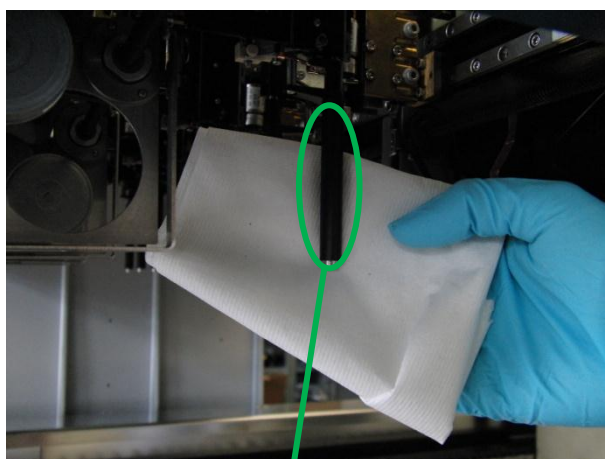
6. The procedure continues with the tightness check of the pipetting channels. The pipetting arm will travel to the right hand side to pick up the maintenance needles.
Two checks are done with the pipetting channels, the over-pressure and the under-pressure check.
7. For the capacitive liquid level (cLLD) check the needles are picked-up again. One channel by the other is checked for the proper functioning of the cLLD.
8. If there is a CVS installed a rinse procedure is started. Refer to Chapter 4.13 "TCC"
9. If there is a needle wash station installed a weekly maintenance procedure is been started. Refer to Chapter 4.8 "Needles".

10. The end of the weekly maintenance program is displayed:



The weekly maintenance process status is saved on the instrument and a report file is created. Refer to [Chapter 4.7 "Printing a Report"](#).

11. Clean the tip eject sleeve (outer part of the pipetting channels) with a lint-free cloth soaked in Deconex® SOLARSEPT.



Sleeve

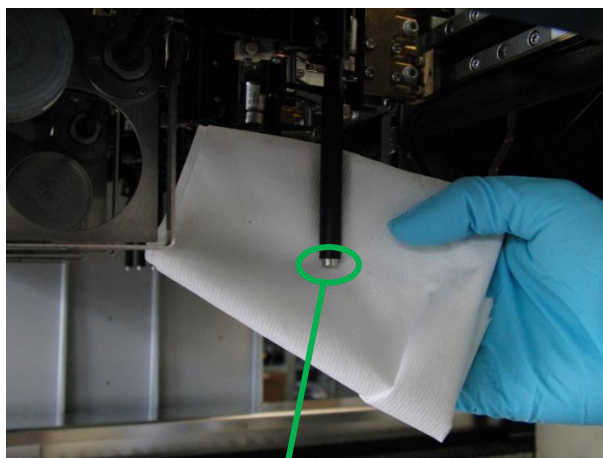


ATTENTION

Take care that no liquid gets inside the tip channel.

Whenever it is necessary to move channels on the x-arm, move them gently by pushing close to their Y-slide. Never force them as this may lead to damage. If possible, switch on the instrument as this will result in a smoother motion when channels have to be moved on the x-arm.

12. Clean the stop disk and the O-rings of the pipetting head (outer part of the pipetting channels) with a lint-free cloth soaked in Deconex® SOLARSEPT.



Stop disc



ATTENTION

Take care that no liquid gets inside the tip channel.

Whenever it is necessary to move channels on the x-arm, move them gently by pushing close to their y-slide. Never force them as this may lead to damage. If possible, switch on the instrument as this will result in a smoother motion when channels have to be moved on the x-arm.

13. Spray the front and side cover with Deconex® SOLARSEPT and wipe dry.
14. Clean the Autoload protecting ribbon with a cloth soaked in Deconex® SOLARSEPT, and wipe without exerting pressure.



ATTENTION

Do not spray directly at the Autoload unit or at electrical boards or connectors.

15. Clean the x-guide shaft behind the upper front cover with a dry cloth at least once a month.



NOTE

Carriers must be completely clean and dry before re-using.

4.7 Printing a Report

The maintenance process status can be printed. To print such a report:

From the “File” menu, select “Open Report”. All maintenance and verification processes which are found in the default “Report Path” are listed.

- If necessary, change the report path using the browse button[<...>].
- Select a report and press the **[Open]** button. The Report Viewer displays the selected report file.
- From the “File” menu, select **[Print]** to print the report file.

4.8 Needles

The recommended procedures for the maintenance of the needles - if the needles are badly stained – are as follows:

Remove the needle(s) with the needle pickup tool (Needle Service Kit, [see Appendix B “Ordering Information”](#))

1. Clean the needles in an ultrasonic bath at 50°C containing Deconex® 61 DR for 15 - 20 minutes
2. Rinse the needles with warm de-ionized water (50°C)
3. Put the needles back into the wash module.



NOTE

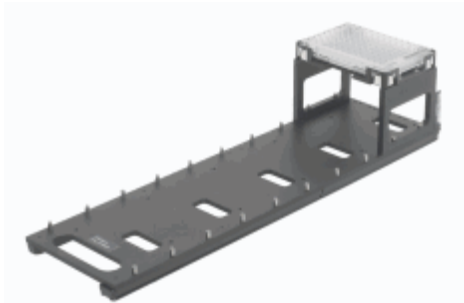
Hamilton recommends replacing steel needles every six months.

4.9 CO-RE 96 Probe Head TADM

This section thoroughly discusses the preparation and execution of the maintenance procedures for the CO-RE 96 MPH TADM 1000µl. For more information about the labware, please refer to the [TADM Manual \(P/N 624031\)](#) or the [Microlab STAR Programmer's Manual](#).

A CO-RE 96 MPH TADM 1000µl must be mounted in an ML STARLine to perform the maintenance procedures indicated in this section.

It is required to have either of the carriers (presented in the table found on the next page) with a defined plate position for the maintenance tool.

188042 MULTIFLEX DWP MODULE	188041 MULTIFLEX MTP MODULE
	
182090 PLT_CAR-L5AC	182365 PLT_CAR_L5MD
	
199199 Maintenance Tool	
	

Preparation

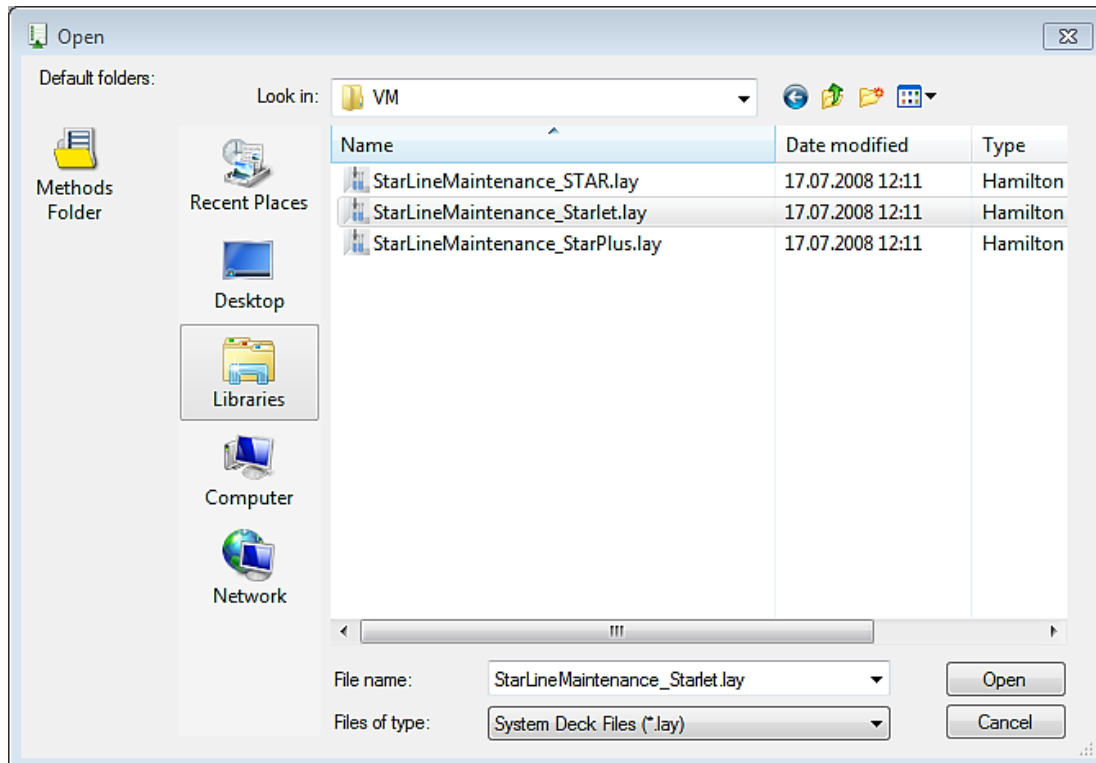
To prepare for the maintenance procedure, create a System Deck Layout. Follow these steps:

1. Start the Method Editor by activating the icon shown below.



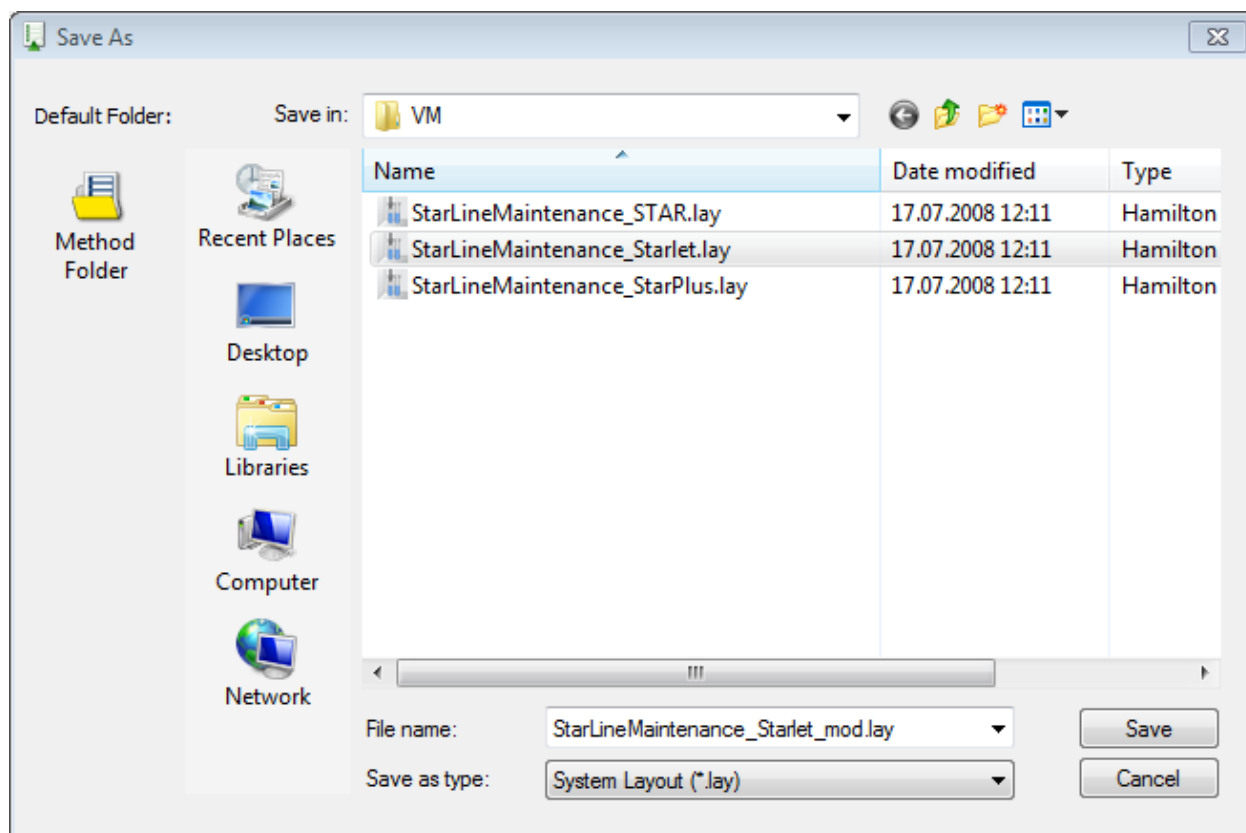
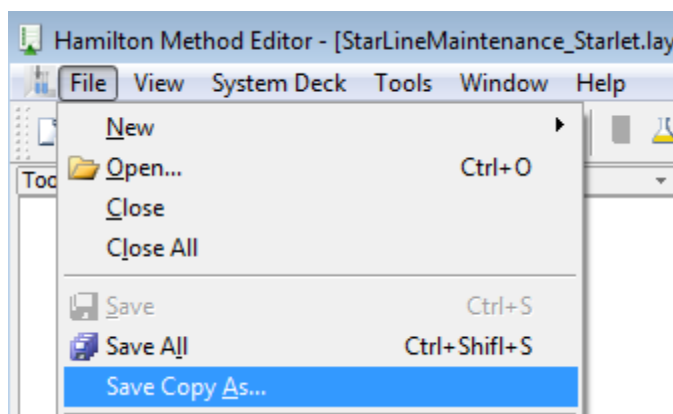
2. Open the existing deck layout found in the directory: \\HAMILTON \Methods \VM

- for ML STAR_{let} → **StarLineMaintenance_Starlet.lay**
- for ML STAR → **StarLineMaintenance_Star.lay**
- for ML STAR_{Plus} → **StarLineMaintenance_Starplus.lay**

**NOTE**

Specifying the "Files of type" to System Deck Files (*.lay) is necessary. This will make the desired *.lay files visible and selectable.

- Now that the file is open, create and save a copy of the file selected. In this case, save a copy of the “StarLineMaintenance_Starlet.lay” file and name it as “StarLineMaintenance_Starlet_mod.lay”. For uniformity purposes and to avoid confusion in the later steps, always use the “_mod” keyword as a prefix to the file copies.

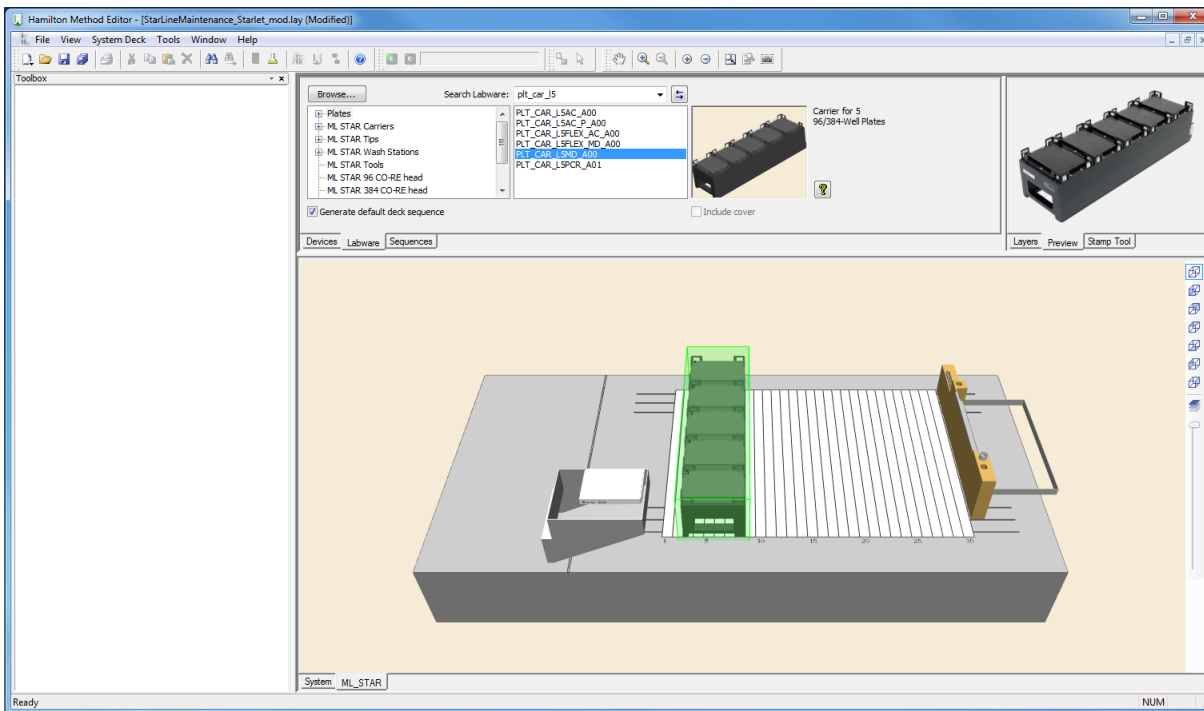


- After having successfully created a duplicate, open the file copy “StarLineMaintenance_Starlet_mod.lay”.
- Add the chosen plate carrier from the table, as instructed earlier. Remember to add the plate carrier onto a track position reachable by the CO-RE 96 Probe Head TADM.

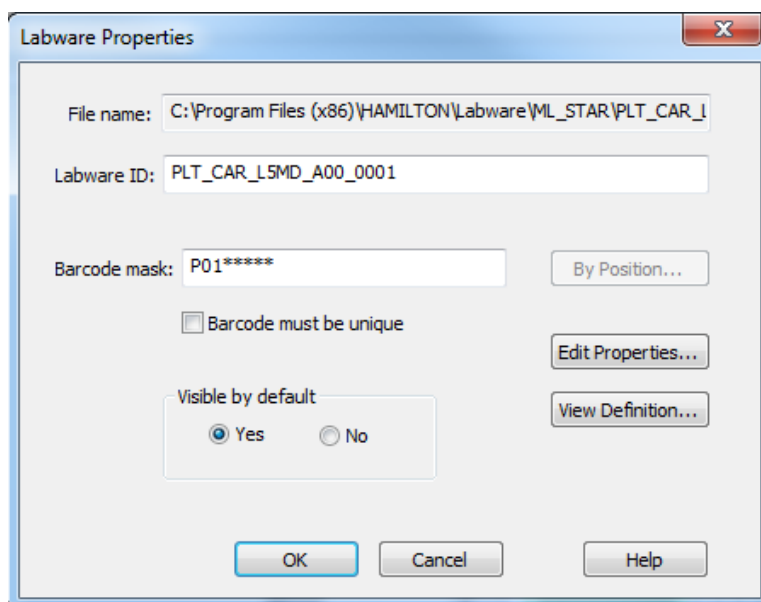
**NOTE**

The deck position has to be available for the maintenance routine.

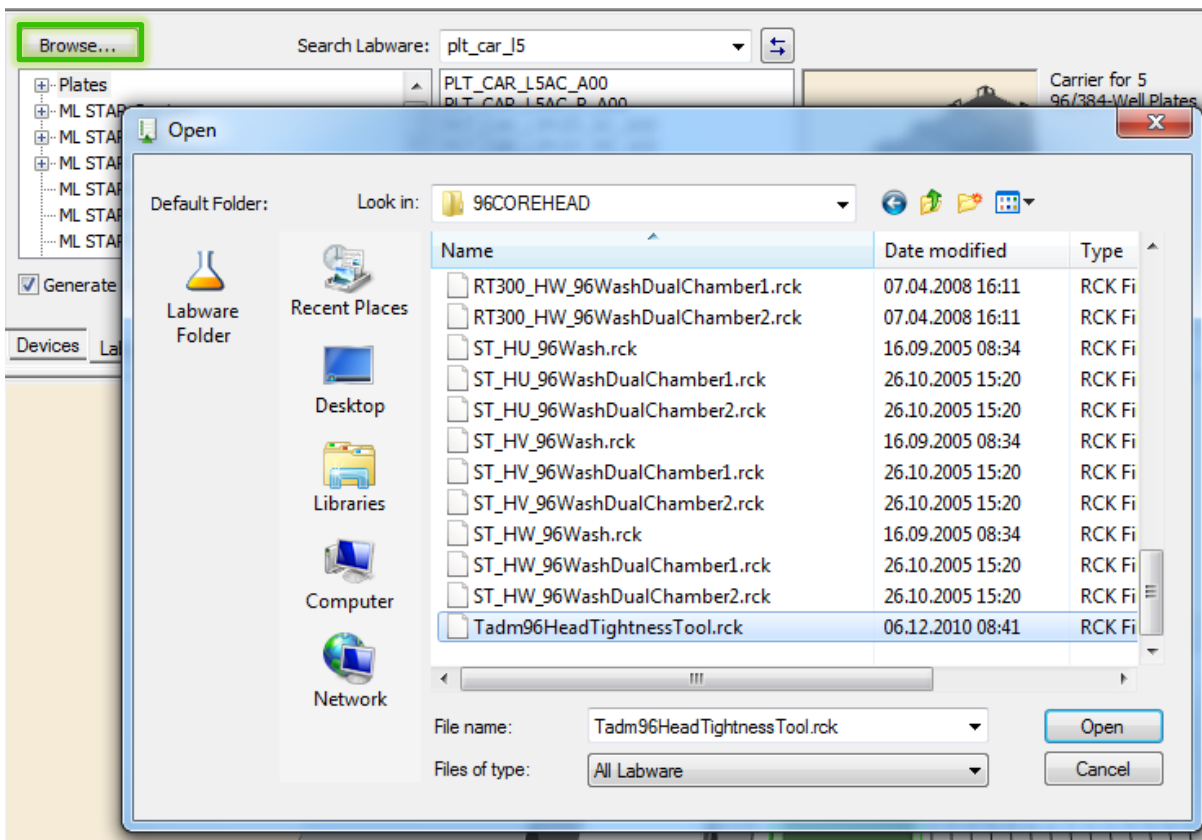
In this example, a PLT_CAR_L5MD (P/N 182365) has been chosen.



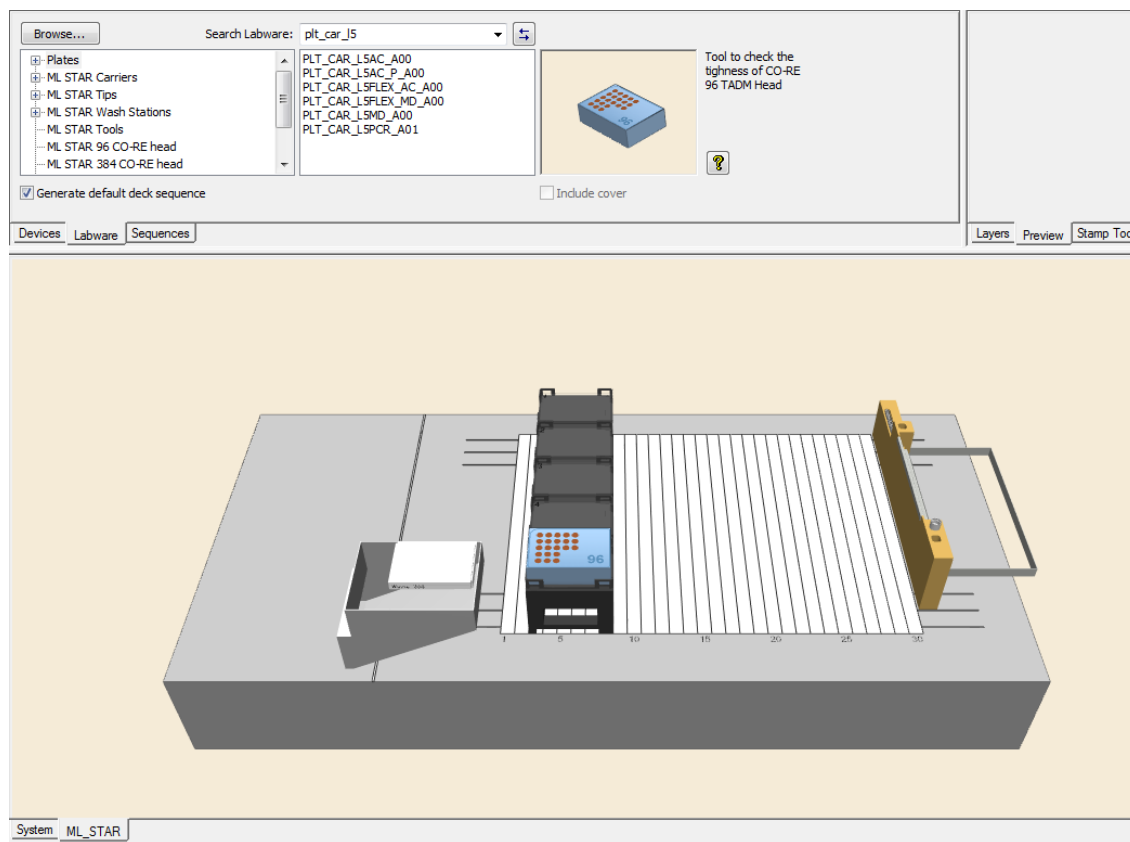
6. Right click on the plate carrier and select Properties. A dialog box with predefined values will prompt, as shown below. Set the Labware Property “Visible by default” to Yes. This will display the carrier in the deck layout editor during the maintenance process. To close the dialog box, click **[OK]**.



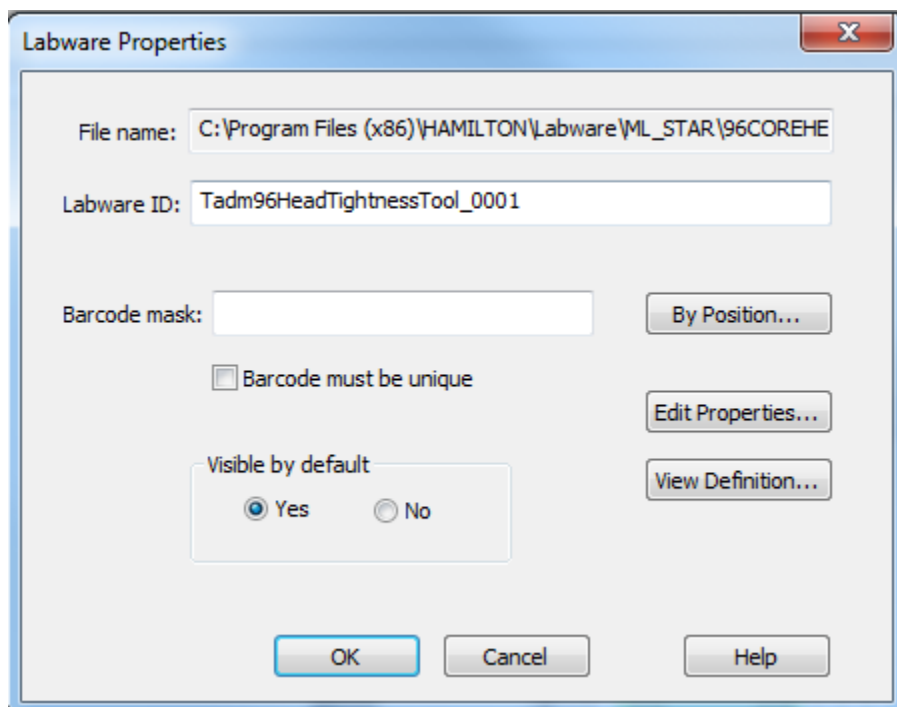
7. To add the Maintenance Tool to the chosen plate carrier, click [Browse] from the Labware Tab of the Method Editor. Locate the file by opening through the folders **Hamilton > LabWare > ML_STAR > 96COREHEAD**.



8. Add the tool onto the plate carrier. Any available position on the plate carrier may be chosen.



9. After having been successfully added the tool onto the plate carrier, right click on it and select Properties. A dialog box with predefined values will prompt, as shown below. Set the Labware Property “Visible by default” to Yes. This will display the tool in the deck layout editor during the maintenance process. To close the dialog box, click **[OK]**.



10. Finish by saving the deck layout and closing the Method Editor.

Daily Maintenance

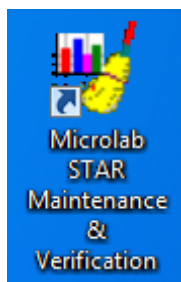
**ATTENTION**

Be aware of moving parts (i.e. X-Arm with its CO-RE 96 Probe Head TADM).

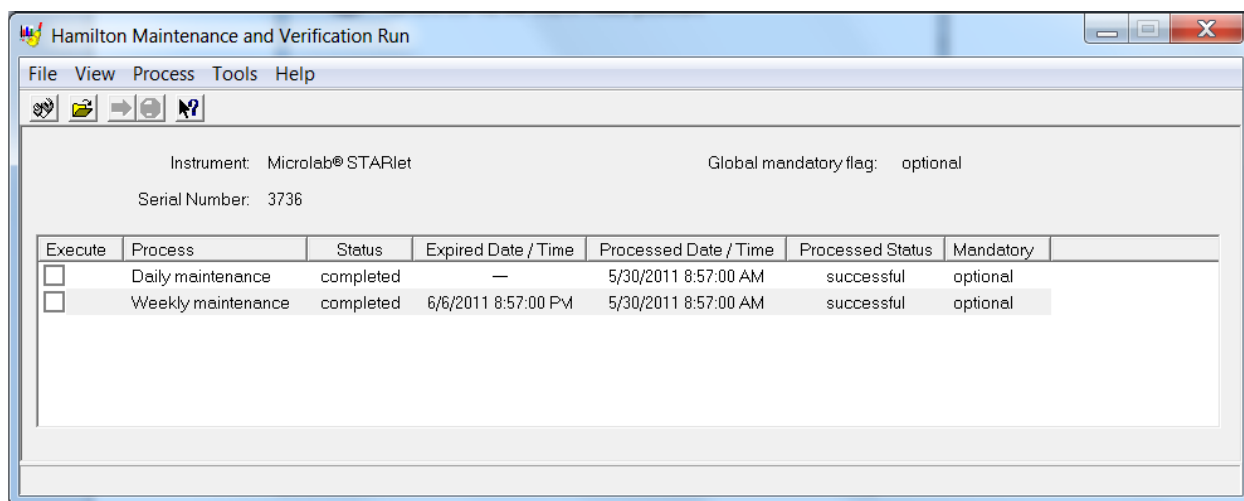
An instruction to place the carrier and the CO-RE 96 MPH Maintenance Tool onto the deck will be prompted. This step requires opening the front cover. After doing so, make sure to close the front cover. This applies to both placing and removing a carrier and a tool.

The following steps should be followed to perform the daily maintenance for the CO-RE 96 Probe Head TADM.

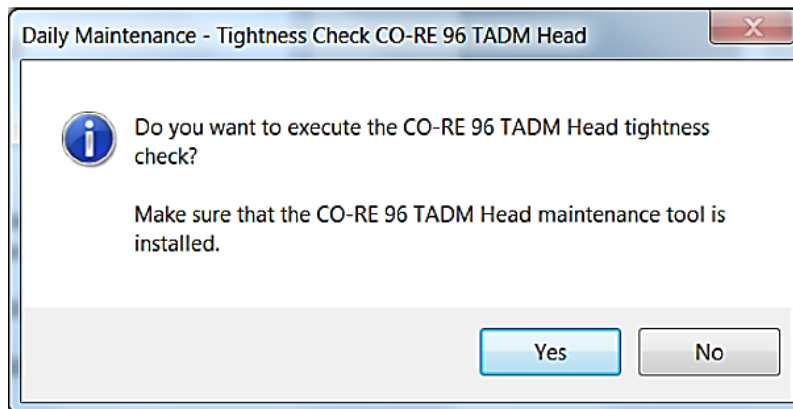
1. Start by activating the Microlab STAR Maintenance & Verification icon shown below.



2. Now that the Hamilton Maintenance and Verification is running, select “Daily Maintenance” from the choices and start. See image below.



- Follow the instructions and when prompted with the dialog box shown below, select **[YES]**. This will execute the CO-RE 96 Probe Head TADM tightness check.

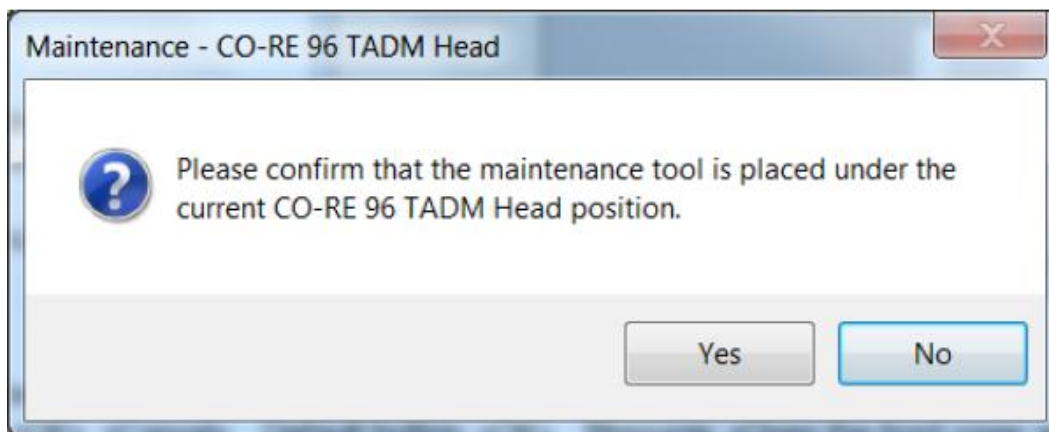


- Place the carrier with the tool onto the deck. Make sure that the tool when placed on the carrier has a "Maintenance" label engraved on it, as shown in the photo below. The position of the carrier may vary depending on how the system deck has been defined, as described in the Preparation section.

**NOTE**

During this step, the front cover may be opened. Once finished, make sure to close the cover. Only then continue to the next procedure.

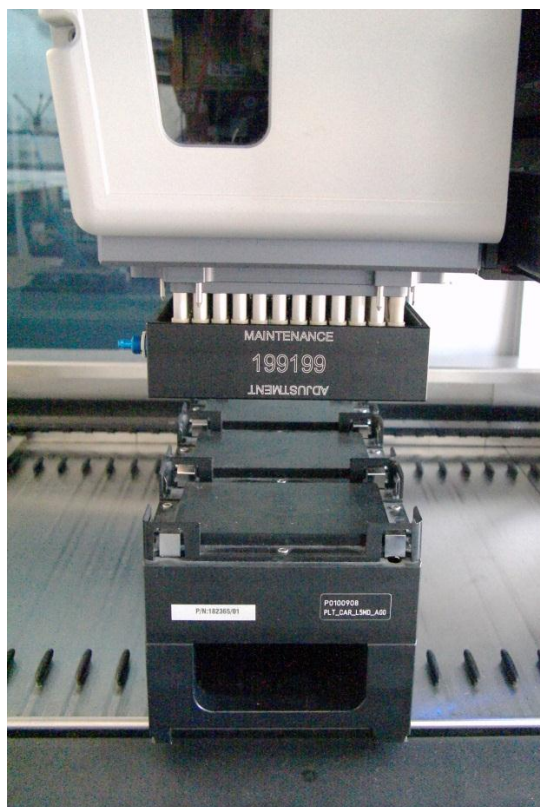
5. The CO-RE 96 Probe Head TADM will be pre-positioned above the tool. A dialog box to confirm the correct position will be prompted.



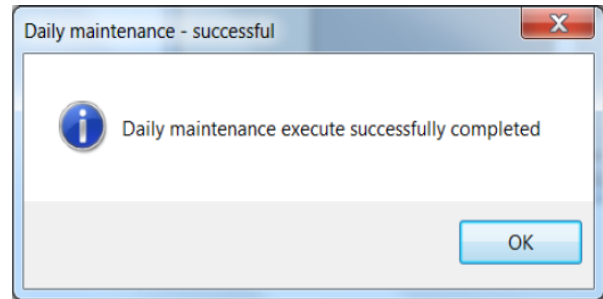
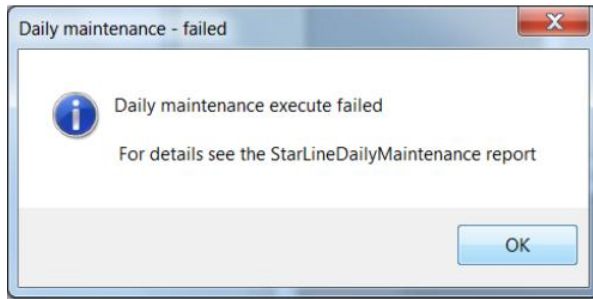
ATTENTION

*Choosing **[YES]** even if the position is incorrect may lead to a crash and can damage the CO-RE 96 Probe Head TADM and the Tool.*

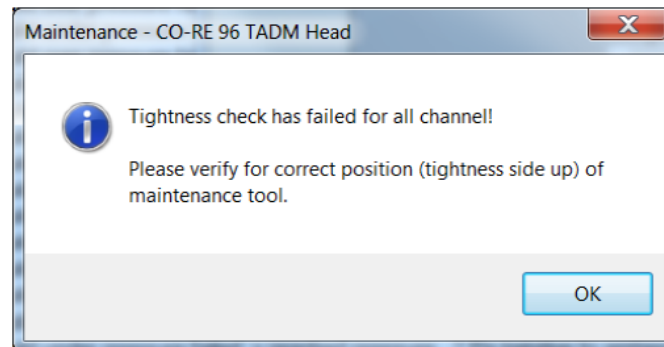
The CO-RE 96 Probe Head TADM will then pick up the Tool and the tightness check will then be executed.



Upon the completion of the tightness check, either of the following prompts will be displayed:



OR



Now, remove the carrier and the maintenance tool. The CO-RE 96 Probe Head TADM Daily Maintenance is completed.



ATTENTION

Ensure that the carrier and the tool are removed from the deck after the maintenance has been finished.

Weekly Maintenance



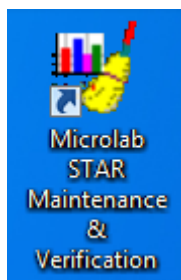
ATTENTION

Be aware of moving parts (i.e. X-Arm with its CO-RE 96 Probe Head TADM).

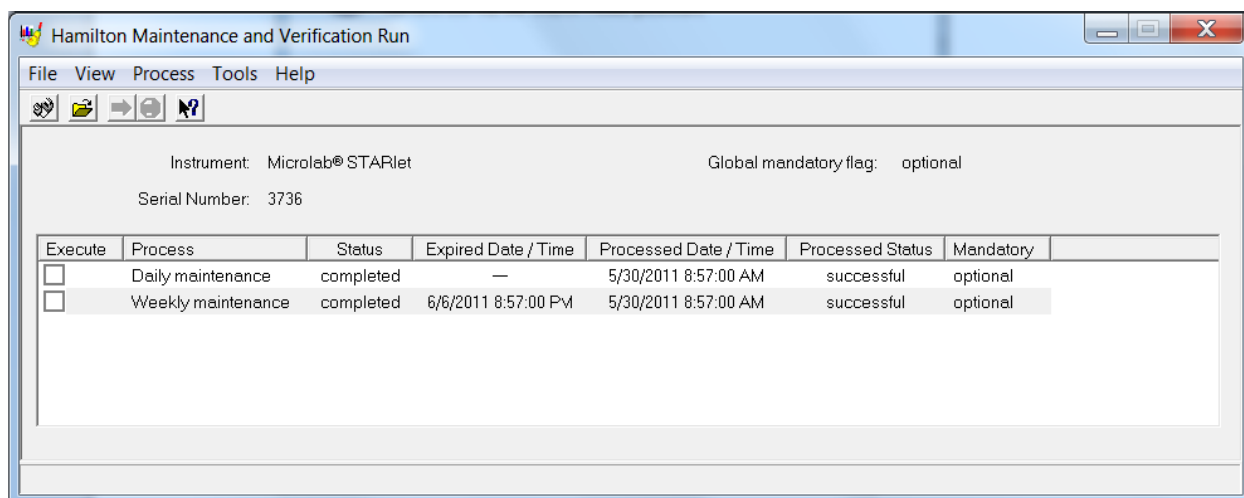
An instruction to place the carrier and the CO-RE 96 MPH Maintenance Tool onto the deck will be prompted. This step requires opening the front cover. After doing so, make sure to close the front cover. This applies to both placing and removing a carrier and a tool.

The following steps should be followed to perform the weekly maintenance for the CO-RE 96 Probe Head TADM.

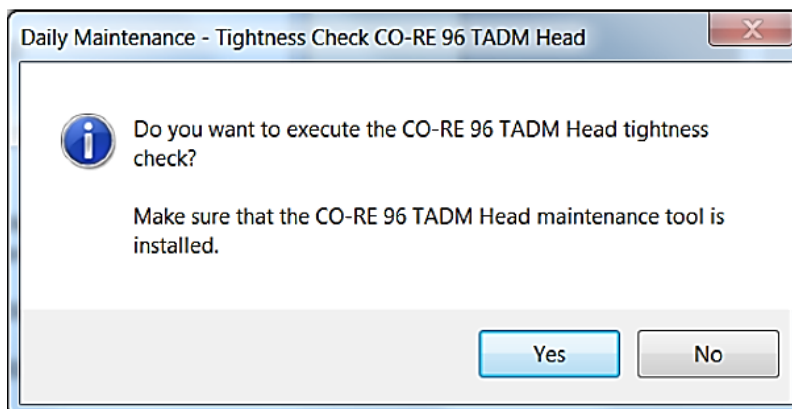
1. Start by activating the Microlab STAR Maintenance & Verification icon shown below.



2. Now that the Hamilton Maintenance and Verification is running, select “Weekly Maintenance” from the choices and start. See image below.



1. Follow the instructions and when prompted with the dialog box shown below, select **[YES]**. This will execute the CO-RE 96 Probe Head TADM tightness check.

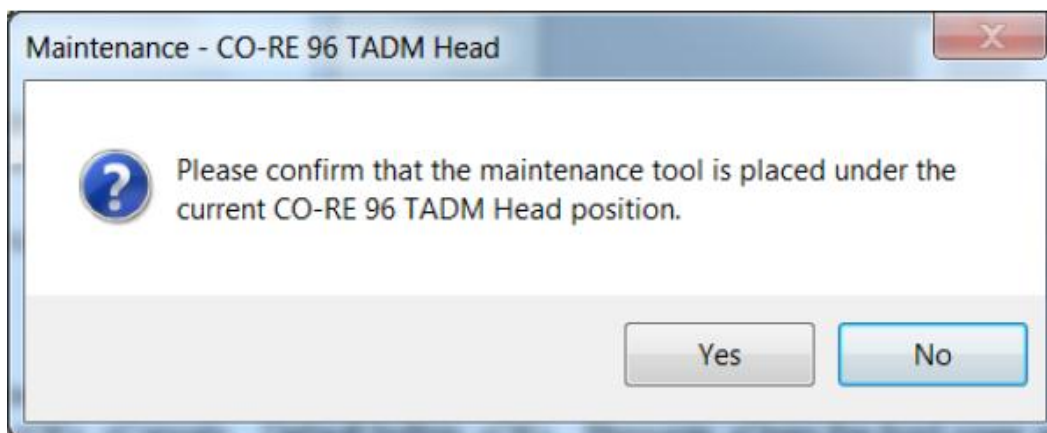


2. Place the carrier with the tool onto the deck. Make sure that the tool when placed on the carrier has a "Maintenance" label engraved on it, as shown in the photo below. The position of the carrier may vary depending on how the system deck has been defined, as described in the Preparation section.

**NOTE**

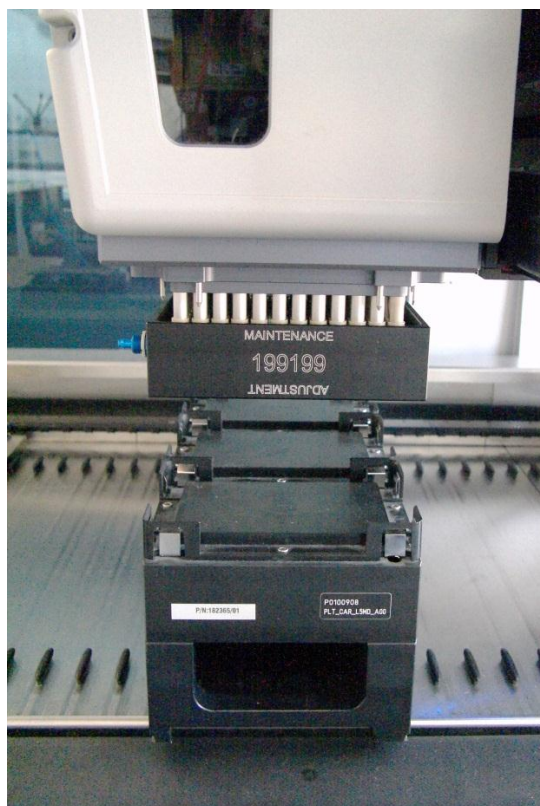
During this step, the front cover may be opened. Once finished, make sure to close the cover. Only then continue to the next procedure.

3. The CO-RE 96 Probe Head TADM will be pre-positioned above the tool. A dialog box to confirm the correct position will be prompted.

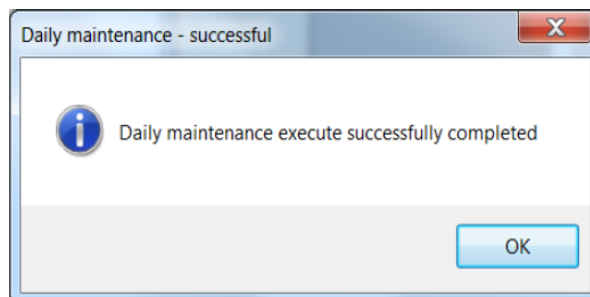
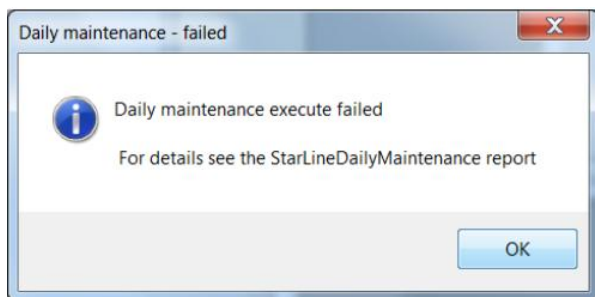
**ATTENTION**

Choosing **[YES]** even if the position is incorrect may lead to a crash and can damage the CO-RE 96 Probe Head TADM and the Tool.

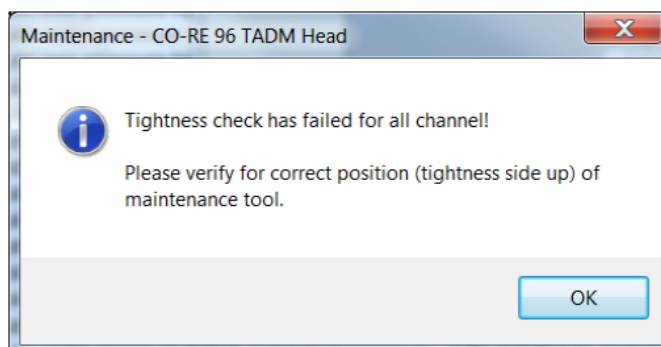
The CO-RE 96 Probe Head TADM will then pick up the Tool and the tightness check will then be executed.



Upon the completion of the tightness check, either of the following prompts will be displayed:



OR



Now, remove the carrier and the maintenance tool. The CO-RE 96 Probe Head TADM Weekly Maintenance is completed.



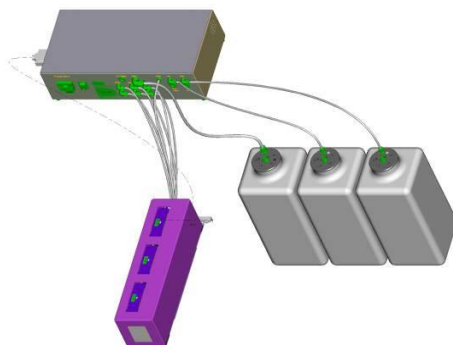
ATTENTION

Ensure that the carrier and the tool are removed from the deck after the maintenance has been finished.

4.10 CR Needle Wash Station

Hamilton recommends the following daily rinse routine and the weekly maintenance procedure to maintain the functionality of the needle wash station. Excellent washing results can be achieved only with periodic maintenance.

If so configured in the VENUS Software, daily and weekly maintenance are required.

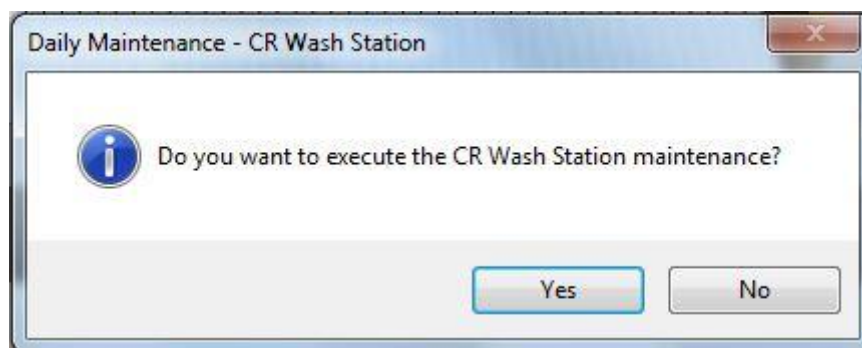


Before starting the procedure make sure the needles to be washed are placed in the Needle Wash Station.

4.10.1 Daily Maintenance

The daily maintenance procedure for the CR needle wash station takes 15 minutes. The purpose of this procedure is to rinse the fluid path of the wash station. With this procedure, deposits inside the fluid path can be minimized. If the wash station is seriously soiled, the operator should start the weekly instead of the daily maintenance.

Empty any remaining liquid of the wash containers and the waste container. Partially fill the wash solution containers with de-ionized water. With the cap facing upwards, shake the containers in lengthwise direction for a few moments. Empty the containers.



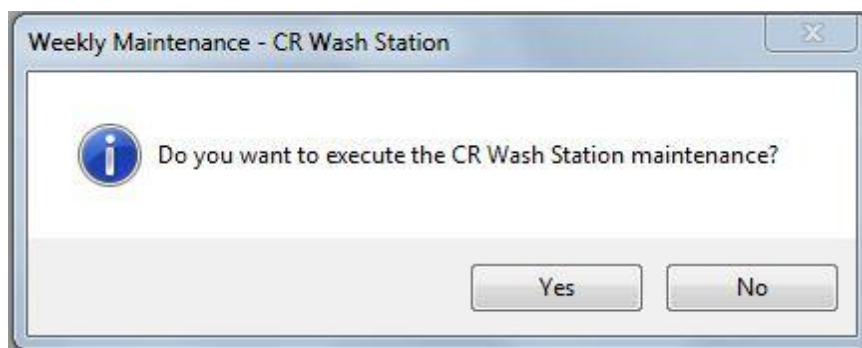
1. Once the maintenance procedure for the needle wash station has been started, the software advises the operator to fill Wash Container 1 (red dot on the container lid) with 4 liters of de-ionized water and 40ml of Deconex® 61 DR (P/N 281242).
2. 1x wash step (wash solution; Rinse time 60s, soak time 1s, flow rate 12ml/s, draining time 12s).

3. The rinse procedure is interrupted by the maintenance program, to advise the operator to empty wash container 1 (red dot on the container lid) and to fill it up with 3 liters of de-ionized water.
4. 2x rinse step (de-ionized water; Rinse time 60s, soak time 1s, flow rate 12ml/s, draining time 12s).
5. At the end of the daily maintenance procedure, wash container 1 and the waste container need to be emptied.

4.10.2 Weekly Maintenance

The weekly maintenance procedure for the CR needle wash station takes 40 minutes. The purpose of this procedure is to carry out periodical in-depth maintenance. The fluid path of the wash station and the needles are soaked in a special cleaning solution.

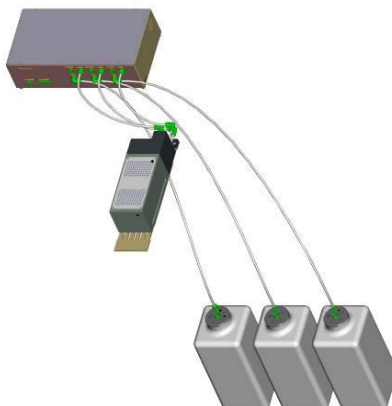
Empty any remaining liquid of the wash containers and the waste container. Partially fill the wash solution containers with de-ionized water. With the cap facing upwards, shake the containers in lengthwise direction for a few moments. Empty the containers.



1. Once the maintenance procedure for the needle wash station has been started, the software advises the operator to fill wash container 1 (red dot on the container lid) with 6 liters of de-ionized water and 60ml of Deconex® 61 DR (P/N 281242).
2. 1x wash step (wash solution; Rinse time 60s, soak time 300s, flow rate 12ml/s, draining time 12s).
3. 1x wash step (wash solution; Rinse time 60s, soak time 1s, flow rate 12ml/s, draining time 12s).
4. The procedure is interrupted by the maintenance program to advise the operator to empty wash container 1 (red dot on the container lid) and to fill it up with 6 liters of de-ionized water. Partially fill the wash solution container with de-ionized water. With the cap facing upwards, shake the container in lengthwise direction for a few moments. Empty the container.
5. Fill up wash container 1 with 3 liters of de-ionized water.
6. 2x rinse step (de-ionized water; Rinse time 60s, soak time 1s, flow rate 12ml/s, draining time 12s).
7. At the end of the weekly maintenance procedure, wash container 1 and the waste container need to be emptied and let them dry.

4.11 Wash Station 96/384 dual

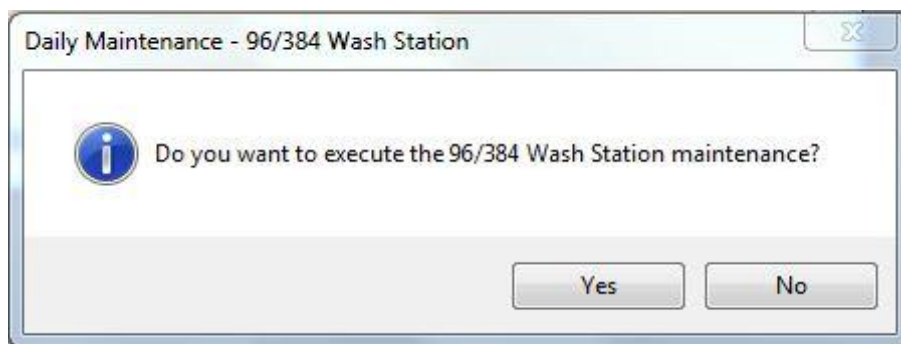
Hamilton recommends the following daily rinse routine and the weekly maintenance procedure to maintain the functionality of the wash station 96/384. Excellent washing results can be achieved only with periodic maintenance.



4.11.1 Daily Maintenance

The daily maintenance procedure for the wash station 96/384 takes 10 minutes. The purpose of this procedure is to rinse the fluid path of the wash station. With this procedure, deposits inside the fluid path can be minimized. If the wash station is seriously soiled, the operator should carry out the weekly instead of the daily maintenance.

Empty any remaining liquid of the wash container and the waste container. Partially fill the wash solution containers with de-ionized water. With the cap facing upwards, shake the container in lengthwise direction for a few moments. Empty the containers.



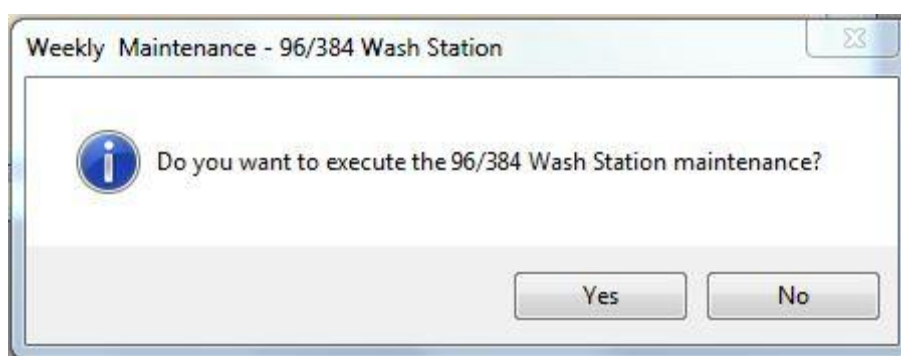
1. Fill the wash containers with 2 liters of de-ionized water and 20ml of Deconex[®] 61 DR (P/N 281242).
2. Empty the waste container.
3. 1x priming (initialize washer, empty)
4. 2x rinse step for both wash chambers (fill washer, empty)
5. Empty the wash containers. Partially fill the wash containers with de-ionized water. With the cap facing upwards, shake the containers in lengthwise direction for a few moments. Empty the wash containers. Fill the wash containers with 3 liters of de-ionized water again.

6. 2x rinse step for both wash chambers (fill washer, empty)
7. Empty the wash containers and waste container
8. Empty any remaining wash solution. Partially fill the containers with de-ionized water. With the cap facing upwards, shake the containers in lengthwise direction for a few moments. Empty the containers and let them dry.

4.11.2 Weekly Maintenance

The weekly maintenance procedure for the wash station 96/384 takes 20 minutes. To guarantee excellent wash efficiency, carry out the following steps at least once a week.

Empty any remaining liquid of the wash container and the waste container. Partially fill the wash solution containers with de-ionized water. With the cap facing upwards, shake the container in lengthwise direction for a few moments. Empty the containers.



1. Fill the wash containers with 2 liters of de-ionized water and 20ml of Deconex® 61 DR (P/N 281242).
2. Empty the waste container.
3. 1x Priming (Initialize washer, empty)
4. 1x Rinse step for both wash chambers (fill washer)
5. Soaking: Timer of 5 minutes
6. 1x Rinse step for both wash chambers (empty washer, refill after empty)
7. 1x Empty washer
8. Empty the wash containers. Partially fill the wash containers with de-ionized water. With the cap facing upwards, shake the containers in lengthwise direction for a few moments. Empty the wash containers. Fill the wash containers with 3 liters of de-ionized water again.
9. 2x rinse step for both wash chambers (Fill washer, empty)
10. 1x Empty washer
11. Empty any remaining wash solution of the wash and the waste containers. Partially fill the containers with de-ionized water. With the cap facing upwards, shake the containers in lengthwise direction for a few moments. Empty the containers and let them dry.

4.12 Hamilton Heater Shaker

Hamilton recommends cleaning the HHS regularly to prolong its life-span. In case of contamination or spillages clean the HHS immediately. The HHS must be turned off and cooled down for the maintenance procedure. Follow the maintenance procedure described in the ML STAR Line Operator's Manual. There are no further actions required.



ATTENTION

Do not clean the HHS when it is still hot!

Wait until the HHS has cooled down to room temperature before starting the cleaning procedure. Do not soak the manifold top.

4.13 Crystal Vacuum System

The CVS has to be connected to a Hamilton waste bottle, or to a suitable in-house liquid disposal system. Hamilton recommends maintaining the CVS by following the daily and weekly maintenance procedures described below. Only with a periodical maintenance a long lifespan of the CVS can be achieved.

4.13.1 Activating CVS Maintenance

Activation of the CVS maintenance requires modification of the two source files namely "**HslStarLineMaintMetConst.hs_**" and "**HslStarLineMaintMetConst.stp**". They are available in the folder ...\\HAMILTON\\Library.

The files can be modified by following the instructions below:

Make a copy of both HSL source files namely "**HslStarLineMaintMetConst.hs_**" and "**HslStarLineMaintMetConst.stp**". After renaming the files, modify the value of the constant "**isBVSInstalled(hslFalse);**" to "**isBVSInstalled(hslTrue);**". Save the files. Rename the original files to "**old_ HslStarLineMaintMetConst.hs_**" and "**old_ HslStarLineMaintMetConst.stp**". Finally rename the modified files generated with the original names.

4.13.2 Daily Maintenance

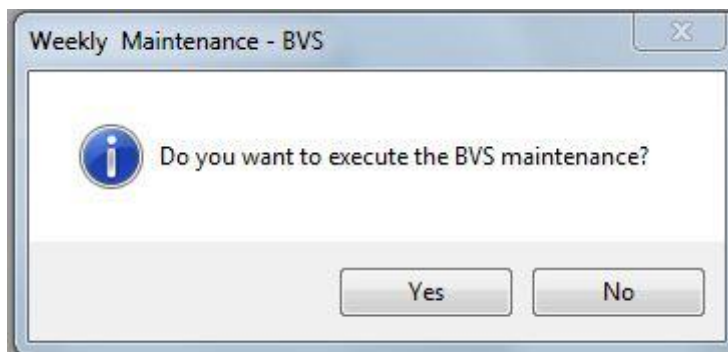
The daily maintenance procedure for the CVS takes 10 minutes. The purpose of this procedure is to rinse the tubing and the vacuum chamber. With this procedure, deposits inside the fluid path can be minimized. If the CVS is seriously soiled, the operator should start the weekly instead of the daily maintenance.



1. It is recommended to detach the Manifold Top from the vacuum module. It shall be cleaned separately to avoid water spill.
2. Prepare approximately 350ml of de-ionized water with 3.5ml of Deconex[®] 61 DR (P/N 281242). Pour the mixture into the vacuum chamber of the CVS carrier until it is full. Please do not overspill. Use a soaking time of 1 minute.
(Please keep in mind the ratio of 100ml de-ionized water:1ml Deconex[®])
3. Vacate the vacuum chamber.
4. Pour in de-ionized water into the vacuum chamber of the CVS carrier until it is full. Please do not overspill.
5. Vacate the vacuum chamber
6. Empty the waste bottle. Fill the waste bottle with de-ionized water to a degree. Ensure that the waste bottle is securely closed. Shake the container in a semi-inclined position for a few moments. Empty the waste bottle.
7. Clean the Manifold Top with a lint-free cloth soaked in Deconex[®] SOLARSEPT. Rinse afterwards with de-ionized water and let dry before using it again.

4.13.3 Weekly Maintenance

The weekly maintenance procedure for the CVS takes 15 minutes time. The purpose of this procedure is to rinse the tubing and the vacuum chamber. With this procedure, deposits inside the fluid path can be minimized.



1. It is recommended to detach the Manifold Top from the vacuum module. It shall be cleaned separately to avoid water spill.
2. Prepare approximately 350ml of de-ionized water with 3.5ml of Deconex[®] 61 DR (P/N 281242). Pour the mixture into the vacuum chamber of the CVS carrier until it is full. Please do not overspill. Use a soaking time of 5 minutes.
(Please keep in mind the ratio of 100ml de-ionized water:1ml Deconex[®])
3. Vacate the vacuum chamber.
4. Pour in de-ionized water into the vacuum chamber of the CVS carrier until it is full. Please do not overspill.
5. Vacate the vacuum chamber
6. Empty the waste bottle. Fill the waste bottle with de-ionized water to a degree. Ensure that the waste bottle is securely closed. Shake the container in a semi-inclined position for a few moments. Empty the waste bottle.
7. Clean the Manifold Top with a lint-free cloth soaked in Deconex[®] SOLARSEPT. Rinse afterwards with de-ionized water and let dry before using it again.



ATTENTION

Do not soak the manifold top.

4.14 Temperature Controlled Carrier (TCC)

Here is the recommended procedure for the daily/weekly maintenance of the TCC:

Clean all surfaces with a cloth soaked with Deconex[®] SOLARSEPT (see [Appendix B "Ordering Information"](#)).

5 ML STAR Instrument Decontamination

In general, for decontamination GLP must be observed. Here is the recommended procedure for decontaminating the ML STAR instrument:

Spray the front and side cover with Deconex® SOLARSEPT (see [Appendix B "Ordering Information"](#)).

Open the front cover and spray the deck with Deconex SOLARSEPT (see [Appendix B "Ordering Information"](#)).

Remove the tip eject plate of the tip waste station and clean it.

Spray Deconex® SOLARSEPT directly onto the surface of the tip waste station.

Remove the frame that holds the plastic bag in place, and discard the plastic bag in the laboratory's contaminated waste. Put the tip eject plate back in place.

Clean the tip eject sleeve (outer part of the pipetting channels) with a lint-free cloth soaked in Deconex® SOLARSEPT (see [Appendix B "Ordering Information"](#)).

The use of disinfectant liquids is recommended for decontamination of Carriers. Clean all carriers with Deconex® SOLARSEPT (see [Appendix B "Ordering Information"](#)) and leave them to dry. If they are heavily soiled, soak them afterwards in a solution of Deconex® 61 DR (see the [product data sheet](#) for further information).

Perform needle maintenance (see [Chapter 4.8 "Needles"](#)).

Perform washer maintenance (see [Chapter 4 "Maintenance" and the following sections](#)).

Autoclaving, i.e. using superheated steam under pressure, may only be used for decontamination followed by final disposal of disposables and racks; they will be destroyed by the process. Autoclaving cannot be used for instrument components or accessories (channels, transport tools, wash station components, heaters, shakers, carriers).

In some cases other decontamination procedures may be desirable, e.g. for reliably destroying infectious materials or DNA/RNA. Many of these decontamination procedures are very aggressive and can cause damage to the ML STAR instrument. Please follow the guideline given below. If using other decontamination procedures not listed here, be aware that they may increase service and maintenance requirements and may make shorter maintenance intervals necessary.

5.1 Surface Decontamination using Liquids

Preferably, use the procedure using Deconex® 61 DR or Deconex® SOLARSEPT described above. Other procedures have not been tested by Hamilton. Use Deconex® 61 DR at a maximum temperature of 50°C.

If using other liquids or sprays for surface decontamination, follow the manufacturer's instructions. Pay particular attention to potential corrosiveness, e.g. acidic or alkaline solutions and oxidizing agents. Use of such agents may increase service and maintenance requirements (O-rings exchange, greasing of spindles etc.) and may make shorter maintenance intervals necessary. Do not use disinfecting materials which contain hypochlorite (Javel water, Chlorox) or bleaching fluids. If possible, use non corrosive, neutral liquids.

When the instrument deck and carriers are cleaned using enzyme solutions such as DNase and RNase, make sure to thoroughly remove any remainders afterwards by wiping deck and carriers with deionized water in order to avoid aggressive substances to be left on the surface.

5.2 Decontamination using Gases

Should decontamination by fumigation be necessary, Hamilton recommends using ethylene oxide. Be aware that ethylene oxide fumigation may increase service and maintenance requirements (O-rings exchange, greasing of spindles etc.) and may make shorter maintenance intervals necessary. Hamilton does not carry out such fumigation procedures, please use a 3rd party contractor for such service.

Fumigation using hydrogen peroxide (H₂O₂) is possible, but leads to bleaching or discoloration of many instrument materials. Be aware that hydrogen peroxide fumigation may increase service and maintenance requirements (O-rings exchange, greasing of spindles etc.) and may make shorter maintenance intervals necessary. Hamilton does not carry out such fumigation procedures, please use a 3rd party contractor for such service.

Do not use formaldehyde fumigation or chlorine oxides (chemical compounds of chlorine and oxygen such as bleach). They are not suitable for the ML STAR instrument because of chemical reaction and corrosion.

5.3 Decontamination using UV Light

Hamilton offers a UV option for ML STAR Line instruments. We recommend UV irradiation for 15 minutes and moving the arm left and right while the UV-lamp is on.

If using other UV irradiation options, use wavelength, intensity and duration according to manufacturer's instructions.

Be aware that UV irradiation causes many synthetic materials to embrittle. This may increase service and maintenance requirements and may make shorter maintenance intervals necessary. The instrument cover should be made of Makrolon[®], not of standard acrylic glass, due to its better resistance to UV irradiation.

6 Verification

6.1 General Description

To perform the verification of the ML STAR instrument “Field Verification 2” is used. The Field Verification 2 can only be performed by trained personnel.

Verification specifications (as applicable in the Microlab STAR Line Field Verification 2 User Manual) are different from the specifications given in the technical specifications of this user manual.

Especially the volume verification specifications (as applicable in the Microlab STAR Line Field Verification 2 User Manual) are different from the pipetting specifications for disposable tips given in the technical specifications of this user manual. The field verification contains validated procedures and equipment defined by Hamilton to demonstrate and to verify the correct function of the instrument according to specifications given by Hamilton suitable for the field. The field verification is therefore a reference defined by Hamilton to compare the instrument’s performance according to given procedures valid for a broad operating range.

Based on that, specifications as applicable in the technical specifications will be achieved by maintaining defined environmental conditions in the laboratory, by keeping the operating range as small as possible, by optimizing the methods such as adapting the liquid classes, knowing the sample liquids and the characteristics of used labware, etc. See also Chapter 1 “General Information”.

For the 1000µl-channels, the CO-RE 96 Probe Head and the CO-RE 384 Probe Head, a dye-pipetting procedure followed by gravimetric and photometric analysis is used to verify the trueness and precision. The 5ml-channels are verified by a gravimetric approach.

Devices such as Heater Shaker, Cooler, barcode reader and cover safety can also be verified with the “Field Verification 2”.



Note

This is an additional feature of the ML STAR Line, which is not included in the VENUS two Software.

Further information can be found in Appendix B “Ordering Information” of this manual; for additional information consult a local Hamilton representative.

7 Technical specifications

7.1 Basic ML STAR Line Specifications

Instrument dimensions	(for detailed information on configurations, see Appendix C “Instrument Dimensions”)		
STARlet	width: 1124mm (1450mm with 96/384 Probe Head), height: 903mm, depth: 795 mm (Autoload: 1006 mm)		
STAR	width: 1664mm (1990mm with 96/384 Probe Head), height: 903mm, depth: 795 mm (Autoload: 1006mm)		
STARplus	width: 2160mm (2486mm with 96/384 Probe Head), height: 903mm, depth: 795mm (Autoload: 1006mm)		
Work area dimensions	Width (x):	Height (z) ¹⁾ :	Depth (y):
STARlet	675mm	145mm	497mm
STAR	1215mm	145mm	497mm
STARplus	1705mm	145mm	497mm
Weight	8 channels	96/384 Probe Head and 8 individual channels	
STARlet	135 kg	150 kg	
STAR	145 kg	160 kg	
STARplus	205 kg	220 kg	
Deck capacity			
STARlet	30 tracks (T) allow combinations of: maximum of 30 tube carriers (1 T) holding 24 or 32 tubes per carrier maximum of 5 carriers (6 T) holding 5 tip racks or 5 plate positions per carrier		
STAR	54 tracks (T) allow combinations of: maximum of 9 carriers (6 T) holding 5 tip racks or 5 plate positions per carrier		
STARplus	71 tracks (T) allow combinations of: maximum of 11 carriers (6 T) holding 5 tip racks or 5 plate positions per carrier		
Modal precision	x-y-z positional accuracy of 0.1mm		
Tip sizes	low volume: 10µl, intermediate volume 50µl, standard volume: 300µl, high volume: 1000µl		
Needle sizes	low volume: 10µl, standard volume: 300µl, high volume: 1000µl, needles available only for 1000µl single channels		
Power consumption	Standby power consumption: 100 VA Maximum power consumption: 600 – 1000 VA, depending on configuration		

¹⁾ The maximum height for labware which can be used on the deck is 140mm.

7 Technical Specifications

Pipetting specifications for disposable tips*	Disposable tip size	Volume	Trueness R (%)	Precision CV (%)
individual 1000µl-channels	10µl	0.5µl	10.0%	6.0%
	10µl	1µl	5.0%	4.0%
	10µl	5µl	2.5%	1.5%
	10µl	10µl	1.5%	1%
	50µl	0.5µl	10.0%	6.0%
	50µl	1µl	5.0%	4.0%
	50µl	5µl	2.5%	1.5%
	50µl	50µl	2.0%	0.75%
	300µl	10µl	5.0%	2.0%
	300µl	50µl	2.0%	0.75%
	300µl	200µl	1.0%	0.75%
	1000µl	10µl	7.5%	3.5%
	1000µl	100µl	2.0%	0.75%
	1000µl	1000µl	1.0%	0.75%
	For pipetting of less than 10µl Hamilton recommends 10µl/ 50µl volume disposable tips to achieve highest pipetting precision.			
Pipetting specifications for needles*	Needle size	Volume	Trueness R (%)	Precision CV (%)
	10µl	1µl	5.0%	8.0%
	10µl	5µl	2.5%	2.0%
	10µl	10µl	1.5%	1.0%
	300µl	5µl	8.0%	8.0%
	300µl	50µl	2.0%	2.0%
	300µl	200µl	1.0%	1.0%
	1000µl	50µl	5.0%	3.0%
	1000µl	100µl	3.0%	2.0%
	1000µl	1000µl	2.0%	1.0%
Pipetting specifications for 5ml disposable tips*	Disposable tip size	Volume	Trueness R (%)	Precision CV (%)
	5ml	50µl	5.0%	2.5%
	5ml	500µl	2.0%	1.5%
	5ml	1000µl	1.5%	1.0%
	5ml	5000µl	2.0%	1.5%

7 Technical Specifications

Pipetting specifications for disposable tips*	Disposable tip size	Volume	Trueness R (%)	Precision CV (%)
1000µl CORE 96 Probe Head	Maximum pipetting volume: 1000µl	10µl	1µl	5.0%
		10µl	5µl	2.5%
		10µl	10µl	1.5%
		50µl	1µl	5.0%
		50µl	5µl	2.5%
		50µl	50µl	1.5%
		300µl	10µl	3.0%
		300µl	50µl	1.5%
		300µl	300µl	1.0%
		1000µl	10µl	7.5%
		1000µl	100µl	2.0%
		1000µl	1000µl	1.0%
*Test criteria available upon request	For pipetting of less than 10µl Hamilton recommends 10µl/ 50µl volume disposable tips to achieve highest pipetting precision.			
Pipetting specifications for disposable tips*	Disposable tip size	Mode:	Volume	Precision CV (%)
50µl CORE 384 Probe Head	Maximum pipetting volume: 50µl	Surface:	0.1µl	8.0%
		Surface:	0.5µl	6.0%
		Surface:	1µl	3.5%
		Jet:	1µl	15.0%
		Surface:	5µl	3.0%
		Jet:	5µl	4.0%
		Surface:	10µl	2.0%
		Jet:	10µl	3.0%
		Surface/Jet:	50µl	2.0%
Using the 50µl CORE 384 Probe Head as a 96 Probe Head	Maximum pipetting volume: 300µl	300µl Rocket	2µl	4.0%
		300µl Rocket	5µl	2.0%
		300µl Rocket	10µl	2.0%
		300µl Rocket	100µl	2.0%
		300µl Rocket	300µl	2.0%
Operating data for tips	Temperature range +15°C – +25°C			
Storage data for tips	Temperature Max. +55°C			

Liquid Level Detection	Individual channels: Capacitive Liquid Level Detection (cLLD) and pressure (pLLD) on aspiration, cLLD on dispense, minimum volume 10µl, depending on container type 96 Probe Head: Capacitive Liquid Level Detection (cLLD) 384 Probe Head: Capacitive Liquid Level Detection (cLLD)
Throughput	8 channels: To fill one 96-well Microtiter Plate with 100µl samples (new tip for each sample): 320s. Aliquot reagent to a 96-well Microtiter Plate (<90µl per well): 60s 96 Probe Head: Replication of one 96-well plate, 100µl, with cLLD on aspiration: 35s (incl. new tips) Reformatting of four 96-well plates to one 384-well plate, 50µl, new tips, with cLLD on aspiration: 140s 384 Probe Head: Replication of one 384-well plate, 20µl, with cLLD on aspiration: 35s (incl. new tips) Reformatting of four 384-well plates to one 1536-well plate, 10µl, new tips, with cLLD on aspiration: 140s
Labware	All SBS standard plate types up to 1536 wells and most commercially available tube types
Carriers	For all standard labware formats and according to customer requirements.

Operating data	Maximum power consumption	600 VA or 1000 VA (depending on configuration)
	Voltage	115 VAC / 230 VAC
	Frequency	50 / 60 Hz $\pm$ 5%
	Delayed action fuse	600 VA: 115V~: 6.3A (T6.3AL250) 230V~: 3.15A (T3.15AL250) 1000 VA: 115V~: 10A (T10AL250) 230V~: 5A (T5AL250)
	Installation category	II
	Pollution degree	2
	Temperature range	15°C – 35°C
	Relative humidity	30% – 85% (non-condensing, indoors)
	Noise level	< 65 dBA (regarding EN27779) < 46 dBA in standby mode
	Altitude	max. 2000m above sea level
	Heat: the power consumed will be transferred to head	e.g. 600 or 1000 Watts of Heat = 600 or 1000 Joules/ Second
	Indoor use only	
Storage and transportation	Temperature range	-25°C – +70°C
	Relative humidity	10% – 90% (non-condensing, indoors)
Lifespan	7 years based on 8h/day, 5 days/week	
Materials used in ML STAR Line instruments and accessories	Steel: 1.4301, 1.4305, 1.4310, 1.4435 Aluminum: AA 5083.0, various anodizing Bronze Brass Brazing solder: Zinc Plastics: PE, PP, PTFE, PEEK, FFPM, EPT, EPDM, FPM, NBR, POM, PCDF, FFKM, Hypalon, SI, Kapton, Polyamide PCBs	
Recommended PC	Intel Core 2 Duo, $\geq$ 4 GB RAM, 500 GB Hard Drive, 16x DVD +/-RW, DirectX 250 MB graphic card, MS Windows 7 Professional. (not included in shipment)	
Communication	USB or RS232 with dual processor board	
	Ethernet or USB with LAN dual processor board	

7.2 Hamilton standard PCs Specifications

	Desktop computer Windows 7
Configuration (overview)	OptiPlex XE DT: N Series Base Desktop Intel Core 2 Duo E8400 (3GHz, 1333 MHz, 6MB) 4096 MB (2x 2048 MB) 1333MHz DDR3 Dual Channel RAM 16x DVD+/-RW 2x 250 GB, RAID1, HDD 2x RS232, 6x USB, 2x Ethernet, 1x Digital Video Port Flat Screen 21.5" (1920 x 1080) Webcam Mouse and Dell Soundbar Country specific Keyboard MS Windows 7 Ultimate 64 Bit MS Office 2013 for Home and Business Recovery DVD 2Yr Dell warranty (next business day support)

7.3 Touch Screen Monitor Specifications for the ML STAR Line

Screen size	21,5"	Technology	AccuTouch Five-Wire Resistive
Resolution	1920 x 1080	Colors	Up to 16,2 million
Interfaces	1x VGA HD D-Sub 1x USB Type A 1x serial RS-232	Power	120/230V (50/60Hz)
		Consumption	50W
		# of touches	35 millions
Brightness	211 cd/m ²	Certification	FCC Class B, CE, IC Class B, UL, TUV GS, VCCI Class B ITE, cUL, C-Tick Class B, RoHS
Contrast	550:1	Weight	7,9 kg
Viewing angle	+/- 70 horizontal +75/-60 vertical		

7.4 CR Needle Wash Station Specifications

Specification	Needle capacity	24 per Wash Station 8 per chamber
	Wash time (typ.)	45 sec/cycle
	Liquid consumption (typ.)	100 ml/cycle
	Carry over (typ.)	1000µl needle, 100µl: $< 3 \times 10^{-6}$ 300µl needle, 100µl: $< 2 \times 10^{-6}$ 10µl needle, 10µl: $< 5 \times 10^{-6}$
	Deck capacity	6 tracks
	Max. units	2 per instrument
Operating data	Maximum power consumption	140VA
	Nominal Voltage	100 VAC (-15%) 240 VAC (+ 10%)
	Nominal Frequency	50 / 60Hz $\pm 5\%$
	Delayed action fuse	115V~: 3.15A (T3.15L250) 230V~: 1.6A (T1.6AL250)
	Installation category	II
	Pollution Degree	2
	Temperature range	15°C – 35°C
	Relative humidity	30% – 85% (non-condensing, indoors)
	Altitude	max. 2000m above sea level
	Indoor use only	
Storage and transportation	Temperature range	-25°C – +70°C
	Relative humidity	10% – 90% (non-condensing, indoors)

7.5 96/384 Wash Station Dual Specifications

Specification	Wash time (typ.)	67 sec/ 2 cycles (96x10 tip) 71 sec/ 2 cycles (96x300 tip) 73 sec/ 2 cycles (384x30 tip)
	Liquid consumption (typ.)	880 ml/ 2 cycles
	Carry over (typ.)	2 cycles 10µl Tips, 10µl: $< 3 \cdot 10^{-6}$ 300µl Tips, 300µl: $< 5 \cdot 10^{-6}$ 30µl Tips: $< 1.8 \cdot 10^{-5}$
		3 cycles 30µl Tips: $< 1 \cdot 10^{-6}$
	Deck capacity	6 tracks
	Max. units	3 per Instrument
Operating data	Power consumption	41 V / 100VA (max.) supplied by the ML STAR Line
	Temperature range	15°C – 35°C
	Relative humidity	30% – 85% (non-condensing, indoors)
	Altitude	max. 2000m above sea level
	Indoor use only	
Storage and transportation	Temperature range	-25°C – +70°C
	Relative humidity	10% – 90% (non-condensing, indoors)

7.6 iSWAP Specifications

Plate format	Microtiter footprint plate height ≤ 43 mm		
Absolute positioning	Accuracy	x,y,z = 0.5mm	
	Reproducibility	x,y,z = 0.25mm	
Movement range (on a STAR 8/iSWAP instrument)	Min. absolute position	Max. absolute position	Remarks
x	-206mm	+1578mm	at $x_{\min}$ 58mm space between plate and deck at $x_{\max}$ half plate on deck
y	-185mm	+605mm	
z	+100mm	+282mm	: iSWAP 182600 rev. 00 – 02
	+0mm	+282mm	: iSWAP 182600 rev. 03 and iSWAP landscape 190220
Gripper opening	72mm	108mm	: iSWAP 182600 rev. 00 – 03
	72mm	132mm	: iSWAP landscape 190220
Gripping force	5 N – 16 N (default 9 N) : iSWAP 182600 rev. 00 – 03		
	5 N – 16 N (default 9 N) : iSWAP landscape 190220		
Transport mass	300g filled Deep Well Plate		
No restriction of random access range for 4-, 8-, 12-, and 16-channel ML STAR Line's.			
Operating data	Temperature range	15°C – 35°C	
	Relative humidity	30% – 85% (non-condensing, indoors)	
	Altitude	2000m above sea level	

7.7 Tube-gripper Specifications

Tube sizes	Tube diameter from 8 mm to 20 mm Tube height ≤ 120 mm		
Modal precision	x-, y-, z- positional accuracy of 0.1mm (measured on tube-gripper)		
Movement range	x	Reaches the same X-coordinate as the single channels (e.g. on a STAR 8x 1000µl-channel instrument)	
	y	Reaches all tube positions of the sample carriers: SMP-CAR-32 and SMP-CAR-24 (on a ML STAR instrument with 6x 1000µl-channel)	
	z	Min. absolute position	Max. absolute position
		+54.2mm	+254.2mm
			Remarks
			Measured from deck to the gripping point on a tube
Gripper opening		min. absolute position	max. absolute position
		5.5mm	22mm
Transport mass	200g		
Operating data	Temperature range	15°C – 35°C	
	Relative humidity	30% – 85% (non-condensing, indoors)	
	Altitude	2000m above sea level	

7.8 CO-RE Gripper 1000µl Specifications

Labware format	Microtiter footprint plate height ≤ 43 mm		
Absolute positioning	Accuracy x,y,z = 0.5mm Reproducibility x,y,z = 0.25		
Movement range	x Track 1 – n (depending on instrument type) y Depending on # of channels and used front channel z Lowest position = 15mm over metal deck sheet		
Gripper opening	Min. opening 9mm 18mm	Max. opening Dependent upon travel range on arm Dependent upon travel range on arm	Arm Modular pipetting arm Pipetting arm MPH/iSWAP, iSWAP arm
Gripping force	5 N – 16 N (default 9 N)		
Transport mass	300g filled Deep Well Plate		

7.9 CO-RE Gripper 5ml Specifications

Labware format	Microtiter footprint plate height ≤ 43 mm		
Absolute positioning	Accuracy x,y,z = 0.5mm Reproducibility x,y,z = 0.25		
Movement range	x Track 1 – n (depending on instrument type) y Depending on # of channels and used front channel z Lowest position = 46mm over metal deck sheet		
Gripper opening	Min. opening 18mm 36mm	Max. opening Dependent upon travel range on arm Dependent upon travel range on arm	Arm Modular pipetting arm Pipetting arm MPH/iSWAP, iSWAP arm
Gripping force	5 N – 16 N (default 9 N)		
Transport mass	300g filled Deep Well Plate		

7.10 Autoload Option: Barcode and Reader Specifications

Carriers, containers, racks and tip racks can be identified by a barcode, which a reader, mounted on the Autoload slide, scans. The system must allow specification of ranges (barcode mask) for plausibility checking of barcode information.

7.10.1.1 Barcode Symbolologies

The following barcode symbolologies can be detected by the system:

- ISBT standard
- Code 128 (subset B and C)
- Code 39
- Codabar
- Code 2 of 5 Interleaved
- UPC A

For the highest reading safety Hamilton recommends:

1. To use barcode type Code128 (subset B and C)
2. Disabling the unused barcode types in the configuration editor of the User Software (refer to the [Microlab STAR Programmer's Manual](#))
3. Defining a barcode mask via the labware editor of the User Software (refer to the [Microlab STAR Programmer's Manual](#))

7.10.1.2 Reading Accuracy

The rate of inaccurate readings of sample plates and container bar codes is less than 1 ppm.

The above-mentioned specification is valid under the following conditions:

- Bar code symbology module: ISBT standard
- Code density: 0.0065" (0.1651 mm)
- Print Quality see [Chapter 7.10.1.3 "Barcode Specifications"](#)
- Recognized errors are defined as an accurate reading.

7.10.1.3 Barcode Specifications

Type	Black bars and white background	
Length of string	Maximum 20 characters excluding start, stop and check characters, depending on the code length (see label dimensions).	
Code density, tolerance	Minimum module width (x dimension) including a print tolerance: $\geq 0.0065''$ (0.1651mm) Maximum module width (x dimension) including a print tolerance: $\leq 0.02''$ (0.508mm) Best reading performance with x dimension $\geq 0.01''$ (0.254mm)	
Check character	ISBT standard	One character
	Code 128	One character
	Code 39	None
	Codabar	None
	Code 2 of 5 Interleaved	None
	UPC A	One character
Quiet zone	≥ 10 times the x dimension, but at least 3 mm.	
Print quality	The barcode print must be of a high quality. A printed barcode with an ANSI/ CEN/ ISO grade A or B is required. Offset, typographic, intaglio and flexographic printing are suitable. Mechanical dot matrix and thermo matrix printing are <u>not</u> suitable. The surface may be treated, sealed or plastic-coated.	

7.10.1.4 Sample Barcodes

Barcode specifications

For general barcode specification see [Chapter 7.10.1.3](#)

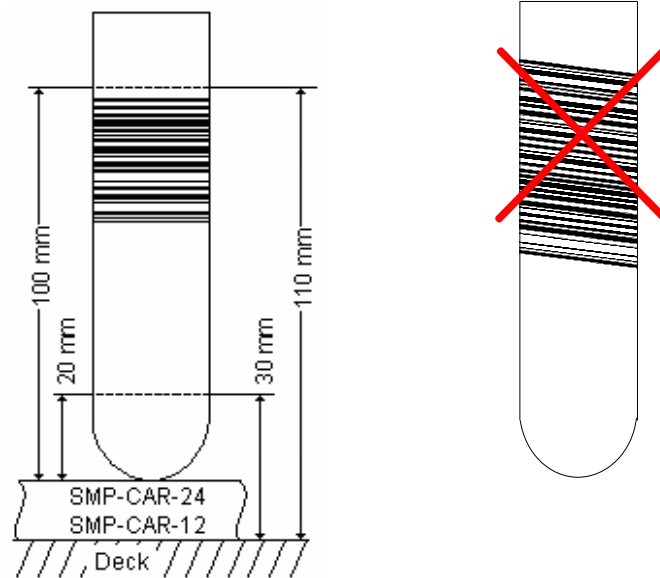


Dimension		Min.	Max.
A	Label length	-	80mm
B	Code length	-	74mm
C	Quiet zone	3mm	
D	Label width	12mm	-
E	Code width	12mm	-
F	Distance from code to label edge	-	1mm

Positioning barcode labels

The label must be glued within a range of between 20mm to 100mm from the bottom of the tube.

The label must fit tightly at an angle of approximately 90° to the tube.



The label must fit tightly over its whole length.

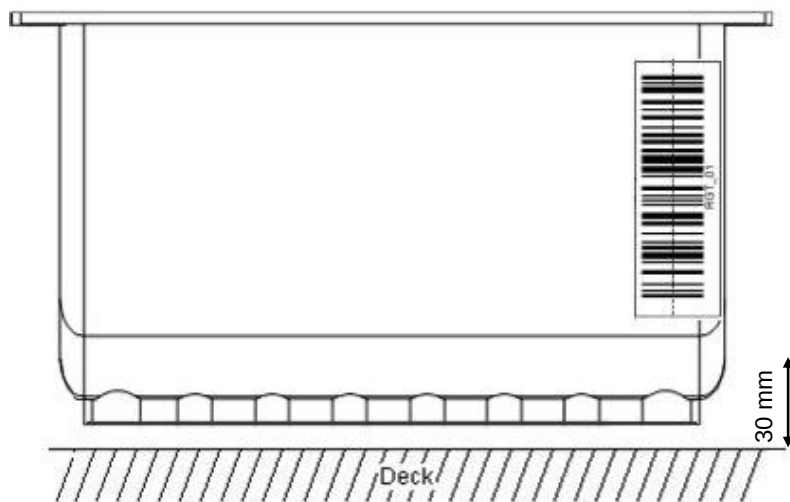
7.10.1.5 Reagent Barcodes

Proposed barcode mask: RGT_mm	RGT: Reagent _: Separator (underline) mm: Reagent number 1.....99		
Label specification For general barcode specification see Chapter 7.10.1.3			
	Dimension		Min Max
	A	Label length	- 66 mm
	B	Code length	- 60 mm
	C	Quiet zone	3 mm -
	D	Label width	15 mm -
	E	Code width	12 mm -
	F	Distance from code to edge of label	- 1 mm

Positioning of barcode labels:

The label must be positioned on the upper edge, in the middle of the container.

The label must fit tightly over its whole length.

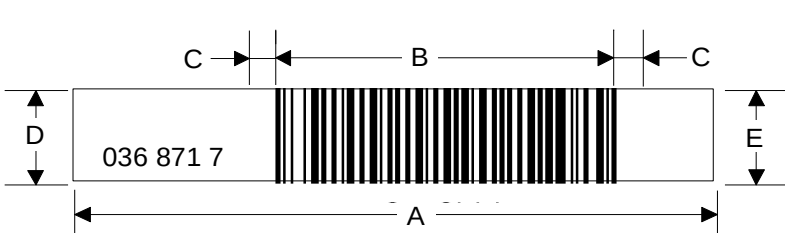
Positioning of barcode label on reagent trough 50ml:

The label must be glued within a range of between 30mm to 110mm from the deck.

7.10.1.6 Plate Barcodes

Label specification

For general barcode
specification see [Chapter
7.10.1.3](#)

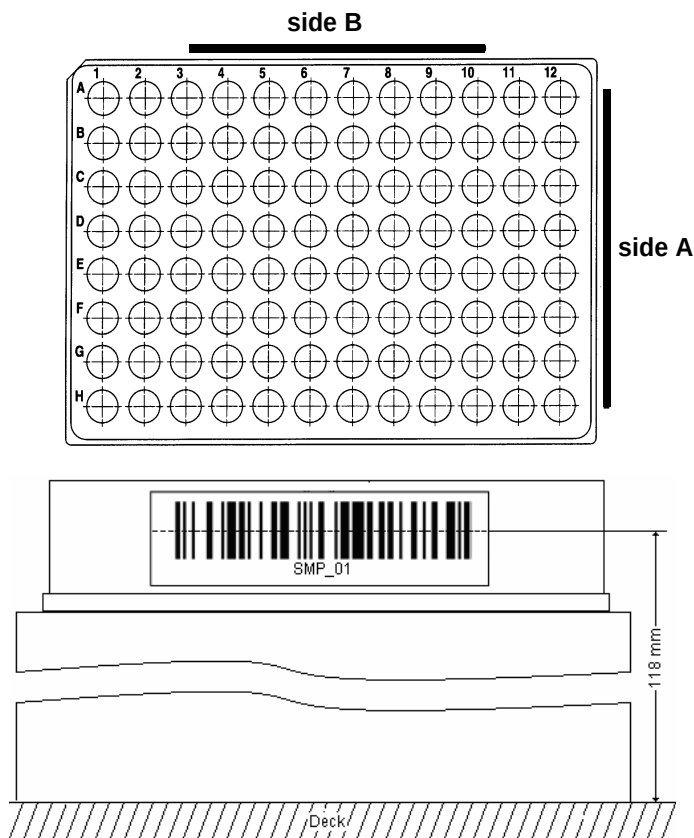


Dimension		Min	Max
A	Label length	-	66mm
B	Code length	-	30mm
C	Quiet zone	3mm	-
D	Label width	10mm	-
E	Code width	7mm	-
	Distance from code to label edge (if necessary)	-	1mm

Positioning barcode labels

The plate barcode must fit on side A or side B of the plate.

The barcode must be positioned in the middle of the plate.



The barcode label must be centered and parallel to the edge of the plate.



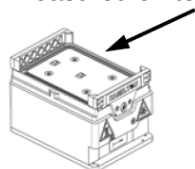
The barcode label must not protrude above or below the edge of the plate.



7.10.2 Hamilton Heater Shaker Specifications

Instrument dimensions	Width (x)	Height (z)		Depth (y)
	150 mm	90 mm		105 mm
Weight	2500 g			
Labware	Standard Microtiter Plates with adapter Standard Deep Well Plates with adapter Maximum weight (incl. adapter): 500 g			
Temperature specifications*	Temperature	Ramp from 25°C	Deviation compared to target temp	Tolerance band for measurement and control: Deviation on plate (middle to edge position)
	37°C	in 3 min	36.0°-38.0°C	35.0°-39.0°C
	60°C	in 10 min	58.5°-61.5°C	57.0°-63.0°C
	90°C	in 20 min	88.0°-92.0°C	86.0°-94.0°C
	100°C	in 30 min	97.5°-102.5°C	95.0°-105.0°C
	105°C	In 35 min	102.5°-107.5°C	100°-110°C
Features of the Heater Shaker	Temperature control		From 5°C above ambient temperature to 105°C	
	Shaking directions		Controlled by two sensors (located in the middle and at the edge of the adapter plate)	
	Acceleration		clockwise and counterclockwise	
	Deceleration		2.0s (from 0 to max rpm)	
	Shaking orbits: (peak to peak)		2.0s (from max rpm to 0)	
			1.5mm	
			2.0mm	
			3.0mm	

*Measured on top of the MTP flat bottom plate



Features of the Heater Shaker (cont.) ¹	Recommended shaking speed in rpm depending on labware and shaking orbit				
	Orbit	MTP	DWP	Sarstedt	customized
	1,5 mm	Not available	2000	1800	Not available
	2,0 mm	2500	2000	Not available	2000
	3,0 mm	2400	1800	Not available	1800
Operating data	Power Consumption		41 V / 140 W (max.) supplied by ML STAR Line instrument or HSB		
	Installation category		II		
	Pollution degree		2		
	Temperature range		15°C - 35°C		
	Relative humidity		15% - 85%		
	Noise level		<65 dBA with max. speed		
	Altitude		Max. 2000 m above sea level		
	Life time		6 years		
	Indoor use only				
Storage and transportation	Temperature range		-25°C - +70°C		
	Relative humidity		10% - 90% (non-condensing)		
Communication	CAN via TCC connector or via USB				

¹ The Heater Shaker can always be adjusted from 100 to 2500 rpm independent of the version.

7.10.3 Hamilton Heater Shaker Box (HSB) Specifications

Instrument dimensions	Width (x)	Height (z)	Depth (y)
	270 mm	90 mm	210 mm
Weight	2500 g		
Operating Data	Voltage	100-240 VAC	
	Frequency	47-63 Hz	
	Power consumption	1000 VA (max.)	
Operating Data	Installation category	II	
	Pollution degree	2	
	Temperature range	15°C - 35°C	
	Relative humidity	15% - 85%	
	Noise level	<65 dBA	
	Altitude	Max. 2000 m above sea level	
	Life time	6 years	
	Indoor use only		
Storage and Transportation	Temperature range	-25°C - +70°C	
	Relative humidity	10% - 90% (non-condensing)	
HHS connectors	8		

7.10.4 Crystal Vacuum System (CVS) Specifications

CVS carrier dimensions	Width (x)	Height (z)	Depth (y)
	155mm	112mm	555mm
Vacuum pump dimensions	Width (x)	Height (z)	Depth (y)
	242mm	245mm	257mm
Waste container	Diameter: 155		Height: 380mm
Weight of CVS	4 kg		
Weight of vacuum pump	13.5 kg		
Operating data	Maximum power consumption	530 W (230 VAC)	
	Voltage	100-120 VAC 50/60 Hz 180-254 VAC 50/60 Hz	
	Temperature Range	10°C - 40°C	
	Relative humidity	30% - 85%	
	Maximum HHS Shaker Speed	2000 rpm	

7.10.5 Temperature Controlled Carrier (TCC) Specifications

TCC dimensions	Width (x)	Height (z)	Depth (y)
	132mm	120mm	555mm
External heat exchanger dimensions	Width (x)	Height (z)	Depth (y)
	400mm	220mm	500mm
Operating data	Maximum power consumption		120 VA
	Voltage		41 V
	Temperature range		Ambient minus 22°C to 60°C $\pm$ 2°C
	Minimal temperature		4°C
	Temperature accuracy		$\pm$ 2°C

7.10.6 Multiflex Heating Module Specifications

Dimensions (without carrier base)	Width (x)	Height (z)	Depth (y)
	135mm	120mm	190mm
Weight	0.5 kg		
Operating data	Maximum power consumption		50 W
	Voltage		41 V
	Temperature range		Ambient plus 5°C to 65°C
	Temperature accuracy at 30°C - 65°C		< -0.5/+1.5°C
	Temperature accuracy		$\pm$ 1°C

7.10.7 Multiflex Cooling Module Specifications

Dimensions (without carrier base)	Width (x)	Height (z)	Depth (y)
	135mm	120mm	190mm
Weight	1.2 kg		
Operating data	Maximum power consumption		50 W
	Voltage		41 V
	Temperature range		15°C to 0°C
	Temperature accuracy		$\pm$ 1°C

APPENDICES

A. Chemical Compatibility

Abbreviations of metals and polymers used in the following tables

1.4310	X10CrNi18-8 steel	PE	Polyethylene
1.4435	X2CrNiMo18-14-3 steel	PEEK	Polyetheretherketone
AA 5083 O	Aluminum	PMMA	Polymethyl-methacrylate
EPDM	Ethylene-propylene-elastomer	POM	Polyoxymethylene
FPM	Fluoroelastomer	PP	Polypropylene
FFKM	Kalrez	PTFE	Polytetrafluorethylene
FFPM	Per-Fluor-elastomer	PVC	Polyvinylchloride
FKM	Viton	PVDF	Polyvinylidene fluoride
NBR	Acrylonitril-butadiene-rubber	SI	Silicone

The tables for chemical compatibility are based on information from different manufacturers. The results refer to laboratory tests with raw materials. The outcomes with these materials are often liable to effects which cannot be observed under laboratory conditions (e.g. temperature, pressure, tension, chemical influences of substances, design features etc.). The results listed may be considered only as a guideline. In case of doubt we recommend significant tests. The chemical resistance is not sufficient for an evaluation of a particular material for a product. Particular regulations, e.g. explosion prevention in the case of flammable liquids, have to be taken into account.

Information resources:

- Otto Bürkle GmbH, <http://www.buerkle.de/en/> (search for "chemical compatibility")
- Ahlborn-Kunststoffe, http://www.ahlborn-kunststoffe.at/gehr/chem_d.exe

Chemical resistance of CO-RE Heads

Chemical	Materials													Overall resistance		
	1.4034	1.4301	1.4305	1.4404	1.4435	PE	PP	PTFE	PEEK	FKM	FFKM	EPT	ZrO ₂	CO-RE	96	384
Acetic acid, 20%	2	1	1	1	1	1	1	1	1	2	1	1	0	1	1	2
Acetic acid, glacial	2	1	1	1	1	1	1	1	1	4	1	1	0	1	1	4
Acetone	1	1	1	1	1	2	1	1	1	4	1	1	0	1	1	4
Acetonitrile	1	1	1	1	1	1	3	1	0	2	0	3	0	3	3	2
Ammonium hydroxide, 5%	1	1	1	1	1	1	1	1	1	1	1	1	0	1	1	1
Chloroform	1	1	1	1	1	3	3	1	1	1	1	4	0	4	4	1
Deionized water	1	1	1	1	1	1	1	1	1	1	1	1	0	1	1	1
Dimethyl formamide	1	1	1	1	1	1	1	1	1	3	1	1	0	1	1	3
Dimethyl sulfoxide	1	1	1	1	1	1	1	1	0	0	0	1	0	1	1	1
Ethyl acetate	1	1	1	1	1	2	1	1	1	4	1	1	0	1	1	4
Hexane	1	1	1	1	1	3	2	1	1	1	1	4	0	4	4	1
Hydrochloric acid, 5%	4L	2L	3L	2L	2L	1	1	1	1	1	1	1	1	1	1	1
Hydrochloric acid, 20%	4L	3L	3L	2L	2L	1	1	1	1	1	1	1	1	1	1	1
Hydrogen peroxide, 10%	1	1	1	1	1	2	2	1	1	2	2	2	1	2	2	2
Isopropyl alcohol	1	1	1	1	1	1	1	1	1	1	1	1	0	1	1	1
Methanol	1	1	1	1	1	1	1	1	1	2	1	1	0	1	1	2
Methylene chloride	1	1	1	1	1	4	3	1	2	2	1	4	0	4	4	2
Nitric acid, 5-10%	1	1	1	1	1	1	1	1	1	1	1	3	1	3	3	1
Nitric acid, 70%	1	1	1	1	1	3	4	1	1	2	1	3	1	3	3	2
Phosphate buffer	1	1	1	1	1	1	1	1	0	1	1	1	0	1	1	1
Phosphoric acid, 85%	3	2	3	2	2	1	1	1	0	1	1	1	1	3	3	2
Potassium hydroxide conc.	3	1	2	1	1	1	1	1	1	3	1	1	1	2	2	3
Sodium acetate	1	1	1	1	1	1	1	1	0	4	1	1	0	1	1	4
Sodium borate	1	1	1	1	1	1	1	1	0	1	1	1	0	1	1	1
Sulfuric acid, 1-75%	4	2	3	2	2	1	1	1	2	1	1	1	1	3	3	2
Urine	1	1	1	1	1	1	1	1	1	1	1	1	0	1	1	1
Triethylamine	1	1	1	1	1	0	4	1	0	4	0	4	0	4	4	4
Toluene	1	1	1	1	1	3	3	1	1	1	1	4	0	4	4	1
Sodium hydroxide 5%	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
Formic acid 5%	3	1	2	1	1	1	1	1	1	2	1	1	0	2	2	2
Sodium hypochloride 10%	3L	2L	2L	1L	1L	1	1	1	0	1	1	1	0	1	1	1
Ethanol	1	1	1	1	1	1	1	1	1	1	1	1	0	1	1	1

Effects (key to codes in above table):

- 1 = no effect, little or no noticeable change
- 2 = slight corrosion or discoloration
- 3 = moderate corrosion or other change in physical properties or dimensions; not recommended for continuous contact
- 4 = severe corrosion or physical change; prolonged contact not recommended
- 0 = no data
- L = danger of pitting corrosion (a localized form of corrosion that leads to the creation of small holes in the metal)

Assigned materials

CO-RE Head consist of 1.4305, EPDM, PEEK, ZrO₂ and PTFE
96 Head consist of 1.4305, EPDM, PEEK, ZrO₂ and PTFE
384 Head consist of 1.4435, Viton and ZrO₂

Chemical resistances of CR needle wash station

Chemical	1.4435	1.4310	AA 5083 0	FFKM	PE	PTFE	PEEK	PP	FFPM	Over all resistance
Acetic acid, 20%	1	1	2	1	1	1	1	1	1	1
Acetic acid, glacial	1	1	2	4	1	1	1	1	1	1
Acetone	1	1	1	1	2	1	1	1	1	2
Acetonitrile	1	1	0	0	1	1	0	3	0	(3)
Ammonium hydroxide, 5%	1	1	0	0	1	1	1	1	1	1
Chloroform	1	1	1	1	3	1	1	3	1	3
Deionized water ²	1	1	1	1	1	1	1	1	1	1
Dimethyl formamide	1	1	0	1	1	1	1	1	1	1
Dimethyl sulfoxide max. 30%	1	1	0	1	1	1	0	1	0	1
Ethyl acetate	1	1	0	1	2	1	1	1	1	2
Hexane	1	1	1	1	3	1	1	2	1	3
Hydrochloric acid, 20%	2L	2L	2	1	1	1	1	1	1	4
Isopropyl alcohol	1	1	1	1	1	1	1	1	1	1
Methanol	1	1	1	1	1	1	1	1	1	1
Methylene chloride	1	1	1	1	4	1	2	3	1	4
Nitric acid, 5-10%	1	1	2	1	1	1	1	1	1	1
Nitric acid, 70%	1	1	2	1	3	1	1	4	1	4
Phosphate buffer	1	1	0	1	1	1	0	1	1	(1)
Phosphoric acid, 85%	2	2	0	1	1	1	0	1	1	(2)
Potassium hydroxide conc.	1	1	4	1	1	1	1	1	1	1
Sodium acetate	1	1	0	1	1	1	0	1	1	(1)
Sodium borate	1	1	0	1	1	1	0	1	1	(1)
Sulfuric acid, 1-75%	1L	2L	0	1	1	1	0	1	0	2
Urine	2	2	3	1	1	1	2	1	1	1
Triethylamine	1	1	0	1	0	1	1	1	1	4
Toluene	1	1	0	0	3	1	0	4	0	3

Effects (key to codes in above table):

- 1 = no effect, little or no noticeable change
- 2 = slight corrosion or discoloration
- 3 = moderate corrosion or other change in physical properties or dimensions; not recommended for continuous contact
- 4 = severe corrosion or physical change; prolonged contact not recommended
- 0 = no data
- L = danger of pitting corrosion (a localized form of corrosion that leads to the creation of small holes in the metal)

Chemical resistances of Wash Station

Chemical	1.4310	PE	PTFE	PEEK	PP	PVDF	FPM	SI	PVC	EPDM	NBR	POM
Acetic acid, 20%	1	1	1	1	1	1	3	2	1	2	3	1
Acetic acid, glacial	1	1	1	1	1	1	4	2	4	4	4	4
Acetone	1	2	1	1	1	3	4	3	0	1	4	1
Acetonitrile	1	1	1	0	3	1	3	0	0	3	4	3
Ammonium hydroxide, 5%	1	1	1	1	1	2	2	1	0	1	2	1
Chloroform	1	3	1	1	3	1	3	4	4	4	4	4
Deionized water ²⁾	1	1	1	1	1	1	1	1	1	1	1	1
Dimethyl formamide	1	1	1	1	1	4	4	2	4	2	4	1
Dimethyl sulfoxide max. 30%	1	1	1	0	1	3	3	0	4	3	4	1
Ethyl acetate	1	2	1	1	1	3	4	2	4	3	4	1
Hexane	1	3	1	1	2	1	1	4	4	4	1	1
Hydrochloric acid, 20%	4	1	1	1	1	1	1	3	1	1	4	4
Isopropyl alcohol	1	1	1	1	1	1	1	1	4	1	3	1
Methanol	1	1	1	1	1	1	3	1	3	1	3	1
Methylene chloride	1	4	1	2	3	1	3	4	4	4	4	3
Nitric acid, 5-10%	1	1	1	1	1	1	1	2	1	2	4	4
Nitric acid, 70%	1	3	1	1	4	1	2	4	4	4	4	4
Phosphate buffer	1	1	1	0	1	1	1	4	0	1	1	1
Phosphoric acid, 85%	2	1	1	0	1	1	1	3	1	3	4	4
Potassium hydroxide conc.	1	1	1	1	1	2	4	3	0	1	3	3
Sodium acetate	1	1	1	0	1	1	3	4	3	1	3	1
Sodium borate	1	1	1	0	1	1	1	1	1	1	3	1
Sulfuric acid, 1-75%	2	1	1	2	1	1	1	3	1	4	4	4
Urine	1	1	1	1	1	1	1	1	1	1	1	1
Triethylamine	1	0	1	0	4	3	3	4	0	4	3	1
Toluene	1	3	1	1	3	1	1	4	4	4	4	1

1 = no effect, little or no noticeable change

2 = slight corrosion or discoloration

3 = moderate corrosion or other change in physical properties or dimensions; not recommended for continuous contact

4 = severe corrosion or physical change; prolonged contact not recommended

0 = no data

Chemical resistance of Wash Station 96/384

Chemical	1.4310	PE	PP	PTFE	PEEK	FFPM	Hypalon	Over all resistance
Acetic acid, 20%	1	1	1	1	1	1	2	2
Acetic acid, glacial	1	1	1	1	1	1	3	3
Acetone	1	2	1	1	1	1	3	3
Acetonitrile	1	1	3	1	0	0	0	(3)
Ammonium hydroxide, 5%	1	1	1	1	1	1	3	3
Chloroform	1	3	3	1	1	1	4	4
Deionized water	1	1	1	1	1	1	1	1
Dimethyl formamide	1	1	1	1	1	1	3	3
Dimethyl sulfoxide, max. 30%	1	1	1	1	0	0	0	(1)
Ethyl acetate	1	2	1	1	1	1	4	4
Hexane	1	3	2	1	1	1	2	3
Hydrochloric acid, 20%	4	1	1	1	1	1	2	4
Isopropyl alcohol	1	1	1	1	1	1	1	1
Methanol	1	1	1	1	1	1	1	1
Methylene chloride	1	4	3	1	2	1	4	4
Nitric acid, 5-10%	1	1	1	1	1	1	2	2
Nitric acid, 70%	1	3	4	1	1	1	4	4
Phosphate buffer	1	1	1	1	0	1	1	(1)
Phosphoric acid, 85%	2	1	1	1	0	1	2	(2)
Potassium hydroxide conc.	1	1	1	1	1	1	2	2
Sodium acetate	1	1	1	1	0	1	2	(2)
Sodium borate	1	1	1	1	0	1	1	(1)
Sulfuric acid, 1-75%	2	1	1	1	2	1	2	2
Urine	1	1	1	1	1	1	1	1
Triethylamine	1	0	4	1	0	0	3	4
Toluene	1	3	3	1	1	1	1	3

1 = no effect, little or no noticeable change

2 = slight corrosion or discoloration

3 = moderate corrosion or other change in physical properties or dimensions; not recommended for continuous contact

4 = severe corrosion or physical change; prolonged contact not recommended

0 = no data

Chemical resistance of CVS

Chemical	PE	PP	PVDF	PMMA	AA6023 TTG	Remarks
N-Hexan	2/3	2/3	1/1	(3)	1/1	flammable
Ethyl-Acetate	1/3	1/3	3/3	0	1/1	flammable
Methanol	1/1	1/1	1/1	(3)	1/0	flammable
Acetonitrile (100%)	1/1	3/4	1/1	(3/4)	(1)	flammable
Methylene Chloride	0/0	(3)	(2)	(3)	1/1	flammable
Formic Acid (0.1%)	1/1	1/2	1/1	(3)	(3)	
Ammonia (0.1%)	1/1	1/1	(2)	2/2	1/1	

*PMMA may cause stains to the labware.

Resistance:

Two values are given for each medium:

left number = value at +20 ° C /

right number = value at +50 ° C.

0 = No Data Available / No Statement Possible

1 = Very Stable / Suitable

2 = Good Resistance / Suitable

3 = Partially Resistant

4 = Not Resistant

K = No General Specifications Possible

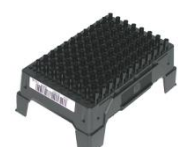
L = Danger of Pitting Corrosion

() = Estimated

B. Ordering Information

Part number	Description	
STANDARD TIPS		
5000µl / 4000µl (filter tips) standard tips		
184020	5ml CO-RE tips without filter Conductive Case of 720 tips (blister pack)	
184022	5ml CO-RE tips without filter Conductive, sterilized, Case of 720 tips (blister pack)	
194050	5ml CO-RE tips without filter Conductive Case of 96 tips, 4 tips individually wrapped (blister pack)	
184021	4ml CO-RE tip with filter Conductive Case of 720 tips (blister pack)	
184023	4ml CO-RE tip with filter Conductive, sterilized Case of 720 tips (blister pack)	
194053	4ml CO-RE tip with filter Conductive Case of 96 tips, 4 tips individually wrapped (blister pack)	
0.5-10µl standard tips		
235900	Low vol. CO-RE tips without filter Conductive Case of 5760 tips (blister pack)	
235901	Low vol. CO-RE tips with filter Conductive Case of 5760 tips (blister pack)	
235935	Low vol. CO-RE tips without filter Conductive, sterilized Case of 5760 tips (blister pack)	
235936	Low vol. CO-RE tips with filter Conductive, sterilized Case of 5760 tips (blister pack)	
235949	Low vol. CO-RE tips without filter Conductive, in a Nested Tip Box Case of 11520 tips (NTR)	
235983	Low vol. CO-RE tips without filter Conductive, sterilized Case of 11520 tips (NTR)	

Part number	Description	
235971	Low vol. CO-RE tips without filter Clear tip Case of 11520 tips (NTR)	
1-50µl standard tips		
235966	50µl vol. CO-RE tips without filter Conductive Case of 5760 tips (blister pack)	
235948	50µl vol. CO-RE tips with filter Conductive Case of 5760 tips (blister pack)	
235978	50µl vol. CO-RE tips without filter Conductive, sterilized Case of 5760 tips (blister pack)	
235948	50µl vol. CO-RE tips with filter Conductive Case of 5760 tips (blister pack)	
235979	50µl vol. CO-RE tips with filter Conductive, sterilized Case of 5760 tips (blister pack)	
235947	50µl vol. CO-RE tips without filter Conductive Case of 11520 tips (NTR)	
235987	50µl vol. CO-RE tips without filter Conductive, sterilized Case of 11520 tips (NTR)	
235964	50µl vol. CO-RE tips without filter Clear tip Case of 11520 tips (NTR)	
10-300µl standard tips		
235902	Standard vol. CO-RE tips without filter Conductive Case of 5760 tips (blister pack)	
235903	Standard vol. CO-RE tips with filter Conductive Case of 5760 tips (blister pack)	
235937	Standard vol. CO-RE tips without filter Conductive, sterilized Case of 5760 tips (blister pack)	

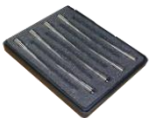
Part number	Description	
10-300µl standard tips		
235938	Standard vol. CO-RE tips with filter Conductive, sterilized Case of 5760 tips (blister pack)	
235985	Standard vol. CO-RE tips without filter Conductive, sterilized Case of 11520 tips (NTR)	
235950	Standard vol. CO-RE tips without filter Conductive Case of 11520 tips (NTR)	
235965	Standard vol. CO-RE tips without filter Clear tip Case of 11520 tips (NTR)	
10-1000µl standard tips		
235904	High vol. CO-RE tips without filter Conductive Case of 3840 tips (blister Pack)	
235905	High vol. CO-RE tips with filter Conductive Case of 3840 tips (blister pack)	
235939	High vol. CO-RE tips without filter Conductive, sterilized Case of 3840 tips (blister pack)	
235940	High vol. CO-RE tips with filter Conductive, sterilized Case of 3840 tips (blister pack)	
0.5-50µl standard tips for CO-RE 384 Probe Head		
235989	384 Probe Head CO-RE tips 50µl without filter Conductive Case of 7680 tips, 384 tips per rack (NTR)	
235446	384 Probe Head CO-RE tips 50µl without filter Clear tips Case of 7680 tips, 384 tips per rack (NTR)	
235993	384 Probe Head CO-RE tips 50µl without filter Conductive Case of 1920 tips, 96 tips per rack (NTR)	
235447	384 Probe Head CO-RE tips 50µl without filter Clear tips Case of 1920 tips, 96 tips per rack (NTR)	

Part number	Description	
Rocket tips for CO-RE 384 Probe Head		
235974	300µl Rocket tips without filter Conductive, rack, containing 96 tips (4 channels to 1) Case of 4800 tips (blister pack)	
SPECIAL TIPS		
10-300µl extra-long and slim tips		
235806	Slim tips 300µl without filter Conductive Case of 3840 tips (blister pack)	
235648	Slim tips 300µl without filter Conductive, sterilized Case of 3840 tips (blister pack)	
235647	Slim tips 300µl with filter Conductive Case of 3840 tips (blister pack)	
235646	Slim tips 300µl with filter Conductive, sterilized Case of 3840 tips (blister pack)	
2-250µl / 2-150µl (filter tips) robotic piercing tips		
235805	Piercing tips 250µl without filter Conductive Case of 3840 tips (blister pack)	
235659	Piercing tips 250µl without filter Conductive, sterilized Case of 3840 tips (blister pack)	
235658	Piercing tips 150µl with filter Conductive Case of 3840 tips (blister pack)	
235649	Piercing tips 150µl with filter Conductive, sterilized Case of 3840 tips (blister pack)	

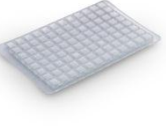
Part number	Description	
10-300µl wide bore tips		
235452	Wide bore tips, 0.71mm orifice with filter Conductive Case of 5760 tips (blister pack)	
235688	Wide bore tips, 0.71mm orifice without filter Conductive Case of 5760 tips (blister pack)	

Part number	Description	
235451	Wide bore tips, 1.55mm orifice without filter Conductive Case of 5760 tips (blister pack)	
235449	Wide bore tips, 1.55mm orifice with filter Conductive Case of 5760 tips (blister pack)	
235679	HVT, 1.2mm Conductive Case of 3840 tips (blister pack)	
235678	HVT Sterile, 1.2mm Conductive Case of 3840 tips (blister pack)	
10-1000µl wide bore HIGH tips		
235444	Wide bore tips, 3.20mm orifice without filter Conductive Case of 3840 tips (blister pack)	
235541	Wide bore tips, 3.20mm orifice with filter Conductive Case of 3840 tips (blister pack)	

Disposable waste bags		
Part number	Description	
199201	PLASTIC CHUTE “BIOHAZARD” WITH BIOHAZARD LABELING (roll of 10 pcs), 700 x 600mm, PE This chute is used with Waste Container (P/N 281520)	
199202	WASTE BAG WITHOUT BIOHAZARD LABELING (roll of 25 pcs), 460 x 500mm, PE Waste bags are used with Waste Hoop of ML STAR	
199203	WASTE BAGS “BIOHAZARD” WITH BIOHAZARD LABELING (roll of 25 pcs), 460 x 500mm, PE Waste bags with Biohazard Labeling	
185319	WASTE CHUTE WITHOUT BIOHAZARD LABELING (roll of 10 pcs), 460 x 700mm, PE This chutes are used with Waste Container (P/N 281520)	
281520	WASTE CONTAINER “BIOHAZARD” (Pack of 10 boxes), combustible This box is used with Waste Chute (P/N 199201 and P/N 185319)	

Steel needles for 1000µl single channels		
Part number	Description	
235930	1000µL NEEDLE SET CR Set of 8x 1000µl needles, CR Wash Station	
235931	300µL NEEDLE SET CR Set of 8x 300µl needles, CR Wash Station	
235932	10µL NEEDLE SET CR Set of 8x 10µl needles, CR Wash Station	
187290	NEEDLE SERVICE KIT Contains a Needle pick-up tool to remove needles individually from the wash station.	
182136	SET OF 8 TEACHING NEEDLES Used for the maintenance to check the pressure tightness of the 1000µl-channels	
182176	TEACHING NEEDLE 1 needle for 1000µl-channels, used for labware teaching (Set of 8 -> 182136)	
184184	TEACHING NEEDLE 1 needle for 5ml-channels, used for labware teaching	

Reagent containers		
Part number	Description	
56694-01	REAGENT CONTAINER 60ml Set of 28 reagent troughs 60ml, self-supporting with lid, for carrier RGT_CAR_5R60 (194057 1T)	
56694-02	REAGENT CONTAINER 60ml Set of 28 reagent troughs 60ml, black, self-supporting with lid, for carrier RGT_CAR_5R60 (194057 1T)	
194051	REAGENT CONTAINER 60ml Set of 28 reagent troughs 60ml, self-supporting, barcodes included in box, w/o lid, individually wrapped for carrier RGT_CAR_5R60 (194057 1T)	
182703	REAGENT CONTAINER 120ml Set of 12 reagent troughs 120ml for carrier RGT_CAR_3R (185290 1T)	
194052	REAGENT CONTAINER 120ml Set of 12 reagent troughs, self-supporting, barcodes included in box, 120ml for carrier RGT_CAR_3R (194058 1T)	
56695-01	REAGENT CONTAINER 200ml Set of 10 reagent troughs 200ml, self-supporting with lid, for carrier RGT_CAR_4R200 (185436)	
56695-02	REAGENT CONTAINER 200ml Set of 10 reagent troughs 200ml, black, self-supporting with lid, for carrier RGT_CAR_4R200 (185436)	
Deep Well Plates		
Part number	Description	
6471-01	DEEP WELL BLOCK 1.2 ML, PP 96 well block, polypropylene, 1.2ml volume per well, round wells. W: 126.3mm; D: 84.7mm; H: 40.6mm 32 pcs/case, 4 plates are wrapped	
6472-01	DEEP WELL BLOCK 1.2 ML, PS 96 well block, polystyrene, transparent, 1.2ml volume per well, round wells. W: 127.5mm; D: 85.5mm; H: 41mm 32 pcs/case, 4 plates are wrapped	

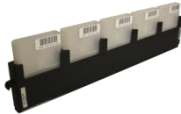
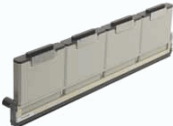
Part number	Description	
235655	DEEP WELL BLOCK 1.2 ML, PS 96 well block, polystyrene, transparent, with barcode 1.2ml volume per well, round wells. W: 127.5mm; D: 85.5mm; H: 41mm 32 pcs/case, 4 plates are wrapped	
6473-01	DEEP WELL BLOCK 2.2 ML, PP 96 well block, polypropylene, 2.2ml volume per well, square wells, round bottom. W: 126.3mm; D: 84.7mm; H: 40.6mm 32 pcs/case, 4 plates are wrapped	
235656	DEEP WELL BLOCK 2.2 ML, PP 96 well block, polypropylene, with barcode, 2.2ml volume per well, square wells, round bottom. W: 126.3mm; D: 84.7mm; H: 40.6mm 32 pcs/case, 4 plates are wrapped	
6474-01	SEALING MAT Sealing mat for 2.2ml blocks, ethylenevinylacetate. W: 121.5mm; D: 78.8mm; H: 4.68mm 50 pcs/case, 10 pcs/bag	

Reagent tubes		
Part number	Description	
235454	FLIPTUBES 1.5ml reagent tubes with a lid attached which can close the tube tightly.	

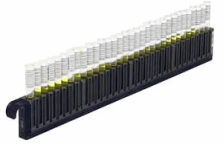
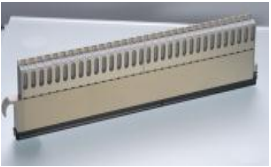
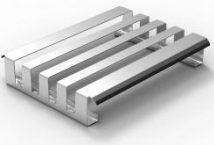
Liquids for maintenance		
Part number	Description	
289014	HEAT EX. SOLUTION FOR TCC Color-coded heat exchange liquid for TCC cooler	
281242	MICROLAB DETERGENT & DISINFECTANT KIT contents: 1 beaker 100ml 3x 1l bottles Deconex® 61 DR	
281243	MICROLAB DISINFECTANT SPRAY KIT contents: 4x 500ml Deconex® SOLARSEPT spray bottles	
281245	MICROLAB DISINFECTANT STARTER KIT contents: 1 Beaker 100ml 2x 500ml Deconex® SOLARSEPT spray bottles 1x 1l bottle Deconex® 61 DR	

Plate carrier		
Part number	Description	
182035	PLT_CAR_L5PCR384 Carrier for 5x 384 PCR plates	
182065	PLT_CAR_P3AC Carrier for 3x Deep Well Plates, portrait orientation (6T)	
182070	PLT_CAR_L5PCR Carrier for 5x 96-well PCR plates (6T)	
182075	PLT_CAR_P3MD Carrier for 3x 96/384 well plates, portrait orientation (6T)	
182090	PLT_CAR-L5AC Carrier for 5 Deep Well Plates or for 5x 384 tip racks (e.g.384HEAD_384TIPS_30µl) (6T)	
191287	PLT-CAR-L4HD Carrier for 4x 1536-well plates (6T)	
182190	PLT-CAR-P3HD Carrier for 3 x 1536-well plates, portrait orientation (6T)	
182365	PLT_CAR_L5MD Carrier for 5x 96/384-well plates (6T)	
185295	PLT_CAR_L5AC PINNED Carrier for 5x Hamilton DWP (6T)	

Plate carrier		
Part number	Description	
185330	PLT_CAR_L4ST (4x8 MTP) Stacker carrier for 4 x 8 MTP (7T)	
185340	PLT_CAR_L4ST (4x5 MTP) Stacker carrier for 4 x 5 MTP (7T)	
182735	PLT_CORE_COVER Lid to cover MTP on standard carriers	
187223	ANTI-EVAPORATION LID Lid for minimizing evaporation of liquid in MTP. Typically used for protein crystallization plate preparation. Can be handled by SWAP, iSWAP or CO-RE gripper. For the use with P/N 188054APE PLT-CAR-L4-Crystal or protein crystallization base package (PN 187007) (not included).	
182712	FRAME FOR FILTER PLATE To place filter plates on archive carriers	

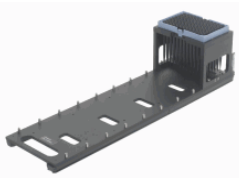
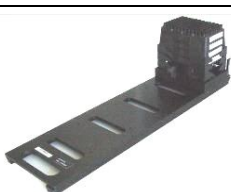
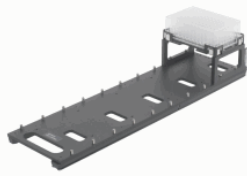
Reagent carrier		
Part number	Description	
194057	RGT_CAR_5R60 Carrier for 5x reagent troughs 60ml (194051 1T)	
187239	RGT_CAR_4R100 Carrier for 4x reagent troughs 100ml (1T)	
182080	RGT_CAR_12R Carrier for 12x reagent troughs 100ml (137257 6T)	
185290	RGT_CAR_3R Carrier for 3x reagent troughs 120ml (182703 1T)	
194058	RGT_CAR_3R120 Carrier for 5x reagent troughs 120ml (194052 1T)	
185436	RGT_CAR_4R200 Carrier for 4x reagent troughs 200ml (56695-01/56695-02 2T)	

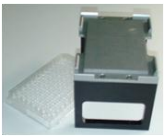
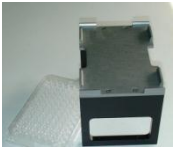
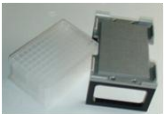
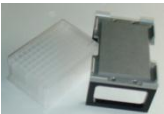
Tip carrier		
Part number	Description	
182060	TIP_CAR_288 Carrier for 3x 96 tip racks portrait for 12-channel (1000µl) Microlab STAR Line or 3x 24 tip racks portrait for 6-channel (5ml) Microlab STAR Line (4T)	
182085	TIP_CAR_480 Carrier for 5x 96 tip (10µl, 50µl, 300µl, 1000µl) racks or 5x 24 tip (5ml) racks (6T)	
182390	TIP_CAR_L384_A00 Carrier for 4x 96 tip racks for 16-channel (1000µl) Microlab STAR Line or 3x 24 tip racks portrait for 8-channel (5ml) Microlab STAR Line (6T)	
182074	TIP_CAR_NTR_A00 Carrier for Nestable Tip Racks (NTR) 6T; for 1 pack tray containing 1920x 10µl, 50µl, 300µl CO-RE tips (5 stacks x 4 tip racks x 96 tips) or 7680x 50µl CO-RE tips for 384 Probe Head (5 stacks x 4 tip racks x 384 tips)	
182040	ADAPTER FOR TIP_CAR_480 Intermediate storage position for tips allows pick-up of a single tip/row/column using the CO-RE 96 Probe Head	
191055	384 TIP SUPPORT Intermediate storage position for tips allows pick-up of a single tip/row/column using the CO-RE 384 Probe Head The tip support fits into Multiflex tip module (P/N 188160) or tip carrier (P/N 182085)	
191056	ROCKET TIP SUPPORT Intermediate storage position for tips allows pick-up of a single tip/row/column using the CO-RE 384 Probe Head The tip support fits into Multiflex Tip Module (P/N 188160) or tip carrier (P/N 182085)	
182041	DRIP PAN FOR TIP CARRIER	

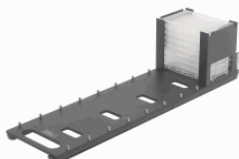
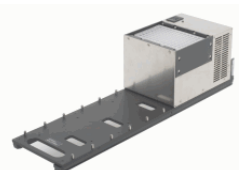
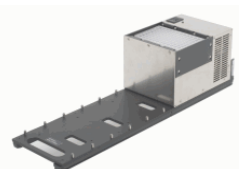
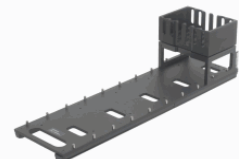
Sample tube carrier and inserts		
Part number	Description	
173400	4 SMP-CAR-24 Set of 4x carrier for 24 tubes 14.5x60 – 18x120mm (1T)	
173410	3 SMP-CAR-32 Set of 3x carrier for 32 tubes 11x60 – 14x120mm (1T)	
182045	SMP-CAR-12 Carrier for 12 Falcon tubes 50ml (2T)	
809030	SMP_CAR_32FT Carrier for 32 FlipTubes (1T)	
185346	TUBE CARRIER FILLING RACK Holds up to four 1T sample carriers for loading and filling of tubes in carriers. Compatible with 1T sample carriers 24, 32.	
190070	RACK HOLDER Holds up to four 1T sample carriers for loading and filling tubes in carriers and transporting up to four sample carriers to and from a ML STAR line workstation. Compatible with 1T sample carriers 24, 32.	
182238	EPI-INS-32L Set of 32 inserts for 1.5ml Eppendorf cups in SMP-CAR-32	
182239	EPI-INS-32S Set of 32 inserts for 0.5ml Eppendorf cups in SMP-CAR-32	
187142	24-FALCON-INS-15ML Set of 24 inserts for 15ml Falcon tubes in SMP-CAR-24	
187350	8 INSERTS FOR 1.5ML EPPENDORF CUPS IN SMP-CAR-32 Set of 8 inserts for 1.5ml Eppendorf cups in SMP-CAR-32	

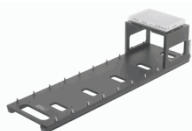
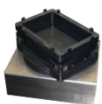
Sample tube carrier and inserts		
Part number	Description	
185393	SET OF INSERTS 24/16.5MM 24 Set of 24 inserts for tubes with outer diameter < 16.5mm in SMP-CAR-24	

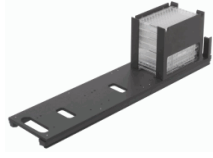
There are more inserts available. Please consult a local Hamilton representative.

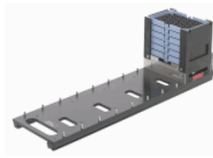
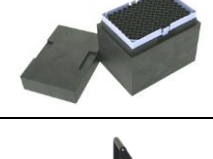
Multiflex carrier		
Part number	Description	
188039	MULTIFLEX CARRIER BASE (LANDSCAPE ORIENTATION) Labware carrier base for up to 5 Multiflex Modules	
188160	MULTIFLEX TIP MODULE Module to position a high-, standard-, low volume or 5ml tip rack (but not a 384 tip rack) The Multiflex carrier base is not included.	
191420	MULTIFLEX NTR 96 MODULE Module to position a Nestable Tip Rack (NTR) with standard (300µl), low volume (10µl) or 50µl tips The Multiflex carrier base is not included.	
196371	MULTIFLEX NTR 384 MODULE Module to position a Nestable Tip Rack (NTR) with 384 50µl tips The Multiflex carrier base is not included.	
191425	MULTIFLEX NTR4 MODULE Module to position a stack of 4 Nestable Tip Racks (NTR) with standard (300µl), low volume (10µl) or 50µl tips The Multiflex carrier base is not included.	
188041	MULTIFLEX MTP MODULE Module to position 96-/384-well plates in SBS format / or flat reagent troughs The Multiflex carrier base is not included.	
188042	MULTIFLEX DWP MODULE Module to position a Deep Well Plate / tube racks (MATRIX or MICRONICS) / NUNC reagent trough The Multiflex carrier base is not included.	

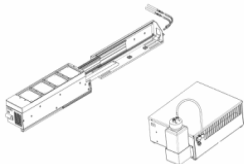
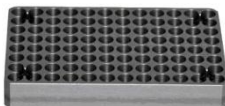
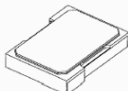
Multiflex carrier		
Part number	Description	
188094APE	MULTIFLEXMTPNESTCONTBASED Module to hold a Microtiter Plate by supporting the wells on a Multiflex Carrier. The Multiflex carrier base is not included. Alternatively the module can be mounted on a PLT_CAR_L4_SHAKER base plate, P/N 187001. This requires P/N 188133APE MULTIFLEXMODULEBRACKET7T for fixation.	
188228APE	MULTIFLEXMTPNESTRACKBASED Module to hold the frame of a Microtiter Plate on a Multiflex Carrier. The Multiflex carrier base is not included. Alternatively the module can be mounted on a PLT_CAR_L4_SHAKER base plate, P/N 187001. This requires/N 188133APE MULTIFLEXMODULEBRACKET7T for fixation.	
188293	MULTIFLEXXLDWPNESTRACKBASED Module to position any SBS format plate, including Deep Well Plates up to H=90mm / or flat reagent containers. Plates are held on their frame. The Multiflex carrier base is not included. Alternatively the module can be mounted on a PLT_CAR_L4_SHAKER base plate, P/N 187001. This requires/N 188133APE MULTIFLEXMODULEBRACKET7T for fixation.	
188180APE	MULTIFLEXDWPNESTCONTBASED Module to hold a Deep Well Plate by supporting the wells on a Multiflex Carrier. The Multiflex carrier base is not included. Alternatively the module can be mounted on a PLT_CAR_L4_SHAKER base plate, P/N 187001. This requires/N 188133APE MULTIFLEXMODULEBRACKET7T for fixation.	
188043	CO-RE GRIPPER ON MULTIFLEX MODULE Plate handling tool for plate transfer on the deck using two pipetting channels. Includes module with parking position for CO-RE gripper. The Multiflex carrier base is not included.	

Multiflex carrier		
Part number	Description	
188044	MULTIFLEX STACKER MODULE (LANDSCAPE) Module to use as passive plate hotel. Depending on plate height, up to 10 plates can be stacked on one position. The Multiflex carrier base is not included.	
188045	MULTIFLEX HEATING MODULE Heating module (up to 60°C) including one adapter for one labware type. Temperature: Room temperature up to 60°C Temperature gradient: $\pm 1^{\circ}\text{C}$ The Multiflex carrier base is not included.	
188046	MULTIFLEX COOLING MODULE Cooling module (4 - 15°C) including one adapter for one labware type. Temperature: 4 – 15 °C. Temperature gradient: $\pm 1^{\circ}\text{C}$ The Multiflex carrier base is not included.	
808440	MULTIFLEX HEATING COOLING MODULE A combined module that cools or heats plates. Temperature: 0°C - 99°C Dry mode feature. The Multiflex carrier base is necessary but is not included.	
188047	MULTIFLEX REAGENT TROUGH MODULE Module to hold six 50ml troughs The Multiflex carrier base is not included.	
188048	MULTIFLEX TUBE / CUP MODULE Module to hold EPPENDORF, SARSTEDT, NUNC tubes 0.5/ 1.5 / 2.0 ml with or without snap-lid in an passively cooled adapter The Multiflex carrier base is not included.	
809300	MULTIFLEX ADAPTER FOR FLIPTUBES Module to hold 24 FlipTubes in a passively cooled adapter. The Multiflex carrier base is not included.	

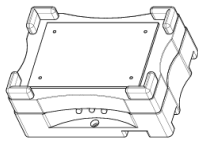
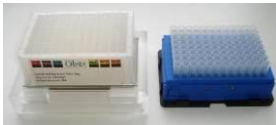
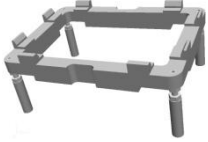
Multiflex carrier		
Part number	Description	
188313	FIXATION FRAME FOR 1.4ML MATRIX TUBES Standard fixation frame for 1.4ml matrix tubes used in piercing applications. Requires a Multiflex MTP Module (P/N 188041, not included) and a Multiflex carrier base (not included).	
188049	MULTIFLEX PCR PLATE MODULE 96 Module to position a 96-well PCR plate. The Multiflex carrier base is not included.	
188052	MULTIFLEX PCR PLATE MODULE 384 Module to position a 384-well PCR plate The Multiflex carrier base is not included.	
188053	MULTIFLEX CARRIER BASE (PORTRAIT ORIENTATION) Labware carrier base for up to 3 Multiflex modules	
188055APE	PLATE TURNTABLE Automated device to turn plates from landscape into portrait orientation or vice-versa The Multiflex carrier base is not included.	
188313	MULTIFLEXMATRIXDOWNHOLDER Module to hold down sealed Matrix tubes in a 96well tube rack for piercing using needles or tips. Supports one pierceable SBS format 96well tube rack. The Multiflex carrier base and piercing needles are not included.	
188095APE	MULTIFLEXMTPFIXATIONFRAME Module to hold down sealed plates for piercing using needles or tips. Supports one pierceable SBS format plate. The Multiflex carrier base and piercing needles are not included.	
188058APE	MULTIFLEX LID PARKING MODULE Module to park the lid of the cooling or heating module. The Multiflex carrier base is not included.	

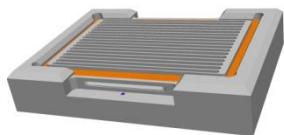
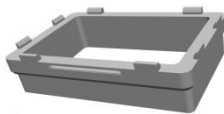
Multiflex carrier		
Part number	Description	
188059	MULTIFLEX PLATE STACKER MODULE (PORTRAIT) Module to use as passive plate hotel. Depending on plate height, up to 10 plates can be stacked on one position. The Multiflex carrier base is not included.	
188114APE	MULTIFLEX LIQUID DISPENSER TROUGH 8 Module to automatically refill a trough on the deck with fresh reagent. Compatible with 8-channel instruments. The Multiflex carrier base is not included.	
188061APE	MULTIFLEX TILT MODULE Module to tilt plates on the y-axis (in landscape orientation) The Multiflex carrier base is not included.	

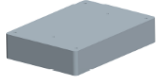
Multiflex carrier		
Part number	Description	
188062	MULTIFLEX TIP STACKER MODULE Module to hold 4 standard volume or 6 low volume tip racks on one position The Multiflex carrier base is not included.	
188063APE	MULTIFLEX SEESAW MODULE Module to shake bead-, cell-, or reagent solutions at a predefined speed in troughs or plates. Angle is adjustable up to 15°. 16-69 rpm adjustable in 4 steps. The Multiflex carrier base is not included.	
188115APE	MULTIFLEX LIQUID DISPENSER TROUGH 96 Module to automatically refill a trough on the deck with fresh reagent. Compatible with 96-channel instruments. The Multiflex carrier base is not included.	
188078APE	MULTIFLEX STERILE TIP BOX MODULE Module to store sterile tips on the deck.	
182774	STAR SHELF 4MTP/2AC PLATES Shelving Unit for 4 MTP or 2 archive plates	

Temperature controlled carrier (TCC)		
Part number	Description	
182400	TCC FOR ML STAR/PLATES Temperature Controlled Carrier for MTP (no DWP)	
182436	TCC-RGT-TUB 100ml reagent tub with lid for TCC	
182532	TCC-PCR-PLT-ADAPTERKIT 96 Set of 4 TCC-adapter for 96-well PCR plates	
182673	TCC-PCR-PLT-ADAPTERKIT 384 Set of 4 TCC-adapter for 384-well PCR plates	
182734	TCC-LID-ADAPTERKIT Set of 4 lids to cover plates on the TCC	

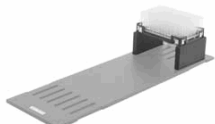
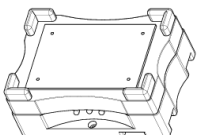
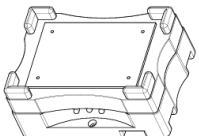
CVS Vacuum System		
Part number	Description	
199020 190021	CVS VACUUM SYSTEM INCLUDING PUMP (230V) CVS VACUUM SYSTEM INCLUDING PUMP (115V) Carrier with a vacuum box, one park position for manifold top and two plate positions including Vacuubrand Pump ME 4C VARIO with controller, pressure sensor and air-bleed valve (Waste bottles are not included. Please select P/N 281539 or 281540)	
187143 187149	TELESHAKER 220V FOR DWP TELESHAKER 115V FOR DWP Maximum one shaker per CVS carrier.	

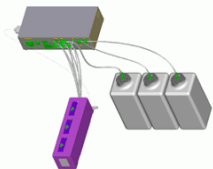
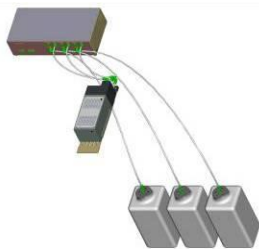
CVS Vacuum System		
Part number	Description	
187788	SHAKER HEATER CAT SH10 FOR DWP (110/220V) Shaker heater (Shaking: 200 - 1200 1/min, orbital: 2mm, temperature: from RT+5°C - 90°C) for standard 96 and Deep Well Plates Maximal one shaker heater per CVS carrier.	
281539	BVS 2-LITRE WASTE BOTTLE Glass waste bottle for the BVS / CVS, including connectors and LDPE Bucket 11L	
281540	BVS 4-LITRE WASTE BOTTLE PP waste bottle for the BVS / CVS with confectioned cap, including connectors and LDPE Bucket 11L	
190034	DWP KIT FOR BVS Manifold Top for BVS / CVS and insert to adapt the BVS / CVS to the used collecting plate	 
186320	BVS INSERT KIT Adjustable insert to adapt the BVS / CVS to the used collecting plate	
190035	BVS INSERT KIT FOR MTP Insert to adapt the BVS / CVS to the used collecting plate	
190036	BVS INSERT KIT FOR DWP Insert to adapt the BVS / CVS to the used collecting plate	

CVS Vacuum System		
Part number	Description	
190037	BVS GRID I FOR MILLIPORE Manifold Top for BVS / CVS incl. Grid I for Millipore Kits "Montage Plasmid Miniprep96 Kit" and "Montage PCR96 Cleanup Kit"	
186321	BVS GRID II FOR MILLIPORE Manifold Top for BVS / CVS incl. Grid II for Millipore Kits with 96/384 SBS plates with one filter	
186303	BVS ADAPTER FOR MN DNA KIT Insert to adapt the BVS / CVS to the used collecting plate	
182712	FRAME FOR FILTER PLATE To place filter plates on archive carriers	

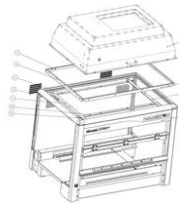
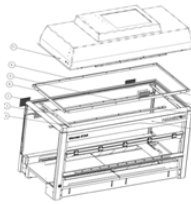
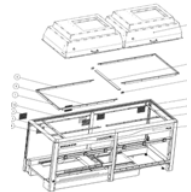
Shaker		
Part number	Description	
187001	PLT_CAR_L4_SHAKER Template carrier with 4 positions for Hamilton Heater Shaker (optional: Shaker H+P, Shaker Heater CAT) and plate bases (7T)	
190755	HEATER SHAKER BOX Needed if more than two Hamilton Heater Shakers are integrated (or the connectors "TCC1", "TCC2" on the ML STAR are occupied by other devices). One heater shaker module is connected to a USB port of the PC and serves as master module for up to seven additional HHSs. The modules are connected to the external heater shaker box (HSB), which serves as power supply and as signal distributor.	
199013	HEATER SHAKER SUPPORT BLOCK FOR MTP Hamilton Heater Shaker support block for MTP. Used to raise the heater shaker if only MTPs are processed.	

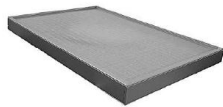
Shaker		
Part number	Description	
199027	HEATER SHAKER 1.5MM SARSTEDT 48 X 1.5ML Heater Shaker with 1.5mm shaking orbit and fitted adapter for plates (shaking speed: 100 -1800 rpm, temperature control: RT+5°C - 105°C, max. loading: 300mm) Sarstedt tubes 1.5ml: sterile Sarstedt Cat. No. 72.687.772 non-sterile Sarstedt Cat. No. 72.687 http://www.Sarstedt.com	
199033	HEATER SHAKER 2.0 MTP FLAT BOTTOM Heater Shaker with 2.0mm shaking orbit and flat bottom adapter (shaking speed: 100 -2500 rpm, temperature control: RT+5°C - 105°C, max. loading: 300mm)	
199034	HEATER SHAKER 3.0MM FLAT BOTTOM Heater Shaker with 3.0mm shaking orbit and flat bottom adapter (shaking speed: 100 -2400 rpm, temperature control: RT+5°C - 105°C, max. loading: 300mm)	
199037	HEATER SHAKER 1.5MM NUNC DWP 96 2ML Heater Shaker with 1.5mm shaking orbit and fitted adapter for plates (shaking speed: 100 -2000 rpm, temperature control: RT+5°C - 105°C, max. loading: 300mm)	
199038	HEATER SHAKER 2.0 NUNC DWP 96 2ML Heater Shaker with 2.0mm shaking orbit and fitted adapter for plates (shaking speed: 100 -2000 rpm, temperature control: RT+5°C - 105°C, max. loading: 300mm)	
199039	HEATER SHAKER 3.0 NUNC DWP 96 2ML Heater Shaker with 3.0mm shaking orbit and fitted adapter for plates (Shaking speed: 100 -1800 rpm, Temperature control: RT+5°C - 105°C, max. loading: 300mm)	
188318	HEATER SHAKER 2.0MM MTP FLAT BOTTOM APE Heater Shaker with 2.0mm shaking orbit and the option to be equipped with a customized adapter. The labware has to be sent to Hamilton Bonaduz AG. (Shaking speed depends on used adapter, temperature control: RT+5°C - 105°C, max. loading: 300mm)	

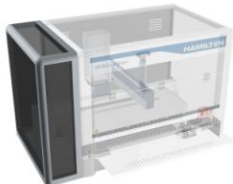
Shaker		
Part number	Description	
188319	HEATER SHAKER 3.0MM MTP FLAT BOTTOM APE Heater Shaker with 3.0mm shaking orbit and the option to be equipped with a customized adapter. The labware has to be sent to Hamilton Bonaduz AG. (shaking speed depends on used adapter, temperature control: RT+5°C - 105°C, max. loading: 300mm)	
809031	HHS ADAPTER FOR FLIPTUBES Module to hold and fix 24 FlipTubes on a Hamilton Heater Shaker. A HHS is not included.	
187143 187149	TELESHAKER 220V FOR DWP TELESHAKER 115V FOR DWP Shaker (shaking: 100 - 2000 1/min, orbital: 2mm) for standard 96 and Deep Well Plates on the PLT_CAR_L4_shaker (7T)	
187295 187296	TELESHAKER 220V FOR MTP TELESHAKER 115V FOR MTP Shaker (Shaking: 100 - 2000 1/min, Orbital: 2mm) for standard 96-well plates (no DWP) on the PLT_CAR_L4_shaker (7T)	
187144	DWP BASE FOR SHAKER CAR Position for standard and Deep Well Plates on the PLT_CAR_L4_shaker (7T)	
187292	MTP BASE FOR SHAKER CAR Positions for standard well plates (no DWP) on the PLT_CAR_L4_shaker (7T)	
187788	SHAKER HEATER CAT SH10 FOR DWP (110/220V) Shaker heater (shaking: 200 - 1200 1/min, orbital: 2mm, temperature: from RT+5°C - 90°C) for standard 96 and Deep Well Plates on the PLT_CAR_L4_shaker (7T)	
187789	SHAKER HEATER CAT SH10 FOR MTP (110/220V) Shaker heater (shaking: 200 - 1200 1/min, orbital: 2mm, temperature: from RT+5°C - 90°C) for standard 96-well plates (no DWP) on the PLT_CAR_L4_shaker (7T)	

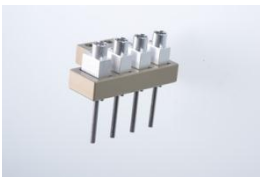
Wash Stations		
Part number	Description	
186360	CR NEEDLE WASH STATION Wash Station for all needle types, two wash liquids, needle wash in parallel to the pipetting	
190248	96/384 WASH STATION DUAL Wash Station for 96/384 disposable tips, CO-RE 384 Probe Head, 2 wash chambers	
281107	WASH / WASTE CONTAINER Wash and waste container for wash stations	

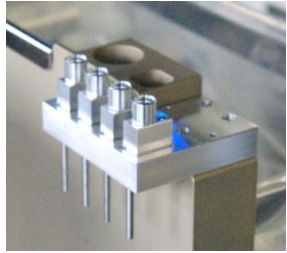
HEPA Filter Hoods		
Part number	Description	
55500-01	HOOD FLOW STARLET Hood mounting on top of a ML STARlet; supplies filtered clean air to the inside of a ML STARlet workstation. Flow is laminar. Flow volume is controllable by software. Adds 321 mm to height of workstation. A seal kit for the ML STARlet (P/N 188232) is required when ordering this item. It has to be ordered separately.	
55501-01	HOOD FLOW STARLET UV Hood mounting on top of a ML STARlet; supplies filtered clean air to the inside of a ML STARlet workstation. Flow is laminar. Flow volume is controllable by software. Includes UV lamps. Adds 321 mm to height of workstation. A seal kit for the ML STARlet (P/N 188232) is required when ordering this item. It has to be ordered separately. UV Light requires UV resistant covers which have to be ordered separately. Please consult a Hamilton representative.	
55495-01	HOOD FLOW STAR Hood mounting on top of a ML STAR; supplies filtered clean air to the inside of a ML STAR workstation. Flow is laminar. Flow volume is controllable by software. Adds 321 mm to height of workstation. A seal kit for the ML STAR (P/N 188233) is required when ordering this item. It has to be ordered separately.	
55496-01	HOOD FLOW STAR UV Hood mounting on top of a ML STAR; supplies filtered clean air to the inside of a ML STAR workstation. Flow is laminar. Flow volume is controllable by software. Includes UV lamps. Adds 321 mm to height of workstation. A seal kit for the ML STAR (P/N 188233) is required when ordering this item. It has to be ordered separately. UV Light requires UV resistant covers which have to be ordered separately. Please consult a Hamilton representative.	

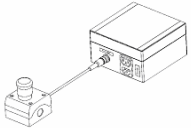
Part number	Description	
55506-01	HOOD STARPLUS Hood mounting on top of a ML STARplus; supplies filtered clean air to the inside of a ML STARplus workstation. Flow is laminar. Flow volume is controllable by software. Adds 321 mm to height of workstation. A seal kit (P/N 188234) is required when ordering this item. It has to be ordered separately.	
55507-01	HOOD STARPLUS UV Hood mounting on top of a ML STARplus; supplies filtered clean air to the inside of a ML STARplus workstation. Flow is laminar. Flow volume is controllable by software. Includes UV lamps. Adds 321 mm to height of workstation. A seal kit (P/N 188234) is required when ordering this item. It has to be ordered separately. UV Light requires UV resistant covers which have to be ordered separately. Please consult a Hamilton representative.	
188232	STARLET HEPA-HOOD KIT Seal kit required for the HEPA Filter Hood of HOOD FLOW STARLET, P/N 55500-01, HOOD FLOW STARLET UV, P/N 55501-01	
188233	STAR HEPA-HOOD KIT Seal kit required for the HEPA Filter Hood of HOOD FLOW STAR, P/N 55495-01, HOOD FLOW STAR UV, P/N 55496-01	
188234	STARPLUS HEPA-HOOD KIT Seal kit required for the HEPA Filter Hood of HOOD FLOW STAR, P/N 55506-01-01, HOOD FLOW STAR UV, P/N 55507-01	
51015-01	FILTER, PLEATED, 14X18X1	

Part number	Description	
51011-01	FILTER, HEPA, 18X24X3	
51476-01	LAMP UV, STARLINE Length: 45.7cm, 17Watt	

Housing Extension		
Part number	Description	
188122	HOUSING EXTENSION MPH This housing extension can be mounted on either side of ML STAR Line instruments. If used on the left hand side on instruments with modular arm and MPH (96 or 384), the following waste has to be ordered separately; either: 188124APE WASTE MPH L REAR or, 188125APE WASTE MPH L FRONT If used on the left hand side on instruments with 'thin-' arm and MPH (96 or 384) the following waste has to be ordered separately; either: 188124APE WASTE MPH L REAR or, 188125APE WASTE MPH L FRONT Please note: Slide waste for MPH's (96 or 384) will not fit. In conjunction with one of the following HEPA hoods, a seal kit (P/N 188235) is required when ordering this item. It has to be ordered separately. P/N 55500-01 HOOD FLOW STARLET, P/N 55501-01 HOOD FLOW STARLET UV, P/N 55495-01 HOOD FLOW STAR, P/N 55496-01 HOOD FLOW STAR UV, P/N 55506-01 HOOD STARPLUS, P/N 55507-01 HOOD STARPLUS UV UV Light requires UV resistant covers which have to be ordered separately. Please consult a Hamilton representative.	
188235	HOUSING EXTENSION MPH HEPA-HOOD KIT Seal kit required for the HEPA Filter Hood of HOUSING EXTENSION MPH, P/N 188122	

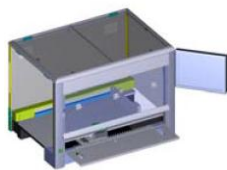
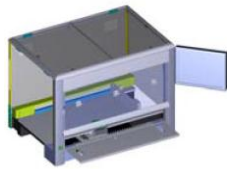
CO-RE tools		
Part number	Description	
188066	CO-RE GRIPPER WITH ATTACHMENT FOR WASTE BLOCK Plate handling tool for plate transfer on the deck using two 1000µl-channels. Includes parking position for attachment to waste block (waste block not included). Note: Order this gripper if the Microlab STAR Line instrument is equipped with 1000µl-channels only.	
184089	CO-RE GRIPPER WITH ATTACHMENT FOR WASTE BLOCK (INSTRUMENTS WITH 1000µL- AND 5ML-CHANNELS) Plate handling tool for plate transfer on the deck using two 1000µl-channels. Includes parking position for attachment to waste block (waste block not included). Note: Order this gripper if the Microlab STAR Line instrument is equipped with 1000µl- and 5ml-channels.	
184099	CO-RE GRIPPER WITH ATTACHMENT FOR WASTE BLOCK (INSTRUMENTS WITH 5ML-CHANNELS) Plate handling tool for plate transfer on the deck using two 5ml-channels. Includes parking position for attachment to waste block (waste block not included). Note: Order this gripper if the Microlab STAR Line instrument is equipped with 5ml-channels.	
186100	CO-RE GRIPPER Gripper tool for plate transport with pipetting channels, including parking position. Note: This gripper can be installed by removing a plate position of plate carrier PLT_CAR_L5MD, P/N 182365	
809032	FLIPTUBE TOOL	
809305	FLIPTUBE TOOL PARK STATION ON MFX CARRIER	

CO-RE tools		
Part number	Description	
809306	FLIPTUBE TOOL PARK STATION ON WASTE	
188082APE	CORE-LID TOOL (CLT) Suction cup that can be picked up by a CO-RE channel to move around lids	

Emergency stop box		
Part number	Description	
186060	EMERGENCY STOP BOX Category 0 Emergency Stop Button	

Hamilton standard computers			
Part number		Description	
		WINDOWS 7 ULTIMATE COMPUTERS	
396258	STANDARD PC WIN7 CH	Swiss German	
396259	STANDARD PC WIN7 FR	French	
396260	STANDARD PC WIN7 US	US English	
396261	STANDARD PC WIN7 ES	Spanish	
396262	STANDARD PC WIN7 IT	Italian	
396263	STANDARD PC WIN7 GE	German	
396264	STANDARD PC WIN7 GB	UK English	
		(Main power cable not included)	

Mains power cables			
Part number	Description		
355234	POWER CABLE EU, 2.50m C13		
355235	POWER CABLE CH, 2.50m C13		
355236	POWER CABLE US, 2.50m C13		
355237	POWER CABLE UK, 2.50m C13		

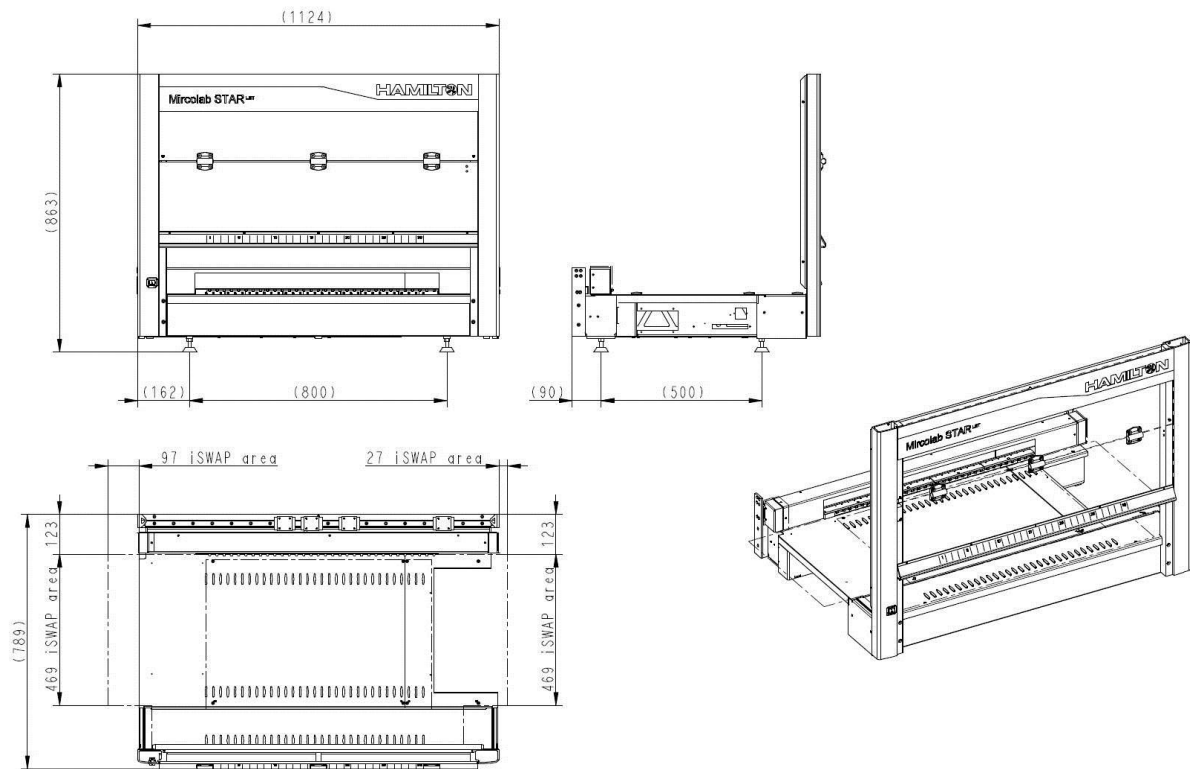
Touch screen monitor for the Microlab STAR Line			
Part number	Description		
188250	19" TOUCH SCREEN KIT SHADED (Mains power cable not included)		
188260	19" TOUCH SCREEN KIT CLEAR (Mains power cable not included)		
355010	MAINS POWER CABLE DE		
355020	MAINS POWER CABLE CH		
355009	MAINS POWER CABLE US		
235062	MAINS POWER CABLE GB		

APPENDIX B Ordering Information

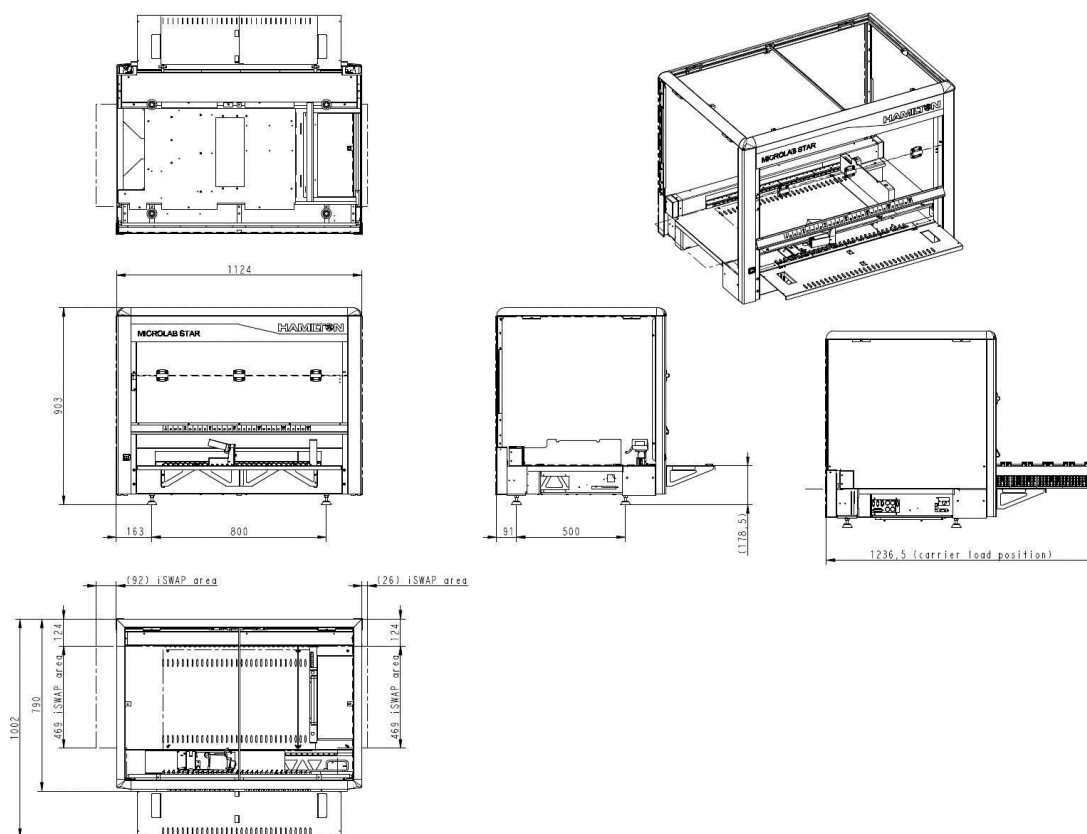
Software	
Part number	Description
911186	VENUS THREE BASE PACK V4.4
173992	UPGRADE KIT V4.4 (consists of the following: 911186/00 VENUS Three Base Pack 4.4, 624043 ML STAR Programmer's Manual, 610766 ML STAR Operator's Manual)
911095	VENUS ONE DYNAMIC SCHEDULER 5.1 (This package is compatible to VENUS three)
911099	VENUS ONE TADM FEATURE 5.1 (This package is compatible to VENUS three)
911122	VENUS ONE DATA BASE PLUS 1.1 (This package is compatible to VENUS three)
910111	VENUS TWO BASE PACKAGE 4.3 FOR MICROLAB SWAP
624668	ML STAR OPERATOR'S MANUAL
624669	ML STAR PROGRAMMER'S MANUAL

C. Instrument Dimensions

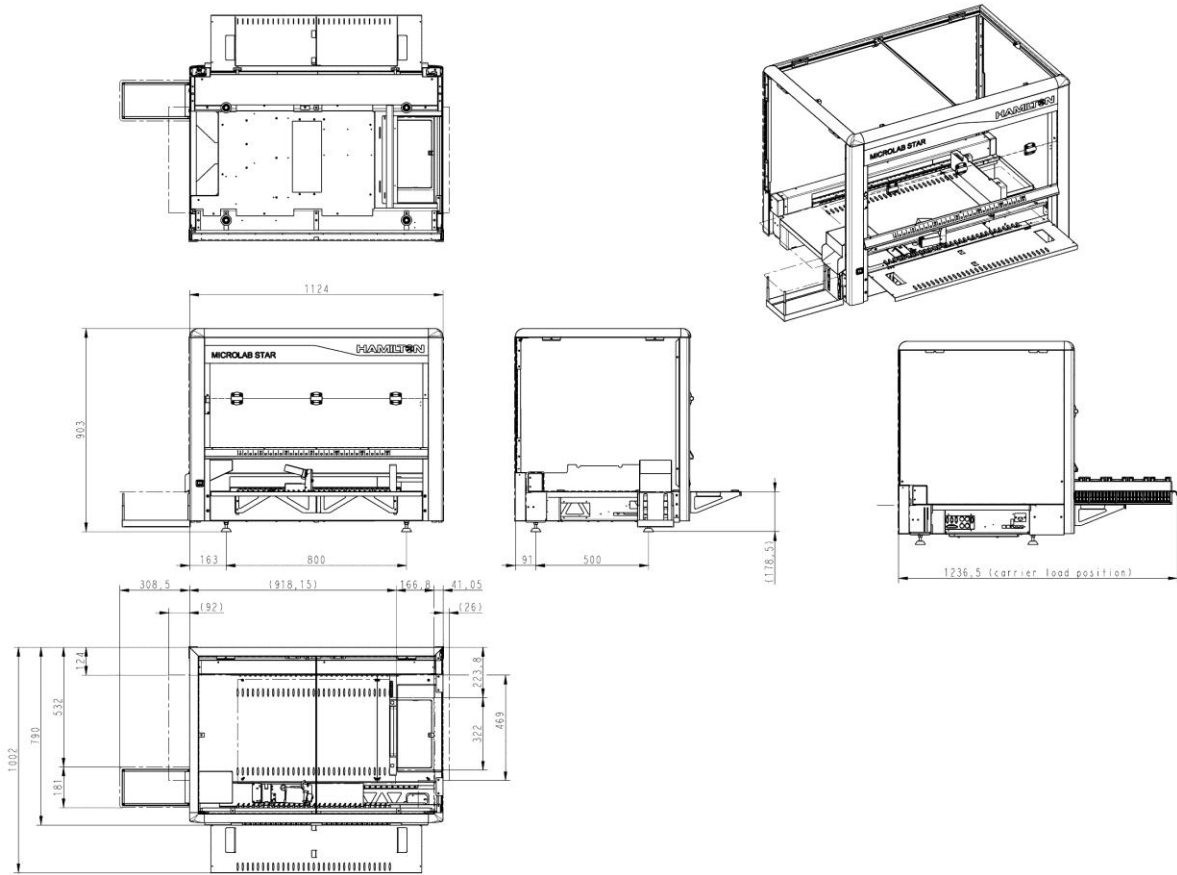
ML STARlet, manual load, shown without housing



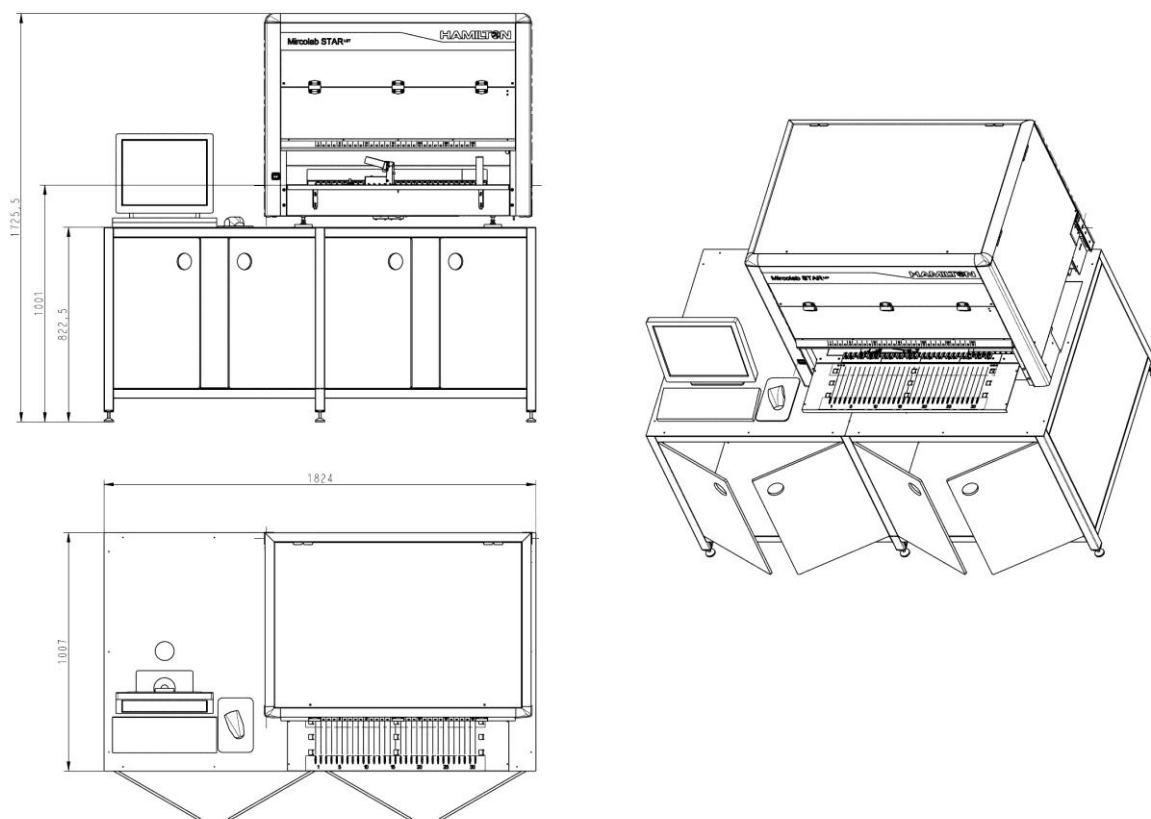
ML STARlet, Autoload



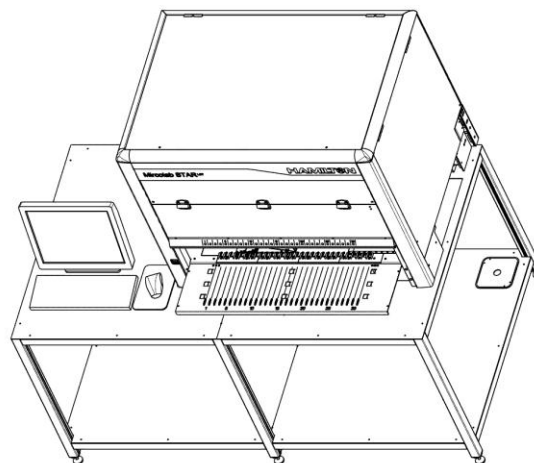
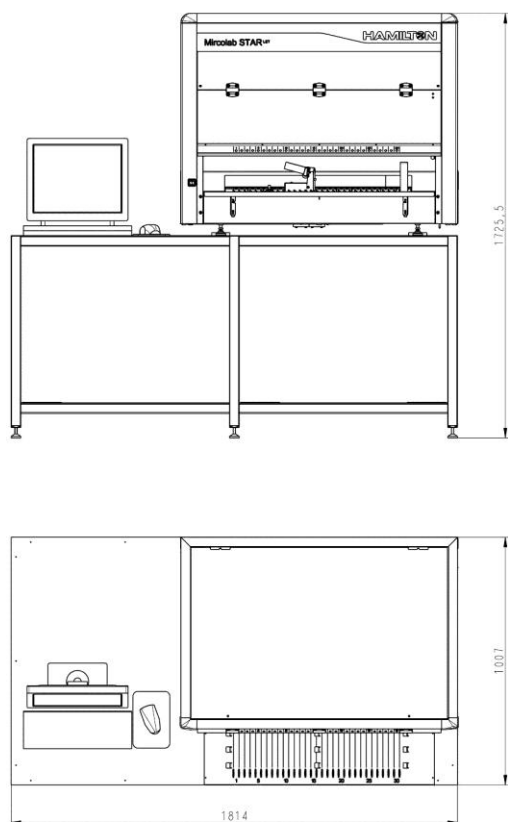
ML STARlet, Autoload, MPH



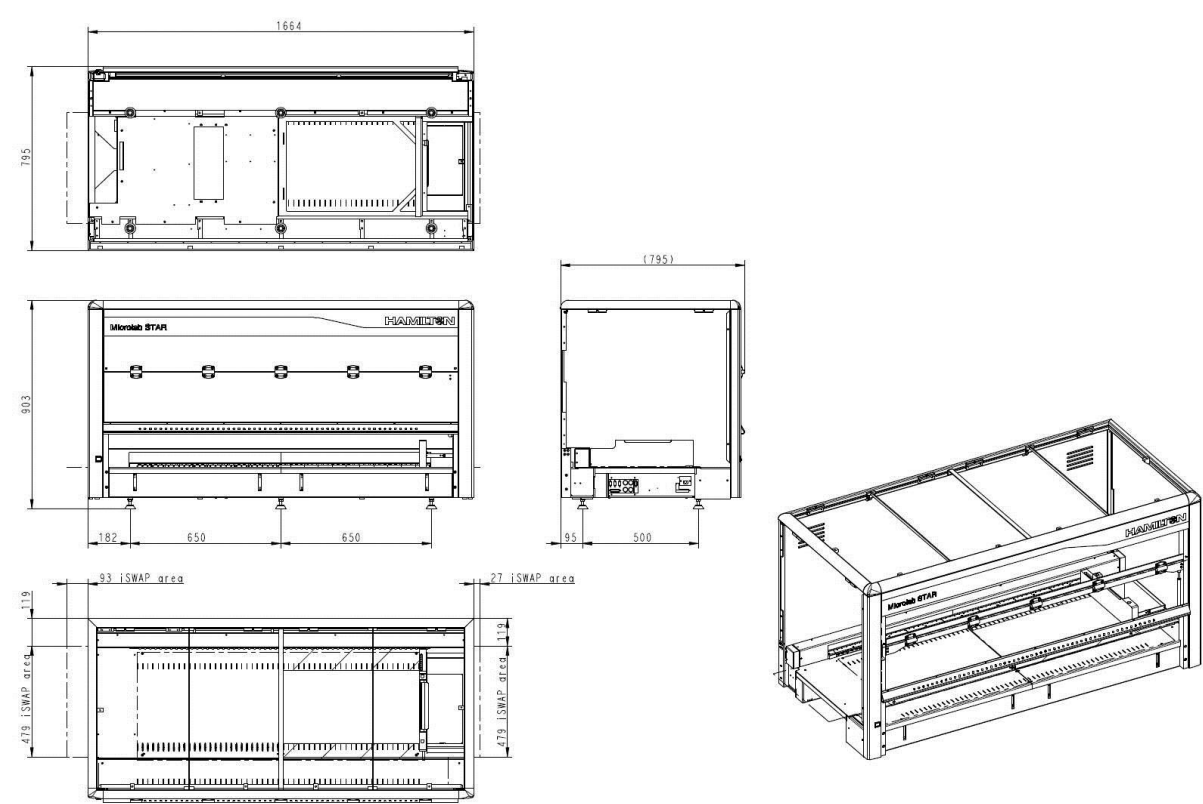
ML STARlet, on bench with doors and PC



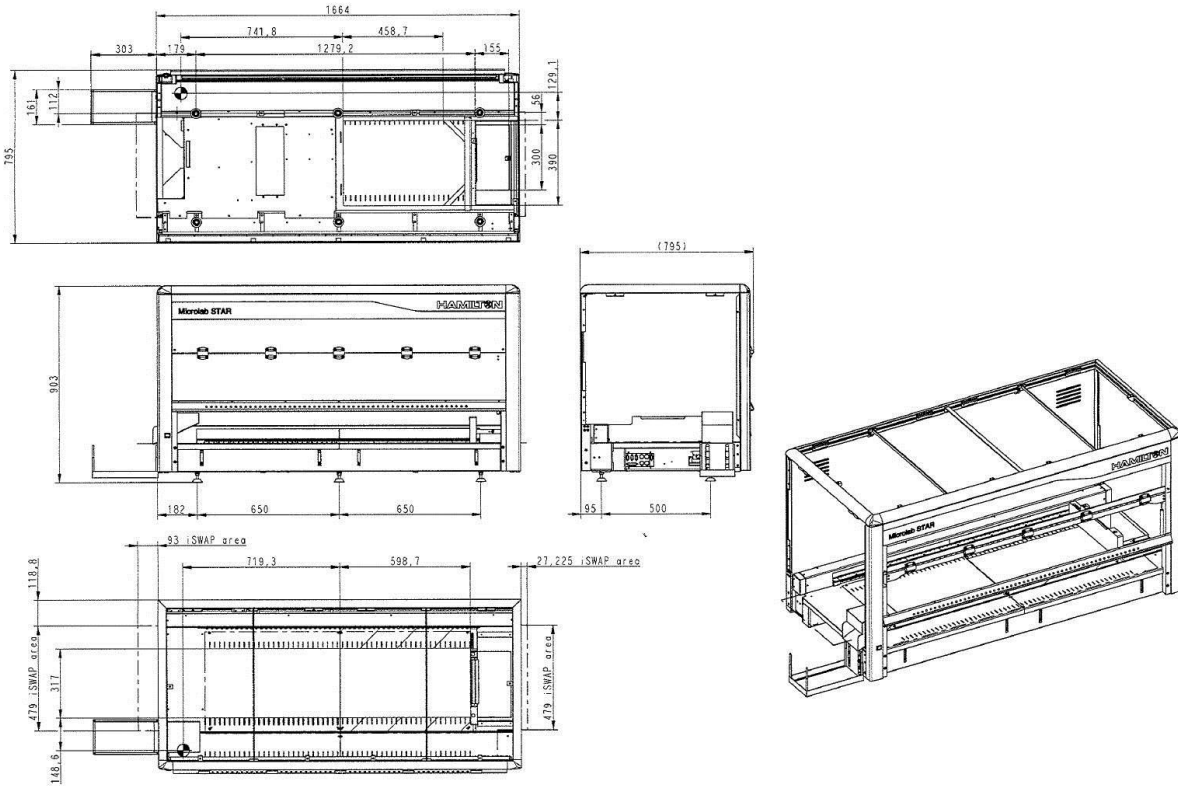
ML STARlet, on table with PC



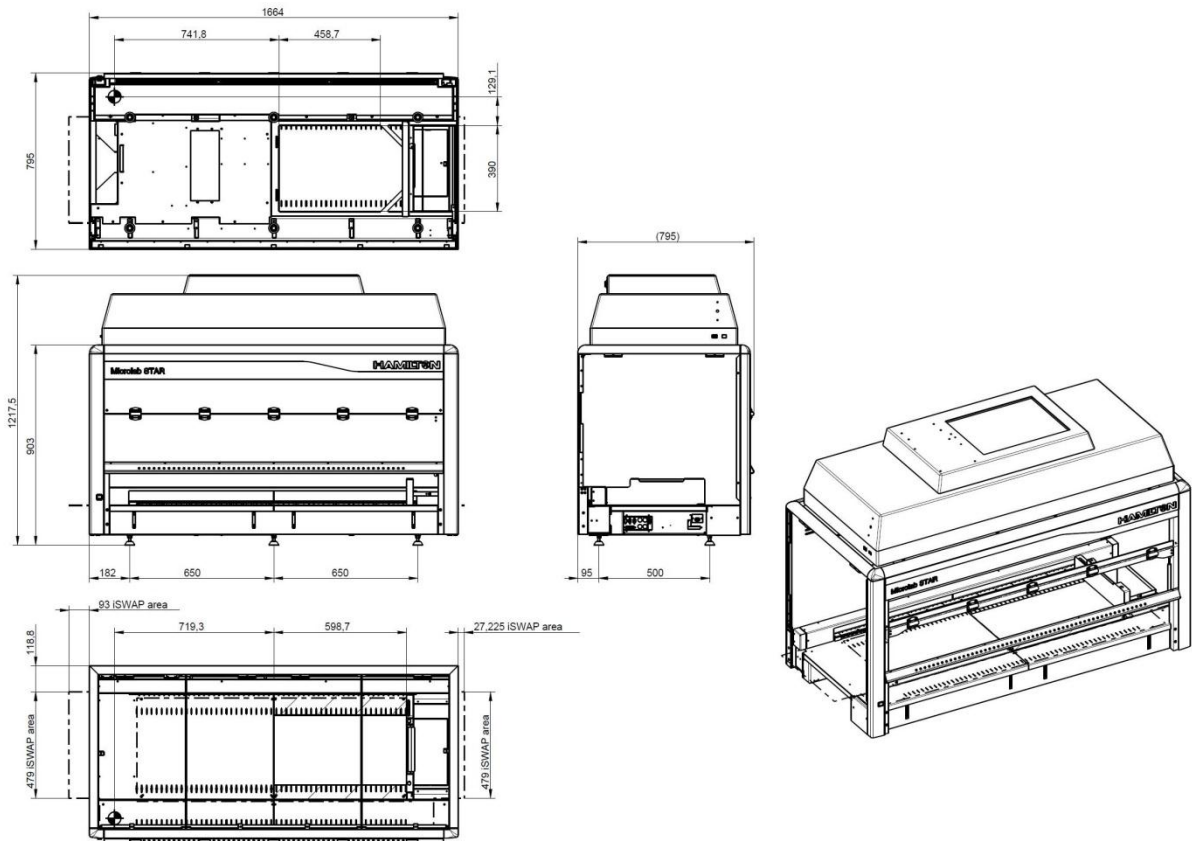
ML STAR, manual load



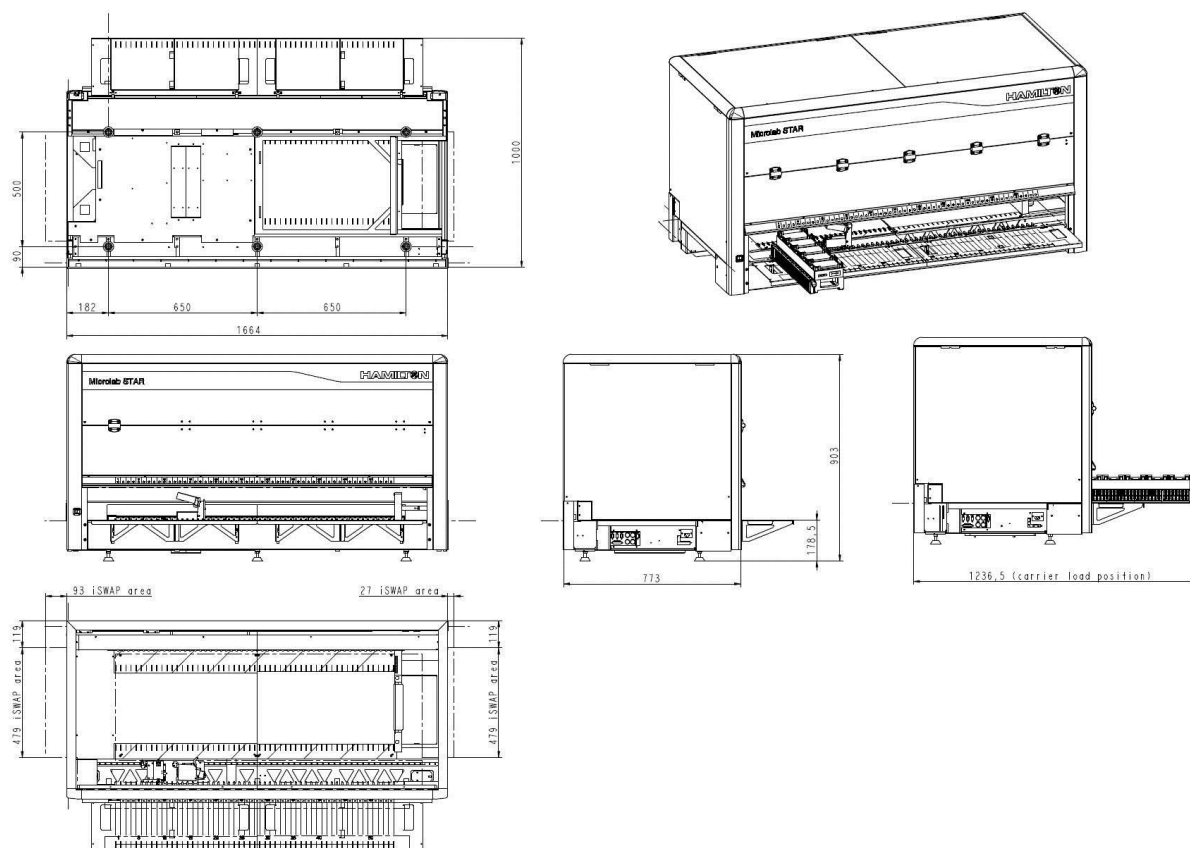
ML STAR, manual load, MPH



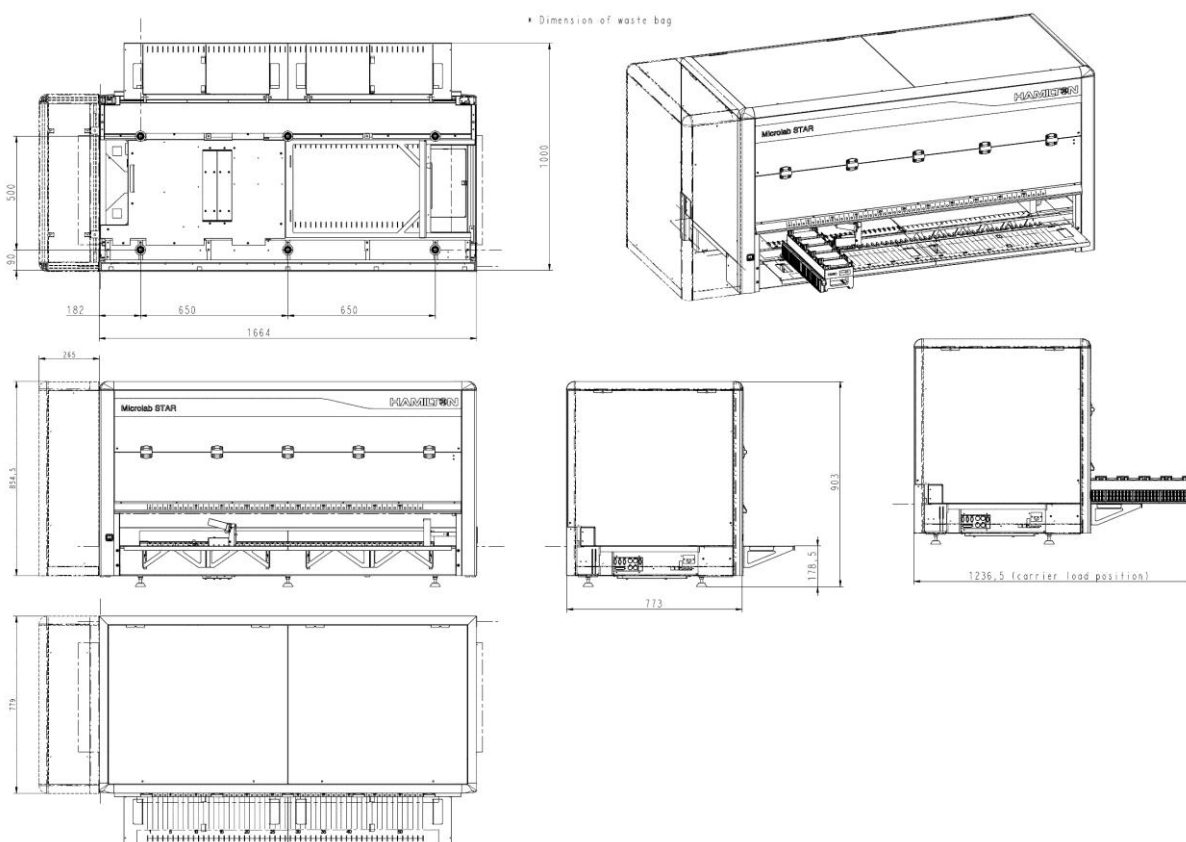
ML STAR, manual load with HEPA hood



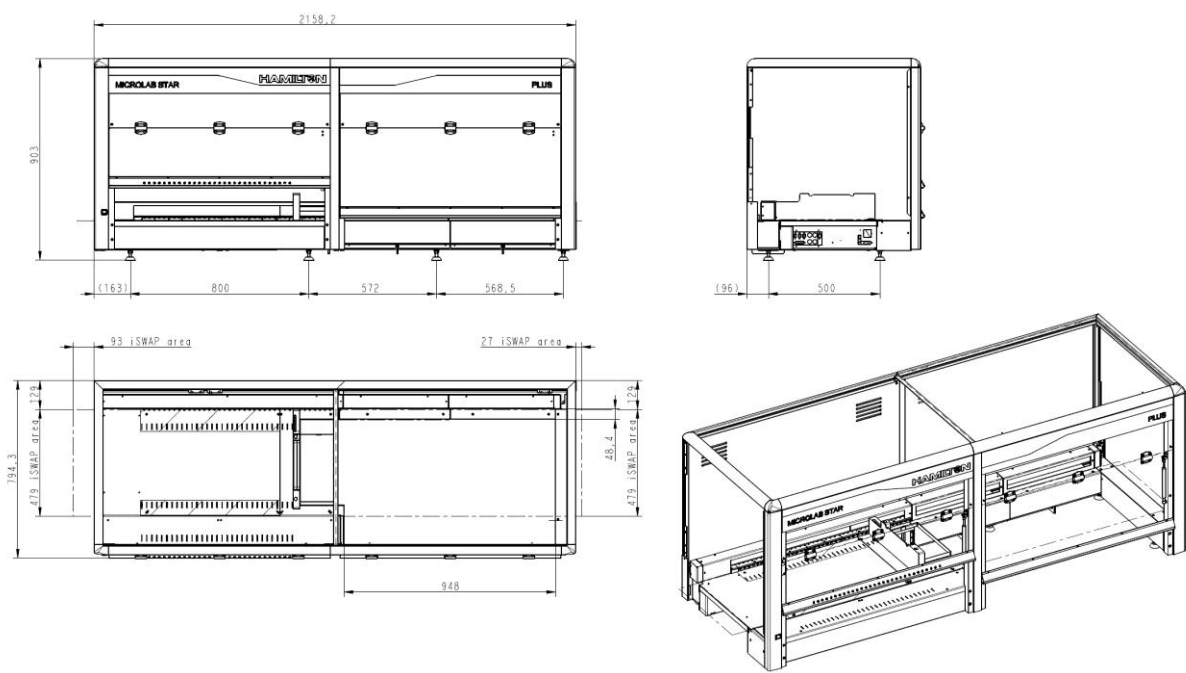
ML STAR, Autoload



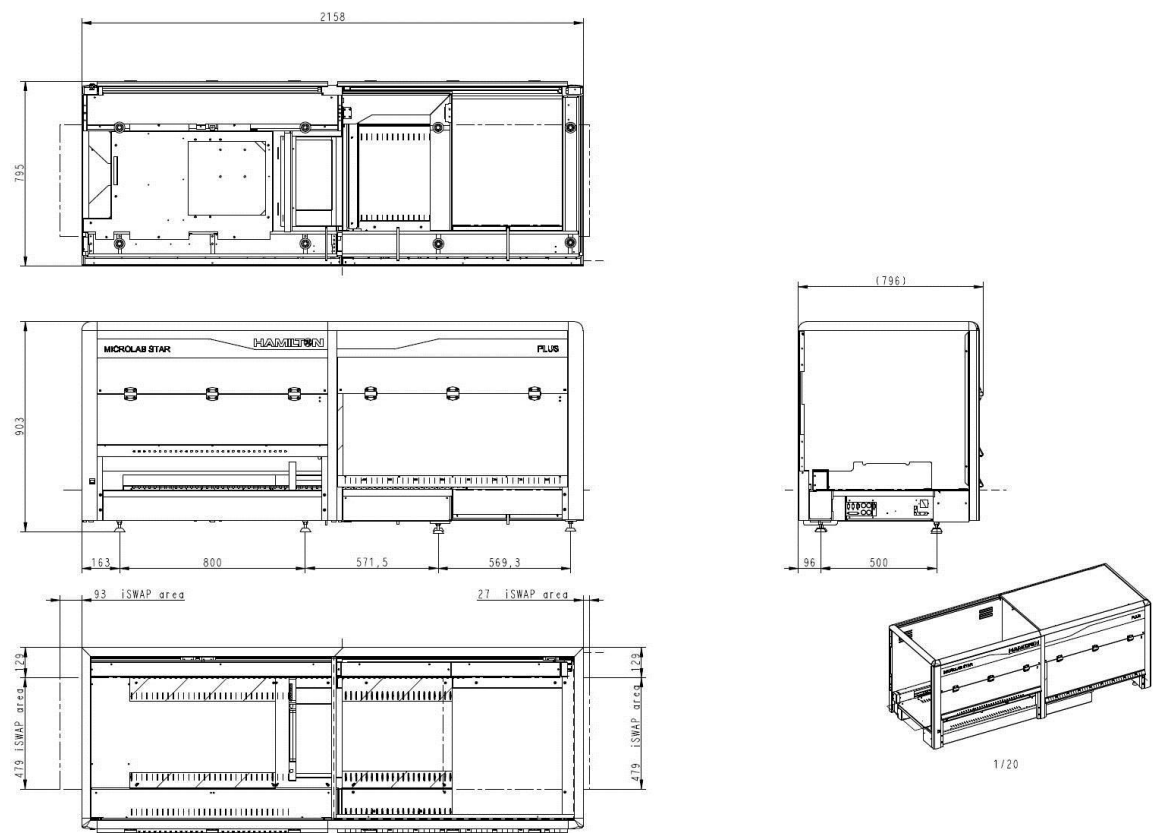
ML STAR, Autoload, extension housing left



ML STARplus, Manual load, 30T

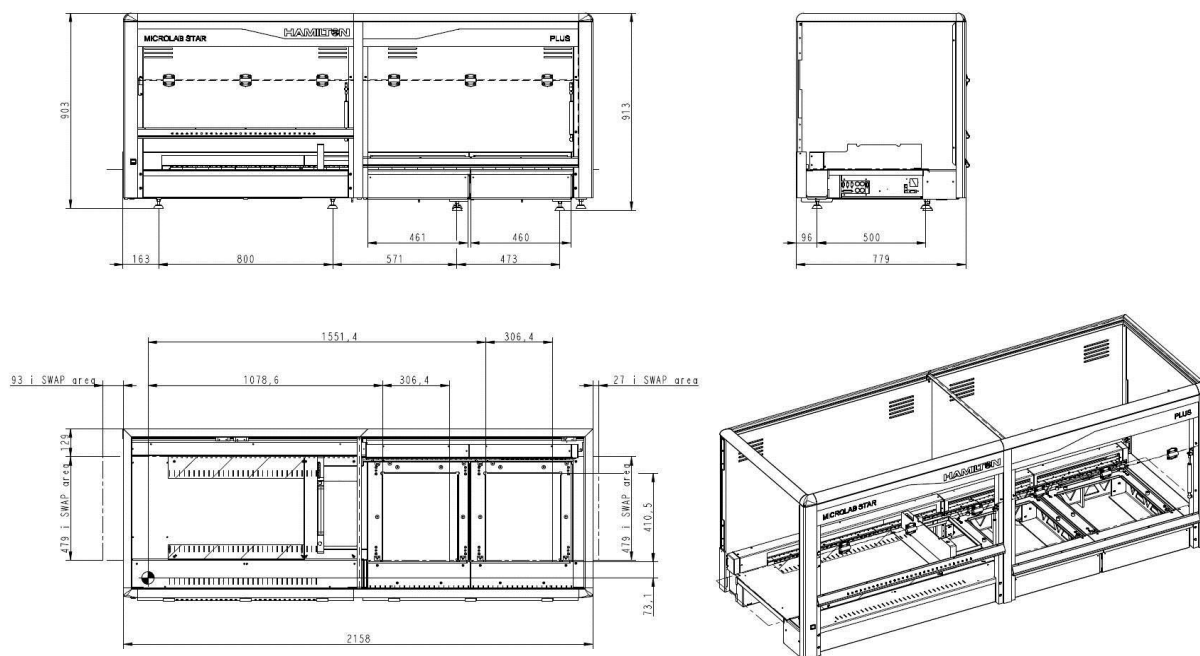


ML STARplus, manual load, 50T

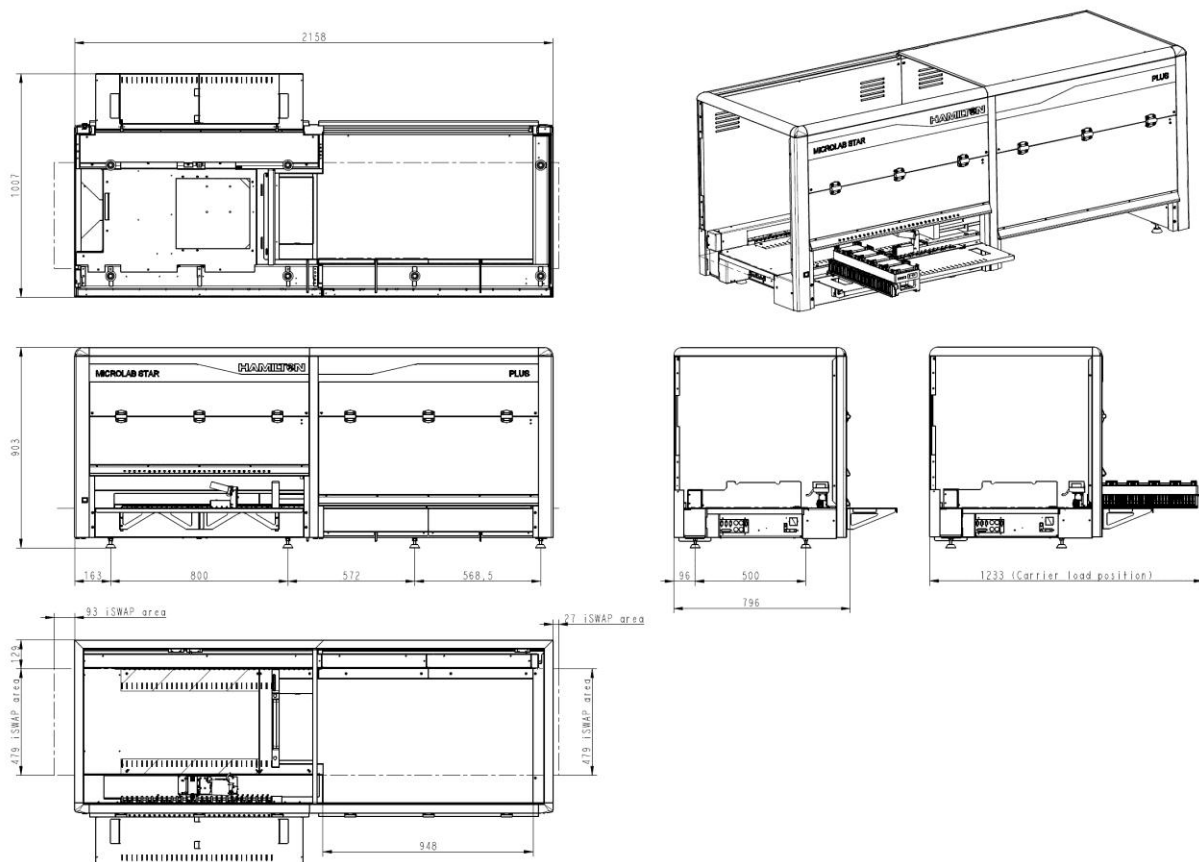


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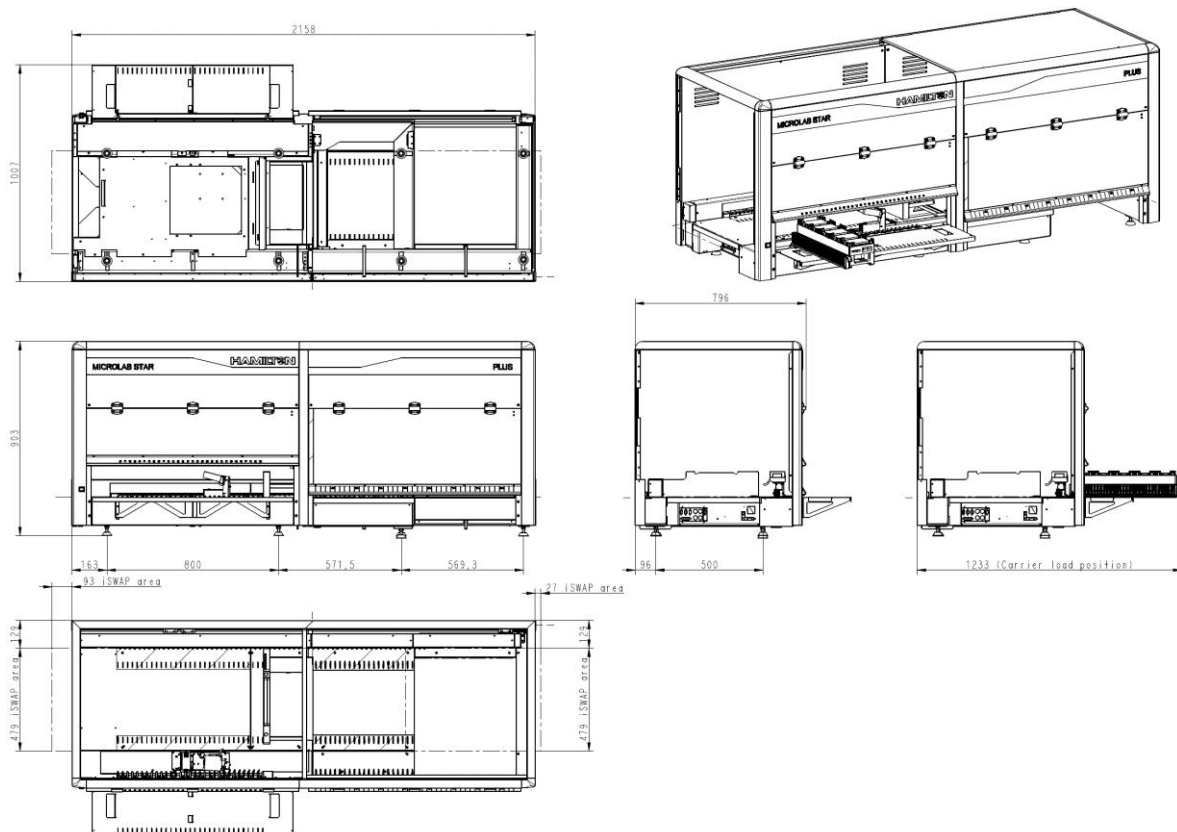
ML STARplus, manual load, 71T



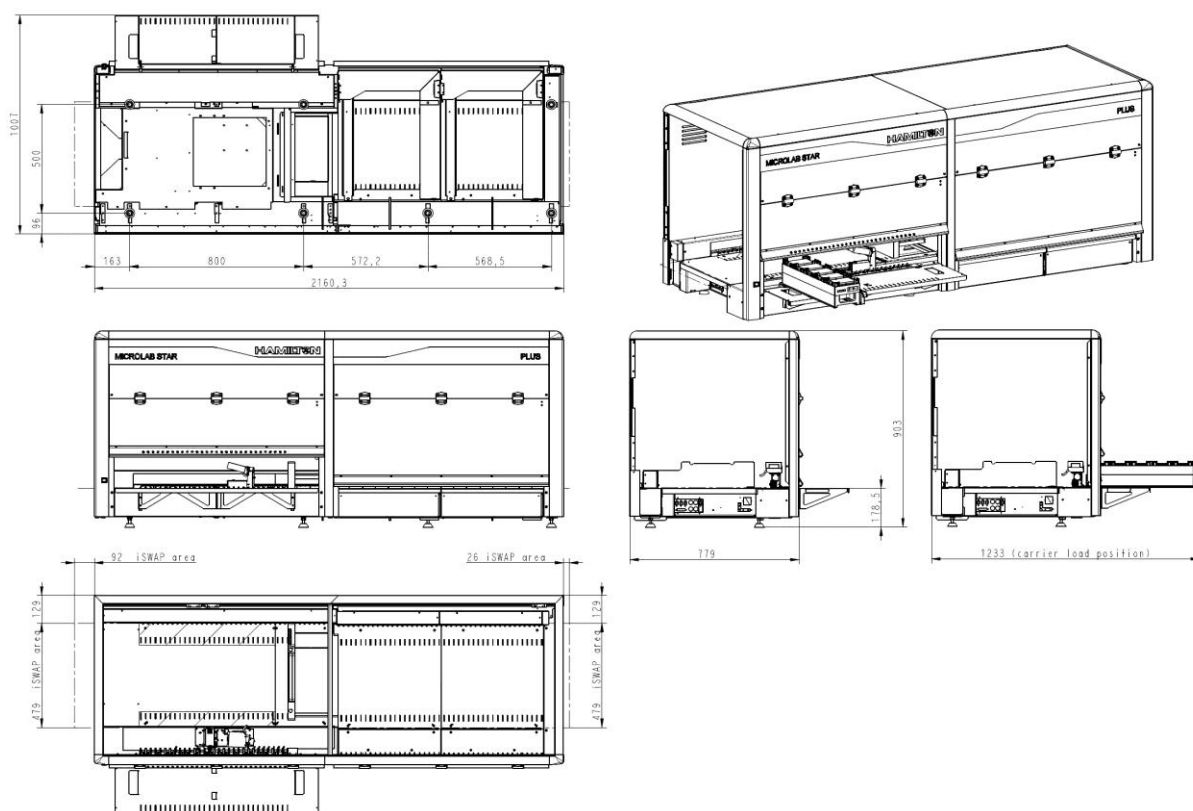
ML STARplus, Autoload, 30T



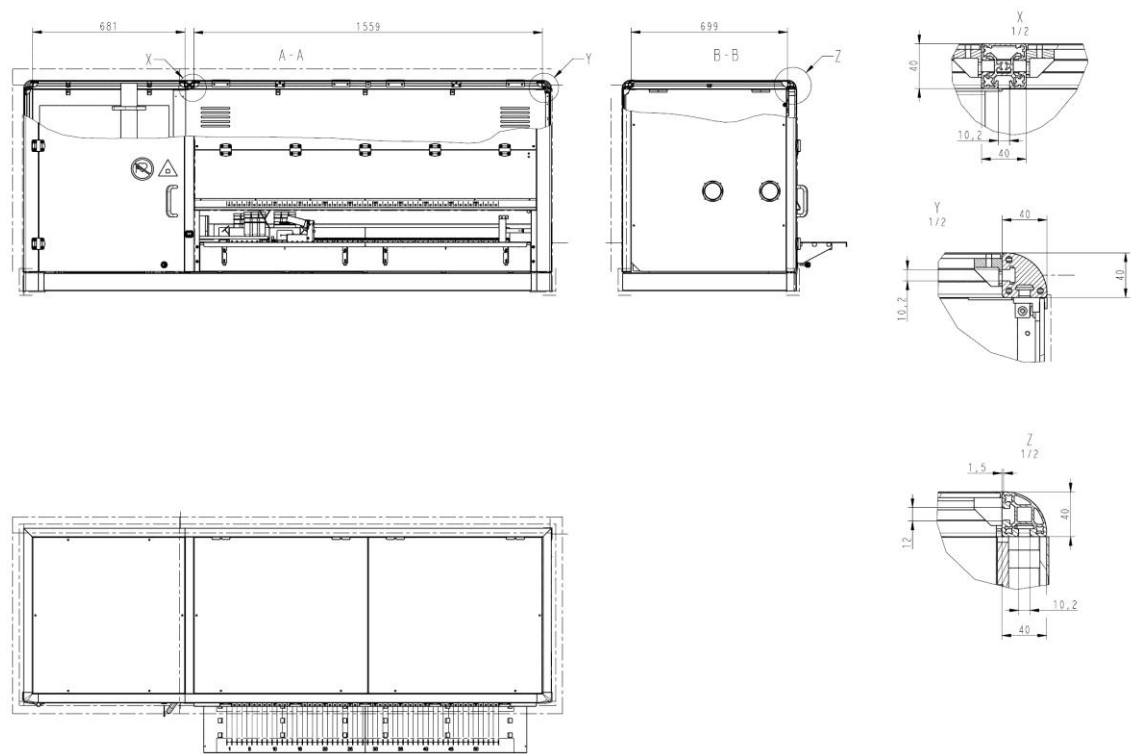
ML STARplus, Autoload, 50T



ML STARplus, Autoload, 71T



Chemagic STAR



D. Regulatory Affairs

CE, CSA and UL conformity are maintained for Microlab STAR Line. See the Declaration of Conformity for the instrument reproduced on the next page.

Radio Interference (USA and Canada)

This equipment has been tested and found to comply with the limits for a Class A digital device, pursuant to both Part 15 of the FCC Rules and the radio interference regulations of the Canadian Department of Communications. These limits are designed to provide reasonable protection against harmful interference when the equipment is operated in a commercial environment. This equipment generates, uses and can radiate radio frequency energy and, if not installed and used in accordance with the present user manual, may cause harmful interference to radio communications. Operation of this equipment in a residential area is likely to cause harmful interference, in which case the user will be required to correct the interference at his own expense.

Pursuant to the Canadian Radio Interference Regulations, ICES-001 Notice for Industrial, Scientific and Medical Radio Frequency Generators, this ISM apparatus meets all requirements of the Canadian Interference-Causing Equipment Regulations. Please note that this requirement is only for generators which operate at over 10,000 Hz.

***In vitro* Diagnostics**

The ML STAR Line is classified as a general laboratory instrument and not as an *in vitro* diagnostic device. There are ML STAR Line based instruments for *in vitro* diagnostic use.

The following text defines an *in vitro* diagnostic device [from: Directive 98/79/EC of the European Parliament and of the Council of 1998-10-27 on *in vitro* diagnostic medical devices]:

'[...] *in vitro* diagnostic medical device' means any medical device which is a reagent, product, calibrator, control material, kit, instrument, apparatus, equipment, or system, whether used alone or in combination, intended by the manufacturer to be used *in vitro* for the examination of specimens, including blood and tissue donations, derived from the human body, solely or principally for the purpose of providing information:

- concerning a physiological or pathological state, or
- concerning a congenital abnormality, or
- to determine the safety and compatibility with potential recipients, or
- to monitor therapeutic measures.

Specimen receptacles are considered to be *in vitro* diagnostic medical devices. 'Specimen receptacles' are those devices, whether vacuum-type or not, specifically intended by their manufacturers for the primary containment and preservation of specimens derived from the human body for the purpose of *in vitro* diagnostic examination.

Products for general laboratory use are not *in vitro* diagnostic medical devices unless such products, in view of their characteristics, are specifically intended by their manufacturer to be used for *in vitro* diagnostic examinations; [...]

There are ML STAR Line based instruments for *in vitro* diagnostic use.

Declaration of Conformity - EN ISO 9001:2008



Certificate

The Certification Body of
TÜV Rheinland LGA Products GmbH

hereby certifies that the organization

Hamilton Bonaduz AG
Via Crusch 8
7402 Bonaduz
Schweiz

has established and applies a quality management system
for the following scope:

**Design and development, manufacturing, distribution
and servicing of products for analytical,
biotechnical and industrial laboratories**

Proof has been furnished that the requirements specified in

EN ISO 9001:2008

are fulfilled. The quality management system is subject to yearly surveillance.

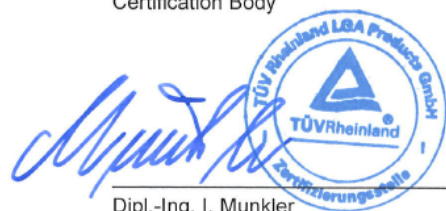
Certificate Registration No.: SY 60040271 0001

An audit was performed. Report No.: 21167491 001

This Certificate is valid until: 04.08.2016

Certification Body

Date 05.08.2011



Dipl.-Ing. I. Munkler

TÜV Rheinland LGA Products GmbH - Tillystraße 2 - 90431 Nürnberg

Tel.: +49 221 806-1371 Fax: +49 221 806-3935 e-mail: cert-validity@de.tuv.com <http://www.tuv.com/safety>

Declaration of Conformity - EN ISO 13485:2003 + AC 2009



Certificate

The Certification Body of
TÜV Rheinland LGA Products GmbH

hereby certifies that the organization

Hamilton Bonaduz AG
Via Crusch 8
7402 Bonaduz
Schweiz

has established and applies a quality management system for medical devices
for the following scope:

**Design and development, manufacturing, distribution
and service of products for diagnostic laboratories.
Contract manufacturing of medical devices.**

Proof has been furnished that the requirements specified in

EN ISO 13485:2003 + AC:2009

are fulfilled. The quality management system is subject to yearly surveillance.

Certificate Registration No.: SX 60040270 0001

An audit was performed. Report No.: 21167491 001

This Certificate is valid until: 04.08.2016



Akkreditiert durch
Zentralstelle der Länder
für Gesundheitsschutz
bei Arzneimitteln
und Medizinprodukten
ZLG-ZQ-995.00.01-46

Date 05.08.2011

Certification Body



Dipl.-Ing. I. Munkler

TÜV Rheinland LGA Products GmbH - Tillystraße 2 - 90431 Nürnberg

Tel.: +49 221 806-1371 Fax: +49 221 806-3935 e-mail: cert-validity@de.tuv.com <http://www.tuv.com/safety>

The following declarations are only templates. These templates will only be filled out according to the specifications after the order has been made.

Declaration of Conformity: CE, CSA

Each individual instrument includes a Declaration of Conformity to European and American standards.

Declaration of Conformity

We Hamilton Bonaduz AG, CH-7402 Bonaduz/Switzerland confirm that the following product

Product name **Microlab STAR Line**

Product type: ☐ Microlab STAR
☐ Microlab STARlet
☐ Microlab STARplus

S/N

meets the following EC directives (including all applicable amendments):



EMC Directive 2004/108/EC

Low Voltage Directive 2006/95/EC

Applied harmonised standards:

Safety: EN 61010-1 (2001)
 EN 61010-2-081 (2002)
 EN 60825-1 (2003)

EMC: Emission EN 61326 -1 (2006)
 Immunity EN 61326 -1 (2006)

Additional Information:



Canada, USA

CAN/CSA-C22.2 No. 1010.1-92
 UL Std. No. 3101-1
 FCC, Part 15, class A

Hamilton Bonaduz AG

[Signature]
 Andreas Wieland
 CEO

Bonaduz, 30.10.08

608399/05

HAMILTON Bonaduz AG
 Via Crusch 8
 CH-7402 Bonaduz/Switzerland

Tel.: +41 81 660 60 60
 Fax: +41 81 660 60 70
www.hamiltoncompany.com



Reg. No. SY 2111892 01

1 / 1

Declaration of Quality

Declaration of Quality

We Hamilton Bonaduz AG, CH-7402 Bonaduz/Switzerland confirm that the following product passed the manufacturer's final inspection and was within specifications.

Product: **Microlab STAR Line**

Product type: ☐ Microlab STAR 16 channel
☐ Microlab STARlet 16 channel
☐ Microlab STARplus 16 channel

S/N

The following dosage specifications have been measured:

Test No	Test volume [µl]	Tip size [µl]	No of measurements per channel	Standard deviation [µl]	Precision [%]	
					Spec.	Actual
1	2	10	10		< 6.0	
2	5	10	10		< 2.0	
3	10	300	10		< 2.5	
4	100	300	10		< 1.0	
5	500	1000	10		< 1.0	
6	1000	1000	10		< 1.0	

Method: gravimetical
Medium: 0.1% sodium chloride
Balance: Mettler MX5

Test equipment: EPA 617912 Rev.

Environmental conditions: Ambient temperature:°C

Date: Personal No: Signature:

608584/00

HAMILTON Bonaduz AG
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CH-7402 Bonaduz/Switzerland

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Reg. No. SY 2111892 01

1 / 2

Applied Statistics

$$\text{Precision} = \frac{s}{\bar{x}} 100\%$$

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$

$$s = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2}$$

n: number of measurements
x_i: volume dispensed in measurement i [μl]
s: standard deviation [μl]
 $\bar{x}$: mean [μl]

608584/00

HAMILTON Bonaduz AG
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Tel.: +41 81 660 60 60
Fax: +41 81 660 60 70
www.hamiltoncompany.com



Reg. No. SY 2111892 01

2 / 2

Declaration of Cross Contamination Free Processing



Declaration of cross contamination free processing on the ML STAR

We Hamilton Bonaduz AG, CH-7402 Bonaduz/Switzerland, confirm that with following features and products a cross contamination free process on the ML STAR is given.

- **HAMILTON CO-RE Disposable Filter Tips**
Tips are manufactured in clean room and are free of DNA-contamination, PCR-Inhibition, DNase, RNase, ATP contamination and Endotoxin.
With filter tips you will have no contamination of the channel.
- **HAMILTON CO-RE Technology**
This technology allows to pick up tips with a stable lock –and-key fit. The ML STAR requires no vertical force for tip attachment or tip ejection, which eliminates aerosol production upon tip ejection.
For more information see: <http://www.hamiltonrobotics.com/liquid-handling-technology/tip-attachment-co-re.html>
- **HAMILTON TADM Technology (including drop control)**
This feature allows you to control the whole pipetting process. You will detect clots, bubbles, foam, blocked tip, leaky system.
For more information see: <http://www.hamiltonrobotics.com/liquid-handling-technology/process-monitoring-tadm.html>
The drop control feature will eliminate any droplets between the aspirate and dispense step.
For more information see: <http://www.hamiltonrobotics.com/liquid-handling-technology/anti-droplet-control-adc.html>

Hamilton Bonaduz AG


Rainer Beckbissinger
Quality Manager Robotics

Bonaduz, 2009-08-17

HAMILTON Bonaduz AG
Via Crusch 8
CH-7402 Bonaduz/Switzerland

Tel.: +41 81 660 60 60
Fax: +41 81 660 60 70
www.hamiltoncompany.com



Reg. No. SY 2111892 01

1 / 1

WEEE Declaration

Recycling of a Hamilton's ML STAR Line Instruments in accordance to EC directive WEEE

The European Community requires from manufacturers to organize the disposal and waste of electrical and electronic equipment (WEEE). For this reason, Hamilton Bonaduz AG took part in an initiative to organize the disposal of ML STAR Line products through a European disposal network called RENE. RENE is the largest recycling network for the disposal of electronic equipment in Europe. The mission of RENE is a European-wide, WEEE-compliant, high quality recycling for electrical and electronic equipment through a dense network with both innovative and SMB-sized partner companies. As a consequence, Hamilton Bonaduz AG gets a turn-key-solution that includes all processes from treatment of incoming orders over collection, logistics and recycling down to reporting and the according management of material flows.

Hamilton offers a WEEE process in collaboration with Toolpoint and RENE AG:

- Request for the collection of the Hamilton instrument via Toolpoint Home Page(www.toolpoint.ch)
- Completion of the decontamination confirmation form
- Preparation for transport: packing
- Activation of the recycling order
- Archiving of the decontamination confirmation
- Disposal of instrument

Responsibilities	
Ordering party	Decontamination Preparation for transport Note: The cost for decontamination and preparation for shipment is paid by the ordering party. On request HAMILTON offers to take care for that part of the recycling process.
RENE	Transport Disposal
Toolpoint	Registration Invoice the disposal to HAMILTON
HAMILTON	Organize the disposal in accordance of the WEEE directive

Recycling process**1. Request for disposal of the instrument**

Access to the order registration is given by the toolpoint homepage www.toolpoint.ch

☒ Recycling ☒ Order registration form

2. Completion of the Decontamination form

Once the form has been completed the request for disposal is automatically activated and transferred to Toolpoint. The confirmation of the order will be sent to the registered contact person.

3. Decontamination

The responsibility for decontamination remains at the ordering party. It is mandatory to sign the decontamination form and send an electronically copy to Toolpoint. Toolpoint forwards the documentation to RENE who is in charge with the disposal of the instrument.

4. Packing / Preparation for shipment

For the transportation instruments with a weight of over 30kg need to be fixed on a euro pallet. Instruments below 30kg can be packed in a cardboard or plastic box. A signed copy of the decontamination form needs to be added to the outer part of the shipping box or instrument.

Please also refer to the Technical Note "Recycling of a Hamilton's ML STAR Line Instruments in accordance to EC directive WEEE".

RoHS Compliance

Since S/N 4610 (June 2011) ML STARLine instruments are in compliance with Directive 2002/95/EC of the European Parliament and of the Council of 27 January 2003, on the restriction of the use of certain hazardous substances in electrical and electronic equipment (RoHS) and its amendments.

21 CFR Part 11 Compliance

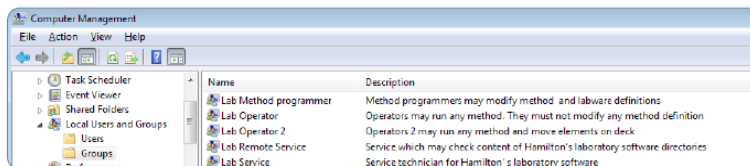
The FDA issued regulations covering agency criteria for maintenance, security and submittal of electronic records by publication of Final Rule 21 CFR Part 11 Electronic Records; Electronic Signatures in the Federal Register of March 20, 1997. Laboratory implementation and compliance with the 21 CFR Part 11 regulation requires a program combining GLP (Good Laboratory Practice) with compliant instrument software and secure LIMS database management. Hamilton's Vector/VENUS software contains the tools necessary for 21 CFR Part 11 compliant operation of the Microlab robotic instruments (shown on the back of this document). This chapter describes the specific features of Microlab Vector / VENUS software enabling compliant instrument operation along with areas of laboratory responsibility.

Requirements	Vector/VENUS Software	21 CFR Part 11.10 section
Controlled system access	Microlab Vector/VENUS uses the security tools provided in Windows 7 for five defined user groups Vector/VENUS functional protections	a, d d, g
Files accessible and printable in a human readable form	Files can either be printed as text or from the correct viewer	b
Electronic records must be protected and maintained throughout the records retention period	User id and date/time stamps are applied to every electronic record when it is saved Changes to records are monitored by checksum Hamilton maintains backward compatibility when reasonable Vector/VENUS is compatible with database software programs for version control for complete audit trails	g g b, c e
Documentation	Hamilton documentation for robotic instruments are controlled through an ISO-9001 compliant change control process consistent with 21 CFR Part 11 regulation	k (1)
Training	Hamilton provides ISO-9001- compliant training with certification of training for users	i

Summary of Microlab Vector/VENUS software compliance. The subsections of 21 CFR Part 11.10 are addressed individually below. The letters in the third column refer to the subsection relating to each point.

What do I need to do to be 21 CFR Part 11 compliant with my Hamilton Microlab instrument and Vector/VENUS software?

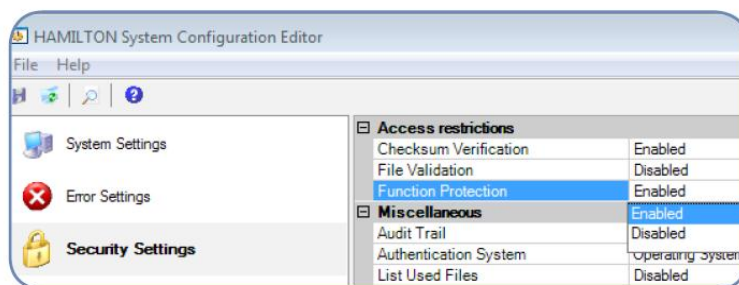
1. Define Windows user groups and assign users to the groups with unique user id/passwords. Define accessible directories for the user groups along with privileges in those directories, such as read-only or full access.



Windows environment for defining user groups and adding users to the groups.

The Windows operating systems have security tools to restrict user access to desktops and directories by user id/password protection and user group definitions. Each user group is given different access and privilege to Hamilton files as is appropriate for their functions. Methods can be protected by Windows tools by saving all validated methods in limited-access directories, while saving non-validated methods in other restricted directories. Low level users can be allowed to access the directory with validated methods while blocking these user groups from the non-validated methods directory. Barcode and worklist files are not protected by Vector/VENUS software. These files may be created and used by programs outside of Hamilton's control. Further protection of these files for full compliance must be accomplished by laboratory practices.

2. Enable the function protections in Vector/VENUS software. When these protections are enabled, Vector/VENUS checks the user group of the logged in user and allows functional privileges based on the user group.



Environment for enabling Vector/VENUS function protection.

The function protection in Vector/VENUS, along with the user groups established in Windows, provide authority-based access to the software for user level authentication and access. All files are additionally protected by checksum and are date/time stamped with the user id when the files are saved or created.

3. Define the SOPs and practices for complete laboratory compliance.

Laboratories are responsible for defining SOPs for reaching full compliance when using Vector/VENUS software with Hamilton instrumentation. These SOPs include: maintaining archives of old methods and software revisions throughout the records maintenance period; implementing practices or additional software sufficient for a complete audit trail of records; and maintaining internal documentation on any Hamilton instrumentation. Recommendations for necessary SOPs are addressed inside.

Hamilton Electronic Records

Operation of a Hamilton instrument involves multiple file types or electronic records, which need to be protected under 21 CFR Part 11. These files are either input to be used during the method development and runtime or output files, which are created by the software during runtime.

Input Files:

- Labware Definitions
- Liquid Class Definitions
- Method Files
- System configuration files

Output Files:

- Trace files

Barcode and worklist files are not protected by Vector/VENUS software. These files may be created and used by programs outside of Hamilton's control. Further protection of these files for full compliance must be accomplished by laboratory practices.

Controls for closed systems

Hamilton robotic instruments are controlled by closed system computer systems. A closed system is one to which access is controlled by persons responsible for the content of the electronic records on that system. Below each section of 21 CFR Part 11.10 is followed by an explanation of Hamilton's Vector/VENUS software compliance approach and the implementation responsibilities of the end user.

Controls for Closed Systems - 21 CFR Part 11.10

Persons who used closed systems to create, modify, maintain, or transmit electronic records shall employ procedures and controls designed to ensure the authenticity, integrity, and when appropriate, the confidentiality of electronic records, and to ensure that the signer cannot readily repudiate the signed record as not genuine. Such procedures and controls shall include the following:

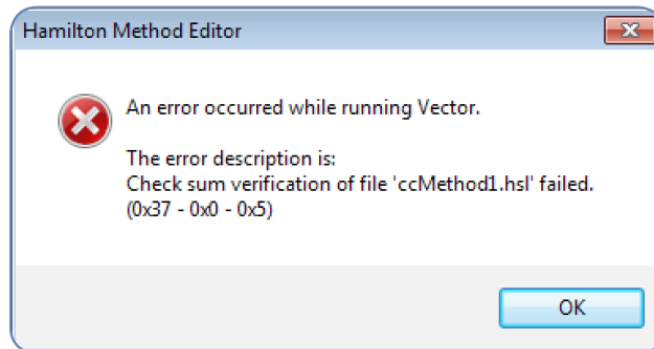
This section describes the requirements to define an electronic record as protected for closed systems. This applies to all records created on a closed system, whether or not it is validated or formally recognized by the organization. The following sections address more specific requirements.

Validation - 21 CFR Part 11.10 (a)

Validation of systems to ensure accuracy, reliability, consistent intended performance, and the ability to discern invalid or altered records.

This refers to the validation of the electronic signature system to be used. The electronic signature system must be able to restrict access to various levels of functionality, identify when a change is made to an electronic record, by whom, and whether these changes were validated. Microlab Vector/ VENUS software uses the security tools of the Windows 7 operating system.

This system allows for different user groups within an organization to have different levels of access to the software. The files are user id and date/time stamped whenever they are saved. In addition, these files are checksum protected which protects files from inappropriate modification.

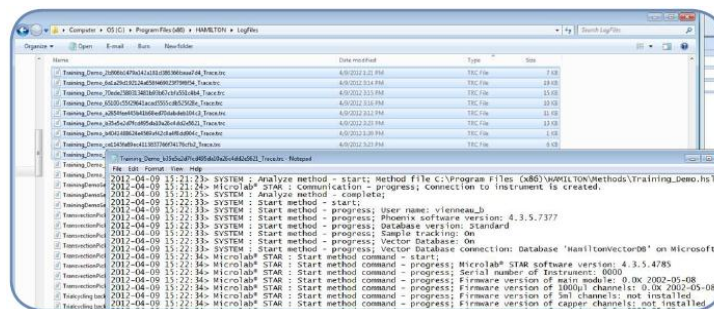


Error message seen when a checksum-protected file has been inappropriately modified. This message will appear when the user attempts to reopen the file

Readability - 21 CFR Part 11.10 (b)

The ability to generate accurate and complete copies of records in both human readable and electronic form suitable for inspection, review, and copying by the agency.

All records must be readily generated in either electronic or human readable form, along with the procedures used to create the records. This means all trace files must be retrievable along with the method files used to create these trace files (Figure 5). Organizations need to maintain electronic records as defined by agency-defined retention periods. Obsolete hardware and software versions do not need to be maintained, as long as all the records are retrievable with the change in technologies.



A trace file (.trc) documenting run time events for a particular method. A unique trace file is generated for each run time event with date/time and user id information located at the end of the file.

Vector/VENUS software files are either binary or ASCII text. ASCII is easily printed in human readable form with a text program. Other file types can be viewed and printed with the correct viewer or converted to ASCII and printed. Hamilton maintains backward compatibility of method files between software revisions when reasonable. In the event a software revision no longer supports an archived method, the appropriate software version must be maintained to support the archived record.

Archived Record Protection - 21 CFR Part 11.10 (c)

Protection of records to enable their accurate and ready retrieval throughout the records retention period.

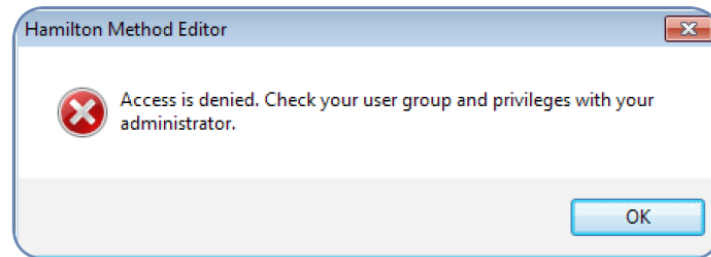
This is similar to the above section requiring maintained electronic files throughout the data retention period. However, this rule is not limited to retrieval of data. The method must be able to be processed as originally done. Hamilton maintains backward compatibility of method files between software revisions when reasonable. Even if the current software version on the instrument does not support the archived method, records associated with the archived methods can still be viewed and are protected.

System Security - 21 CFR Part 11.10 (d)

Limiting system access to authorized individuals.

System access can be limited by current bilateral identification systems, such as user id and passwords, and/or by a single biometric identification. User access must be defined and limited to various levels of the software. This security can be applied by through Windows 7 security tools.

System access for Vector/VENUS software is controlled through the Windows 7 operating system. Software allows access to various levels of the software based on the user group membership of the logged in user. If a user with an insufficient access level attempts to enter restricted sections of the software, an error message is displayed and the user is blocked from inappropriate access.



Error message shown when a user with insufficient access attempts to open a restricted file or directory.

Audit Trail - 21 CFR Part 11.10 (e)

Use of secure, computer-generated time-stamped audit trails to independently record the date and time of operator entries and actions that create, modify or delete electronic records. Record changes shall not obscure previously recorded information. Such audit trail documentation shall be retained for a period at least as long as that required for the subject electronic records shall be available for agency review and copying.

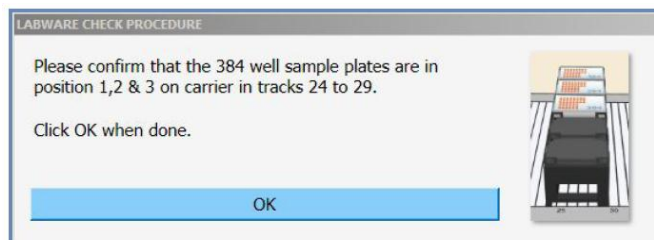
Complete audit trail documentation is outside the scope of Vector/VENUS software and must be maintained by laboratory practices.

Sequencing - 21 CFR Part 11.10 (f)

Use of operational system checks to enforce permitted sequencing of steps and events, as appropriate.

The purpose of operational system checks is to verify that operations are not performed out of sequence as defined by the method. It is the agency's intent that such checks be performed by

the computer system. Method checkpoints can be written into the program to direct the user and ensure that certain events have taken place before continuing with the method, such as all labware being in position. However, fulfilling this is a laboratory procedure and is not enforced within the Vector/VENUS software.

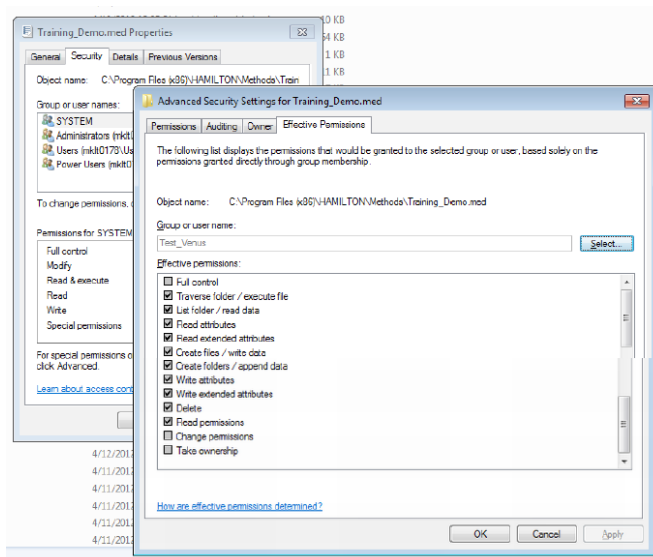


Messages to the operator can be programmed into the method to guide the operator in the correct sequence of events.

Authority - 21 CFR Part 11.10 (g)

Use of authority checks to ensure that only authorized individuals can use the system, electronically sign a record, access the operation or computer system input or output device, alter a record, or perform the operation at hand.

Five user groups have been defined and each group has a distinct level of access to the Vector/VENUS software (Table 2). Methods can be controlled by saving validated methods in an access-controlled directory. Lab Operator and Lab Operator 2 user groups should be denied access to directories containing non-validated methods while having read-only access to validated method directories (Figure 8). Additional protection is supplied by the checksum system for all records, which protects files from inappropriate modification. The user id and time/date stamp are applied to every electronic record when it is saved.



User groups can be given different levels of access to files and directories with Windows 7 security. For example, the Lab Method Programmer will have full control to modify, execute, read and write method files, while the Lab Operator will have read-only access to method files.

User group	Allowed function
Lab Operator	Allowed to run validated methods
Lab Operator 2	Allowed to run validated methods and move labware using the Layout Editor
Lab Method Programmer	Allowed to change methods, labware and sequences
Lab Service	Laboratory service engineers with full access in order to install and service the system
Lab Remote Service	Allowed to only read data

These are the user groups recognized by Vector/VENUS software for the function protection features. Vector/VENUS software checks the user's user group identity to appropriately restrict their functional access to the software.

Location Checks - 21 CFR Part 11.10 (h)

Use of device (e.g. terminal checks) to determine, as appropriate, the validity of the source of data input or operational instruction.

This requirement is applicable to a source of data (e.g. plate reader) that can receive commands from more than one system, such as on a network. The source of data instrument would have to question the source of the command to ensure that only the authorized workstation is the actual source of the commands. Hamilton workstations have hardwired connections to the computer. For unique instruments, location checks are unnecessary. In the event of identical instruments under control by the same computer, the node identity of the second instrument must be changed from the default setting during installation, creating a unique address for each of the identical instruments. This differentiates the systems to the computer. Correct instrument identification is verified during instrument installation.

Education/Training - 21 CFR Part 11.10 (i)

Determination that persons who develop, maintain, or use electronic record/electronic signature systems have the education, training, and experience to perform their assigned tasks.

Hamilton supplies training and application support as necessary to assist in developing and maintaining application performance. After training with a Hamilton representative, each trainee receives a certificate of training completion. This certificate can be used to document adequate training for use of a Hamilton instrument. Internal training programs and documentation need to be developed and documented by the end user laboratory.



Certificate for user training. Training certificates must be maintained by laboratory SOPs.

Written Policies - 21 CFR Part 11.10 (j)

The establishment of, and adherence to, written policies that hold individuals accountable and responsible for actions, initiated under their electronic signatures, in order to deter record and signature falsification. This is a laboratory procedure, separate from Hamilton instrumentation.

Document Controls/Audit Trails - 21 CFR Part 11.10 (k (1))

Adequate controls over the distribution of, access to, and use of documentation for system operation and maintenance.

This rule applies to the system documentation, which describes how a system operates and is maintained. System documentation includes, but is not limited to, operation manuals, help files, SOPs, access information, operating system manuals and privilege logs. This rule only pertains to documentation that can be changed by individuals within an organization. Hamilton instrumentation documentation such as operation manuals or help files can only be changed by Hamilton. Laboratories do not need to control these documents. Even though laboratories do not need a change control procedure for these documents, they are responsible for archiving and making them accessible to the appropriate users. Laboratories are also responsible for protecting their own documents regarding Hamilton instrumentation. Hamilton regulates the distribution of, access to and use of system documentation by ISO-9001 compliant procedures.

Hamilton Company Controlled Documents	
Software Manuals	
Hardware Manuals	
Technical Bulletins	
Service Bulletins	
Help Files	

These documents are controlled by Hamilton's change-control process. Laboratories do not need to maintain change documentation on these documents. However, the laboratory is responsible for archiving and making these documents available to the appropriate users.

Document Controls/Audit Trails - 21 CFR Part 11.10 (k (2))

Revision and change control procedures to maintain an audit trail that documents time-sequenced development and modification of systems documentation.

This rule is similar to the above in that it refers to the same documents mentioned above. Hamilton maintains an internal change control process for documentation by procedures that are consistent with the 21 CFR Part 11 guidelines. Laboratories are responsible for the change control process of their internal documents.

Terms and definitions

Agency:	Food and Drug Administration
Bilateral Identification:	A method of verifying an individual's identity based on a user id and password system.
Biometric Identification:	A method of verifying an individual's identity based on measurement of the individual's physical feature(s) or repeatable actions where those features and/or actions are both unique to that individual and measurable.
Checksum:	A system by which the authenticity of a file can be checked, based on the binary code. This protects files from being improperly modified. The checksum value is updated every time a file is legitimately saved. If the checksum doesn't match when the file is reopened, the file has been improperly modified and the file is blocked from use.
Closed System:	An environment in which system access is controlled by persons who are responsible for the content of electronic records that are on the system.
Electronic Records:	Any combination of text, graphics, data, audio, pictorial, or other information represented in digital form that is created, modified, maintained, archived, retrieved, or distributed by a computer system.
Electronic Signature:	A computer data compilation of any symbol or series of symbols executed, adopted, or authorized by an individual to be the legally binding equivalent of the individual's handwritten signature.
Method Files:	A method in Vector/VENUS software is defined by the .med, .stp, .lay, .sub, .res and .hsl files. These six files are necessary for running a method in Vector/VENUS software.
SOP:	Standard operating procedure
Trace Files:	The file generated by Vector/VENUS software for every runtime event. The trace file contains all the events of the run, date, time and user id information.
Validation:	Confirmation by examination and provision of objective evidence that the system specifications conform to user needs and intended uses, and that all requirements can be consistently fulfilled.

Vector/VENUS Software: Vector/VENUS software is the software designed for Hamilton robotic instrumentation control. The software provisions for 21 CFR Part 11 apply to all instruments controlled by the software including the Microlab STAR Line instruments.

E. Glossary

ADC

Anti-Droplet Control

Adjustment

Detailed positional setting for the hardware.

Air displacement tip

Commercial pipetting tip.

Aliquot

Aliquots are identical small volumes of liquid.

APE

Application engineering.

Autoload

Option and hardware assembly that enables automatic loading of the Microlab STAR Line. It consists of a loading head movable in y direction, which draws the carriers into the Microlab STAR Line and can read the barcodes on them.

Autoload tray

Hardware unit. The carriers can be placed on it and held outside the Microlab STAR Line. The loading tray is attached to the Microlab STAR Line, to support the automatic loading and unloading process.

Barcode Mask

The barcode mask defines the basic structure of a barcode. It is a pattern to which a barcode must conform. The assignment of a specific labware item can be done this way. The barcode mask can require a barcode to contain specific strings at fixed positions. It can contain wildcards, too.

Barcode Reader

Device for reading sample/plate barcodes. Part of the Autoload option.

Basic Microlab STAR Line

Basic parts of the Microlab STAR Line with pipetting arm and deck, to which the loading unit and the options can be added on.

BVS

Basic Vacuum System for automation of vacuum based filtration kits. Predecessor of the CVS.

Carrier

Unit for loading plates, tubes and tips on the Microlab STAR Line deck. Can be used manually or, if possible, by the Autoload option.

cLLD

Capacitive Liquid Level Detection.

CLT

MULTIFLEX CORE-LID TOOL; Suction cup that can be picked up by a CO-RE channel to move around lids.

Container

A container defines a tube, vessel or a single well of a plate.

Container identification

Barcode for the identification of a container. Serves for unambiguous identification of the vessel, e.g. a sample test tube.

Continuous loading

Refers to the loading of elements that can be manipulated onto the Microlab STAR Line after processing has been started.

CVS

Crystal Vacuum System for automation of vacuum based filtration kits.

Deck

The work surface of the Microlab STAR Line. It presents at the same time the greatest possible area, cf. *Work Area*. The placing of the carriers on it is defined by the tracks, as long as they are in the operating range of the pipetting area.

Deck Layout

A collection of Labware placed on a Deck.

Dispense

To distribute quantities of liquid from a pipetting device.

Docking station

The long bar at the back of the Microlab STAR Line Instrument for guiding the cables and tubing of accessories, as Wash Stations, TCC, etc.

DWP (Deep Well Plate)

Micro titer plate with large well volume, used when higher volumes of sample need to be stored or collected, e. g. chemistry libraries, for cell culture or filtration applications. In general we assume a plate with 96 wells (8 x 12) 9 mm wide, standard SBS format. There are also DWPs with higher or lower well number.

Firmware

Lower Level program code that is carried out on the processors of the Microlab STAR Line Instrument.

Front Cover

Protective covering for the Microlab STAR Line Instrument, featuring a hinged front window made of transparent acrylic glass. With this option and assembly, the work surface of the Microlab STAR Line is covered in such a way that it is shielded from user intervention and other outside influences (such as dust). At the same time, it protects the user from the movements of the Microlab STAR Line.

Good Laboratory Practices

Also written as GLP, are set of appropriate laboratory behaviors which should be observed.

Hardware error

Type of error that is caused by a technical problem of the hardware.

HHS

Hamilton Heater Shaker.

HSB

Heater Shaker Box.

HSL

Hamilton Standard Language.

Instrument

Hardware of the Microlab STAR Line (mechanics, electronics, and firmware).

Instrument commands

The commands made available by the firmware for controlling the Microlab STAR Line.

Labware

Refers to movable items to be placed on the Microlab STAR Line deck, such as carriers, containers, or racks.

LIMS

Higher-level data processing system, generally known as Laboratory Information Management System, also LMS.

Liquid

Includes all kinds of liquids, among which are included reagents, controls, standards, wash fluids.

LLD (Liquid Level Detection)

Positive detection of liquid which may be achieved either by pressure, or capacitive signal detection and transfer.

Loading, unloading

The process by which plate, tube and tip carriers are brought on and off the Microlab STAR Line deck. This can happen automatically by means of the Autoload Option, or manually.

MAD

Monitored Air Displacement: aspiration monitoring feature. During the aspiration process, the pressure within the pipetting channel is measured in real time.

Method

The method contains all instructions as to how the content of the source container is to be processed.

MFX

Multiflex: A multiple-use carrier base concept.

Microlab STAR Line VENUS Software

Software to run the Microlab STAR Line.

MPH

Multiprobe head, a pipetting device consisting of an array of pipetting channels (96 or 384).

MTP (Microtiter Plate)

In general we assume a plate with 96 wells (8 x 12) 9 mm wide, standard SBS format.

There are also plates with 384 wells (16 x 24 / 4.5 mm), or others with a different size.

NTR

Nestable tip rack

Pause

Interruption of processing. The current processing steps are completed.

Orbit / amplitude (of Hamilton Heater Shaker)

The orbit (rotation distance) is defined as peak to peak distance in one direction, e.g. distance between extreme positions in the Y-Direction of the plate measured in millimeters [mm].

The amplitude is defined as the distance from the center of the shaking movement and it is 50% of the peak to peak distance.

Pipetting

Transfer of liquids, usually a defined volume, from one container to another.

Pipetting arm

Assembly equipped with the pipetting tool and/or plate handler, as well as the common X-drive.

Pipetting channel

Hardware including the function of picking up a tip, aspirating, dispensing, tip eject, Liquid Level Detection and the Y/Z- movements.

Pipetting module

Firmware (-processor-program) which controls a pipetting channel, in which category are included the Y and Z pipetting movement, and the LLD.

pLLD

Pressure-based Liquid Level Detection.

Pooling

Pipetting of different liquids in one well:

1,2,3 to n, and n to 1,2,3,....

Processing step

Defines what must be carried out on the Microlab STAR Line Instrument, as well as the location where it must be carried out and possible interaction with other system components or labware. The action is defined in accordance with the methods, the loading, and the tasks.

Pump station

Part of the needle/tip wash station. Its function is to pump wash liquid to and from the wash station.

Rack

Grouping of containers, as DWP, MTP, etc.

Rack identification

Barcode for Rack identification.

Random access

Means that every channel can access any place on the work area.

Run

Execution of the processing steps defined in the relevant method with the aim of processing one or more liquids and containers (e.g. MTP). The run is a series of timed commands, in order to carry out processing on the Microlab STAR Line according to the processing plan. The run can be interrupted to load more elements. Then processing goes on according to a newly calculated processing plan, with the run being started again. Loading is not a part of the run.

Run abort

Cancelled run by the user or by the Microlab STAR Line.

Run visualization

Visualization of the current run, reporting the status of the Microlab STAR Line.

Sample

Refers to a liquid in an unambiguously identified container which is to be processed.

SBS format

Standard format for micro titer plates, defined by the Society for Biomolecular Screening.

Stacker

Storage unit for racks

T

Abbreviation used for “track”. The ML STAR Line instruments have equal partitions of 22.5mm, equivalent to 1-T. Labware carriers are adapted to those partitions.

TADM

Total Aspiration and Dispensing Monitoring: The pressure inside each individual pipetting channel is monitored, during aspiration and dispensing.

TCC

Temperature controlled carrier

Tip

Disposable tip for pipetting

Tip rack

Frame that hold the tips.

Tip waste

Container for ejecting tips.

Touch-off

Type of dispensing whereby the tip or needle approaches the bottom of the empty container so close as to allow the dispensed droplet to have simultaneous contact with the tip or needle and the container bottom.

Trace

Record of the status during processing.

Tube

A container for liquid, usually having a circular cross-section, and a cylindrical longitudinal section.

User

User of the software. Access rights for different types of users can be defined, such as operators, laboratory managers, etc.

Verification kit

Balance, liquid, disposable tips to verify the function (volume check) of the Microlab STAR Line pipetting heads.

Waste container

A device on the Microlab STAR Line deck to collect used disposable tips.

Well

The individual container of a MTP or DWP.

Well type

Geometrical shape of the well, such as U, V, or flat shape.

Wick side of container

Type of dispensing whereby the tip or the needle touches the side of a container and thus releases the droplet. This function is not available for Microlab STAR Line instruments.

Work area

The area on the Microlab STAR Line to which access is provided during processing. Elements to be pipetted or handled can be placed in this area.

Worklist

Information sent from outside the system, as to what method(s) is (are) to be executed on the Microlab STAR Line, and with what liquid.